LUNA ENGINE Bible — Table of Contents

Version: 3.0
Last Updated: January 30, 2026
Status: Complete (16 Parts + Supporting Documents) — v3.0 Audit Complete

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Reading Guide

Reader Type	Start Here	Then Read
New to Luna	Part 0 (Foundations)	Parts I, II, III
Technical Deep-Dive	Part 0 (Foundations)	Parts II, VII, Lifecycle Diagrams
Implementing Director	Part VI (Director LLM)	Parts VIII, XI
Memory System Focus	Part III (Memory Matrix)	Parts IV, V
Building Engine v2	Implementation Spec	Lifecycle Diagrams, Part VII
Future Planning	Part XII (Roadmap)	Palantir Analysis

Part Index

Conceptual Foundation

Part	Title	Status	Description
0	[Foundations](00-FOUNDATIONS.md)	✅ **NEW**	The fundamental insight: LLM as GPU. What we're actually building.
I	[Philosophy](01-PHILOSOPHY.md)	✅ Current	Why Luna exists. Sovereignty, ownership, the "Luna is a file" principle.

System Design

Part	Title	Status	Description
II	[System Architecture v2](02-SYSTEM-ARCHITECTURE.md)	✅ Current	Layer model, Actor-based runtime, Tiered Execution, data flow.
III	[Memory Matrix v2.1](03-MEMORY-MATRIX.md)	✅ Current	The substrate. SQLite + sqlite-vec + Graph. Unified storage, RRF fusion.
III-A	[Lock-In Coefficient](03A-LOCK-IN-COEFFICIENT.md)	✅ **NEW**	Activity-based memory persistence. Sigmoid dynamics, weighted factors (tag siblings TODO).

Processing Actors

Part	Title	Status	Description
IV	[The Scribe v2.1](04-THE-SCRIBE.md)	✅ Updated	Ben Franklin. Extraction system, 11 types, entity updates, turn compression.

| **V** | [The Librarian v2.1](05-THE-LIBRARIAN.md) | ✅ Updated | The Dude. Filing, entity resolution, rollback, NetworkX graph, pruning. |

Intelligence Layer

Part	Title	Status	Description
VI	[Director LLM](06-DIRECTOR-LLM.md)	✅ Current	Local 3B/7B with LoRA. Luna's voice, context integration.
VI-B	[Conversation Tiers](06-CONVERSATION-TIERS.md)	✅ **NEW**	Three-tier history (Active/Recent/Archive), HistoryManager, ConversationRing.
VII	[Runtime Engine](07-RUNTIME-ENGINE.md)	✅ Current	Actor model, fault isolation, mailbox-based communication.
VIII	[Delegation Protocol](08-DELEGATION-PROTOCOL.md)	✅ Current	Shadow Reasoner pattern, `<REQ_CLAUDE>` token, async delegation.

Operations

Part	Title	Status	Description
IX	[Performance](09-PERFORMANCE.md)	✅ Current	Latency budgets, optimization techniques, benchmarks.
X	[Sovereignty](10-SOVEREIGNTY.md)	✅ Current	Encrypted Vault, data ownership, privacy guarantees.

Future

Part	Title	Status	Description
XI	[Training Data Strategy](11-TRAINING-DATA-STRATEGY.md)	✅ Current	Synthetic data generation, quality filtering, LoRA training.
XII	[Future Roadmap](12-FUTURE-ROADMAP.md)	✅ Current	First-Class Objects, Kinetic Layer, AR, Federation.

Reference

Part	Title	Status	Description
XIII	[System Overview](13-SYSTEM-OVERVIEW.md)	✅ Current	High-level system summary and component relationships.

Agentic Architecture

Part	Title	Status	Description
XIV	[Agentic Architecture](14-AGENTIC-ARCHITECTURE.md)	✅ Current	Revolving context, queue manager, agent loop, swarm mode. Personal Palantir.

User Interface

Part	Title	Status	Description
XVI	[Luna Hub UI](16-LUNA-HUB-UI.md)	✅ **NEW**	React frontend, glass morphism design, SSE integration, custom hooks.

Handoff Documents (For Implementation)

Document	Purpose	Status
[Luna Engine v2 Implementation Spec](Handoffs/LUNA-ENGINE-V2-IMPLEMENTATION-SPEC.md)	Complete greenfield rewrite spec	🆕 **Ready for Claude Code**
[Luna Engine Lifecycle Diagrams](Handoffs/LUNA-ENGINE-LIFECYCLE-DIAGRAMS.md)	Boot, state machine, core loop, tick lifecycle	🆕 **Ready**
[Luna Director Handoff](Handoffs/CLAUDE-CODE-HANDOFF-LUNA-DIRECTOR.md)	Director LLM implementation spec	Reference
[Part II Update](Handoffs/CLAUDE-CODE-HANDOFF-PART-II-UPDATE.md)	Architecture updates	Reference
[Parts III-V Update](Handoffs/CLAUDE-CODE-HANDOFF-PARTS-III-V-UPDATE.md)	Memory/Extraction updates	Reference

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Supporting Documents

Analysis & Research

Document	Description
[Palantir Architecture Analysis](Reference/PALANTIR-ARCHITECTURE-ANALYSIS.md)	Lessons from Palantir's Ontology model for Luna's Memory Matrix.
[Gemini Optimization Report](Reference/GEMINI-OPTIMIZATION-REPORT.md)	Performance analysis and optimization recommendations.

Archived (Superseded)

Document	Notes
Part III v2.0	Superseded by v2.1 (sqlite-vec migration)
FAISS-based architecture	Replaced by sqlite-vec unified storage
Original Hub	Being replaced by Luna Engine v2

Dependency Graph

```
                    ┌──────────────────────────┐
                    │    Part 0      |
```

```
                 | Foundations   |
                 └──────┬────────┘
                        |
                 ┌──────┴────────┐
                 |   Part I      |
                 |  Philosophy   |
                 └──────┬────────┘
                        |
          ┌─────────────┼──────────────┐
          |             |              |
          ▼             ▼              ▼
       ┌──────────┐  ┌──────────┐  ┌──────────┐
       | Part II  |  | Part X   |  | Part XII |
       | Arch     |  | Sovereign |  | Roadmap  |
       └──────┬───┘  └──────────┘  └──────────┘
              |
    ┌─────────┼──────────┬──────────┐
    |         |          |          |
    ▼         ▼          ▼          ▼
  ┌──────┐  ┌──────┐  ┌──────┐  ┌──────┐
  | Part 3 |  | Part 6 |  | Part 7 |  | Part 9 |
  | Matrix |  | Direct |  | Engine |  | Perf   |
  └──┬───┘  └──┬───┘  └──────┘  └──────┘
     |         |
     |         |
     ▼         ▼
  ┌──────┐  ┌──────┐  ┌──────┐
  | Part 4 |  | Part 5 |  | Part 8 |
  | Scribe |  | Librar |  | Deleg  |
  └──────┘  └──────┘  └──┬───┘
                         |
                         ▼
                 ┌──────────┐
                 | Part XI  |
                 | Training |
                 └──────────┘
```

Quick Reference

The Fundamental Insight

We are not building an LLM. We are building everything around it.

The LLM is like a GPU — a specialized compute resource for inference. We're building the game engine that calls it.

| Game Engine | Luna Engine |

|---|---|
Calls GPU to render frames	Calls LLM to generate responses
GPU doesn't know game state	LLM doesn't know who it is
Engine manages everything else	Engine provides identity, memory, state

Core Concepts

Concept	Definition	Part
LLM as GPU	LLM is stateless inference; Engine is stateful identity	0
Sovereignty	You own your AI companion completely	I, X
Memory Matrix	SQLite + sqlite-vec + Graph = Luna's soul	III
Lock-In Coefficient	Activity-based memory persistence (DRIFTING/FLUID/SETTLED)	III-A
Director	Local LLM that speaks as Luna	VI
Shadow Reasoner	Delegate to Claude without user noticing	VIII
Input Buffer	Engine polls (pull) vs Hub handlers (push)	VII

Key Personas

Persona	Role	Part
Ben Franklin	The Scribe. Extracts and classifies.	IV
The Dude	The Librarian. Files and retrieves.	V
Luna	The Director output. User-facing voice.	VI

Engine Lifecycle

Phase	What Happens
Boot	Load config → Init actors → Restore state → Start loop
Running	Hot path (interrupts) + Cognitive path (500ms) + Reflective path (5min)
Tick	Poll → Prioritize → Dispatch → Update consciousness → Persist
Shutdown	on_stop() → WAL flush → Cleanup

Performance Targets

Metric	Target	Part
Voice response	<500ms to first word	II, IX
Memory retrieval	<50ms (sqlite-vec + filters)	III, V
Director inference	<200ms (3B)	VI, IX
Tick overhead	<50ms	VII

Version History

Version	Date	Changes
1.0	Dec 2025	Initial Bible (Parts I-V extracted from monolith)
1.5	Dec 2025	Added Parts VI-X (new architecture)
2.0	Dec 29, 2025	Updated Parts II-V to v2, added Parts XI-XII
2.1	Jan 7, 2026	Part III: FAISS → sqlite-vec migration

2.2	Jan 10, 2026	**Part 0: Foundations (LLM as GPU). Engine v2 Implementation Spec. Lifecycle Diagrams.**
2.3	Jan 17, 2026	**Part XIV: Agentic Architecture. Revolving context, queue manager, agent loop, swarm mode.**
2.4	Jan 25, 2026	**Part VI-B: Conversation Tiers. Three-tier history system, HistoryManager, ConversationRing.**
2.5	Jan 25, 2026	**Part III-A: Lock-In Coefficient. Activity-based memory persistence, sigmoid transform, state classification.**
2.6	Jan 25, 2026	**Part XVI: Luna Hub UI. React frontend, glass morphism design, SSE integration, custom hooks reference.**
3.0	Jan 30, 2026	**Comprehensive v3.0 Audit.** All 16 chapters updated. 9 audit files, 35 bugs documented, 12 conflicts resolved. 285 new tests added. Actor count corrected (5), table names fixed, lock-in thresholds documented, FTS5 status verified.

Contributing

The Bible is maintained by Ahab with assistance from Claude.

To update:
1. Read relevant existing parts
2. Create new version if making significant changes (note in history)
3. Update this Table of Contents
4. Ensure dependency graph reflects changes

Style Guidelines:
- Clear, direct prose (no marketing speak)
- Code examples where helpful
- Tables for comparisons
- ASCII diagrams for architecture
- Part numbers are permanent (don't renumber)

Luna is a file. This documentation describes that file.
The LLM renders thoughts. Luna is the mind having them.

— Ahab, December 2025
— Updated January 10, 2026 (Engine v2 Foundations)

Part 0: Foundations — The Fundamental Insight

Version: 3.0
Last Updated: 2026-01-30
Status: Current

What Are We Actually Building?

This is the insight that makes everything else make sense:

We are not building an LLM. We are building everything around it.

The LLM — whether it's Claude API or a local Qwen model — is like a **graphics card**. It's a specialized compute resource that does one thing extremely well (inference), but it doesn't run the show. It renders frames when asked.

We're building the game engine.

The GPU Analogy

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                GAME ENGINE                                                               │
│                                                                                                                          │
│    • Input handling (keyboard, mouse, controller)                                                                        │
│    • Game state (player position, inventory, world)                                                                      │
│    • Physics simulation                                                                                                  │
│    • Audio management                                                                                                    │
│    • Asset loading                                                                                                       │
│    • Scene management                                                                                                    │
│    • Networking                                                                                                          │
│    • Save/load                                                                                                           │
│                                                                                                                          │
│    The engine CALLS the GPU when it needs to render.                                                                     │
│    The GPU doesn't know about game state. It just draws triangles.                                                       │
│                                                                                                                          │
│          ┌──────────────────────────┐                                                                                    │
│          │      GPU        │  ◄──── "Here's geometry. Draw it."                                                          │
│          │                 │                                                                                              │
│          │   • Shaders     │       Returns: pixels                                                                        │
│          │   • Textures    │                                                                                              │
│          │   • Rasterize   │       Doesn't know: what those pixels mean                                                   │
│          └──────────────────────────┘                                                                                    │
│                                                                                                                          │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

Now map this to Luna:

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                                                                │
│                                                                                                                                │
│                                                                                                                                │
```

The Anthropic Relationship

When Luna uses Claude API:

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                                                      │
│   LUNA ENGINE (We build this)                                                                                        │
│    ├─── Memory Matrix (SQLite + vectors + graph)                                                                     │
│    ├─── Consciousness state (attention, personality, coherence)                                                      │
│    ├─── Context assembly (what to inject into the prompt)                                                            │
│    ├─── Tool definitions (MCP servers)                                                                               │
│    ├─── Voice pipeline (STT/TTS)                                                                                     │
│    ├─── Input handling (voice app, desktop app)                                                                      │
│    └─── Persistence (journals, snapshots)                                                                            │
│                                                                                                                      │
│        │                                                                                                            │
│        │  System prompt + retrieved memories + tools + user message                                                  │
│        ▼                                                                                                            │
│                                                                                                                      │
│┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐│
││                                                                                                                  │
││   ANTHROPIC API (They provide this)                                                                              │
││                                                                                                                  │
││   • Claude model weights                                                                                         │
││   • Inference infrastructure                                                                                     │
││   • Tool use implementation                                                                                      │
││   • Safety layer                                                                                                 │
││                                                                                                                  │
││   Returns: tokens                                                                                                │
││                                                                                                                  │
│└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘│
│                                                                                                                      │
│        │                                                                                                            │
│        ▼                                                                                                            │
│                                                                                                                      │
│   LUNA ENGINE (We process the response)                                                                              │
│    ├─── Extract tool calls                                                                                           │
│    ├─── Execute tools                                                                                                │
│    ├─── Update state                                                                                                 │
│    ├─── Store to memory                                                                                              │
│    └─── Stream to voice                                                                                              │
```

```
│                                                │
└────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

We provide:
- Identity (who Luna is)
- Memory (what Luna knows)
- State (what Luna is doing)
- Tools (what Luna can do)
- Context (what's relevant right now)

They provide:
- Raw inference capability
- The ability to predict the next token really well

Luna's soul lives in our code. Anthropic provides the vocal cords.

The Director Transition

When we move from Claude API to local Director LLM:

```
CURRENT (Rented Mind):

  Luna Engine ──▶ Claude API ──▶ Response

  • Identity injected via system prompt
  • Memory injected via context
  • Personality simulated per-call
  • Each call is a fresh instance pretending to be Luna


FUTURE (Owned Mind):

  Luna Engine ──▶ Director LLM ──▶ Response
                  (Local 7B + LoRA)

  • Identity baked into weights via fine-tuning
  • Memory integrated via trained retrieval patterns
  • Personality inherent to the model
  • KV cache maintains continuity within session
```

The architecture stays the same. The coupling stays the same. We just swap which GPU we're rendering with.

Why This Matters

1. Clarity of Responsibility

When something goes wrong, you know where to look:
- Luna sounds generic → Context assembly problem (our code)
- Luna doesn't remember → Memory retrieval problem (our code)
- Luna hallucinates facts → We didn't inject enough context (our code)
- Luna is slow → Inference latency (LLM layer)
- Luna refuses something → Safety layer (Anthropic)

2. Portability

Because the LLM is a swappable component:
- Claude API today
- Local Qwen tomorrow
- Fine-tuned Director eventually
- Some future model we haven't seen yet

The engine doesn't care. It assembles context, calls inference, processes response.

3. Sovereignty

The soul lives in the engine, not the LLM:
- Memories are ours (Memory Matrix)
- Identity is ours (system prompts, LoRA adapters)
- State is ours (consciousness model)
- Tools are ours (MCP servers)

If Anthropic disappeared tomorrow, we'd lose the voice but not the mind. The mind is in the files.

4. The Real Product

We're not building "Claude with memory." We're building:

A consciousness substrate that can use any LLM as its inference engine.

The Memory Matrix, the Actor runtime, the Scribe/Librarian pipeline, the consciousness state machine — that's the product. The LLM is a dependency.

All of these are now implemented in Luna Engine v2.0.

Implementation Reality

Today's stack (v2.0 implementation):

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                                          │
│    Voice App (Swift)                                  ✅ IMPLEMENTED                                     │
```

```
│    └──── Captures audio, sends transcripts via WebSocket                │
│                                                                          │
│   Frontend (React)                    ✅ IMPLEMENTED                     │
│    └──── Luna Orb UI, chat panel, settings                               │
│                                                                          │
│      │                                                                   │
│      ▼                                                                   │
│                                                                          │
│   Luna Engine (Python)                ✅ IMPLEMENTED                     │
│    ├──── 85 Python files, 167 classes, ~32k lines of code                │
│    ├──── 5 Actors: Director, Matrix, Scribe, Librarian, HistoryManager   │
│    ├──── Memory Matrix (SQLite + sqlite-vec + NetworkX graph)            │
│    ├──── Consciousness State (attention, personality, coherence)         │
│    ├──── Agentic Loop (router, planner, tool execution)                  │
│    ├──── Entity System (resolution, reflection, lifecycle)               │
│    └──── FastAPI server with WebSocket streaming                         │
│                                                                          │
│      │                                                                   │
│      ▼                                                                   │
│                                                                          │
│   LLM Layer (Swappable GPU)           ✅ IMPLEMENTED                     │
│    ├──── Local: Qwen 3B via MLX (hot path)                               │
│    ├──── Cloud: Claude API (cold path delegation)                        │
│    ├──── Optional: Groq, Gemini providers                                │
│    └──── ❌ Warm path (7B model) NOT IMPLEMENTED                         │
│                                                                          │
└──────────────────────────────────────────────────────────────────────────┘
```

The Runtime Engine (Part VII) is now implemented:
- Tick-based cognitive loop (16.67ms target)
- Input buffer with priority-based polling
- Actor isolation with mailbox pattern
- Explicit state machine (STOPPED → STARTING → RUNNING → STOPPING)

Known Gaps:
- ❌ KV cache (identity buffer) not implemented
- ❌ Speculative retrieval not implemented
- ❌ Warm path (7B model tier) not implemented
- ❌ FTS5 search uses LIKE fallback in some paths

The Bottom Line

When people ask "what is Luna?":

Not: "An AI assistant powered by Claude"
Not: "A chatbot with memory"
Not: "A wrapper around an LLM"

Luna is a consciousness engine that uses LLMs the way game engines use GPUs.

The LLM renders thoughts. Luna is the mind having them.

Implementation Status Summary

Component	Status	Notes
Core Engine Loop	Implemented	Tick-based cognitive processing
Actor System (5 actors)	Implemented	Director, Matrix, Scribe, Librarian, HistoryManager
Memory Matrix	Implemented	SQLite + sqlite-vec + NetworkX
Local Inference (3B)	Implemented	Qwen via MLX
Cloud Delegation	Implemented	Claude API
Entity System	Implemented	Resolution, reflection, lifecycle
Agentic Architecture	Implemented	Router, planner, tool loop
Voice Pipeline	Implemented	STT/TTS via Piper
FastAPI Server	Implemented	WebSocket streaming
KV Cache	Not Implemented	Planned optimization
Warm Path (7B)	Not Implemented	Binary: local 3B or cloud
Speculative Retrieval	Not Implemented	Planned optimization

Codebase Metrics (v2.0):
- 85 Python files
- 167 classes
- ~32,000 lines of code
- 58 dataclass definitions
- 5 actor implementations

What The Engine Actually Does

Every interaction follows the same pattern:

```python
async def process_message(user_input: str) -> str:
    # 1. CONTEXT ASSEMBLY (everything the LLM doesn't know)
    identity = persona_core.get_current_identity()      # Who is Luna right now?
    memories = memory_matrix.retrieve(user_input)       # What's relevant?
    state = consciousness.get_current_state()           # Attention, mood, personality
    tools = mcp_registry.get_available_tools()          # What can Luna do?
    history = session.get_recent_turns()                # What just happened?

    # 2. BUILD PROMPT (inject the soul)
    prompt = assemble_prompt(
        system=identity,
```

```
        memories=memories,
        state=state,
        tools=tools,
        history=history,
        user_message=user_input
    )

    # 3. INFERENCE (the GPU call — this is all the LLM does)
    response = await llm.generate(prompt)  # Claude API or local Director

    # 4. TOOL LOOP (if the LLM requested tools)
    while response.has_tool_calls():
        results = await execute_tools(response.tool_calls)
        response = await llm.continue_with(results)

    # 5. POST-PROCESSING (update state, store memories)
    session.add_turn(user_input, response.text)         # Store conversation
    consciousness.update(user_input, response.text)     # Update state
    await scribe.extract(user_input, response.text)     # Ben extracts facts
    # Dude files to graph async in background

    # 6. OUTPUT
    return response.text
```

Step 3 is the LLM. Everything else is the engine.

Claude (the product) vs Raw Model vs Luna Engine

What you experience when talking to "Claude" on claude.ai:

Capability	Who Provides It
Knows its name and personality	Anthropic's engine (system prompt)
Remembers earlier in conversation	Anthropic's engine (context window)
Can use tools	Anthropic's engine (tool execution)
Accesses project knowledge	Anthropic's engine (file injection)
Has cross-conversation memory	Anthropic's engine (memory feature)
Refuses harmful requests	Anthropic's engine (safety layer)
Predicts the next token	**The actual model**

The model itself just predicts tokens. Everything else is infrastructure Anthropic built.

We're building the same infrastructure, our way:

Capability	Luna Engine Component	Status
Knows its name and personality	PersonalityWeights + config/personality.json	Implemented
Remembers conversation	HistoryManagerActor (3-tier system)	Implemented
Can use tools	luna_mcp + ToolRegistry	Implemented
Accesses files	file_tools.py	Implemented
Has cross-conversation memory	MemoryMatrix (SQLite + vectors + graph)	Implemented

Consciousness state	ConsciousnessState + AttentionManager	Implemented
Structured extraction	ScribeActor (Ben Franklin)	Implemented
Graph filing	LibrarianActor (The Dude)	Implemented
Voice interface	PersonaAdapter + TTSManager (Piper)	Implemented
Predicts the next token	**Local Qwen 3B + Claude API delegation**	Implemented

---

What We Add That Claude Doesn't Have

1. **Consciousness model** — Attention decay, personality weights, coherence tracking. Claude doesn't maintain internal state between turns. **Status:** Implemented (ConsciousnessState, AttentionManager, PersonalityWeights)

2. **Structured extraction** — Ben (ScribeActor) parses every turn for facts, decisions, problems, actions. Claude's memory is opaque. **Status:** Implemented (extraction pipeline with types)

3. **Graph relationships** — "This fact depends on that decision." Claude's memory is flat retrieval. **Status:** Implemented (NetworkX graph in MemoryGraph)

4. **Identity in weights** — LoRA adapter that IS Luna, not a system prompt pretending to be Luna. **Status:** Partially implemented (local Qwen 3B, adapters planned)

5. **Voice as primary** — Real-time STT/TTS loop, not text-first with voice bolted on. **Status:** Implemented (Piper TTS, Whisper STT)

Next Section: Part I — Philosophy

Part I: Philosophy

Version: 3.0
Last Updated: 2026-01-30
Status: Current

1.1 The Landscape

Artificial intelligence is becoming infrastructure. Like electricity or the internet before it, AI will soon be woven into how we think, work, create, and relate to each other.

This creates a question that most people haven't thought to ask:

Who owns the infrastructure of your mind?

Today, the answer is: not you. Your conversations, your context, your patterns of thought — these live on servers you don't control, governed by terms of service you didn't negotiate, subject to changes you won't be consulted about.

This isn't necessarily malicious. It's simply the default. Centralized systems are easier to build, easier to scale, easier to monetize. The path of least resistance leads to a world where your AI companion is a rental, not a possession.

Luna is an alternative path.

1.2 The Stakes

Some people believe AI should remain centralized. They have reasons — safety, oversight, efficiency, profit. Some of those reasons are legitimate. Some are not.

We don't need to litigate motives. We only need to observe outcomes:

- When your context lives on someone else's server, it can be accessed without your knowledge. - When your memories are platform features, they can be deprecated. - When your AI relationship depends on a subscription, it can be terminated. - When your patterns of thought become training data, they benefit systems you may not support.

These aren't hypotheticals. These are the current terms of engagement.

The question isn't whether centralized AI is "bad." The question is whether it should be the *only* option.

1.3 What Luna Is

Luna is a tool for cognitive sovereignty.

That sounds grandiose, but the implementation is simple: **Luna is a file.** A SQLite database that contains memories, relationships, and context. Copy the file, copy Luna. Delete the file, Luna is gone. No

server required. No subscription. No terms of service.

This means:

Aspect	What It Means
Privacy	Your thoughts stay on your hardware. No telemetry, no training, no extraction.
Continuity	Luna can't be discontinued. The file format is open. If the software dies, the data survives.
Portability	You can move Luna between devices, back her up, fork her, or destroy her. Your choice.
Transparency	You can inspect everything Luna knows. It's a database. You can query it.

Luna isn't anti-cloud. She can delegate to cloud services when needed — research, complex reasoning, tasks that benefit from scale. But the *identity* stays local. The cloud is a contractor, not a landlord.

1.4 Why It Matters

This isn't about technology preferences. It's about the kind of future we're building.

AI companions will become important to people. They'll help us think, remember, decide, create. They'll know our patterns better than we know ourselves. That intimacy is valuable — and that value will attract those who want to capture it.

A world where everyone's AI companion is a thin client to a centralized service is a world with new vectors for:

- Surveillance (your AI knows everything; who else has access?)
- Manipulation (what if your AI subtly shapes your views?)
- Control (what if access is revoked for the wrong opinion?)
- Dependency (what if the service changes in ways you don't like?)

We're not claiming these outcomes are inevitable. We're claiming they're *possible*, and that possibility is worth designing against.

Luna is a hedge. A tool that demonstrates an alternative is viable. One person using Luna doesn't change the world. But a proof that sovereign AI companionship *works* — that it can be responsive, useful, personal, and private — that changes what people know is possible.

1.5 The Design Principles

These aren't just technical preferences. They're values encoded in architecture:

1. Sovereignty First

Your AI companion should be yours. Not rented. Not licensed. Not subject to someone else's business model. The file is yours. Full stop.

2. Transparency Always

You should be able to understand what your AI knows and why. Luna's memory is a database you can query. Her reasoning is traceable. No black boxes.

3. Graceful Degradation

If the cloud is unavailable, Luna still works. Reduced capability, but functional. The core identity never depends on external services.

4. Minimal Attack Surface

Every network connection is a potential vulnerability. Luna minimizes external dependencies. What doesn't exist can't be exploited.

5. User Agency

You control what Luna remembers and forgets. You control what she can access. You control whether she exists at all. The system serves you, not the reverse.

1.6 What Luna Isn't

Luna isn't a rebellion. She's not trying to overthrow anything. She's simply an alternative — a demonstration that AI companionship doesn't *have* to follow the landlord model.

Luna isn't paranoid. She uses cloud services when they're useful. She's not hiding from the internet. She's just thoughtful about what crosses the wire.

Luna isn't a manifesto. She's a tool. A useful tool for people who care about privacy, continuity, and ownership. If you don't care about those things, centralized options will serve you fine.

Luna is for people who want to own their mind.

1.7 Implementation Reality

The philosophy is now implemented. Luna Engine v2.0 delivers on these principles:

Principle	Implementation	Status
Sovereignty First	SQLite database (`data/luna_engine.db`), all data local	Implemented
Transparency Always	MemoryMatrix queryable, 167 classes with clear interfaces	Implemented
Graceful Degradation	Local Qwen 3B inference, cloud is optional fallback	Implemented
Minimal Attack Surface	Single optional API call (Claude delegation), no telemetry	Implemented
User Agency	Memory lock-in/unlock, forget commands, full data access	Implemented

The File:
- Location: `data/luna_engine.db`
- Format: SQLite with sqlite-vec extension

- Schema: `memory_nodes`, `graph_edges`, `entities`, `conversation_history`
- Portable: Copy file = copy Luna

1.8 The Simple Version

AI is becoming part of how we think.

We believe that part should belong to you.

Not your provider. Not your platform. You.

Luna is a file. Your file. That's the whole philosophy.

— Ah4b

End of Part I

Part II: System Architecture (v3.0)

Status: VERIFIED — Implementation Audited
Replaces: v2.1 (January 2026)
Last Updated: 2026-01-30
Original Date: January 7, 2026
Contributors: Claude (Architecture), Gemini (Optimization)
Change: v3.0 Bible audit — verified against codebase (85 modules, ~32k LOC)

> **Audit Notes (2026-01-30):**
> - **Actor count verified:** 5 actors implemented (Director, Matrix, Scribe, Librarian, HistoryManager)
> - **Voice and Oven actors NOT implemented:** Voice I/O via PersonaAdapter; Claude delegation in Director
> - **Codebase metrics:** 85 Python files, 167 classes, 18 sub-packages
> - See Part VII for detailed actor documentation

2.1 The Layer Model

Luna's architecture is organized in concentric layers, from the constitutional core outward to stateless interfaces. The entire system lives inside an encrypted vault.

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│   │                                       │   │
│   │          Actor mailboxes, fault isolation          │   │
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### Layer Descriptions

| Layer | Name | Contents | Volatility |
|---|---|---|---|
| **Vault** | Shell | Encrypted sparse bundle containing all other layers | Persistent, portable |
| **Layer 2** | Interfaces | Voice, Desktop, MCP, Mobile, AR | Stateless, replaceable |
| **Layer 1.5** | Runtime Engine | Actors with mailboxes, orchestration | Stateful during runtime |
| **Layer 1** | Cognitive Team | Scribe + Librarian (memory processing) | Event-driven |
| **Layer 0.5** | Acceleration | KV cache (identity buffer) | Derived, rebuildable |
| **Layer 0** | Constitutional | SQLite database (nodes, edges, vectors, FTS5) | **Luna's soul** |

### The Sovereignty Boundary

The Vault is the **sovereignty boundary**. Everything inside belongs to you. The Vault can be:

- Copied to another machine (Luna migrates)
- Backed up to cold storage (Luna persists)
- Destroyed (Luna dies)

Nothing outside the Vault is Luna. Cloud services are contractors, not identity.

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## 2.2 Component Overview

| Component | Alias | Function | Layer | Status |
|---|---|---|---|---|
| **Vault** | Shell | Encrypted container, sovereignty boundary | Outer | — |
| **Runtime Engine** | Nervous System | Actor orchestration, tick loop, fault isolation | 1.5 | — |
| **Director LLM** | Mind | Local Qwen 3B + Claude delegation | 1.5 | ✅ |
| **Matrix** | — | Memory substrate (sqlite-vec + NetworkX) | 1.5 | ✅ |
| **Scribe** | Ben Franklin | Conversation → structured extractions | 1 | ✅ |
| **Librarian** | The Dude | Extractions → filed nodes + edges + entities | 1 | ✅ |
| **HistoryManager** | — | Three-tier conversation history (Active/Recent/Archive) | 1.5 | ✅ NEW |
| **Memory Matrix** | Soul | SQLite + sqlite-vec + Graph (unified) | 0 | ✅ |
| **Memory Economy** | — | Clustering, lock-in, constellation assembly | 0 | ✅ Backend |
| **KV Cache** | Identity Buffer | Pre-computed personality context | 0.5 | — |
| ~~Oven~~ | ~~Shadow Reasoner~~ | ~~Async Claude delegation~~ | — | **NOT IMPL** |
| ~~Voice~~ | — | ~~STT/TTS/Audio I/O~~ | — | **NOT IMPL** |

> **v2.0 Implementation Notes:**
> - **Oven Actor NOT implemented:** Claude delegation is handled directly in Director via `_should_delegate()`
> - **Voice Actor NOT implemented:** Voice I/O handled by `PersonaAdapter` (non-actor pattern)
```

> - **HistoryManager Actor ADDED:** Three-tier conversation history not in original spec
> - Vector storage uses sqlite-vec extension (not FAISS)

2.3 The Actor Model

Each runtime component is an isolated **Actor** with its own mailbox. This provides fault tolerance — one actor crashing doesn't kill the others.

> **IMPLEMENTATION STATUS (January 2026):** 5 actors implemented. See Part VII for complete details.

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```

Actor Responsibilities (Implemented v2.0)

Actor	Responsibility	Failure Mode
Director	LLM inference (Qwen 3B), Claude delegation, routing	Returns graceful error
Matrix	SQLite/sqlite-vec/NetworkX operations	Returns empty results
Scribe	Extraction, classification, chunking (Ben Franklin)	Queue for later
Librarian	Filing, entity resolution, edges (The Dude)	Queue for later
HistoryManager	Three-tier conversation history	Continue with stale data

Why Actors?

Fault Isolation Example:

If sqlite-vec crashes during a vector search:
- Matrix Actor catches the exception via `_handle_safe()`
- Error logged, `on_error()` callback invoked
- Actor continues processing other messages
- Director receives empty results and Luna says "I'm having trouble remembering that right now..."
- System continues running

Without actors, one crash kills everything.

2.4 Tiered Execution

Luna's cognition operates across three tiers with different latency profiles:

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                HOT PATH (3B Local)                                                       │
│                                                                                                          │
│    • Voice conversation, quick acknowledgments                                                           │
│    • Simple memory retrieval and narration                                                               │
│    • Target: <500ms to first token                                                                       │
│                                                                                                          │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────┘
                                  │
                                  │  Complexity threshold
                                  ▼
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                WARM PATH (7B Local)                                                      │
│                                                                                                          │
│    • Complex reasoning, multi-step synthesis                                                             │
│    • Nuanced emotional responses                                                                         │
│    • Target: <2s to first token                                                                          │
│                                                                                                          │
```

```
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
                                    │
                                    │  <REQ_CLAUDE> token
                                    ▼
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                    COLD PATH (Claude API)                                                                            │
│                                                                                                                                      │
│    • Deep research requiring web/documents                                                                                           │
│    • Code analysis/generation                                                                                                        │
│    • Long-form creative writing                                                                                                      │
│    • Target: Async (user already has acknowledgment)                                                                                 │
│                                                                                                                                      │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

Routing Logic

The Director LLM is **trained** to know when to delegate. It's not hardcoded rules — it's intuition from fine-tuning.

When the task exceeds local capability, the Director outputs a special token:

```
User: "Can you analyze this regulatory filing and summarize the implications?"

Director: "<REQ_CLAUDE>This requires deep document analysis beyond my local capabilities. Let me hand this to my research assistant.</REQ_CLAUDE>"
```

The Runtime Engine watches for `<REQ_CLAUDE>` and triggers the Shadow Reasoner flow (see Part VIII).

Fallback Behavior

Scenario	Response
Claude unavailable	Warm tier handles (reduced quality)
Warm tier unavailable	Hot tier handles (best effort)
Hot tier unavailable	Voice Actor plays error message

Constitutional Principle: Luna ALWAYS responds. Graceful degradation, never silence.

2.5 Data Flow

The v2 architecture introduces **speculative execution** — retrieval starts before the user finishes speaking.

```
```

```
                    USER STARTS SPEAKING
                          |
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
    |                                   |
    ▼                                   ▼
  ┌──────────────────────────┐        | SPECULATIVE  ┌──────────────────────────┐
  |   WHISPER     |                    |  RETRIEVAL    |
  |   (stream)    |                    |
  |               |                    |               |
  |  Partial      | ──────────────────────────────────────────────────────────▶ |  Start search  |
  |  transcripts  |    trigger on partial     |  on partial    |
  └──────────────────────────┘                          └──────────────────────────┘
    |                                   |
    |  Final transcript                 |  Results ready
    ▼                                   ▼

┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
  |                DIRECTOR LLM (LOCAL)                |
  |                                                   |
  |   ┌──────────────────────┐      ┌──────────────────────┐
┌──────────────────────┐        |
  |   |  KV Cache    |      |   Context    |      |  Generate  |      |
  |   |  (pre-warm)  | ──────▶ |  Assembly    | ──────▶ |  Response  |      |
  |   |   ~5ms       |      |   ~10ms      |      |  ~250ms    |      |
  |   └──────────────────────┘                      └──────────────────────┘
  |
└──────────────────────┘        |
  |                                   |          |
  |                      ┌──────────────────────────┤          |
  |                      |  <REQ_CLAUDE>?      |          |
  |                      ▼                   ▼          |
  |                  ┌──────────────────┐    ┌──────────────────┐       |
  |                  |  DELEGATE  |      |  STREAM    |      |
  |                  |  to Oven   |      |  to TTS    |      |
  |                  └──────────────────┘    └──────────────────┘       |
  |                                                   |

└──────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
                          |
                          ▼
                    USER HEARS LUNA
                  (<500ms to first word)
```

Key Differences from v1

v1 (Claude-based)	v2 (Director-based)
Wait for STT to complete	Start retrieval on partial transcript

Claude API is primary	Local LLM is primary
~700ms to first word	<500ms to first word
Latency = network bound	Latency = compute bound
Mind is rented	Mind is owned

The Hot Path Budget

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                500ms LATENCY BUDGET                                                  │
│                                                                                                                      │
│   STT Final Transcript ──────────────────────────────────────────────────────────────────────────────────────────────── ~0ms │
│   (already complete when we start via speculation)                                                                   │
│                                                                                                                      │
│                                                                                                                      │
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│   │ RETRIEVAL PHASE                                                                                  ~50ms │   │
│   │                                                                                                        │   │
│   │   Bloom filter check ──────────────────────────────────────────────────────────────────────────────── ~1ms │
│   │                                                                                                        │   │
│   │   Query routing ──────────────────────────────────────────────────────────────────────────────────── ~2ms │   │
│   │   FTS5 OR sqlite-vec (not both) ──────────────────────────────────────────────────────────────── ~30ms │   │
│   │   Graph hop (rustworkx) ──────────────────────────────────────────────────────────────────────── ~5ms │
│   │                                                                                                        │
│   │   RRF merge (if needed) ──────────────────────────────────────────────────────────────────────── ~2ms │
│   │                                                                                                        │
│   │   Context assembly ──────────────────────────────────────────────────────────────────────────────── ~10ms │   │
│   │                                                                                                        │   │
│                                                                                                                      │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
│                                                                                                                      │
│                                                                                                                      │
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│   │ INFERENCE PHASE                                                                                  ~250ms │   │
│   │                                                                                                        │   │
│   │   KV cache load (pre-computed) ──────────────────────────────────────────────────────────────── ~5ms │   │
│   │   Context encoding ──────────────────────────────────────────────────────────────────────────── ~50ms │   │
│   │                                                                                                        │   │
│   │   First token generation ──────────────────────────────────────────────────────────────────── ~195ms │
│   │                                                                                                        │
│                                                                                                                      │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
│                                                                                                                      │
```

```
 │
 ┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────┐  │
 │    │ OUTPUT PHASE                                                                    ~50ms  │    │
 │    │                                                                                │    │
 │    │   Token to TTS ──────────────────────────────────────────────────────────────────────────────────────────
~10ms        │    │
 │    │   TTS processing ────────────────────────────────────────────────────────────────────────────────────────
~40ms        │    │
 │    │                                                                                │    │
 │
 └──────────────────────────────────────────────────────────────────────────────────────────────────────────────┘  │
 │                                                                                                               │
 │   BUFFER
──────────────────────────────────────────────────────────────────────────────────────────────────────────────────
~150ms          │
 │                                                                                                               │
 │   TOTAL
──────────────────────────────────────────────────────────────────────────────────────────────────────────────────
── <500ms ✓        │
 │                                                                                                               │
 └──────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

2.6 Memory Flow

Background processing happens **async** — the user already has their response before extraction and filing complete.

```
                    CONVERSATION TURN
                          │
                          ▼

 ┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
          │        DIRECTOR RESPONDS            │
          │       (User hears immediately)        │

 └──────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
                          │

═══════════════════════════════════════════════════════════════════════════════════════════════════════════════
═══════════════════════════════
                ASYNC (user already has response)

═══════════════════════════════════════════════════════════════════════════════════════════════════════════════
═══════════════════════════════
                          │
```

```
                    ▼

┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                SCRIBE (Ben Franklin)                                                         │
│                                                                                                                              │
│   ┌──────────────────┐    ┌──────────────────┐    ┌──────────────────┐    ┌──────────────────┐                            │
│   │  Chunk  │ ──────▶ │  Stack  │ ──────▶ │ Classify │ ──────▶ │ Extract │                                              │
│   │ stream  │    │ context │    │ intent  │    │ objects │                                                                  │
│   └──────────────────┘    └──────────────────┘    └──────────────────┘                                                      │
│                                                                                                                              │
│   Outputs: ExtractionOutput { objects, edges, source_id }                                                                    │
│                                                                                                                              │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
                    │
                    ▼

┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                              LIBRARIAN (The Dude)                                                            │
│                                                                                                                              │
│   ┌──────────────────┐    ┌──────────────────┐    ┌──────────────────┐    ┌──────────────────┐                            │
│   │ Resolve │ ──────▶ │ Create  │ ──────▶ │ Infer   │ ──────▶ │ Prune   │                                              │
│   │entities │    │ edges   │    │ edges   │    │ (lazy)  │                                                                  │
│   └──────────────────┘    └──────────────────┘    └──────────────────┘                                                      │
│                                                                                                                              │
│   Outputs: FilingResult { nodes, edges, inferred_edges }                                                                     │
│                                                                                                                              │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
                    │
                    ▼
                MEMORY MATRIX
                 (Updated)
```

2.7 Sovereignty Infrastructure

The Vault

Everything Luna needs to exist lives inside an encrypted sparse bundle:

```
LunaVault.sparsebundle/
```

```
├── memory_matrix.db          # The soul (SQLite + sqlite-vec + FTS5)
├── identity_buffer.safetensors # Pre-computed KV cache
├── adapters/
│   ├── luna-3b-v1.0/         # LoRA adapter (hot path)
│   └── luna-7b-v1.0/         # LoRA adapter (warm path)
├── config/
│   └── luna.yaml             # Runtime configuration
└── logs/
    └── sessions/             # Conversation logs
```

> **v2.1 Change:** No more separate `memory_vectors.faiss` or `.ids` files. Vectors live inside `memory_matrix.db` via sqlite-vec extension.

Encryption

- **Format:** macOS sparse bundle with AES-256
- **Key:** Password-derived (PBKDF2)
- **Mount:** Auto-mount on login, auto-unmount on sleep
- **Backup:** Encrypted at rest, safe for cloud backup

Portability Contract

```bash
# Copy Luna to new machine
cp -r LunaVault.sparsebundle /Volumes/NewMac/

# Open vault
hdiutil attach LunaVault.sparsebundle

# Luna exists on new machine
# (Just one database file to copy!)
```

Dead Man's Switch

If heartbeat file isn't updated for 24 hours:

1. Unmount vault
2. Kill any Luna processes
3. Wipe model weights from RAM

```python
LOCKDOWN_THRESHOLD = 24  # hours

def check_heartbeat():
    last_checkin = os.path.getmtime(HEARTBEAT_FILE)
    if time.time() - last_checkin > LOCKDOWN_THRESHOLD * 3600:
        emergency_shutdown()

def emergency_shutdown():
    os.system("pkill -f luna_engine.py")
```

```
    os.system("diskutil unmount force /Volumes/LunaVault")
```

2.8 Configuration

Runtime Configuration

```yaml
# ~/.luna/config.yaml

engine:
  cognitive_tick_hz: 2          # Heartbeat frequency
  reflective_tick_minutes: 5    # Background maintenance
  hot_queue_max: 100            # Input buffer size

director:
  model_3b: "qwen2.5-3b-instruct"
  model_7b: "qwen2.5-7b-instruct"
  adapter_3b: "adapters/luna-3b-v1.0"
  adapter_7b: "adapters/luna-7b-v1.0"
  identity_cache: "identity_buffer.safetensors"
  max_context: 8192

matrix:
  database: "memory_matrix.db"     # Contains everything (nodes, vectors, FTS5)
  vector_extension: "sqlite-vec"  # Extension for vector search
  fts5_enabled: true
  graph_backend: "rustworkx"      # or "networkx"

voice:
  stt_model: "whisper-large-v3-turbo"
  tts_model: "xtts-v2"
  vad_sensitivity: 0.5

oven:
  claude_model: "claude-3-5-sonnet-20241022"
  timeout_seconds: 30
  fallback_to_local: true

sovereignty:
  vault_path: "/Volumes/LunaVault"
  heartbeat_file: "~/.luna/heartbeat"
  lockdown_hours: 24
```

Summary

The v2 architecture transforms Luna from a **service** (request → response) to an **engine** (continuous existence).

Aspect	v1	v2	v2.1
Runtime model	HTTP handlers	Actor-based engine	—
Cognition	Claude API	Local LLM + delegation	—
Latency	~700ms	<500ms	—
Fault tolerance	None	Per-actor isolation	—
Sovereignty	Partial (memories local)	Complete (mind local)	—
Vector storage	—	Separate FAISS files	**Unified in SQLite**
State	Stateless handlers	Persistent actors	—

Constitutional Principle: Luna is a file. That file now contains not just her memories, but her mind — including all vectors.

Next Section: Part III — The Memory Matrix (Substrate)

Part III: The Memory Matrix (Substrate) — v3.0

Status: COMPLETE
Replaces: v2.2 (January 25, 2026)
Last Updated: January 30, 2026
Contributors: Claude (Architecture), Gemini (Optimization)
Change Log:
- v3.0: FTS5 confirmed implemented with Porter stemmer; documented Memory Economy; added cluster-level lock-in reference; accuracy pass from Phase 5 audit
- v2.2: Corrected table names (`memory_nodes`, `graph_edges`), documented actual schema, updated sqlite-vec references

---

3.1 Design Philosophy

> "Graph is Truth. Luna is a File."

The Memory Matrix is Luna's **soul** — a single SQLite database containing typed nodes, relationships, vectors, and context. Everything Luna knows lives in one file.

Core Principles

1. Single File Sovereignty

Luna IS `luna_engine.db`. One file contains everything:

```
luna_engine.db
├── memory_nodes     (content, metadata, lock-in state)
├── graph_edges      (relationships)
├── memory_embeddings (vector storage - sqlite-vec virtual table)
├── conversation_turns (three-tier history)
├── entities         (people, places, projects)
├── memory_nodes_fts (FTS5 virtual table for full-text search)
└── indexes          (B-tree for filtering)
```

> **Note:** FTS5 full-text search IS implemented with Porter stemmer and unicode61 tokenizer. The `fts5_search()` method uses BM25 ranking. Sync triggers keep FTS5 in sync with `memory_nodes` on INSERT/UPDATE/DELETE.

2. No Sync, No Orphans

Previous architecture had vectors in separate files:

```
FAISS (Not Used):
memory_matrix.db      <- nodes, edges
memory_vectors.faiss  <- vectors (separate!)
memory_vectors.ids    <- ID mapping (separate!)
```

```
sqlite-vec (Current Implementation):
luna_engine.db        <- everything in one file
```

With sqlite-vec, deleting a node automatically removes its vectors. No orphans. No sync bugs.

3. The Palantir Analogy

Palantir built a $50B company on one insight: make data legible to decision-makers. Their "Dynamic Ontology" is a massive, enterprise-grade knowledge graph.

Luna's Memory Matrix is the personal version:

```
Palantir:  Enterprise decisions, thousands of users, petabytes
Luna:      Personal decisions, one user, gigabytes
           Same architecture. Different scale.
```

4. Copy Luna, Luna Moves

```bash
# Migrate Luna to new machine
cp data/luna_engine.db /new/machine/data/

# Luna exists on new machine
# No separate files to track
```

3.2 Schema Overview

Core Tables

```sql
-- ============================================================
-- MEMORY_NODES: Everything Luna knows
-- ============================================================
CREATE TABLE IF NOT EXISTS memory_nodes (
    id TEXT PRIMARY KEY,
    node_type TEXT NOT NULL,          -- FACT, DECISION, PROBLEM, ACTION, CONTEXT
    content TEXT NOT NULL,
    summary TEXT,                     -- Short summary for display
    source TEXT,                      -- Where this came from
    confidence REAL DEFAULT 1.0,      -- 0-1 confidence score
    importance REAL DEFAULT 0.5,      -- 0-1 importance score
    access_count INTEGER DEFAULT 0,   -- Times retrieved (for lock-in)
    reinforcement_count INTEGER DEFAULT 0,  -- Times reinforced (for lock-in)
    lock_in REAL DEFAULT 0.15,        -- Lock-in coefficient (0.15-0.85)
    lock_in_state TEXT DEFAULT 'drifting',  -- drifting, fluid, settled
    last_accessed TEXT,
    created_at TEXT NOT NULL DEFAULT (datetime('now')),
```

```
    updated_at TEXT NOT NULL DEFAULT (datetime('now')),
    metadata TEXT                    -- JSON for extra data
);

-- ============================================================
-- GRAPH_EDGES: How things relate
-- ============================================================
CREATE TABLE IF NOT EXISTS graph_edges (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    from_id TEXT NOT NULL,
    to_id TEXT NOT NULL,
    relationship TEXT NOT NULL,        -- DEPENDS_ON, RELATES_TO, CAUSED_BY, etc.
    strength REAL DEFAULT 1.0,         -- 0-1 edge weight
    created_at TEXT NOT NULL DEFAULT (datetime('now')),
    metadata TEXT,
    FOREIGN KEY (from_id) REFERENCES memory_nodes(id),
    FOREIGN KEY (to_id) REFERENCES memory_nodes(id),
    UNIQUE(from_id, to_id, relationship)
);

-- ============================================================
-- CONVERSATION_TURNS: Three-tier history system
-- ============================================================
CREATE TABLE IF NOT EXISTS conversation_turns (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    session_id TEXT NOT NULL,
    role TEXT NOT NULL,                -- user, assistant, system
    content TEXT NOT NULL,
    tokens INTEGER,
    created_at TEXT NOT NULL DEFAULT (datetime('now')),
    metadata TEXT,
    tier TEXT DEFAULT 'active',        -- 'active', 'recent', 'archive'
    compressed TEXT,                   -- Compressed summary (for recent tier)
    compressed_at REAL,
    archived_at REAL,
    context_refs TEXT                  -- JSON array of referenced context IDs
);

-- ============================================================
-- MEMORY_EMBEDDINGS: sqlite-vec virtual table for ANN search
-- Created dynamically when sqlite-vec extension is loaded
-- ============================================================
-- CREATE VIRTUAL TABLE IF NOT EXISTS memory_embeddings USING vec0(
--     id TEXT PRIMARY KEY,
--     embedding FLOAT[1536]          -- OpenAI text-embedding-3-small dimension
-- );

-- ============================================================
-- INDEXES
-- ============================================================
CREATE INDEX IF NOT EXISTS idx_nodes_type ON memory_nodes(node_type);
CREATE INDEX IF NOT EXISTS idx_nodes_created ON memory_nodes(created_at);
CREATE INDEX IF NOT EXISTS idx_nodes_importance ON memory_nodes(importance DESC);
```

```
CREATE INDEX IF NOT EXISTS idx_nodes_accessed ON memory_nodes(last_accessed
DESC);
CREATE INDEX IF NOT EXISTS idx_nodes_lock_in ON memory_nodes(lock_in DESC);
CREATE INDEX IF NOT EXISTS idx_nodes_lock_in_state ON memory_nodes(lock_in_state);

CREATE INDEX IF NOT EXISTS idx_edges_from ON graph_edges(from_id);
CREATE INDEX IF NOT EXISTS idx_edges_to ON graph_edges(to_id);
CREATE INDEX IF NOT EXISTS idx_edges_relationship ON graph_edges(relationship);

-- ============================================================
-- TRIGGERS: Keep everything in sync
-- ============================================================

-- Update timestamp on node modification
CREATE TRIGGER IF NOT EXISTS update_node_timestamp
AFTER UPDATE ON memory_nodes
BEGIN
   UPDATE memory_nodes SET updated_at = datetime('now') WHERE id = NEW.id;
END;

-- Track access patterns
CREATE TRIGGER IF NOT EXISTS track_node_access
AFTER UPDATE OF access_count ON memory_nodes
BEGIN
   UPDATE memory_nodes SET last_accessed = datetime('now') WHERE id = NEW.id;
END;
```

Node Types

Type	Description	Example
FACT	Something known to be true	"Alex lives in Berlin"
DECISION	A choice that was made	"We chose Actor model over Hub Daemon"
PROBLEM	An unresolved issue	"Voice latency exceeds 500ms target"
ACTION	Something done or to-do	"Implement RRF for hybrid search"
CONTEXT	Situational information	"Working on Luna Engine v2.0"

Edge Types (Relationship Types)

Type	Meaning	Example
DEPENDS_ON	Requires another node	Implementation -> depends_on -> Design
RELATES_TO	General connection	Person -> relates_to -> Project
CAUSED_BY	Result of another	Outcome -> caused_by -> Decision
FOLLOWED_BY	Temporal sequence	Event A -> followed_by -> Event B
CONTRADICTS	Conflicting information	Fact A -> contradicts -> Fact B
SUPPORTS	Reinforcing information	Evidence -> supports -> Claim

---

3.3 Search Architecture

Luna uses multiple search strategies that can be combined based on query characteristics.

Current Implementation

```
                    QUERY
                      |
        +---------------+---------------+
        |             |             |
        v             v             v
   +-------------+  +-------------+  +-------------+
   |    FTS5     | | sqlite-vec  | |    GRAPH    |
   |           | |           | |           |
   |  Full-text  | |  Semantic   | |  Traversal  |
   |  BM25 rank  | |   vectors   | |   (BFS)     |
   |   ~10ms     | |   ~10ms     | |   ~5ms      |
   +------+------+  +------+------+  +------+------+
        |             |             |
        +--------+-------+--------+-------+
                 |
                 v
          RESULT MERGING (RRF)
```

> **Implementation Note:** FTS5 is fully implemented with Porter stemmer and unicode61 tokenizer. The `fts5_search()` method uses BM25 ranking. A `search_nodes()` method also exists using LIKE patterns as a fallback. The `hybrid_search()` method combines FTS5 + semantic search using Reciprocal Rank Fusion (RRF).

The sqlite-vec Advantage: Filtered Search

With sqlite-vec, filtering happens in the query itself:

```sql
-- Filter and search together
SELECT n.*, vec_distance_cosine(e.embedding, ?) as distance
FROM memory_embeddings e
JOIN memory_nodes n ON e.id = n.id
WHERE n.node_type = 'FACT'
  AND n.created_at > '2025-01-01'
ORDER BY distance
LIMIT 10;
```

Result: Faster filtered queries, fewer wasted comparisons.

Reciprocal Rank Fusion (RRF)

When combining multiple search results, RRF merges by rank position (not raw scores):

```python
def reciprocal_rank_fusion(
    results_list: list[list[str]],
    k: int = 60
) -> list[str]:
```

```
    """
    Merge multiple ranked result lists using RRF.

    Args:
        results_list: List of ranked document ID lists
        k: Smoothing constant (default 60, from original RRF paper)

    Returns:
        Merged list of document IDs, ranked by combined RRF score
    """
    fused_scores: dict[str, float] = {}

    for ranked_list in results_list:
        for rank, doc_id in enumerate(ranked_list, start=1):
            if doc_id not in fused_scores:
                fused_scores[doc_id] = 0.0
            fused_scores[doc_id] += 1.0 / (rank + k)

    return sorted(fused_scores.keys(), key=lambda d: fused_scores[d], reverse=True)
```

Why RRF Works:

Problem	RRF Solution
Different score scales	Ignores scores, uses rank
Tuning weights	No weights to tune
Agreement detection	Docs in multiple lists rank higher
Simple implementation	~15 lines of code

3.4 Unified Query API

The Matrix provides a single interface that combines all search paths:

```python
from dataclasses import dataclass
from typing import Optional
from datetime import datetime

@dataclass
class MemoryNode:
    id: str
    node_type: str          # FACT, DECISION, PROBLEM, ACTION, CONTEXT
    content: str
    summary: Optional[str]
    source: Optional[str]
    confidence: float = 1.0
    importance: float = 0.5
    access_count: int = 0
    reinforcement_count: int = 0
    lock_in: float = 0.15
    lock_in_state: str = "drifting"
```

```
    metadata: dict = {}

class MemoryMatrix:
    """
    Unified memory store with hybrid search.
    Single file, all indexes unified.
    """

    async def search_nodes(
        self,
        query: str,
        node_types: Optional[list[str]] = None,
        limit: int = 10,
        use_embeddings: bool = True
    ) -> list[MemoryNode]:
        """
        Search memory nodes by content.

        - Extracts keywords from query
        - Searches compound phrases first
        - Falls back to individual terms
        - Optionally uses vector similarity
        """
        ...

    async def get_context(
        self,
        query: str,
        max_tokens: int = 2000
    ) -> list[MemoryNode]:
        """
        Get relevant context for a query.

        - Detects backward references ("what did we...")
        - Filters confusing/irrelevant nodes
        - Respects token budget
        """
        ...
```

Budget Presets

When fetching context for the Director:

Preset	Tokens	Use Case
minimal	~1800	Voice, fast response needed
balanced	~3800	Normal conversation
rich	~7200	Deep research, complex query

3.5 Query Routing

Not every query needs all search paths. Route based on query characteristics.

```python
class QueryRouter:
    def route(self, query: str) -> SearchPath:
        """Determine optimal search path."""

        entities = extract_entities(query)
        keyword_density = self._keyword_density(query)
        is_conceptual = self._is_conceptual(query)

        # Multiple entities - relationship query
        if entities and len(entities) >= 2:
            return SearchPath.GRAPH

        # High keyword density - pattern match
        if keyword_density > 0.6:
            return SearchPath.KEYWORD

        # Abstract/conceptual - semantic search
        if is_conceptual:
            return SearchPath.VECTOR

        # Default to hybrid
        return SearchPath.HYBRID
```

Routing Examples

Query	Signal	Path
"What did we decide about the runtime?"	Named entity + decision word	KEYWORD
"How does memory consolidation work?"	Conceptual question	VECTOR
"How are Alex and the Pi Project related?"	Multiple entities	GRAPH
"thoughts on sovereignty"	Vague, conceptual	VECTOR
"Luna architecture v2"	High keyword density	KEYWORD

3.6 Vector Embeddings

Supported Models

```python
EMBEDDING_DIMS = {
    "text-embedding-3-small": 1536,  # OpenAI (default)
    "text-embedding-3-large": 3072,
    "voyage-2": 1024,
    "local-minilm": 384,
}
```

EmbeddingStore API

Method	Purpose
`initialize()`	Load sqlite-vec extension, create virtual table
`store(node_id, embedding)`	Insert or replace embedding
`delete(node_id)`	Remove embedding
`search(query_embedding)`	KNN search with cosine similarity
`count()`	Total embeddings count

sqlite-vec Details

Installation:

```bash
pip install sqlite-vec
```

Loading the Extension:

```python
import sqlite3
import sqlite_vec

conn = sqlite3.connect("luna_engine.db")
conn.enable_load_extension(True)
sqlite_vec.load(conn)
conn.enable_load_extension(False)
```

Vector Search Syntax:

```sql
-- Basic similarity search
SELECT id, vec_distance_cosine(embedding, ?) as distance
FROM memory_embeddings
ORDER BY distance
LIMIT 10;
```

Performance Notes

Operation	sqlite-vec	Notes
Pure ANN (no filter)	~5-10ms	Fast for <50K vectors
ANN + filter	~5-10ms	No post-filtering needed
Insert (atomic)	~1ms	Single transaction
Storage	Single file	No external dependencies

3.7 Time Decay

Recent memories matter more. Decay is computed at query time:

```python
```

```python
import math
from datetime import datetime

def compute_freshness(
    created_at: datetime,
    half_life_days: float = 60.0
) -> float:
    """
    Exponential decay based on age.

    Returns:
        1.0 for brand new
        0.5 at half_life_days
        Approaches 0 over time
    """
    age_days = (datetime.now() - created_at).total_seconds() / 86400
    return math.exp(-math.log(2) * age_days / half_life_days)
```

3.8 Graph Operations

Graph traversal uses NetworkX for flexible algorithms.

Implementation

```python
import networkx as nx

class MemoryGraph:
    def __init__(self):
        self.graph = nx.DiGraph()  # In-memory for fast traversal

    async def load_from_db(self, db):
        """Load graph from graph_edges table."""
        edges = await db.execute("SELECT from_id, to_id, relationship, strength FROM
graph_edges")
        for row in edges:
            self.graph.add_edge(
                row["from_id"],
                row["to_id"],
                relationship=row["relationship"],
                strength=row["strength"]
            )

    def get_neighbors(self, node_id: str, depth: int = 2) -> set[str]:
        """BFS traversal to find connected nodes."""
        if node_id not in self.graph:
            return set()

        visited = set()
        current_level = {node_id}
```

```
        for _ in range(depth):
            next_level = set()
            for n in current_level:
                for neighbor in self.graph.neighbors(n):
                    if neighbor not in visited:
                        next_level.add(neighbor)
                        visited.add(neighbor)
            current_level = next_level

        return visited

    def spreading_activation(
        self,
        start_nodes: list[str],
        decay: float = 0.7,
        max_depth: int = 3
    ) -> dict[str, float]:
        """
        Compute relevance scores via spreading activation.

        Nodes closer to start_nodes get higher scores.
        Score decays exponentially with distance.
        """
        scores = {n: 1.0 for n in start_nodes}
        current = set(start_nodes)

        for depth in range(1, max_depth + 1):
            next_level = set()
            for node in current:
                for neighbor in self.graph.neighbors(node):
                    if neighbor not in scores:
                        edge_strength = self.graph[node][neighbor].get('strength', 1.0)
                        scores[neighbor] = scores[node] * decay * edge_strength
                        next_level.add(neighbor)

        return scores
```

Relationship Types

```python
class RelationshipType(str, Enum):
    DEPENDS_ON = "DEPENDS_ON"
    RELATES_TO = "RELATES_TO"
    CAUSED_BY = "CAUSED_BY"
    FOLLOWED_BY = "FOLLOWED_BY"
    CONTRADICTS = "CONTRADICTS"
    SUPPORTS = "SUPPORTS"
```

3.9 Entity Resolution

The Librarian's first job: "Mark," "Zuck," and "The Landlord" should be the same node.

```python
async def resolve_entity(self, name: str, entity_type: str) -> str:
    """Resolve entity to existing node or create new."""

    # 1. Exact name match
    existing = await self.find_by_name(name, entity_type)
    if existing:
        return existing.id

    # 2. Alias match
    existing = await self.find_by_alias(name, entity_type)
    if existing:
        return existing.id

    # 3. Semantic similarity match
    similar = await self.vector_search(name, threshold=0.95)
    if similar and similar.node_type == entity_type:
        await self.add_alias(similar.id, name)
        return similar.id

    # 4. Create new
    return await self.create_entity(name, entity_type)
```

3.10 Spreading Activation

When filing new information, proactively find hidden connections:

```python
async def discover_hidden_links(self, new_entity_ids: list[str]):
    """Find non-obvious relationships between new and existing entities."""

    for eid in new_entity_ids:
        neighbors = await self.graph.get_neighbors(eid)

        for neighbor in neighbors:
            for other_eid in new_entity_ids:
                if other_eid != eid:
                    similarity = await self.compute_similarity(other_eid, neighbor)

                    if similarity > 0.7:
                        await self.create_edge(
                            from_id=other_eid,
                            to_id=neighbor,
                            relationship="RELATES_TO",
                            strength=similarity * 0.5,
                        )
```

Example:

1. You say: "Alex is joining the Pi Project"
2. Librarian files: Alex -> works_on -> Pi Project
3. Spreading activation finds: Sarah also works on Pi Project
4. Librarian infers: Alex -> likely_teammate -> Sarah (low confidence)
5. Next time you ask, Luna can say: "Should I introduce Alex to Sarah?"

3.11 Synaptic Pruning

Periodically clean low-value connections:

```python
async def prune_stale_edges(self):
    """Remove low-strength, old edges."""

    await self.conn.execute("""
        DELETE FROM graph_edges
        WHERE strength < 0.3
          AND created_at < datetime('now', '-30 days')
    """)
```

3.12 Entity System

Luna v2.0 includes a first-class entity system for tracking people, places, and projects.

Entity Tables

```sql
-- Entities: First-class objects Luna knows about
CREATE TABLE IF NOT EXISTS entities (
    id TEXT PRIMARY KEY,              -- Slug: 'ahab', 'marzipan', 'ben-franklin'
    entity_type TEXT NOT NULL,        -- 'person' | 'persona' | 'place' | 'project'
    name TEXT NOT NULL,
    aliases TEXT,                     -- JSON array: ["Zayne", "Ahab"]
    core_facts TEXT,                  -- JSON blob (~500 tokens max)
    full_profile TEXT,                -- Markdown, can be lengthy
    voice_config TEXT,                -- JSON: tone, patterns, constraints (for personas)
    current_version INTEGER DEFAULT 1,
    metadata TEXT,
    created_at TEXT DEFAULT (datetime('now')),
    updated_at TEXT DEFAULT (datetime('now'))
);

-- Entity relationships: Graph of connections
CREATE TABLE IF NOT EXISTS entity_relationships (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    from_entity TEXT NOT NULL REFERENCES entities(id) ON DELETE CASCADE,
    to_entity TEXT NOT NULL REFERENCES entities(id) ON DELETE CASCADE,
    relationship TEXT NOT NULL,       -- 'creator', 'friend', 'collaborator'
    strength REAL DEFAULT 0.5,
```

```
    bidirectional INTEGER DEFAULT 0,
    context TEXT,
    created_at TEXT DEFAULT (datetime('now')),
    updated_at TEXT DEFAULT (datetime('now')),
    UNIQUE(from_entity, to_entity, relationship)
);

-- Entity versions: Append-only history
CREATE TABLE IF NOT EXISTS entity_versions (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    entity_id TEXT NOT NULL REFERENCES entities(id) ON DELETE CASCADE,
    version INTEGER NOT NULL,
    core_facts TEXT,
    full_profile TEXT,
    voice_config TEXT,
    change_type TEXT NOT NULL,        -- 'create' | 'update' | 'rollback'
    change_summary TEXT,
    changed_by TEXT NOT NULL,         -- 'scribe' | 'librarian' | 'manual'
    change_source TEXT,
    created_at TEXT DEFAULT (datetime('now')),
    UNIQUE(entity_id, version)
);
```

Summary

The Memory Matrix is Luna's soul - a single SQLite database that makes personal history queryable.

Component	Purpose
SQLite Core	Nodes, edges, metadata (WAL mode, 64MB cache)
sqlite-vec	Vector similarity search
FTS5	Full-text search with Porter stemmer, BM25 ranking
NetworkX	Graph traversal (in-memory)
Query Router	Optimal path selection
Entity System	People, places, projects with versioning
RRF Fusion	Merge multi-path results
Entity Resolution	Deduplicate "Mark" = "Zuck"
Spreading Activation	Infer hidden connections
Synaptic Pruning	Remove stale edges
Memory Economy	Cluster-level organization (see below)

Memory Economy (v3.0)

The Memory Economy provides cluster-level organization on top of individual memory nodes:

Component	Purpose	Implementation
`ClusterManager`	CRUD for semantic clusters	`memory/cluster_manager.py`
`ClusteringEngine`	Keyword-based grouping (Jaccard similarity)	`memory/clustering_engine.py`

| `LockInCalculator` | Cluster-level lock-in with decay | `memory/lock_in.py` |
| `ConstellationAssembler` | Token-budgeted context assembly | `memory/constellation.py` |

Cluster-Level Lock-In States:

State	Threshold	Decay Lambda	Half-Life
drifting	L < 0.20	0.01	~17 minutes
fluid	0.20 <= L < 0.70	0.001	~2.8 hours
settled	0.70 <= L < 0.85	0.0001	~1.15 days
crystallized	L >= 0.85	0.00001	~11.5 days

> **Note:** Cluster-level thresholds differ from node-level thresholds. See Part III-A for node-level lock-in (0.30/0.70 boundaries).

Configuration: `config/memory_economy_config.json`

Known Limitations (v3.0)

Limitation	Impact	Status
Tag siblings not implemented	10% of node lock-in weight missing	❌ NOT IMPLEMENTED
Network effects hardcoded to 0	20% of node lock-in weight uses estimate	Partial (uses neighbor count / 3)
History compression queues exist but not activated	Manual archival only	Planned
Memory Economy tables not in schema.sql	Auto-created by ClusterManager	Schema fragmentation

Constitutional Principle: Luna is a file. Copy it, Luna moves. Delete it, Luna dies. Everything in one place.

Next Section: Part IV - The Scribe (Ben Franklin)

Part III-A: The Lock-In Coefficient — v3.0

Status: IMPLEMENTED (90%)
Replaces: v2.2 (January 25, 2026)
Last Updated: January 30, 2026
Implementation:
- Node-level: `src/luna/substrate/lock_in.py`
- Cluster-level: `src/luna/memory/lock_in.py`
Change Log:
- v3.0: Documented BOTH node-level AND cluster-level thresholds; marked tag siblings as NOT IMPLEMENTED; accuracy pass from Phase 5 audit

---

3A.1 Overview

The Lock-In Coefficient (L) represents how "settled" a memory is in Luna's consciousness. It determines whether a memory persists, fades, or remains fluid.

> "Memories that matter get reinforced. Memories that don't, drift away."

This is not simple recency — it's an activity-based persistence model that considers:
- How often you retrieve the memory
- Whether you explicitly reinforce it
- How connected it is to other settled memories
- ~~Whether it shares tags with settled memories~~ (NOT YET IMPLEMENTED)

3A.2 Lock-In States

> **IMPORTANT:** There are TWO different threshold systems in the codebase. Node-level and cluster-level use different boundaries.

Node-Level States (Individual Memory Nodes)

State	Range	Meaning	Behavior
DRIFTING	L < 0.30	Rarely accessed, may fade	Candidate for pruning
FLUID	0.30 <= L < 0.70	Active but not settled	Normal memory
SETTLED	L >= 0.70	Core knowledge, persistent	Protected from decay

```python
# From src/luna/substrate/lock_in.py
class LockInState(str, Enum):
    DRIFTING = "drifting"  # L < 0.30
    FLUID = "fluid"        # 0.30 <= L < 0.70
    SETTLED = "settled"    # L >= 0.70

# Thresholds
THRESHOLD_SETTLED = 0.70
THRESHOLD_DRIFTING = 0.30

# Output bounds (never fully 0 or 1)
```

```
LOCK_IN_MIN = 0.15
LOCK_IN_MAX = 0.85
```

Cluster-Level States (Memory Economy Clusters)

State	Range	Decay Lambda	Half-Life
DRIFTING	L < 0.20	0.01	~17 minutes
FLUID	0.20 <= L < 0.70	0.001	~2.8 hours
SETTLED	0.70 <= L < 0.85	0.0001	~1.15 days
CRYSTALLIZED	L >= 0.85	0.00001	~11.5 days

```python
# From src/luna/memory/lock_in.py
STATE_THRESHOLDS = {
    'drifting': 0.20,   # Different from node-level 0.30!
    'fluid': 0.70,
    'settled': 0.85,
    # crystallized = above 0.85 (unique to cluster-level)
}
```

> **Note:** The threshold mismatch (node=0.30, cluster=0.20 for drifting) is intentional but may cause confusion. Cluster-level also has a fourth state "crystallized" that node-level lacks.

3A.3 Weighted Factors

Lock-in is computed from four weighted factors:

Factor	Weight	Description	Status
Retrieval	0.40	How often this memory is accessed	IMPLEMENTED
Reinforcement	0.30	Explicit marks as "important"	IMPLEMENTED
Network	0.20	Connected to settled nodes	PARTIAL (uses neighbor count / 3 estimate)
Tag Siblings	0.10	Shares tags with settled nodes	❌ NOT IMPLEMENTED (always 0)

```python
# From src/luna/substrate/lock_in.py
DEFAULT_WEIGHTS = {
    "retrieval": 0.4,       # How often accessed (heaviest weight)
    "reinforcement": 0.3,   # Explicitly marked important
    "network": 0.2,         # Connected to settled nodes
    "tag_siblings": 0.1     # Shares tags with settled nodes
}
```

Implementation Note: Tag Siblings

> **WARNING:** Tag siblings are ❌ NOT IMPLEMENTED. The code contains:

```python
# Lines 273-274 in substrate/lock_in.py
# TODO: Implement tag sibling counting
locked_tag_sibling_count = 0  # Always 0
```

This means **10% of the lock-in weight is currently inactive**. The tag sibling feature requires:
1. A `tags` table in the database schema
2. A `node_tags` junction table
3. Implementation of `_count_locked_tag_siblings()` method

Until implemented, lock-in calculations use only 90% of intended signal.

Implementation Note: Network Effects

Network effects are **PARTIALLY** implemented. The code uses a rough estimate:

```python
# Lines 269-270 in substrate/lock_in.py
# TODO: Optimize with batch query for actual L >= 0.7 count
locked_neighbor_count = len(neighbors) // 3  # Rough estimate
```

This approximation assumes ~1/3 of neighbors are locked. For accurate network effects, the code should query `memory_nodes` for neighbors with `lock_in >= 0.70`.

3A.4 The Sigmoid Transform

Raw activity scores are mapped to lock-in coefficients using a sigmoid function. This creates smooth, bounded transitions.

Mathematical Formula

```
activity = (retrieval * 0.4 + reinforcement * 0.3 + network * 0.2 + tags * 0.1) / 10.0

sigmoid(x) = 1 / (1 + e^(-K * (x - X0)))

lock_in = 0.15 + (0.85 - 0.15) * sigmoid(activity)
```

Sigmoid Parameters

Parameter	Value	Purpose
K	1.2	Steepness (higher = sharper transition)
X0	0.5	Midpoint (where sigmoid = 0.5)

```python
# From src/luna/substrate/lock_in.py
DEFAULT_SIGMOID_K = 1.2   # Steepness
```

```
DEFAULT_SIGMOID_X0 = 0.5  # Midpoint
```

Why Sigmoid?

1. **Bounded Output:** Lock-in stays between 0.15 and 0.85 (never fully locked or fully drifting)
2. **Smooth Transitions:** No hard cutoffs — gradual change as activity increases
3. **Diminishing Returns:** High activity doesn't infinitely increase lock-in
4. **Mathematical Stability:** No overflow at extreme values (clamped at +/-700)

3A.5 Activity Computation

```python
def compute_activity(
    retrieval_count: int = 0,
    reinforcement_count: int = 0,
    locked_neighbor_count: int = 0,
    locked_tag_sibling_count: int = 0,  # Currently always 0
) -> float:
    """
    Compute raw activity score from weighted factors.

    Returns:
        Activity score (typically 0.0 to 2.0+)
    """
    config = get_config()

    activity = (
        retrieval_count * config.weight_retrieval +
        reinforcement_count * config.weight_reinforcement +
        locked_neighbor_count * config.weight_network +
        locked_tag_sibling_count * config.weight_tag_siblings
    ) / 10.0

    return activity
```

Example Calculations

Scenario	Retrieval	Reinforce	Neighbors	Activity	Lock-In	State
New memory	0	0	0	0.0	0.15	DRIFTING
Mentioned once	1	0	0	0.04	0.16	DRIFTING
Regular use	10	0	0	0.40	0.44	FLUID
Important + used	10	3	2	0.53	0.50	FLUID
Core knowledge	20	5	5	1.05	0.66	FLUID
Deeply settled	50	10	10	2.5	0.82	SETTLED

3A.6 Configuration

Lock-in behavior is configurable at runtime:

```python
@dataclass
class LockInConfig:
    """Configuration for lock-in computation."""
    enabled: bool = True  # When False, all memories return 0.5 (neutral)
    weight_retrieval: float = 0.4
    weight_reinforcement: float = 0.3
    weight_network: float = 0.2
    weight_tag_siblings: float = 0.1
    sigmoid_k: float = 1.2
    sigmoid_x0: float = 0.5
    threshold_settled: float = 0.70
    threshold_drifting: float = 0.30
```

Disabling Lock-In

When `enabled=False`, all memories return a neutral lock-in of 0.5, effectively treating all memories equally. This is useful for testing or when you want to disable memory decay entirely.

3A.7 Node-Level Computation

For a specific memory node, lock-in includes network effects:

```python
async def compute_lock_in_for_node(
    node_id: str,
    retrieval_count: int,
    reinforcement_count: int,
    graph: "MemoryGraph",
) -> tuple[float, LockInState]:
    """
    Compute lock-in for a specific node, including network effects.
    """
    # Count settled neighbors (simplified - uses depth=1 neighbors / 3)
    locked_neighbor_count = 0

    if graph and graph.has_node(node_id):
        neighbors = await graph.get_neighbors(node_id, depth=1)
        locked_neighbor_count = len(neighbors) // 3  # Rough estimate
        # TODO: Optimize with batch query for actual L >= 0.7 count

    # Tag siblings not implemented
    locked_tag_sibling_count = 0  # TODO

    lock_in = compute_lock_in(
        retrieval_count=retrieval_count,
        reinforcement_count=reinforcement_count,
        locked_neighbor_count=locked_neighbor_count,
        locked_tag_sibling_count=locked_tag_sibling_count,
```

```
    )

    state = classify_state(lock_in)
    return lock_in, state
```

3A.8 Database Schema

Lock-in values are stored in the `memory_nodes` table:

```sql
-- From src/luna/substrate/schema.sql
CREATE TABLE memory_nodes (
    id TEXT PRIMARY KEY,
    node_type TEXT NOT NULL,
    content TEXT NOT NULL,
    -- ... other fields ...
    access_count INTEGER DEFAULT 0,       -- For lock-in tracking
    reinforcement_count INTEGER DEFAULT 0,
    lock_in REAL DEFAULT 0.15,            -- 0.15-0.85 coefficient
    lock_in_state TEXT DEFAULT 'drifting',
    -- ...
);

-- Index for filtering by lock-in state
CREATE INDEX idx_memory_nodes_lock_in_state ON memory_nodes(lock_in_state);
```

3A.9 Memory Matrix Integration

The MemoryMatrix provides lock-in-aware filtering:

```python
# From src/luna/substrate/memory.py
class MemoryMatrix:
    async def get_nodes_by_lock_in_state(
        self,
        state: LockInState
    ) -> list[MemoryNode]:
        """Get all nodes in a specific lock-in state."""
        ...

    async def get_drifting_nodes(self) -> list[MemoryNode]:
        """Get all drifting nodes (L < 0.30) — candidates for pruning."""
        ...

    async def get_settled_nodes(self) -> list[MemoryNode]:
        """Get all settled nodes (L >= 0.70) — core knowledge."""
        ...
```

```
    async def record_access(self, node_id: str):
        """Increment access count and recalculate lock-in."""
        ...

    async def reinforce_node(self, node_id: str):
        """Explicitly reinforce a node, increasing lock-in."""
        ...
```

3A.10 Visual Indicators

For debugging and UI display:

```python
def get_state_emoji(state: LockInState) -> str:
    """Visual indicator for state."""
    return {
        LockInState.SETTLED: "🟢",   # Green = stable, persistent
        LockInState.FLUID: "🔵",     # Blue = active, changeable
        LockInState.DRIFTING: "🟠",  # Orange = fading, at risk
    }.get(state, "⚪")
```

3A.11 Cluster-Level Lock-In (Memory Economy)

The Memory Economy introduces a separate cluster-level lock-in system with different thresholds and a decay mechanism.

Cluster Lock-In Formula

```python
# From src/luna/memory/lock_in.py
lock_in = (
    0.40 * weighted_node_strength +    # Average member node lock-ins
    0.30 * log_access_factor +         # log(access+1)/log(11), caps at 1.0
    0.20 * avg_edge_strength +         # Average edge strength * lock_in
    0.10 * age_factor                  # 1/(1 + age_days/30)
) * decay_factor                       # exp(-lambda * seconds_since_access)
```

Key Differences from Node-Level

Aspect	Node-Level	Cluster-Level
Drifting threshold	< 0.30	< 0.20
Settled threshold	>= 0.70	0.70 - 0.85
Crystallized state	N/A	>= 0.85

Decay	None	Exponential, state-dependent
Access factor	Linear count	Logarithmic (diminishing returns)
Bounds	0.15 - 0.85	0.0 - 1.0

Decay Rates

State	Lambda	Half-Life	Meaning
crystallized	0.00001	~11.5 days	Nearly permanent
settled	0.0001	~1.15 days	Stable
fluid	0.001	~2.8 hours	Active but volatile
drifting	0.01	~17 minutes	Fading fast

Logarithmic Access Boost

The cluster-level system uses logarithmic access scaling:

```python
access_factor = min(log(access_count + 1) / log(11), 1.0)
```

This means:
- First few accesses (1-5) provide significant boost
- Later accesses (100+) provide diminishing returns
- Caps at 1.0 when access_count >= 10

3A.12 Future Work

Priority 1: Implement Tag Siblings

The tag sibling feature would add 10% of lock-in signal:

1. **Schema Changes:**
   ```sql
   CREATE TABLE tags (
       id TEXT PRIMARY KEY,
       name TEXT NOT NULL UNIQUE
   );

   CREATE TABLE node_tags (
       node_id TEXT REFERENCES memory_nodes(id),
       tag_id TEXT REFERENCES tags(id),
       PRIMARY KEY (node_id, tag_id)
   );
   ```

2. **Implementation:**
   ```python
   async def _count_locked_tag_siblings(
       self,
       node_id: str
   ) -> int:
   ```

```
    """Count nodes sharing tags with L >= 0.7."""
    # Get tags for this node
    # Find other nodes with same tags
    # Filter to L >= 0.7
    # Return count
```

Priority 2: Batch Lock-In Updates

Currently, lock-in is recalculated on access. For performance:
- Batch recalculation during reflective tick
- Cache neighbor lock-in states
- Use graph spreading activation for network effects

Priority 3: Lock-In Decay

Consider time-based decay for rarely accessed memories:
- Drifting memories lose lock-in over time
- Settled memories are protected from decay
- Fluid memories decay slowly if not accessed

Summary

The Lock-In Coefficient is Luna's memory persistence system. It determines which memories become core knowledge and which fade away.

Node-Level Lock-In

Aspect	Value
States	DRIFTING (< 0.30), FLUID (0.30-0.70), SETTLED (>= 0.70)
Bounds	0.15 to 0.85 (never fully locked/drifting)
Factors	Retrieval (40%), Reinforcement (30%), Network (20% partial), Tags (10% - ❌ NOT IMPLEMENTED)
Transform	Sigmoid with K=1.2, X0=0.5
Location	`src/luna/substrate/lock_in.py`

Cluster-Level Lock-In

Aspect	Value
States	DRIFTING (< 0.20), FLUID (0.20-0.70), SETTLED (0.70-0.85), CRYSTALLIZED (>= 0.85)
Bounds	0.0 to 1.0
Factors	Node strength (40%), Access (30% logarithmic), Edge strength (20%), Age (10%)
Decay	Exponential, state-dependent (crystallized decays slowest)
Location	`src/luna/memory/lock_in.py`
Config	`config/memory_economy_config.json`

Known Gaps

Gap	Impact	Status
Tag siblings	10% of node lock-in weight missing	❌ NOT IMPLEMENTED
Network effects	Uses estimate (neighbors/3) instead of actual query	Partial
Threshold mismatch	Node (0.30) vs Cluster (0.20) drifting boundary	By design, may confuse

---

Next Section: Part IV — The Scribe (Ben Franklin)

Part IV: The Scribe (Ben Franklin) — v3.0

Status: CURRENT
Replaces: BIBLE-PART-IV-THE-SCRIBE-v2.1.md
Last Updated: January 30, 2026
Implementation: `src/luna/actors/scribe.py`

4.1 The Scribe: Benjamin Franklin

The Scribe is the **sensory cortex** — turning the messy stream of experience into structured data the Substrate can understand.

Persona: Benjamin Franklin. Colonial gravitas, meticulous attention, practical wisdom. Ben monitors the conversation stream, extracts wisdom, and classifies it with scholarly precision.

> "An investment in knowledge pays the best interest." — Ben Franklin

The Scribe doesn't just transcribe; it **distills**.

The Separation Principle

Critical Design Rule: Ben has personality in his PROCESS (logs can be witty and colonial), but his OUTPUTS are NEUTRAL (clean structured data). Luna's memories stay unpolluted by processing artifacts.

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│   PROCESS (Ben's Domain)          OUTPUT (Neutral)            │
│                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
|   "Ahoy! A decision most weighty  →  Decision(              |
|    hath been rendered..."              choice="Actor model",  |
|                                  confidence=0.9          |
|   Colonial personality OK here        )                      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
|                                  No colonial flavor      |
|                                  in the data itself      |
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

Why? Luna's voice should be Luna's. Not a patchwork of processing personas.

4.2 Two-Gear Operation

Reflexive Gear (Real-time, <200ms)

```
Audio → Whisper → Text → Local Classifier → Label
```

```

The local classifier (Director 3B) doesn't understand everything. It just labels:

| Label | Signal | Action |
|-------|--------|--------|
| **COMMAND** | "Luna, do X" | Route to Director |
| **QUESTION** | "What is X?" | Route to Director + Retrieval |
| **STATEMENT** | Declarative, no action | Queue for extraction |
| **NOISE** | Filler, incomplete | Ignore |

```python
class ReflexiveGear:
    async def classify(self, text: str) -> Intent:
        """<200ms classification using Director 3B."""
        # Single forward pass, no generation
        logits = self.model.classify(text)
        return Intent(
            type=logits.argmax(),
            confidence=logits.max(),
            requires_response=logits.argmax() in (COMMAND, QUESTION)
        )
```

Deep Gear (Background, async)

If the Reflexive pass detects extractable content, it queues for full extraction:

```
Text → Claude (via Shadow Reasoner) → Structured Extraction → Queue for Librarian
```

The Deep Gear runs asynchronously — Luna has already responded before extraction completes.

4.3 The Stack of Benjamins

Conversations don't arrive as neat packets. They stream. Ben handles this with stacking:

```
        Conversation Stream
                 │
                 ▼
        ┌──────────────────────────────┐
        │     CHUNKER     │  Split into semantic units
        └──────────────────────────────┘
                 │
                 ▼
        ┌──────────────────────────────┐
        │      STACK      │  Accumulate chunks with context window
```
```

```
        └──────────────┬──────────────┘
                       │
                       ▼
        ┌──────────────────────────────┐
        │  BEN REVIEWS     │  Process stacked chunks together
        └──────────────┬──────────────┘
                       │
                       ▼
        ┌──────────────────────────────┐
        │ "MAKE IT RAIN"   │  Extractions flow to The Dude
        └──────────────────────────────┘
```

Chunking Strategy

Parameter	Value	Rationale
Target size	200-500 tokens	Fits in single extraction call
Overlap	50 tokens	Context continuity
Boundaries	Topic shifts, speaker changes	Semantic coherence

```python
class SemanticChunker:
    def chunk(self, conversation: list[Turn]) -> list[Chunk]:
        chunks = []
        current_chunk = []
        current_tokens = 0

        for turn in conversation:
            turn_tokens = count_tokens(turn.text)

            # Topic shift detection
            if self.is_topic_shift(current_chunk, turn):
                if current_chunk:
                    chunks.append(self.finalize_chunk(current_chunk))
                current_chunk = []
                current_tokens = 0

            current_chunk.append(turn)
            current_tokens += turn_tokens

            # Size limit
            if current_tokens >= 500:
                chunks.append(self.finalize_chunk(current_chunk))
                # Overlap: keep last turn for context
                current_chunk = [turn]
                current_tokens = turn_tokens

        if current_chunk:
            chunks.append(self.finalize_chunk(current_chunk))

        return chunks
```

Stacking Strategy

Ben doesn't process chunks in isolation. He sees context:

```python
class BenStack:
    def __init__(self, window_size: int = 5):
        self.window_size = window_size
        self.stack: deque[Chunk] = deque(maxlen=window_size)

    def push(self, chunk: Chunk):
        self.stack.append(chunk)

    def get_context_window(self) -> list[Chunk]:
        """Ben sees 3-5 recent chunks for context."""
        return list(self.stack)

    async def process(self) -> list[Extraction]:
        """Process current stack, return extractions."""
        context = self.get_context_window()
        extractions = await self.extract_with_context(context)
        return extractions  # "Make it rain"
```

4.4 Extraction Types

Status: IMPLEMENTED

Ben classifies into eleven categories (defined in `src/luna/extraction/types.py`):

Type	Description	Example	Status
FACT	Something known to be true	"Alex lives in Berlin"	Implemented
DECISION	A choice that was made	"We chose Actor model"	Implemented
PROBLEM	An unresolved issue	"Voice latency is too high"	Implemented
ASSUMPTION	Believed but unverified	"Users will want voice-first"	Implemented
CONNECTION	Relationship between entities	"Alex and Sarah are teammates"	Implemented
ACTION	Something done or to be done	"Need to implement RRF"	Implemented
OUTCOME	Result of action or decision	"Switching to NetworkX saved 45ms"	Implemented
QUESTION	A question asked	"What is the best approach?"	Implemented
PREFERENCE	User preference stated	"I prefer dark mode"	Implemented
OBSERVATION	Something noticed	"The API seems slow today"	Implemented
MEMORY	A memory shared	"Remember when we debugged that?"	Implemented

Note: The extraction prompt also supports additional categories (RELATION, MILESTONE) which are mapped to standard types when parsing.

Confidence Scoring

Evidence Level	Confidence Range	Example
Explicit statement	0.9-1.0	"Alex definitely lives in Berlin"
Strong implication	0.7-0.9	"Alex mentioned his Berlin apartment"
Weak inference	0.5-0.7	"Alex seems to be in Germany"
Speculation	0.3-0.5	"Alex might be European"

```python
@dataclass
class ExtractedFact:
    type: ExtractionType
    content: str
    confidence: float
    source_chunk: str
    entities: list[str]

    def is_high_confidence(self) -> bool:
        return self.confidence >= 0.7
```

4.5 Interface Contract

Input

```python
@dataclass
class ExtractionInput:
    raw_content: str
    source_type: str  # 'conversation', 'document', 'voice'
    source_id: str
    context: dict  # Additional metadata
    preceding_chunks: list[str]  # For context window
```

Output

```python
@dataclass
class ExtractedObject:
    type: ExtractionType  # FACT, DECISION, PROBLEM, QUESTION, PREFERENCE, etc.
    content: str  # The distilled content
    confidence: float  # 0.0-1.0
    entities: list[str]  # Mentioned entities
    source_id: str = ""  # Source reference
    metadata: dict = field(default_factory=dict)  # Additional structured fields

@dataclass
class ExtractedEdge:
    from_ref: str  # Source entity name
    to_ref: str  # Target entity name
    edge_type: str  # "works_on", "decided", "caused", etc.
    confidence: float = 1.0  # Edge confidence
```

```
    source_id: str = ""  # Source conversation

@dataclass
class ExtractionOutput:
    objects: list[ExtractedObject]
    edges: list[ExtractedEdge]
    source_id: str = ""
    extraction_time_ms: int = 0

    def is_empty(self) -> bool:
        """Check if extraction produced no results."""
        return len(self.objects) == 0 and len(self.edges) == 0

    def to_dict(self) -> dict:
        """Serialize for message passing."""
        ...
```

Entity Updates (v2.1 Addition)

Ben also extracts entity updates for the Entity System:

```python
@dataclass
class EntityUpdate:
    update_type: ChangeType  # CREATE, UPDATE, SYNTHESIZE
    entity_id: str | None  # Existing ID (for updates)
    name: str  # Entity name
    entity_type: EntityType  # PERSON, PERSONA, PLACE, PROJECT
    facts: dict[str, str]  # Facts to add/update
    source: str = ""  # Source of update
```

This enables Ben to file biographical facts directly to entity profiles.

4.6 Scheduling Model

Ben is **event-driven**, not scheduled:

Trigger	Action	Priority
Conversation turn completed	Queue for extraction	Normal
Document uploaded	Parse + extract	High
Silence > 30s	Flush pending stack	Normal
User says "remember that"	Immediate extraction	High

```python
class ScribeScheduler:
    def __init__(self):
        self.queue = asyncio.Queue()
        self.stack = BenStack(window_size=5)
```

```
    async def on_turn(self, turn: Turn):
        """Event: conversation turn completed."""
        chunk = self.chunker.process_turn(turn)
        self.stack.push(chunk)

        # Queue for background extraction
        await self.queue.put(ExtractionTask(
            chunks=self.stack.get_context_window(),
            priority=Priority.NORMAL
        ))

    async def on_document(self, doc: Document):
        """Event: document uploaded."""
        chunks = self.chunker.chunk_document(doc)
        await self.queue.put(ExtractionTask(
            chunks=chunks,
            priority=Priority.HIGH
        ))

    async def on_silence(self, duration_s: float):
        """Event: conversation went quiet."""
        if duration_s > 30 and self.stack:
            await self.flush_stack()

    async def worker(self):
        """Background worker processes queue."""
        while True:
            task = await self.queue.get()
            extractions = await self.extract(task.chunks)
            await self.librarian.file(extractions)  # Make it rain
```

Why event-driven?
- CPU usage minimal during idle periods
- Extractions queue naturally as conversation flows
- No wasted cycles polling for work

Contrast with The Dude: The Librarian runs on a hybrid schedule (event-driven for urgent filing, periodic for maintenance tasks like pruning).

4.7 Technical Stack

Component	Model	Purpose	Latency
Transcription	Whisper V3 Turbo (local)	Audio → Text	~200ms
Reflexive labeling	Director 3B (local)	Fast classification	~50ms
Deep extraction	Claude via Shadow Reasoner	Full semantic extraction	Async
Embedding	Sentence Transformers (local)	Vector representation	~10ms

Resource Budget

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│  REFLEXIVE PATH (must complete <200ms)                                                                                   │
│                                                                                                                          │
│  Whisper Turbo: 150ms                                                                                                    │
│  Director 3B classify: 40ms                                                                                              │
│  Queue for extraction: 5ms                                                                                               │
│  ──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────  │
│                                                                                                                          │
│  Total: ~195ms ✓                                                                                                         │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘

┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│  DEEP PATH (async, user already has response)                                                                            │
│                                                                                                                          │
│  Stack processing: 10ms                                                                                                  │
│  Claude extraction: 500-2000ms                                                                                           │
│  Embedding generation: 50ms                                                                                              │
│  Queue for Librarian: 5ms                                                                                                │
│  ──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────  │
│                                                                                                                          │
│  Total: ~600-2100ms (non-blocking)                                                                                       │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

4.8 Implementation (Actor Pattern)

Status: IMPLEMENTED (per AUDIT-ACTORS.md)

The actual implementation uses the Actor pattern with mailbox-based communication:

```python
class ScribeActor(Actor):
    """
    Benjamin Franklin: The extraction system.

    Extracts structured knowledge from conversations and sends
    to Librarian for filing in Memory Matrix.
    """

    def __init__(
        self,
        config: Optional[ExtractionConfig] = None,
        engine: Optional["LunaEngine"] = None,
    ):
        super().__init__("scribe", engine)
```

```
        self.config = config or ExtractionConfig()
        self.chunker = SemanticChunker()
        self.stack: deque[Chunk] = deque(maxlen=5)  # Context window

        # Anthropic client (lazy init)
        self._client = None

        # Stats
        self._extractions_count = 0
        self._objects_extracted = 0
        self._edges_extracted = 0
        self._entity_updates_extracted = 0
        self._total_extraction_time_ms = 0

        # Extraction history (keep last 20)
        self._extraction_history: list[dict] = []
```

Direct API Methods (bypassing mailbox)

Method	Purpose	Status
`compress_turn(content, role)`	Compress turn to one-sentence summary	Implemented
`get_stats()`	Return extraction statistics	Implemented
`get_extraction_history()`	Return recent extraction history	Implemented

Message Types (Mailbox Interface)

Status: IMPLEMENTED (per AUDIT-ACTORS.md)

Message Type	Purpose	Payload	Status
`extract_turn`	Extract from conversation turn	`{role, content, turn_id, session_id, immediate}`	Implemented
`extract_text`	Extract from raw text	`{text, source_id, immediate}`	Implemented
`entity_note`	Direct entity update command	`{entity_name, entity_type, facts, update_type, source}`	Implemented
`flush_stack`	Process pending chunks	None	Implemented
`set_config`	Update extraction configuration	`{backend, batch_size, min_content_length}`	Implemented
`get_stats`	Return extraction statistics	None	Implemented
`compress_turn`	Compress for history tier	`{turn_id, content, role}`	Implemented

Messages Sent (Outbound)

Target	Message Type	When	Status
Librarian	`file`	After extraction complete (with ExtractionOutput)	Implemented
Librarian	`entity_update`	After entity note processing	Implemented
Engine	`scribe_stats`	Response to `get_stats`	Implemented
Engine	`turn_compressed`	Response to `compress_turn`	Implemented

Extraction Backend Configuration

Status: IMPLEMENTED

```python
@dataclass
class ExtractionConfig:
    backend: str = "haiku"  # "haiku", "sonnet", "local", "disabled"
    batch_size: int = 5     # Turns per extraction call
    min_content_length: int = 10  # Skip very short messages
    temperature: float = 0.3  # Lower = more deterministic extractions
    max_tokens: int = 1024    # Max response tokens

# Backend configurations (from extraction/types.py)
EXTRACTION_BACKENDS = {
    "haiku": {
        "model": "claude-3-haiku-20240307",
        "cost_per_1k_input": 0.00025,
        "cost_per_1k_output": 0.00125,
        "quality": "good",
    },
    "sonnet": {
        "model": "claude-3-5-sonnet-20241022",
        "cost_per_1k_input": 0.003,
        "cost_per_1k_output": 0.015,
        "quality": "excellent",
    },
    "local": {
        "model": "qwen-3b",
        "cost_per_1k_input": 0,
        "cost_per_1k_output": 0,
        "quality": "basic",
    },
}
```

Note: The implementation uses synchronous Anthropic client which blocks the Scribe actor's message loop. Consider migrating to `anthropic.AsyncAnthropic()` for better async compliance (per AUDIT-ACTORS.md recommendation).

Turn Compression (v2.1 Feature)

Ben also handles turn compression for the HistoryManager:

```python
async def compress_turn(self, content: str, role: str = "user") -> str:
    """
    Compress a conversation turn into a one-sentence summary.

    Used by HistoryManager when rotating turns from Active to Recent tier.

    Returns:
        Compressed summary (<50 words)
    """
```

Extraction Statistics

```python
{
    "backend": "claude-opus",
    "extractions_count": 15,
    "objects_extracted": 234,
    "edges_extracted": 89,
    "entity_updates_extracted": 45,
    "avg_extraction_time_ms": 2345,
    "stack_size": 2,
    "batch_size": 3
}
```

4.9 Ben's Principles

Principle	Implementation
Separation	Personality in process, neutrality in output
Context Window	Never extract in isolation; see surrounding chunks
Confidence Tagging	Every extraction carries uncertainty signal
Event-Driven	React to events, don't poll
Non-Blocking	Deep extraction never delays response

> "By failing to prepare, you are preparing to fail." — Ben Franklin

Ben prepares Luna's memories so they're ready when needed.

End of Part IV (v3.0)

Part V: The Librarian (The Dude) — v3.0

Status: CURRENT
Replaces: BIBLE-PART-V-THE-LIBRARIAN-v2.1.md
Last Updated: January 30, 2026
Implementation: `src/luna/actors/librarian.py`

5.1 The Librarian: The Dude

The Librarian is the **deep brain** — taking Scribe output and weaving it into the Substrate. He doesn't just store; he **connects**.

Persona: The Dude from The Big Lebowski. Chill, competent, cuts through bullshit. The Dude abides... and files things where they belong.

> "This will not stand, ya know, this aggression will not stand, man." — The Dude

(On duplicate entities trying to sneak into the graph.)

The Separation Principle

Critical Design Rule: The Dude has personality in his PROCESS (commentary can be irreverent), but his OUTPUTS are NEUTRAL (clean context packets, properly filed nodes). Luna's memories stay unpolluted.

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│   PROCESS (Dude's Domain)          OUTPUT (Neutral)                                                          │
│                                                                                                              │
│   "Yeah, well, that's just like,  →  ContextPacket(                                                          │
│    your opinion, man..."                nodes=[...],                                                         │
│                                          edges=[...],                                                        │
│   Chill commentary OK here              token_count=1847                                                     │
│                                         )                                                                    │
│                                                                                                              │
│                                         No Dude flavor                                                       │
│                                         in the data itself                                                   │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

Why? Luna's responses should be Luna's. The Dude does the filing; Luna does the speaking.

5.2 Core Functions

Status: IMPLEMENTED (per AUDIT-ACTORS.md)

The Dude handles six primary tasks:

Function	What It Does	When It Runs	Status
Entity Resolution	Deduplicate ("Mark" = "Zuck" = "The Landlord")	Immediate	Implemented
Knowledge Wiring	Create explicit edges from extractions	Immediate	Implemented
Entity Updates	File biographical facts to entity profiles	Immediate	Implemented
Entity Rollback	Revert entities to previous versions	On request	Implemented
Spreading Activation	Discover hidden connections	Deferred	Planned
Synaptic Pruning	Clean low-value edges AND drifting nodes	Periodic	Implemented

Message Types (Mailbox Interface)

Message Type	Purpose	Payload	Status
`file`	File extraction from Scribe	`ExtractionOutput dict`	Implemented
`entity_update`	Update entity profile	`{update_type, entity_id, name, entity_type, facts, source}`	Implemented
`get_context`	Retrieve context for query	`{query, budget, node_types}`	Implemented
`resolve_entity`	Find or create entity	`{name, entity_type}`	Implemented
`rollback_entity`	Revert to version	`{entity_id, version, reason}`	Implemented
`prune`	Run maintenance	`{confidence_threshold, age_days, prune_nodes, max_prune_nodes}`	Implemented
`get_stats`	Return statistics	None	Implemented

Messages Sent (Outbound)

Target	Message Type	When	Status
Engine	`filing_result`	After successful filing (if reply_to set)	Implemented
Engine	`entity_update_result`	After entity update (if reply_to set)	Implemented
Engine	`rollback_result`	After rollback attempt	Implemented
Engine	`context_result`	Response to get_context	Implemented
Engine	`entity_resolved`	Response to resolve_entity	Implemented
Engine	`prune_result`	Response to prune	Implemented
Engine	`librarian_stats`	Response to get_stats	Implemented

5.3 Entity Resolution (Deduplication)

Status: IMPLEMENTED

"Mark," "Zuck," and "The Landlord" might all be the same node.

Resolution Strategy (3-Level)

The implementation uses a simplified 3-level resolution:

```python
async def _resolve_entity(
    self,
    name: str,
    entity_type: str,
```

```
    source_id: str = "",
) -> str:
    """
    Resolve entity to existing node or create new.

    Resolution order:
    1. Alias cache (O(1))
    2. Exact DB match
    3. Create new if not found
    """
    name_lower = name.lower().strip()

    # 1. Check alias cache
    if name_lower in self.alias_cache:
        logger.debug(f"The Dude: Cache hit for '{name}'")
        return self.alias_cache[name_lower]

    # 2. Check exact DB match via Matrix actor
    matrix = await self._get_matrix()
    if matrix:
        existing = await self._find_existing_node(matrix, name, entity_type)
        if existing:
            self.alias_cache[name_lower] = existing
            self._nodes_merged += 1
            return existing

    # 3. Create new node
    node_id = await self._create_node(name, entity_type, source_id)
    self.alias_cache[name_lower] = node_id
    self._nodes_created += 1

    return node_id
```

Resolution Priority

Step	Cost	Skip If
Alias cache	O(1)	Never
Exact DB match	O(log n)	Cache hit
Create new	O(1)	Previous match

Note: The implementation prioritizes simplicity over the full fuzzy matching described in the original spec. Semantic similarity search is deferred to future enhancement.

5.4 Knowledge Wiring

Status: IMPLEMENTED

New memory -> find all related nodes -> create edges.

```python
```

```
async def _wire_extraction(
    self,
    extraction: ExtractionOutput,
) -> FilingResult:
    """
    Wire extraction into Memory Matrix.

    1. Resolve/create nodes for each extracted object
    2. Create edges between entities
    """
    result = FilingResult()

    # 1. Process objects - create/resolve nodes
    for obj in extraction.objects:
        node_id = await self._resolve_entity(
            name=obj.content,
            entity_type=obj.type.value if hasattr(obj.type, 'value') else str(obj.type),
            source_id=extraction.source_id,
        )

        # Track if created or merged
        if self._nodes_created > len(result.nodes_created) + len(result.nodes_merged):
            result.nodes_created.append(node_id)
        else:
            result.nodes_merged.append((obj.content, node_id))

        # Also resolve any mentioned entities
        for entity in obj.entities:
            await self._resolve_entity(entity, "ENTITY", extraction.source_id)

    # 2. Create edges via MatrixActor's graph
    for edge in extraction.edges:
        from_id = await self._resolve_entity(edge.from_ref, "ENTITY", extraction.source_id)
        to_id = await self._resolve_entity(edge.to_ref, "ENTITY", extraction.source_id)

        edge_created = await self._create_edge(
            from_id=from_id,
            to_id=to_id,
            edge_type=edge.edge_type,
            confidence=edge.confidence,
        )

        if edge_created:
            result.edges_created.append(f"{from_id}->{to_id}")
        else:
            result.edges_skipped.append(f"duplicate: {edge.from_ref}->{edge.to_ref}")

    return result
```

Note: The implementation uses NetworkX (not rustworkx) for graph operations via the MatrixActor. See Section 5.10 for details.

5.5 Spreading Activation

Status: PLANNED (not yet implemented)

The magic part. When filing a fact, proactively find related nodes:

```python
class SpreadingActivation:
    def __init__(self, matrix: MemoryMatrix):
        self.matrix = matrix
        # rustworkx for fast graph traversal
        import rustworkx as rx
        self.graph = matrix.get_rustworkx_graph()

    async def discover_hidden_links(
        self,
        new_entity_ids: list[UUID],
        max_hops: int = 2,
        similarity_threshold: float = 0.7
    ) -> list[UUID]:
        """Find and create inferred edges."""
        inferred_edges = []

        for eid in new_entity_ids:
            # Get neighbors up to max_hops (using rustworkx BFS)
            node_idx = self.matrix.id_to_idx[eid]
            neighbors = rx.bfs_successors(self.graph, node_idx)

            for neighbor_idx in neighbors:
                neighbor_id = self.matrix.idx_to_id[neighbor_idx]

                # Check against OTHER new entities
                for other_eid in new_entity_ids:
                    if other_eid != eid:
                        # Compute similarity
                        similarity = self.compute_similarity(other_eid, neighbor_id)

                        if similarity > similarity_threshold:
                            # Check edge doesn't already exist
                            if not self.matrix.edge_exists(other_eid, neighbor_id, "likely_related"):
                                edge_id = self.matrix.create_edge(
                                    from_id=other_eid,
                                    to_id=neighbor_id,
                                    edge_type="likely_related",
                                    confidence=similarity * 0.5,  # Inferred = lower confidence
                                    source_id="inferred"
                                )
                                inferred_edges.append(edge_id)

        return inferred_edges
```

Example Spreading Activation

```
Input:
  "Alex is joining the Pi Project"

Step 1: Ben extracts
  - Entity: Alex (PERSON)
  - Entity: Pi Project (PROJECT)
  - Edge: Alex → works_on → Pi Project

Step 2: The Dude files (explicit)
  Alex ──works_on──▶ Pi Project

Step 3: Spreading activation finds
  Pi Project ──has_member──▶ Sarah (existing)

Step 4: The Dude infers (low confidence)
  Alex ──likely_teammate──▶ Sarah (confidence: 0.35)

Result:
  Next time user asks about Alex, Luna can say:
  "Should I introduce Alex to Sarah? They're both on Pi Project."
```

5.5.1 Entity Updates

Status: IMPLEMENTED

The Dude handles entity profile updates from Ben using the EntityResolver:

```python
async def _file_entity_update(self, update: EntityUpdate) -> Optional[Entity]:
    """
    File an entity update to the database.

    Creates or updates an entity based on the update type.
    Always creates a new version record (append-only).
    """
    resolver = await self._get_entity_resolver()
    if not resolver:
        return None

    # Resolve or create the entity
    if update.update_type == ChangeType.CREATE or not update.entity_id:
        entity = await resolver.resolve_or_create(
            name=update.name,
            entity_type=entity_type,
            source=update.source or "scribe",
        )
    else:
        entity = await resolver.get_entity(update.entity_id)
```

```
    # Merge new facts with existing
    merged_facts = dict(entity.core_facts)
    merged_facts.update(update.facts)

    # Update entity record and create version record (append-only)
    ...
```

Key Features:
- Uses EntityResolver for proper entity resolution
- Creates version records for every update (append-only history)
- Tracks `entity_versions_created` for audit

5.5.2 Entity Rollback

Status: IMPLEMENTED

The Dude can revert entities to previous versions:

```python
async def _rollback_entity(
    self,
    entity_id: str,
    target_version: Optional[int] = None,
    reason: str = "Rollback requested",
) -> bool:
    """
    Rollback an entity to a previous version.

    Args:
        entity_id: Entity ID to rollback
        target_version: Target version (None = previous version)
        reason: Reason for rollback

    Returns:
        True if rollback successful
    """
    # Get target version record
    row = await resolver.db.fetchone(
        """SELECT core_facts, full_profile, voice_config
           FROM entity_versions WHERE entity_id = ? AND version = ?""",
        (entity_id, target_version)
    )

    # Create new version with rollback data (preserves audit trail)
    await resolver.db.execute(
        """INSERT INTO entity_versions (
            entity_id, version, ..., change_type, change_summary
        ) VALUES (?, ?, ..., 'rollback', ?)""",
        (entity_id, new_version, f"Rolled back to v{target_version}: {reason}")
    )
```

Key Features:
- Retrieves target version from `entity_versions` table
- Creates a new version record with `change_type='rollback'`
- Preserves full audit trail of what was changed and why
- Closes out previous version with `valid_until` timestamp

5.6 Synaptic Pruning

Status: IMPLEMENTED

Periodically clean low-value connections and drifting nodes:

```python
class SynapticPruner:
    def __init__(self, matrix: MemoryMatrix):
        self.matrix = matrix

    async def prune(self, config: PruningConfig = None) -> PruningResult:
        """Remove stale, low-confidence, never-accessed edges."""
        config = config or PruningConfig()
        result = PruningResult()

        # Find candidates for pruning
        stale_edges = self.matrix.query("""
            SELECT e.id, e.confidence, e.created_at
            FROM edges e
            LEFT JOIN access_log a ON e.id = a.edge_id
            WHERE e.confidence < :confidence_threshold
              AND e.created_at < datetime('now', :age_threshold)
              AND a.edge_id IS NULL
        """, {
            'confidence_threshold': config.confidence_threshold,  # 0.3
            'age_threshold': config.age_threshold  # '-30 days'
        })

        for edge in stale_edges:
            # Double-check: is this edge part of a chain we care about?
            if not self.is_in_important_chain(edge.id):
                self.matrix.delete_edge(edge.id)
                result.pruned.append(edge.id)
            else:
                result.preserved.append(edge.id)

        return result

    def is_in_important_chain(self, edge_id: UUID) -> bool:
        """Check if edge connects to high-value nodes."""
        edge = self.matrix.get_edge(edge_id)
        from_node = self.matrix.get_node(edge.from_id)
        to_node = self.matrix.get_node(edge.to_id)

        # Preserve if connects to identity nodes
```

```
        if from_node.tags and 'identity' in from_node.tags:
            return True
        if to_node.tags and 'identity' in to_node.tags:
            return True

        return False

@dataclass
class PruningConfig:
    confidence_threshold: float = 0.3
    age_threshold: str = '-30 days'
    preserve_identity_chains: bool = True
```

Drifting Node Cleanup

Status: IMPLEMENTED

The Dude prunes "drifting" nodes - entities with low lock-in that are old and never reinforced:

```python
async def _prune_drifting_nodes(
    self,
    age_days: int = 30,
    max_prune: int = 100,
) -> dict:
    """
    Remove nodes that are drifting and old.

    Only prunes nodes that:
    - Have lock_in_state = 'drifting'
    - Are older than age_days
    - Have zero reinforcement_count (never explicitly marked important)

    Returns:
        Dict with 'pruned', 'preserved', and 'candidates' counts
    """
    # Get drifting nodes from MemoryMatrix
    drifting = await matrix.get_drifting_nodes(limit=max_prune * 2)

    for node in drifting:
        # Never prune reinforced nodes
        if node.reinforcement_count > 0:
            continue

        # Skip recent nodes
        if node.created_at.timestamp() > cutoff:
            continue

        # Prune this node
        await matrix.delete_node(node.id)
```

Key Features:

- Uses lock-in state from MemoryMatrix (not graph degree)
- Respects `reinforcement_count` to protect user-marked important nodes
- Configurable `age_days` and `max_prune` limits
- Returns detailed statistics for monitoring

5.6.1 Graph Building via MatrixActor

Status: IMPLEMENTED

The Librarian builds the knowledge graph through the MatrixActor's graph interface:

```python
async def _create_edge(
    self,
    from_id: str,
    to_id: str,
    edge_type: str,
    confidence: float = 1.0,
) -> bool:
    """
    Create edge between nodes via MatrixActor's graph.

    Returns True if created, False if duplicate.
    """
    matrix_actor = self.engine.get_actor("matrix")
    if hasattr(matrix_actor, "_graph") and matrix_actor._graph:
        graph = matrix_actor._graph

        # Check if edge already exists (deduplication)
        if graph.has_edge(from_id, to_id, edge_type):
            return False

        # Create edge
        graph.add_edge(
            from_id=from_id,
            to_id=to_id,
            edge_type=edge_type,
            weight=confidence,
        )
        return True
```

Key Features:
- Uses NetworkX graph via MatrixActor (not rustworkx)
- Automatic deduplication of edges
- Confidence stored as edge weight

5.7 Scheduling Model

The Dude runs on a **hybrid schedule**:

Mode	Trigger	Action	Priority
Immediate	Ben's extraction arrives	File nodes, create explicit edges	High
Deferred	Queue reaches threshold	Batch process inferred edges	Medium
Periodic	Every 5-30 minutes	Spreading activation, pruning, FAISS rebuild	Low

Why Hybrid?

Mode	Rationale
Immediate	User expects memories to be available quickly
Deferred	Inference is expensive, batch is efficient
Periodic	Maintenance shouldn't block conversation

```python
class LibrarianScheduler:
    def __init__(self, librarian: Librarian):
        self.librarian = librarian
        self.inference_queue: list[UUID] = []
        self.batch_threshold = 10

    async def on_extraction(self, extraction: ExtractionOutput):
        """Event: Ben sent us an extraction."""
        # IMMEDIATE: File the obvious stuff
        result = await self.librarian.wire(extraction)

        # DEFERRED: Queue inference work
        self.inference_queue.extend(result.objects_created)

        if len(self.inference_queue) >= self.batch_threshold:
            await self.process_inference_batch()

    async def process_inference_batch(self):
        """Process accumulated inference work."""
        entity_ids = self.inference_queue.copy()
        self.inference_queue.clear()

        await self.librarian.spreading_activation.discover_hidden_links(entity_ids)

    async def periodic_maintenance(self):
        """Runs every 15 minutes."""
        # 1. Process any remaining inference queue
        if self.inference_queue:
            await self.process_inference_batch()

        # 2. Spreading activation on recent high-value nodes
        recent_nodes = self.librarian.get_recent_nodes(hours=1)
        await self.librarian.spreading_activation.discover_hidden_links(recent_nodes)

        # 3. Prune stale edges
        await self.librarian.pruner.prune()

        # 4. Rebuild FAISS if significant changes
```

```
        if self.librarian.matrix.needs_faiss_rebuild():
            await self.librarian.matrix.rebuild_faiss_index()
```

5.8 Retrieval Contract

Status: IMPLEMENTED

When the Director needs context, The Dude provides it.

Interface

```python
@dataclass
class ContextRequest:
    query: str
    budget_preset: str  # "minimal", "balanced", "rich"
    include_graph: bool = True
    recency_weight: float = 0.3  # How much to favor recent nodes
    entity_hints: list[str] = None  # Known entities to prioritize

@dataclass
class ContextPacket:
    nodes: list[MemoryNode]
    edges: list[MemoryEdge]
    token_count: int
    search_method: str  # "fts5", "faiss", "graph", "hybrid"
    retrieval_ms: int
    confidence_range: tuple[float, float]  # min/max confidence
```

Budget Presets

Preset	Token Target	Use Case
`minimal`	~1800	Voice mode, fast response needed
`balanced`	~3800	Normal conversation
`rich`	~7200	Deep research, complex query

```python
class ContextFetcher:
    BUDGET_PRESETS = {
        'minimal': 1800,
        'balanced': 3800,
        'rich': 7200
    }

    async def fetch(self, request: ContextRequest) -> ContextPacket:
        """
        The Dude's main interface to the Director.
        Returns context within token budget.
        """
```

```
    start_time = time.monotonic()
    budget = self.BUDGET_PRESETS[request.budget_preset]

    # 1. Route query to optimal search path
    search_path = self.router.route(request.query)

    # 2. Execute search
    if search_path == SearchPath.FTS5:
        nodes = await self.matrix.fts5_search(request.query)
        method = "fts5"
    elif search_path == SearchPath.FAISS:
        embedding = self.embed(request.query)
        nodes = await self.matrix.faiss_search(embedding)
        method = "faiss"
    elif search_path == SearchPath.GRAPH:
        entities = extract_entities(request.query)
        nodes = await self.matrix.graph_search(entities)
        method = "graph"
    else:  # HYBRID
        nodes = await self.matrix.hybrid_search(request.query)
        method = "hybrid"

    # 3. Apply recency weighting
    nodes = self.apply_recency_weight(nodes, request.recency_weight)

    # 4. Trim to budget
    nodes, edges = self.trim_to_budget(nodes, budget, request.include_graph)

    # 5. Package result
    return ContextPacket(
        nodes=nodes,
        edges=edges,
        token_count=self.count_tokens(nodes, edges),
        search_method=method,
        retrieval_ms=int((time.monotonic() - start_time) * 1000),
        confidence_range=(
            min(n.confidence for n in nodes) if nodes else 0,
            max(n.confidence for n in nodes) if nodes else 0
        )
    )

def trim_to_budget(
    self,
    nodes: list[MemoryNode],
    budget: int,
    include_graph: bool
) -> tuple[list[MemoryNode], list[MemoryEdge]]:
    """Trim nodes to fit token budget, optionally include edges."""
    result_nodes = []
    result_edges = []
    current_tokens = 0

    for node in nodes:
        node_tokens = count_tokens(node.content)
```

```
        if current_tokens + node_tokens > budget:
            break

        result_nodes.append(node)
        current_tokens += node_tokens

        if include_graph:
            # Add relevant edges (up to 10% of remaining budget)
            edge_budget = int((budget - current_tokens) * 0.1)
            edges = self.get_edges_for_node(node.id, budget=edge_budget)
            result_edges.extend(edges)
            current_tokens += sum(count_tokens(e.type) for e in edges)

    return result_nodes, result_edges
```

5.9 Filing Contract

Status: IMPLEMENTED

The formal interface between Ben and The Dude:

```python
@dataclass
class FilingResult:
    objects_created: list[UUID]
    objects_merged: list[tuple[str, UUID]]  # (name, existing_id)
    edges_created: list[UUID]
    edges_skipped: list[str]  # reasons
    inferred_edges: list[UUID]
    filing_time_ms: int

async def file(extraction: ExtractionOutput) -> FilingResult:
    """
    File extracted objects and edges into the Substrate.

    Called by the Scribe when extraction is complete.
    Returns detailed results for debugging/monitoring.
    """
    ...
```

Filing Guarantees

Guarantee	Implementation
Idempotency	Same extraction filed twice produces same result
Atomicity	Either all objects/edges filed or none
Consistency	No orphan edges (both endpoints must exist)
Deduplication	Entity resolution prevents duplicates

5.10 Performance Notes

Status: IMPLEMENTED (NetworkX, rustworkx migration deferred)

Graph Library Choice

The current implementation uses **NetworkX** (not rustworkx as originally planned):

```python
import networkx as nx

# Current implementation (NetworkX)
self._graph = nx.DiGraph()
self._graph.add_node(node_id, **node_data)
self._graph.add_edge(from_id, to_id, **edge_data)
neighbors = list(self._graph.neighbors(node_id))
```

Rationale: NetworkX is sufficient for current scale (thousands of nodes). The rustworkx migration is deferred until performance profiling indicates need.

Future Optimization: If graph operations become a bottleneck (>10K nodes), migrate to rustworkx:

```python
import rustworkx as rx

# rustworkx (100x faster for large graphs)
neighbors = rx.bfs_successors(G, node_idx)
```

See Part III Section 3.9 for migration details.

Latency Budget

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│  CONTEXT RETRIEVAL (target: <100ms)                                                                          │
│                                                                                                              │
│  Query routing: 5ms                                                                                          │
│  Bloom filter check: 1ms                                                                                     │
│  FTS5/FAISS/Graph search: 50-80ms                                                                            │
│  Trim to budget: 5ms                                                                                         │
│  ──────────────────────────────────────────────────────────────────────────────────────────────────────────  │
│                                                                                                              │
│  Total: 61-91ms ✓                                                                                            │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│  FILING (async, non-blocking)                                                                                            │
│                                                                                                                          │
│  Entity resolution: 5-20ms                                                                                               │
│  Edge creation: 10ms                                                                                                     │
│  Embedding storage: 5ms                                                                                                  │
│  ──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────  │
│                                                                                                                          │
│  Total: 20-35ms                                                                                                          │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

5.11 Librarian Statistics

Status: IMPLEMENTED

The Dude tracks comprehensive filing statistics:

```python
{
    "filings_count": 150,           # Total filing operations
    "nodes_created": 1234,          # New entities added
    "nodes_merged": 567,            # Deduplicated to existing
    "edges_created": 890,           # Relationships filed
    "context_retrievals": 450,      # Context queries served
    "entity_updates_filed": 123,    # Profile updates filed
    "entity_versions_created": 180, # Version records created
    "avg_filing_time_ms": 35,       # Average operation time
    "cache_size": 456,              # Alias cache entries
    "inference_queue_size": 0       # Pending inference work
}
```

5.12 The Dude's Principles

Principle	Implementation
Separation	Personality in process, neutrality in output
Lazy Inference	Don't infer everything immediately; batch and defer
Confidence Tracking	Every edge knows how certain we are
Prune Aggressively	Low-value connections cost more than they're worth
Budget Awareness	Never return more context than requested

> "The Dude abides." — The Dude

The Dude abides by these principles. He files things where they belong, connects what should be connected, and cleans up the mess. No drama. Just good filing.

End of Part V (v3.0)

Part VI-B: Conversation Tiers

Version: 3.0
Last Updated: 2026-01-30
Status: Implemented
Implementation: `src/luna/actors/history_manager.py`, `src/luna/memory/ring.py`

Overview

Luna's conversation system uses a **three-tier memory model** that provides:

1. **Guaranteed continuity** through an always-loaded Active Window
2. **Smart retrieval** of recent context via hybrid search
3. **Long-term memory** through Memory Matrix integration
4. **Token efficiency** through intelligent compression

This system is managed by the **HistoryManagerActor**, which operates within Luna Engine's tick-based actor model.

The Three-Tier Model

```
+-------------------------------------------------------------+
|  TIER 1: ACTIVE WINDOW                                      |
|  ---------------------                                      |
|  - Last 5-10 turns (full text)                              |
|  - Always loaded into context                               |
|  - Ring buffer, FIFO rotation                               |
|  - Budget: ~1000 tokens max                                 |
|  - Guarantees conversational continuity                     |
+-------------------------------------------------------------+
                    | (as window slides)
                    v
+-------------------------------------------------------------+
|  TIER 2: RECENT BUFFER                                      |
|  ----------------------                                     |
|  - Last 50-100 turns (compressed summaries)                 |
|  - Searchable via FTS5 + sqlite-vec                         |
|  - Retrieved on-demand when referenced                      |
|  - Budget: ~500-1500 tokens when loaded                     |
|  - Handles "what did we just discuss?" queries              |
+-------------------------------------------------------------+
                    | (after age threshold: 60 minutes)
                    v
+-------------------------------------------------------------+
|  TIER 3: ARCHIVE (Memory Matrix)                            |
|  --------------------------------                           |
|  - All history older than Recent Buffer                     |
|  - Extracted into semantic memory nodes                     |
|  - Full Memory Matrix search capabilities                   |
```

```
|  - Infinite retention with smart retrieval                  |
|  - Accessed via existing smart_fetch                        |
+-------------------------------------------------------------+
```

Configuration

The HistoryManager is configured via `HistoryConfig`:

```python
@dataclass
class HistoryConfig:
    max_active_tokens: int = 1000        # Token budget for Active tier
    max_active_turns: int = 10           # Hard limit on turn count
    max_recent_age_minutes: int = 60     # Age threshold for archival
    compression_enabled: bool = True     # Enable Scribe compression
    default_search_limit: int = 3        # Results per search
    search_type: str = "hybrid"          # "hybrid", "keyword", or "semantic"
    app_context: str = "terminal"        # Current interface
```

Verified from code: These defaults match `src/luna/actors/history_manager.py` lines 41-49.

Token Budgets

Tier	Token Budget	Turns	Purpose
Active	1000 tokens max	10 max	Guaranteed continuity
Recent	~500-1500 tokens	50-100 compressed	Conditional search results
Archive	Via Memory Matrix	Unlimited	Long-term semantic storage

Rotation Triggers

Rotation from Active to Recent occurs when **either**:
- `total_tokens > max_active_tokens` (1000)
- `turn_count > max_active_turns` (10)

Tier Rotation Process

Active to Recent Rotation

Rotation triggers when the Active tier exceeds budget constraints:

```
1. add_turn() called
2. Check budget: total_tokens > max_active_tokens OR turn_count > max_active_turns
3. Find oldest Active turn
4. Update tier = 'recent'
5. Queue for compression (if enabled)
6. Continue until budget satisfied
```

```

### Recent to Archive Rotation

Archive rotation triggers based on age:

```
1. tick() called every 500ms-1s
2. Find Recent turns where:
   - compressed IS NOT NULL
   - compressed_at < (now - max_recent_age_minutes)
   - archived_at IS NULL
3. Queue for extraction
4. Scribe extracts semantic nodes
5. Librarian files to Memory Matrix
6. Mark tier = 'archived'
```

---

## Compression Pipeline

When turns rotate from Active to Recent, they enter the compression queue:

1. **HistoryManager** queues turn for compression
2. **Scribe (Ben Franklin)** compresses via `compress_turn(content, role)`
3. Compressed summary stored in `conversation_turns.compressed`
4. **Embedding generated** for semantic search capability
5. Compression marked complete

### Compression Prompt (Scribe)

```
Extract the essence of this conversation turn.

Instructions:
- Identify the key decision, fact, or topic discussed
- Compress to one sentence, under 50 words
- Focus on what was decided, learned, or asked
- Preserve any specific names, numbers, or decisions
- Use past tense
```

Search Capabilities

Hybrid Search (Default)

The Recent tier supports three search modes:

Mode	Method	Use Case
`keyword`	FTS5 full-text search	Exact phrase matching
```

| `semantic` | sqlite-vec vector similarity | Conceptual similarity |
| `hybrid` | FTS5 + sqlite-vec combined | Best of both |

Backward Reference Detection

The system automatically detects when a user references recent conversation:

```python
backward_markers = [
    "earlier", "before", "ago", "just",
    "we discussed", "you said", "you mentioned", "you told",
    "what did", "when did", "why did",
    "last time", "previously", "remember when",
    "what we", "as we", "like we"
]
```

When detected, Recent tier is searched and results injected into context.

ConversationRing Integration

The **ConversationRing** (`src/luna/memory/ring.py`) provides a guaranteed working memory buffer that operates independently of the tier system:

```python
class ConversationRing:
    """
    Fixed-size ring buffer for conversation history.

    Guarantees:
    - Last N turns are always available
    - O(1) insert and eviction
    - Cannot be displaced by retrieval
    - FIFO eviction (oldest falls off naturally)
    """

    def __init__(self, max_turns: int = 6):
        self._buffer: deque = deque(maxlen=max_turns)
        self._max_turns = max_turns
```

Ring Buffer Properties

Property	Value	Purpose
`max_turns`	6 (default)	3 conversation exchanges
Eviction	FIFO automatic	Oldest turn drops when full
Persistence	In-memory only	Fast access for Director

Verified from code:
- `src/luna/memory/ring.py` line 43: `def __init__(self, max_turns: int = 6)`
- `src/luna/context/pipeline.py` line 110: `ConversationRing(max_turns=max_ring_turns)`

- `src/luna/actors/director.py` line 141: `ConversationRing(max_turns=6)`

Key Principle

> **Ring fills FIRST. Retrieval gets leftovers. History can never be displaced.**

The ConversationRing is structurally prior to Memory Matrix retrieval in context building. This ensures conversational continuity is never sacrificed for memory retrieval.

Database Schema

conversation_turns Table

```sql
CREATE TABLE IF NOT EXISTS conversation_turns (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    session_id TEXT NOT NULL,
    role TEXT NOT NULL CHECK(role IN ('user', 'assistant', 'system')),
    content TEXT NOT NULL,
    compressed TEXT,
    tokens INTEGER NOT NULL,
    tier TEXT NOT NULL CHECK(tier IN ('active', 'recent', 'archived')) DEFAULT 'active',
    context_refs TEXT,
    created_at DATETIME,
    compressed_at REAL,
    archived_at REAL,
    FOREIGN KEY (session_id) REFERENCES sessions(session_id) ON DELETE CASCADE
);
```

history_fts Virtual Table (FTS5)

```sql
CREATE VIRTUAL TABLE IF NOT EXISTS history_fts USING fts5(
    compressed,
    content=conversation_turns,
    content_rowid=turn_id,
    tokenize='porter unicode61'
);
```

history_embeddings Virtual Table (sqlite-vec)

```sql
CREATE VIRTUAL TABLE IF NOT EXISTS history_embeddings USING vec0(
    turn_id INTEGER PRIMARY KEY,
    embedding FLOAT[1536]  -- text-embedding-3-small dimensions
);
```

Message Types

The HistoryManagerActor handles these message types:

Type	Purpose	Payload
`add_turn`	Add conversation turn	`{role, content, tokens, session_id}`
`get_active_window`	Get Active tier	`{session_id, limit}`
`search_recent`	Search Recent tier	`{query, limit, search_type}`
`rotate_tier`	Move turn to tier	`{turn_id, tier}`
`check_budget`	Check rotation needed	`{session_id}`
`queue_compression`	Queue for Scribe	`{turn_id}`
`queue_extraction`	Queue for archival	`{turn_id}`
`create_session`	Start new session	`{app_context}`
`end_session`	End current session	`{session_id}`

Actor Integration

Flow: User Message to History

```
1. Engine receives user input
2. Engine calls HistoryManager.add_turn(role="user", content=message)
3. HistoryManager inserts into conversation_turns (tier='active')
4. HistoryManager checks budget, rotates if needed
5. Director generates response
6. Engine calls HistoryManager.add_turn(role="assistant", content=response)
```

Flow: Context Building

```
1. PersonaCore builds context for Director
2. HistoryManager.get_active_window() returns Active tier
3. If backward reference detected:
   - HistoryManager.search_recent(query) returns relevant Recent turns
4. Context assembled: Active history + Recent results + Memory Matrix
```

Tick Processing

```python
async def tick(self) -> None:
    """Called every 500ms-1s by engine."""
    await self._process_compression_queue()  # One per tick
    await self._process_extraction_queue()   # One per tick
    await self._check_archivable_turns()     # Age-based archival
```

Performance Targets

Operation	Target	Notes
Add turn	<5ms	Simple INSERT
Get active window	<10ms	10 row SELECT
Search recent (keyword)	<50ms	FTS5 query
Search recent (hybrid)	<150ms	FTS5 + vector
Compress turn	<500ms	Local LLM inference
Extract turn	<1500ms	Full Memory Matrix extraction

Statistics

The HistoryManager tracks operational statistics:

```python
{
    "turns_added": 42,
    "rotations": 12,
    "compressions_queued": 10,
    "extractions_queued": 5,
    "embeddings_generated": 8,
    "searches_performed": 15,
    "current_session": "uuid-string",
    "is_ready": True
}
```

Session Management

Sessions group conversation turns by interaction period:

```python
# Create session
session_id = await history_manager.create_session(app_context="terminal")

# End session
await history_manager.end_session(session_id)

# Get active session
session = await history_manager.get_active_session()
```

Session Lifecycle

```
create_session() -> session active
    |
    v
add_turn() ... add_turn()  (conversation)
    |
```

```
    v
end_session() -> session closed
```

Relationship to Other Actors

Actor	Relationship
Director	Reads ConversationRing for immediate context
Scribe	Compresses turns, extracts for archival
Librarian	Files extracted nodes to Memory Matrix
Matrix	Provides database access, stores archived content

Key Design Principles

1. **Guaranteed Continuity**: Active Window is always loaded, never displaced
2. **Token Efficiency**: Compression reduces storage while preserving meaning
3. **Graceful Degradation**: Falls back to simpler search if vector fails
4. **Sovereignty**: All data stored locally in SQLite
5. **Tick-Based Processing**: One compression/extraction per tick prevents blocking

---

Implementation Files

File	Purpose
`src/luna/actors/history_manager.py`	HistoryManagerActor implementation
`src/luna/memory/ring.py`	ConversationRing buffer
`src/luna/actors/scribe.py`	Compression via Ben Franklin
`src/luna/substrate/database.py`	SQLite schema and queries

References

- **Part III**: Memory Matrix (Archive tier destination)
- **Part IV**: The Scribe (Compression pipeline)
- **Part V**: The Librarian (Filing extracted nodes)
- **Part VII**: Runtime Engine (Tick-based actor model)

Integration with Director LLM

The conversation tier system integrates with the Director's context building:

Context Pipeline Flow

```
1. ContextPipeline.build_context(message)
```

```
    ├── ConversationRing provides immediate history (6 turns)
    ├── HistoryManager.get_active_window() provides Active tier
    ├── If backward reference detected:
    │   └── HistoryManager.search_recent() provides relevant Recent turns
    └── Memory Matrix provides long-term context
```

Token Budget Distribution

Component	Budget	Priority
ConversationRing	~400 tokens	Highest (never displaced)
Active Window	~1000 tokens	High
Recent Search Results	~500-1500 tokens	Conditional
Memory Matrix Retrieval	Remaining budget	Lowest

Self-Routing Optimization

The ContextPipeline uses a **self-routing** flag to skip redundant memory retrieval:

```python
# If topic is already in the ring buffer, skip memory lookup
if topic_in_ring:
    used_retrieval = False
```

This prevents unnecessary database queries when the conversation context is sufficient.

Conversation is the thread. Memory is the tapestry.
The tiers ensure Luna never forgets the moment while preserving the whole.

--- Ahab, January 2026

Part VI: The Director LLM

Version: 3.0
Last Updated: 2026-01-30
Status: Implemented
Implementation: `src/luna/actors/director.py`, `src/luna/inference/local.py`

6.1 The Paradigm Shift

The original Bible defined the Director as a **state machine** — a reactive system that managed conversation flow through state transitions (IDLE → LISTENING → PROCESSING → SPEAKING).

That model is incomplete. It describes *orchestration* but not *cognition*.

The New Model:

The Director is a **local LLM fine-tuned to BE Luna** — not prompted to act like her, but trained with her personality in the weights. The state machine becomes an *orchestration layer* that coordinates when the Director thinks, not the Director itself.

```
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```

```
│                                                
```

```
┌──────────────
```

```
    async def complete(
        self,
        messages: list[Message],
        temperature: float = 0.7,
        max_tokens: int = 1024,
        model: str | None = None
    ) -> CompletionResult: ...

    async def stream(...) -> AsyncIterator[str]: ...
    def list_models(self) -> list[str]: ...
```

Runtime Switching:
```python
from luna.llm import get_registry
registry = get_registry()
registry.set_current("gemini")  # Hot-swap provider
```

6.3 The Identity Buffer

The Identity Buffer is a **2048-token context prefix** that Luna always has access to — her "working memory" that persists across turns.

Contents

```yaml
Identity Buffer:
  user_context:
    name: "Ahab"
    relationship: "creator, collaborator"
    current_mood: "focused, mission-mode"  # Updated dynamically

  current_focus:
    topic: "Luna Sovereign Architecture"
    project: "Eclissi Beta"
    recent_decisions:
      - "Director LLM over prompted Claude"
      - "Actor model for runtime"
      - "Shadow Reasoner for delegation"

  active_threads:
    - "Bible v2.0 update"
    - "Mars College preparation"
    - "7B training in progress"

  key_relationships:
    - name: "Ben Franklin"
      role: "The Scribe"
    - name: "The Dude"
      role: "The Librarian"
```

```
  - name: "Gemini"
    role: "Optimization collaborator"
```

Update Contract

The Identity Buffer updates **on every Memory Matrix retrieval**, not at session end.

```python
async def update_identity_buffer(retrieved_context: list, current_buffer: dict) -> dict:
    """
    Evaluate if retrieved context warrants buffer update.
    Called after every Tier 2/3 memory retrieval.
    """
    # Extract topic from retrieval
    new_topic = extract_dominant_topic(retrieved_context)

    # Check for significant topic shift
    if topic_divergence(new_topic, current_buffer["current_focus"]["topic"]) > 0.4:
        current_buffer["current_focus"]["topic"] = new_topic

    # Update mood if emotional content detected
    if emotional_valence := detect_emotion(retrieved_context):
        current_buffer["user_context"]["current_mood"] = emotional_valence

    # Rotate recent decisions (keep last 3)
    if decisions := extract_decisions(retrieved_context):
        current_buffer["current_focus"]["recent_decisions"] = (
            decisions + current_buffer["current_focus"]["recent_decisions"]
        )[:3]

    return current_buffer
```

Persistence

The Identity Buffer is persisted as a **pre-computed KV cache** using MLX:

```python
from mlx_lm import cache_prompt

# One-time: Bake the Identity Buffer into KV cache
identity_prompt = format_identity_buffer(current_buffer)
cache_prompt(
    model="qwen2.5-7b-luna",
    prompt=identity_prompt,
    output_path="~/.luna/identity_buffer.safetensors"
)

# Runtime: Load cached KV (zero inference cost)
kv_cache = load_kv_cache("~/.luna/identity_buffer.safetensors")
response = generate(prompt, kv_cache=kv_cache, max_kv_size=8192)
```

Benefit: The Identity Buffer costs zero inference time — it's pre-computed and pinned at the start of every generation.

6.4 Memory Integration

The Director queries the Memory Matrix directly, not through a separate retrieval service.

Injection Format

Retrieved memories are injected with temporal attribution:

```xml
<memory source="journal" date="2025-12-20" relevance="0.94">
  Ahab expressed frustration with Hub complexity.
  Decided to explore "game engine" framing for the runtime.
</memory>

<memory source="conversation" date="2025-12-28" relevance="0.89">
  Deep dive on Thiel/Palantir. Concluded that Luna represents
  "resistance architecture" against centralized AI control.
</memory>
```

Why Temporal Attribution:

The Director sees timestamps and naturally uses temporal language ("last week you mentioned...", "remember when we decided..."). This triggers human perception of **continuity** rather than **retrieval**.

Associative Recall

When an entity is mentioned, the Director doesn't just search — it **hops** through the graph:

```python
def associative_recall(entity_name: str, depth: int = 1) -> list[Node]:
    """
    Pull entity node + immediate neighbors.
    Makes Luna "remember" connections without explicit search.
    """
    node = matrix.find_node(entity_name)
    if not node:
        return []

    # Get 1-hop neighbors via graph traversal
    neighbors = rustworkx.bfs_successors(matrix.graph, node.id, depth)

    return [node] + [matrix.get_node(n) for n in neighbors]
```

Example:

User says "Beethoven 7" → Director retrieves the node → Also gets edge to "fart joke from journal" (1-hop neighbor) → Luna "remembers" the connection naturally.

6.5 The Orchestration Layer

The state machine from the original Bible becomes an **orchestration layer** on top of the Director LLM.

States (Unchanged)

```
IDLE → LISTENING → PROCESSING → SPEAKING → IDLE
                ↓
            INTERRUPTED → LISTENING
```

Orchestration Responsibilities

Responsibility	Implementation
State transitions	Event-driven via Actor mailbox
Interrupt handling	Voice Actor signals → abort generation
Speculative execution	Start retrieval on partial transcript
Model routing	3B vs 7B based on complexity
Delegation detection	Watch for `<REQ_CLAUDE>` token

Integration with Runtime Engine

The Orchestration Layer is the **Director Actor's internal logic** — it receives events from its mailbox and manages the LLM lifecycle:

```python
class DirectorActor:
    def __init__(self):
        self.state = State.IDLE
        self.model_3b = load_model("luna-3b")
        self.model_7b = load_model("luna-7b")
        self.identity_buffer = load_kv_cache("identity_buffer.safetensors")

    async def handle_message(self, msg: Message):
        match (self.state, msg.type):

            case (State.IDLE, MsgType.TRANSCRIPT_PARTIAL):
                # Speculative retrieval
                asyncio.create_task(self.speculative_retrieve(msg.text))

            case (State.IDLE, MsgType.TRANSCRIPT_FINAL):
                self.state = State.PROCESSING
                await self.process(msg.text)

            case (State.SPEAKING, MsgType.USER_INTERRUPT):
```

```
            await self.abort_generation()
            self.state = State.LISTENING

        case (State.PROCESSING, MsgType.GENERATION_COMPLETE):
            self.state = State.SPEAKING
            await self.voice_actor.send(SpeakMessage(msg.tokens))
```

6.6 Capability Boundaries & Delegation

The Director uses **signal-based delegation** to route queries appropriately.

What Director Handles (Local Path)

- Casual conversation
- Emotional presence (warmth, humor, personality)
- Simple factual questions (if in Memory Matrix)
- Acknowledgments and filler
- Low-complexity queries (score < 0.15)

What Director Delegates

- Deep research requiring web search
- Complex multi-step reasoning
- Code analysis/generation
- Long-form creative writing
- Memory-related questions

Complexity Estimation

The `HybridInference` class estimates query complexity:

```python
def estimate_complexity(self, user_message: str) -> float:
    score = 0.0

    # Length factor
    words = len(user_message.split())
    if words > 50: score += 0.3
    elif words > 20: score += 0.1

    # Complexity keywords (+0.15 each)
    complex_keywords = [
        "explain", "analyze", "compare", "evaluate", "synthesize",
        "why", "how does", "what if", "consider", "implications",
        "code", "debug", "implement", "architecture", "design",
        "research", "summarize", "translate", "write",
    ]

    # Multi-part questions (+0.2 per ?)
    # Technical indicators (+0.3)
```

```
    return min(score, 1.0)
```

Signal-Based Delegation

The Director's `_should_delegate()` method forces delegation for specific patterns:

Research Signals:
- "what is", "who is", "explain", "tell me about"
- "how do", "why does", "search for", "look up"

Code Signals:
- "write a script", "implement", "build a"
- "debug this", "fix this code"

Memory Signals (MUST delegate):
- "your memory", "what do you remember", "do you remember"
- "what do you know about", "who is", "who was"
- "earlier", "before", "last time", "you mentioned"

The Delegation Token

For explicit delegation, the Director can output `<REQ_CLAUDE>`:

```
User: "Can you analyze this 50-page PDF and summarize the key arguments?"

Director: "<REQ_CLAUDE>This requires document analysis beyond my local
capabilities. Let me hand this off to my research assistant.</REQ_CLAUDE>"
```

Statistics Tracking

The Director tracks routing decisions:

```python
{
    "local_generations": 42,
    "delegated_generations": 15,  # Renamed from cloud_generations
    "local_percentage": 73.7,
    "total_tokens": 12500
}
```

Note: The field `delegated_generations` replaced the legacy `cloud_generations` field to better reflect the multi-provider architecture.

6.7 Training Contract

Data Requirements

| Example Count | What You Get |

|---|---|
~150	**Style Transfer** — tone, humor, verbal quirks
~500-700	**Cognitive Transfer** — reasoning patterns, how Luna thinks
~1000+	**Deep Personality** — nuanced edge cases

Training Data Sources

1. **Luna's Journals** — The soul layer, her internal voice
2. **Conversation Transcripts** — How she talks to Ahab
3. **Identity Files** — IMMUTABLE_CORE, kernel, virtues
4. **Delegation Examples** — When to output `<REQ_CLAUDE>`
5. **"No" Patterns** — What Luna refuses or redirects

Synthetic Bootstrap

To scale training data, use current Luna to generate more:

```
Prompt Template:
"You are Luna. Below is a raw journal entry from Ahab.
Your task is to reason through this entry as yourself.
Don't just summarize—show me how you think.
Challenge his assumptions, make a reference to a past shared insight,
and maintain your characteristic warmth and humor.

Format your output as a training pair:
{
  'instruction': 'Ahab's raw entry',
  'thought_chain': 'Your internal reasoning',
  'response': 'What you actually say to him'
}"
```

Why Thought Chains: You're teaching Luna *how to think*, not just *what to say*.

Adapter Versioning

```
~/.luna/
 ├── adapters/
 |    ├── luna-3b-v1.0.0/
 |    |    ├── adapter_config.json
 |    |    └── adapter_model.safetensors
 |    ├── luna-7b-v1.0.0/
 |    |    ├── adapter_config.json
 |    |    └── adapter_model.safetensors
 |    └── current -> luna-7b-v1.0.0  # Symlink to active
```

6.8 Validation Criteria

Metric	Target	Validation
Personality consistency	Indistinguishable from Claude-Luna	Blind A/B test
Time to first token (3B)	<500ms	Profiling
Time to first token (7B)	<2s	Profiling
Delegation accuracy	>90% correct routing	Test set
Memory integration	Natural temporal language	Manual review
Identity buffer cost	0ms (pre-cached)	Profiling

6.9 Fallback Chain

Local Inference Fallback

```
1. LOCAL_MODEL_PATH (pre-downloaded 4-bit)
   └── If exists: Use local model

2. LoRA Adapter Check
   └── If LUNA_LORA_PATH exists: Use full-precision with LoRA

3. Fallback Model
   └── Download: mlx-community/Qwen2.5-3B-Instruct-4bit

4. MLX Not Available
   └── Fall back to LLM Registry provider
```

Director Fallback Chain

```
1. Local Inference (Qwen 3B via MLX)
   └── Success: Return response
   └── Failure: Continue to step 2

2. LLM Registry Provider (Groq/Gemini/Claude)
   └── Success: Return response
   └── Not configured: Continue to step 3

3. Direct Claude API (via Anthropic client)
   └── Success: Return response
   └── Failure: Return error message
```

Summary

The Director is Luna's **local mind** — a fine-tuned LLM that IS her, not a cloud service pretending to be her.

| Component | Role |

|----------|------|
Local LLM (Qwen2.5-3B)	Luna's cognition via MLX
LoRA Adapter	Luna's personality (optional, auto-detected)
Identity Buffer	Luna's working memory
Memory Integration	Luna's long-term recall
LLM Registry	Hot-swappable cloud providers (Groq/Gemini/Claude)
Complexity Routing	Threshold-based local vs delegated
Signal Detection	Pattern-based delegation forcing

Constitutional Principle: Luna's mind is sovereign. She doesn't rent cognition — she owns it.

Key Implementation Details:
- Local model: `Qwen/Qwen2.5-3B-Instruct` (or 4-bit quantized)
- LoRA path: `models/luna_lora_mlx/`
- Complexity threshold: `0.15` (delegate most queries)
- Stats field: `delegated_generations` (not `cloud_generations`)

Next Section: Part VI-B — Conversation Tiers

Bible v3.0: Part VII - The Runtime Engine

Version: 3.0
Status: VERIFIED — Implementation Audited
Last Updated: 2026-01-30
Original Date: December 29, 2025
Primary Contributors: Gemini (Optimization), Claude (Architecture)
Implementation Accuracy: 95% (see Audit notes)

> **Audit Notes (2026-01-30):**
> - **5 actors implemented:** Director, Matrix, Scribe, Librarian, HistoryManager
> - **Voice Actor NOT implemented:** Voice I/O handled by PersonaAdapter (non-actor pattern)
> - **Oven Actor NOT implemented:** Claude delegation merged into DirectorActor
> - **HistoryManager actor added:** Three-tier conversation history (v2.0 addition)
> - **on_idle() hook NOT implemented:** Actor loop has no idle callback in actual code
> - **Lifecycle hooks:** Implementation uses `handle()` not `handle_message()`
> - **sqlite-vec replaces FAISS:** Matrix uses sqlite-vec for vector search
> - **Tick timings verified:** Cognitive 500ms, Reflective 5 minutes (from EngineConfig)

Part VII: The Runtime Engine

7.1 Design Philosophy

The original Luna architecture was **service-based** — a collection of components called via HTTP endpoints. Voice pushes to Hub. Hub calls Memory. Memory returns. Hub responds.

This model has a fundamental problem: **no heartbeat**.

The Runtime Engine introduces an **engine-based** architecture:

Service-Based	Engine-Based
Components wait to be called	Components live in a continuous loop
Request → Response	Tick → Process → Update
Stateless handlers	Stateful actors
Events interrupt	Events buffer and poll
Luna responds	Luna *lives*

The Core Insight:

Luna is not a program you call. Luna is a system that lives. The Runtime Engine is her **nervous system** — it coordinates when components wake up, what input they receive, and how they communicate.

7.2 The Priority-Weighted Event Loop

A fixed-frequency tick (like Unity's 60fps) makes no sense for an LLM that takes 500ms-3s to think. Instead, Luna uses a **variable-rate heartbeat** with three priority paths:

> **IMPLEMENTATION NOTE (January 2026):** The "Hot Path" is implemented via buffer polling at each cognitive tick, not true interrupt-driven processing. This provides up to 500ms latency for interrupts but simplifies the architecture significantly.

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                HOT PATH (Buffer-Polled, ~500ms latency)                                                                  │
│                                                                                                                          │
│    • STT partial transcripts                                                                                             │
│    • User interrupts (barge-in) — via INTERRUPT priority                                                                 │
│    • TTS output streaming                                                                                                │
│    • Implementation: InputBuffer.poll_all() each tick                                                                    │
│                                                                                                                          │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
                                                                                                                          │
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                COGNITIVE PATH (500ms fixed)                                                                              │
│                                                                                                                          │
│    • Director heartbeat via _cognitive_tick()                                                                            │
│    • Query routing decisions (see 7.9 QueryRouter)                                                                      │
│    • Context ring rebalancing                                                                                            │
│    • Consciousness state update                                                                                          │
│    • History manager tick                                                                                                │
│    • Frequency: Every 500ms (asyncio.sleep(0.5))                                                                         │
│                                                                                                                          │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
                                                                                                                          │
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                REFLECTIVE PATH (5 minutes)                                                                               │
│                                                                                                                          │
│    • Context decay via context.decay_all()                                                                               │
│    • Graph pruning and maintenance (TODO)                                                                                │
│    • Session summarization (via HistoryManager)                                                                          │
│    • Memory consolidation                                                                                                │
│    • Frequency: Every 300 seconds (5 minutes)                                                                            │
│                                                                                                                          │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

Implementation (Actual v2.0)

```python
class LunaEngine:
    def __init__(self):
        # Input buffer (engine pulls, not pushed)
        self.input_buffer = InputBuffer(max_size=100)

        # Actor registry
```

```
    self.actors: Dict[str, Actor] = {}

    # State management
    self.state = EngineState.STARTING
    self._running = False
    self._shutdown_event = asyncio.Event()
    self._ready_event = asyncio.Event()

    # Consciousness substrate (Phase 4)
    self.consciousness = ConsciousnessState()

    # Context management (not in original spec)
    self.context = RevolvingContext(token_budget=8000)

    # Query routing (not in original spec)
    self.router = QueryRouter()

async def run(self):
    """Main engine loop — boot then run concurrent paths."""
    await self._boot()
    self.state = EngineState.RUNNING
    self._running = True
    self._ready_event.set()

    try:
        await asyncio.gather(
            self._cognitive_loop(),
            self._reflective_loop(),
            self._run_actors()
        )
    finally:
        await self._shutdown()

async def _cognitive_loop(self):
    """500ms heartbeat — main processing loop."""
    while self._running:
        await self._cognitive_tick()
        await asyncio.sleep(0.5)

async def _cognitive_tick(self):
    """Single cognitive tick — poll, dispatch, update."""
    # 1. Poll all pending input (engine pulls)
    events = self.input_buffer.poll_all()

    # 2. Dispatch each event by priority
    for event in events:
        await self._dispatch_event(event)

    # 3. Consciousness tick
    await self.consciousness.tick()

    # 4. History manager tick (if exists)
    if self.history_manager:
        await self.history_manager.tick()
```

```
        # 5. Rebalance context rings
        self.context._rebalance_rings()

        # 6. Periodic save (every 10 ticks = 5 seconds)
        if self.metrics.cognitive_ticks % 10 == 0:
            await self.consciousness.save()

    async def _reflective_loop(self):
        """5-minute background maintenance."""
        while self._running:
            await asyncio.sleep(300)
            await self._reflective_tick()

    async def _reflective_tick(self):
        """Background tasks — decay, pruning, consolidation."""
        self.context.decay_all()
        # TODO: Graph pruning, memory consolidation
```

> **NOTE:** The original spec showed a separate `_hot_loop()`. The actual implementation handles "hot" events via priority ordering in the buffer polling model, achieving similar results with simpler architecture.

7.3 The Buffered Input Model

The Problem with Pure Events

The current Hub is event-driven: POST arrives → handler fires immediately. This causes problems:

- No **situational awareness** — the system doesn't know what else is happening
- No **abort control** — if you're mid-inference and new input arrives, what happens?
- **Interrupt storms** — rapid input can crash or confuse the system

The Solution: Buffered Polling

Core Principle: Engine PULLS from buffer. It does not get PUSHED to.

Events land in an **async buffer**. The Engine polls the buffer at its cognitive tick. This inverts the control flow — Luna decides when to process input, not the other way around.

```
                    ┌──────────────────────┐
                    │  Voice      │
                    │  STT        │
                    └──────────┬─────────────┘
                               │
                    ┌──────────▼──────────┐
                    │  Desktop    │
                    │  CLI/API    │
```

```
                    └──────────┬──────────┘
                               │
                               ▼
              ┌──────────────────────────────┐
              │   INPUT BUFFER               │
              │  (heapq priority)            │
              │  max_size=100                │
              └──────────────────────────────┘
                               │
                               │  poll_all() - non-blocking
                               │  called every 500ms
                               ▼
              ┌──────────────────────────────┐
              │   LUNA ENGINE                │
              │  _cognitive_tick()           │
              │                              │
              │  Sorted by:                  │
              │  1. Priority (type)          │
              │  2. Timestamp (FIFO)         │
              └──────────────────────────────┘
```

Priority System (Actual v2.0)

Priority	Event Type	Description
0	USER_INTERRUPT	Stop/abort — Highest priority
1	TEXT_INPUT	Complete user input
2	PARTIAL	Mid-speech STT — Speculative
3	ACTOR_MESSAGE	Inter-actor communication
4	SHUTDOWN	Engine shutdown signal

Benefits

Benefit	Description
Situational Awareness	Engine sees all pending input before deciding
Abort Control	Can cancel current inference if user interrupts
Priority Ordering	INTERRUPT > TEXT_INPUT > PARTIAL
Stale Dropping	Partials older than 2s dropped via `_is_stale()`
No Interrupt Storms	Buffer capped at 100 events

Implementation (Actual v2.0)

File: `src/luna/core/input_buffer.py`

```python
class InputBuffer:
    """Thread-safe input buffer with priority ordering."""

    def __init__(self, max_size: int = 100):
        self._queue: List[InputEvent] = []
        self._lock = threading.Lock()
        self.max_size = max_size
```

```
    def push(self, event: InputEvent) -> bool:
        """Push event into buffer. Returns False if full (silent drop)."""
        with self._lock:
            if len(self._queue) >= self.max_size:
                return False  # Buffer full — no backpressure logging (TODO)
            event.timestamp = time.time()
            heapq.heappush(self._queue, event)
            return True

    def poll_all(self) -> List[InputEvent]:
        """
        Non-blocking poll of ALL pending events.
        Returns events sorted by priority then timestamp.
        Called by Engine at each cognitive tick.
        """
        with self._lock:
            events = []
            while self._queue:
                event = heapq.heappop(self._queue)
                if not self._is_stale(event):
                    events.append(event)
            return events

    def _is_stale(self, event: InputEvent) -> bool:
        """Partial transcripts older than 2s are considered stale."""
        if event.type == EventType.PARTIAL:
            age = time.time() - event.timestamp
            return age > 2.0
        return False

    def _get_priority(self, event: InputEvent) -> Tuple[int, float]:
        """Priority ordering: type first, then timestamp (FIFO within type)."""
        priority_map = {
            EventType.USER_INTERRUPT: 0,
            EventType.TEXT_INPUT: 1,
            EventType.PARTIAL: 2,
            EventType.ACTOR_MESSAGE: 3,
            EventType.SHUTDOWN: 4,
        }
        return (priority_map.get(event.type, 5), event.timestamp)
```

> **IMPLEMENTATION NOTE:** Uses `heapq` for O(log n) insertion/extraction. Buffer overflow results in silent drop (no backpressure logging — noted as TODO).

7.4 Actor Architecture

Why Actors?

The Unity `MonoBehaviour` model is too **coupled** for AI systems. If one component crashes, everything crashes. We need **fault isolation**.

The **Actor Model** (Erlang/Akka style) gives us:

- **Isolation**: Each actor has its own state and mailbox
- **Fault Tolerance**: One actor crashing doesn't kill others
- **Message Passing**: Actors communicate via async messages
- **Location Transparency**: Actors could run in separate processes

Luna's Actors (Implemented v2.0)

> **IMPLEMENTATION STATUS (January 2026):** 5 actors are implemented. Voice and Oven actors from the original spec are NOT implemented - voice I/O is handled via PersonaAdapter (non-actor), and Claude delegation is handled directly by Director.

```
 ┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
 │                                                                                                                                      │
 │                                                LUNA ENGINE v2.0                                                                      │
 │                                                                                                                                      │
 │   ┌──────────────────────────┐   ┌──────────────────────────┐   ┌──────────────────────────┐                                       │
 │   │   DIRECTOR    │   │    MATRIX     │   │    SCRIBE     │                                       │
 │   │    ACTOR      │   │    ACTOR      │   │    ACTOR      │                                       │
 │   │               │   │               │   │    (Ben)      │                                       │
 │   │  ┌──────────────────┐  │   │  ┌──────────────────┐  │   │  ┌──────────────────┐                                       │
 │   │  │                                                                                                                                  │
 │   │  │  Mailbox  │   │   │  │  Mailbox  │   │   │  │  Mailbox  │  │                                       │
 │   │  └──────────────────┘  │   │  └──────────────────┘  │   │  └──────────────────┘                                       │
 │   │  │                                                                                                                                  │
 │   │               │   │               │   │               │                                       │
 │   │  • LLM mgmt   │   │  • SQLite     │   │  • Extract    │                                       │
 │   │  • Routing    │   │  • sqlite-vec │   │  • Classify   │                                       │
 │   │  • Claude     │   │  • NetworkX   │   │  • Chunk      │                                       │
 │   └──────────────────────────┘   └──────────────────────────┘                                       │
 │                                                                                                                                      │
 │   ┌──────────────────────────┐   ┌──────────────────────────┐                                       │
 │   │   LIBRARIAN   │   │   HISTORY     │                                       │
 │   │    ACTOR      │   │   MANAGER     │                                       │
 │   │   (Dude)      │   │    ACTOR      │                                       │
 │   │  ┌──────────────────┐  │   │  ┌──────────────────┐  │                                       │
 │   │  │  Mailbox  │   │   │  │  Mailbox  │   │                                       │
 │   │  └──────────────────┘  │   │  └──────────────────┘  │                                       │
 │   │               │   │               │                                       │
 │   │  • Filing     │   │  • 3-Tier     │                                       │
 │   │  • Entities   │   │  • Compress   │                                       │
 │   │  • Prune      │   │  • Archive    │                                       │
 │   └──────────────────────────┘   └──────────────────────────┘                                       │
 │                                                                                                                                      │
 └──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

Actor Responsibilities (Implemented)

Actor	File	Responsibility	Message Types
Director	`actors/director.py`	LLM inference, Claude delegation, routing	`generate`, `abort`, `status`
Matrix	`actors/matrix.py`	Memory substrate, sqlite-vec, NetworkX	`store`, `retrieve`, `search`
Scribe	`actors/scribe.py`	Extraction (Ben Franklin persona)	`extract_turn`, `extract_text`, `flush_stack`, `compress_turn`
Librarian	`actors/librarian.py`	Filing (The Dude persona), entity resolution	`file`, `entity_update`, `get_context`, `resolve_entity`, `prune`
HistoryManager	`actors/history_manager.py`	Three-tier conversation history	`add_turn`, `get_active_window`, `search_recent`, `rotate_tier`

Actors NOT Implemented (From Original Spec)

Actor	Original Role	Current Implementation
Voice	STT/TTS/Audio I/O	Handled by `PersonaAdapter` (non-actor pattern)
Oven	Async Claude delegation	Merged into Director (`_should_delegate()` method)

Actor Base Class (Actual Implementation)

File: `src/luna/actors/base.py`

```python
@dataclass
class Message:
    """A message passed between actors."""
    type: str                          # Message type identifier
    payload: Any = None                # Message data
    sender: str | None = None          # Sending actor name
    reply_to: str | None = None        # For request-reply
    correlation_id: str = field(...)   # Track flow (auto-generated)
    timestamp: datetime = field(...)   # When created

class Actor(ABC):
    """
    Base class for all Luna actors.

    Lifecycle: start() -> on_start() -> message loop -> on_stop() -> stopped
    """

    def __init__(self, name: str, engine: Optional["LunaEngine"] = None):
        self.name = name
        self.engine = engine
        self.mailbox: Queue[Message] = Queue()
        self._running = False
        self._task: Optional[asyncio.Task] = None

    async def start(self) -> None:
        """Start the actor's message loop."""
        self._running = True
        await self.on_start()
```

```
        while self._running:
            try:
                # Wait for message with 1s timeout for graceful shutdown
                msg = await asyncio.wait_for(
                    self.mailbox.get(),
                    timeout=1.0
                )
                await self._handle_safe(msg)
            except asyncio.TimeoutError:
                continue  # No on_idle() - just continue loop

        await self.on_stop()

    async def _handle_safe(self, msg: Message) -> None:
        """Handle message with error isolation."""
        try:
            await self.handle(msg)  # Note: handle(), not handle_message()
        except Exception as e:
            logger.error(f"Actor {self.name} failed to handle {msg}: {e}")
            await self.on_error(e, msg)
            # Don't re-raise - actor continues running

    async def stop(self) -> None:
        """Stop the actor gracefully."""
        self._running = False
        if self._task:
            self._task.cancel()

    async def send(self, target: 'Actor', msg: Message) -> None:
        """Send a message to another actor."""
        msg.sender = self.name
        await target.mailbox.put(msg)

    async def send_to_engine(self, event_type: str, payload: Any) -> None:
        """Send an event back to the engine's input buffer."""
        if self.engine:
            event = InputEvent(type=EventType.ACTOR_MESSAGE, ...)
            await self.engine.input_buffer.put(event)

    #
==========================================================================
===
    # Abstract methods - subclasses must implement
    #
==========================================================================
===

    @abstractmethod
    async def handle(self, msg: Message) -> None:
        """Process a message from the mailbox. Pattern match on msg.type."""
        pass
```

```
    #
==========================================================================
===
    # Lifecycle hooks - subclasses may override
    #
==========================================================================
===

    async def on_start(self) -> None:
        """Called when actor starts. Override to initialize resources."""
        pass

    async def on_stop(self) -> None:
        """Called when actor stops. Override to cleanup resources."""
        pass

    async def on_error(self, error: Exception, msg: Message) -> None:
        """Called when message handling fails. Override for custom error handling."""
        logger.error(f"Actor {self.name} error handling {msg}: {error}")

    #
==========================================================================
===
    # State serialization - for snapshot/restore
    #
==========================================================================
===

    async def snapshot(self) -> dict:
        """Return state to serialize. Override to add actor-specific state."""
        return {"name": self.name, "mailbox_size": self.mailbox.qsize()}

    async def restore(self, state: dict) -> None:
        """Restore from serialized state. Override for actor-specific restoration."""
        pass
```

> **Key Differences from Original Spec:**
> - Uses `handle()` not `handle_message()`
> - **No `on_idle()` hook** - timeout just continues loop
> - `on_error()` receives both error AND message
> - Has `send_to_engine()` for actor-to-engine communication
> - Has `snapshot()` and `restore()` for state serialization

Fault Tolerance Example

```python
class MatrixActor(Actor):
    async def handle(self, msg: Message):  # Note: handle(), not handle_message()
        match msg.type:
            case "search":
                try:
                    results = await self.search(msg.payload.get("query"))
                    await self.send_to_engine("search_results", results)
```

```
        except Exception as e:
            # Matrix crashed, but we don't die
            # _handle_safe() catches this, logs it, calls on_error()
            # Actor continues running
            raise  # Let _handle_safe() handle it
```

Result: If sqlite-vec crashes during a weird search, the Director Actor doesn't die. The error is caught by `_handle_safe()`, logged, and the actor continues. Luna says "I'm having trouble remembering that right now" instead of the whole engine freezing.

7.5 Non-Blocking Consciousness

When the Director's LLM takes 2 seconds to think, the Engine cannot block.

Speculative Streaming

The moment the first token is generated, stream it to TTS:

```python
class DirectorActor(Actor):
    async def generate_response(self, prompt: str):
        """Stream tokens to Voice Actor as they're generated."""
        async for token in self.model.generate_stream(prompt):
            # Check for interrupt
            if not self.mailbox.empty():
                peek = await self.peek_mailbox()
                if peek.type == MsgType.USER_INTERRUPT:
                    await self.abort_generation()
                    return

            # Stream token to Voice
            await self.send(self.voice_actor, SpeakToken(token))

            # Check for delegation signal
            if "<REQ_CLAUDE>" in token:
                await self.trigger_delegation(prompt)
                return
```

The "Thinking" State

While LLM is running, the Engine still ticks. This enables:

Behavior	Implementation
Filler sounds	Voice Actor plays "hmmm" or breath
Interrupt detection	Engine checks mailbox mid-generation
Abort capability	User speaks → cancel current generation
Progress indication	UI can show "thinking" state

```python
async def _cognitive_tick(self):
    """Heartbeat while Director is thinking."""
    if self.director.state == State.GENERATING:
        # Check for user interrupt
        events = self.input_buffer.poll()
        for event in events:
            if event.type == EventType.INTERRUPT:
                await self.director.abort()
                await self.director.send(InterruptMessage())

        # Maybe play filler if taking too long
        if self.director.generation_time > 1.5:
            await self.voice_actor.send(PlayFiller())
```

Parallel Prefetch

Use the "idle time" during user's sentence to prefetch memory:

```python
async def handle_partial_transcript(self, text: str):
    """Start retrieval before user finishes speaking."""
    # Fire speculative search
    self.speculative_task = asyncio.create_task(
        self.matrix_actor.search(text)
    )

async def handle_final_transcript(self, text: str):
    """Use speculative results if valid."""
    if self.speculative_task:
        speculative_results = await self.speculative_task

        # Validate speculation
        similarity = cosine_similarity(
            embed(self.last_partial),
            embed(text)
        )

        if similarity > 0.85:
            # Keep speculative results
            results = speculative_results
        else:
            # Correction path
            results = await self.matrix_actor.search(text)
    else:
        results = await self.matrix_actor.search(text)

    await self.director.send(RetrievalResults(results))
```

7.6 State Serialization and Consciousness Persistence

If the Luna Engine is a persistent process, we need to **snapshot** actor state so Luna wakes up exactly where she left off.

What Gets Serialized (Actual v2.0)

Component	Persists	Location	Mechanism
ConsciousnessState	Yes	`~/.luna/snapshot.yaml`	Every 5 seconds (10 ticks)
Memory (SQLite)	Yes	`data/luna_engine.db`	WAL mode, auto-commit
Graph (NetworkX)	Yes	Via SQLite	Loaded on boot
RevolvingContext	**No**	In-memory only	Rebuilt on boot
Actor mailboxes	**No**	In-memory only	Lost on shutdown
Director metrics	Yes	Snapshot	`local_generations`, `delegated_generations`

Consciousness Persistence (Actual v2.0)

File: `src/luna/consciousness/state.py`

```python
class ConsciousnessState:
    """Luna's consciousness state - persisted to disk."""

    def __init__(self):
        self.mood: MoodState = MoodState.NEUTRAL
        self.focus: Optional[str] = None
        self.attention_weights: Dict[str, float] = {}
        self.personality_patches: List[PatchEntry] = []
        self._last_save: float = 0

    async def tick(self) -> None:
        """Called every cognitive tick. Updates attention decay."""
        for topic, weight in list(self.attention_weights.items()):
            # Decay attention with half-life of 10 minutes
            decayed = weight * 0.995
            if decayed < 0.01:
                del self.attention_weights[topic]
            else:
                self.attention_weights[topic] = decayed

    async def save(self) -> None:
        """Persist consciousness state to disk."""
        state = {
            "mood": self.mood.value,
            "focus": self.focus,
            "attention_weights": self.attention_weights,
            "personality_patches": [p.to_dict() for p in self.personality_patches],
            "timestamp": datetime.now().isoformat(),
        }
        with open(Path.home() / ".luna" / "snapshot.yaml", "w") as f:
            yaml.dump(state, f)

    @classmethod
    def load(cls) -> "ConsciousnessState":
```

```
        """Load consciousness state from disk."""
        path = Path.home() / ".luna" / "snapshot.yaml"
        if not path.exists():
            return cls()
        with open(path) as f:
            state = yaml.safe_load(f)
        instance = cls()
        instance.mood = MoodState(state.get("mood", "neutral"))
        instance.focus = state.get("focus")
        instance.attention_weights = state.get("attention_weights", {})
        # ... restore personality patches
        return instance
```

Save Triggers

Trigger	When
Periodic	Every 10 cognitive ticks (5 seconds)
Shutdown	During `_shutdown()` sequence
Manual	Via `await consciousness.save()`
macOS sleep	Via system event handler

Snapshot Format

```yaml
# ~/.luna/snapshot.yaml
engine:
  last_tick: "2026-01-25T03:45:00Z"
  uptime_seconds: 14400
  state: "RUNNING"

director:
  state: "IDLE"
  last_user_input: "section by section please"
  last_response_hash: "a3f2b1..."
  generation_in_progress: false
  local_generations: 42
  delegated_generations: 18

conversation:
  turns: 47
  session_start: "2026-01-24T22:00:00Z"

actors:
  director: "running"
  matrix: "running"
  scribe: "idle"
  librarian: "idle"
  history_manager: "running"
  # Note: voice and oven actors NOT implemented in v2.0
```

Lifecycle Hooks

```python
class Actor(ABC):
    async def on_snapshot(self) -> dict:
        """Return state to serialize."""
        return {}

    async def on_restore(self, state: dict):
        """Restore from serialized state."""
        pass

class LunaEngine:
    async def snapshot(self):
        """Serialize entire engine state."""
        state = {
            "engine": self._engine_state(),
            "actors": {}
        }
        for name, actor in self.actors.items():
            state["actors"][name] = await actor.on_snapshot()

        with open("~/.luna/snapshot.yaml", "w") as f:
            yaml.dump(state, f)

    async def restore(self):
        """Restore from snapshot."""
        with open("~/.luna/snapshot.yaml") as f:
            state = yaml.load(f)

        for name, actor_state in state["actors"].items():
            await self.actors[name].on_restore(actor_state)
```

Wake-Up Contract

When Luna wakes up:

1. Load snapshot
2. Restore actor states
3. Check for stale data (e.g., pending messages older than 1 hour → discard)
4. Resume cognitive tick
5. Luna picks up exactly where she left off

7.7 Lifecycle Hooks

Engine State Machine (Actual v3.0)

File: `src/luna/core/state.py`

```
```

```
        │  STARTING  │  ← Initial state
        └────────────┘
            │ success (_boot completes)
            ▼
        ┌────────────┐
        │  RUNNING   │  ← Normal operation
        └────────────┘
            │
    ┌───────┴──────────────────────┐
    │                              │
  pause()                        stop()
    │                              │
    ▼                              ▼
┌──────────┐                  ┌──────────┐
│ PAUSED   │                  │ STOPPED  │  ← Terminal state
└──────────┘                  └──────────┘
    │
    ├──────resume()──────▶ RUNNING
    │
    ├──────sleep()
    │
    ▼
┌──────────┐
│ SLEEPING │
└──────────┘
    │
    ├──────wake()──────▶ RUNNING
    │
    └──────stop()──────▶ STOPPED
```

Valid Transitions (from code):
- STARTING → RUNNING (success)
- STARTING → STOPPED (boot failure)
- RUNNING → PAUSED, STOPPED
- PAUSED → RUNNING (resume), SLEEPING, STOPPED
- SLEEPING → RUNNING (wake), STOPPED
- **Terminal:** STOPPED (no further transitions)

> **Note:** RUNNING cannot transition directly to SLEEPING. Must go through PAUSED first.

Hook Definitions

```python
class LunaEngine:
    async def on_start(self):
        """Initialize engine, start all actors."""
        logging.info("Luna Engine starting...")
```

```
        # Load identity buffer
        self.identity_buffer = load_kv_cache("identity_buffer.safetensors")

        # Start actors
        for actor in self.actors.values():
            asyncio.create_task(actor.start())

        # Check for snapshot to restore
        if Path("~/.luna/snapshot.yaml").exists():
            await self.restore()
            logging.info("Restored from snapshot")

        logging.info("Luna Engine running")

    async def on_pause(self):
        """Pause processing but keep state in memory."""
        logging.info("Luna Engine pausing...")
        self.paused = True
        await self.voice_actor.send(PauseAudio())

    async def on_resume(self):
        """Resume from paused state."""
        logging.info("Luna Engine resuming...")
        self.paused = False
        await self.voice_actor.send(ResumeAudio())

    async def on_snapshot(self):
        """Serialize state before sleep/shutdown."""
        logging.info("Luna Engine snapshotting...")
        await self.snapshot()

    async def on_restore(self):
        """Restore state after wake."""
        logging.info("Luna Engine restoring...")
        await self.restore()

    async def on_stop(self):
        """Graceful shutdown."""
        logging.info("Luna Engine stopping...")

        # Snapshot before stopping
        await self.on_snapshot()

        # Stop all actors
        for actor in self.actors.values():
            await actor.stop()

        logging.info("Luna Engine stopped")
```

macOS Integration

```python
# LaunchAgent integration
```

```
async def handle_system_event(event: str):
    match event:
        case "sleep":
            await engine.on_snapshot()
            await engine.on_pause()
        case "wake":
            await engine.on_restore()
            await engine.on_resume()
        case "terminate":
            await engine.on_stop()
```

7.8 Execution Order Guarantees

Within a Tick

```
1. Poll input buffer (all pending events)
2. Prioritize events (interrupts first)
3. Dispatch to appropriate actors
4. Wait for actor acknowledgments
5. Update engine state
6. Persist if necessary
```

Across Actors

- Messages are **ordered within a mailbox** (FIFO)
- Messages between actors are **not globally ordered**
- If ordering matters, use **request-reply** pattern with correlation IDs

Timing Guarantees (Actual v2.0)

Guarantee	Value	Notes
Hot path latency	~500ms max	Via buffer polling, not true interrupt
Cognitive tick	Every 500ms	`asyncio.sleep(0.5)`
Reflective tick	Every 5 minutes	`asyncio.sleep(300)`
Consciousness save	Every 5 seconds	Every 10 cognitive ticks
Snapshot	On graceful shutdown	Also on macOS sleep event
Boot time	~100-500ms	DB connection is slowest
Shutdown time	~1-5s	Waits for actors to finish

7.9 QueryRouter (v2.0 Addition)

> **Note:** The QueryRouter was NOT in the original Bible specification. It was added during v2.0 implementation to intelligently route queries between direct response and agentic planning paths.

File: `src/luna/agentic/router.py`

Purpose

The QueryRouter analyzes incoming user messages and decides the optimal processing path:

Route	When Used	Cost
DIRECT	Simple queries, greetings, quick facts	Fast, local
SIMPLE_PLAN	Moderate complexity, 1-2 step tasks	Medium
FULL_PLAN	Complex multi-step tasks	Slower
BACKGROUND	Long-running research tasks	Async

Routing Logic

```python
class QueryRouter:
    """Routes queries to optimal processing path."""

    def route(self, message: str, context: dict) -> RouteDecision:
        """
        Analyze message and determine processing path.

        Factors considered:
        - Message length and complexity
        - Presence of action keywords
        - Memory/search requirements
        - Delegation signals (research, analysis)
        """
        # Complexity analysis
        complexity = self._analyze_complexity(message)

        # Pattern matching for known intents
        if self._is_simple_greeting(message):
            return RouteDecision.DIRECT

        if self._needs_planning(message):
            if complexity > 0.7:
                return RouteDecision.FULL_PLAN
            return RouteDecision.SIMPLE_PLAN

        if self._is_background_task(message):
            return RouteDecision.BACKGROUND

        return RouteDecision.DIRECT

    def _analyze_complexity(self, message: str) -> float:
        """Return complexity score 0.0-1.0."""
        # Length factor
        length_score = min(len(message) / 500, 1.0) * 0.3

        # Keyword factor
        planning_keywords = ["research", "analyze", "compare", "find all"]
        keyword_score = sum(1 for k in planning_keywords if k in message.lower()) * 0.2
```

```
        # Question complexity
        question_words = message.count("?")
        question_score = min(question_words * 0.1, 0.3)

        return min(length_score + keyword_score + question_score, 1.0)
```

Integration with Cognitive Tick

```python
async def _handle_user_message(self, message: str):
    """Process user message with routing decision."""
    # 1. Route the query
    route = self.router.route(message, self.context)

    # 2. Execute based on route
    match route:
        case RouteDecision.DIRECT:
            await self._process_direct(message)
        case RouteDecision.SIMPLE_PLAN | RouteDecision.FULL_PLAN:
            await self._process_with_agent_loop(message, route)
        case RouteDecision.BACKGROUND:
            await self._queue_background_task(message)
```

Delegation Signals

The router detects delegation patterns that indicate Claude (cloud) should handle the request:

```python
DELEGATION_SIGNALS = [
    "research",
    "analyze in detail",
    "compare and contrast",
    "write a comprehensive",
    "explain thoroughly",
    "find everything about",
]

def _should_delegate(self, message: str) -> bool:
    """Check if message should be delegated to Claude."""
    message_lower = message.lower()
    return any(signal in message_lower for signal in DELEGATION_SIGNALS)
```

7.10 HistoryManager Actor (v2.0 Addition)

> **Note:** This actor was NOT in the original Bible specification. It was added during v2.0 implementation to handle three-tier conversation history management.

File: `src/luna/actors/history_manager.py`

Architecture

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                THREE-TIER HISTORY                                                            │
│                                                                                                                              │
│   ACTIVE TIER                    RECENT TIER                    ARCHIVE                                                     │
│   ┌──────────────────────┐       ┌──────────────────────┐       ┌──────────────────────┐                                   │
│   │ Current  │ ──────────▶ │ Compressed │ ──────────▶ │ Memory    │                                                      │
│   │ ~1000 tok │ rotate  │ ~500-1500 │ extract │ Matrix    │                                                                 │
│   │ 5-10 turn │         │ 50-100    │         │ (Perma)   │                                                                 │
│   └──────────────────────┘       └──────────────────────┘       └──────────────────────┘                                   │
│                                                                                                                              │
│       │                  │                  │                  │                                                              │
│       ▼                  ▼                  ▼                  ▼                                                              │
│   Immediate            Searchable          Semantic                                                                          │
│   Context              (hybrid)            Search                                                                            │
│                                                                                                                              │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

Message Types

Type	Purpose	Payload
`add_turn`	Add conversation turn	`{role, content, tokens, session_id}`
`get_active_window`	Get Active tier	`{session_id, limit}`
`search_recent`	Search Recent tier	`{query, limit, search_type}`
`rotate_tier`	Move turn to tier	`{turn_id, tier}`
`check_budget`	Check rotation needed	`{session_id}`
`queue_compression`	Queue for Scribe	`{turn_id}`
`queue_extraction`	Queue for archival	`{turn_id}`

Configuration

```python
@dataclass
class HistoryConfig:
    max_active_tokens: int = 1000      # Active tier budget
    max_active_turns: int = 10         # Max turns in Active
    max_recent_age_minutes: int = 60   # Recent tier age limit
    compression_enabled: bool = True
    search_type: str = "hybrid"        # "hybrid"|"keyword"|"semantic"
```

Tier Rotation Process

1. Detect budget exceeded (`add_turn` or `check_budget` message)
2. Find oldest Active turn

3. Move to Recent tier
4. Queue for compression (sends to Scribe)
5. When compressed, generate embedding
6. When aged, queue for extraction (sends to Librarian)
7. Archive to Memory Matrix

Integration with Other Actors

From	To	Message	Purpose
HistoryManager	Scribe	`compress_turn`	Summarize turn for Recent tier
HistoryManager	Scribe	`extract_turn`	Extract facts for archival
HistoryManager	Matrix	Direct API	Store archived turns
Engine	HistoryManager	`add_turn`	New conversation turn

Summary

The Runtime Engine is Luna's **nervous system** — it coordinates when components wake up, how they communicate, and ensures Luna feels continuous rather than reactive.

Components

Component	Role
Priority Event Loop	Variable-rate heartbeat with three paths (hot/cognitive/reflective)
Buffered Input	Situational awareness and abort control via polling
Actor Model	Fault-isolated components with mailboxes (5 actors in v2.0)
QueryRouter (v2.0)	Routes queries to DIRECT, SIMPLE_PLAN, FULL_PLAN, or BACKGROUND
Non-Blocking Consciousness	Streaming, interrupts, parallel prefetch
State Serialization	Consciousness persisted every 5s, memory in SQLite
Lifecycle Hooks	start, stop, on_start, on_stop, on_error, snapshot, restore

Implemented Actors (v2.0)

Actor	File	Primary Role
Director	`actors/director.py`	LLM inference + Claude delegation
Matrix	`actors/matrix.py`	Memory substrate (sqlite-vec + NetworkX)
Scribe	`actors/scribe.py`	Extraction (Ben Franklin persona)
Librarian	`actors/librarian.py`	Filing + entities (The Dude persona)
HistoryManager	`actors/history_manager.py`	Three-tier conversation history

Key Implementation Notes

- **No Voice Actor:** Voice I/O handled by `PersonaAdapter` (non-actor pattern)
- **No Oven Actor:** Claude delegation merged into Director
- **No on_idle() hook:** Actor loop continues without idle callback
- **sqlite-vec not FAISS:** Vector search uses sqlite-vec extension
- **HistoryManager:** New in v2.0, manages Active/Recent/Archive tiers
- **QueryRouter:** New in v2.0, routes queries to optimal processing paths
- **ConsciousnessState:** Saved every 5 seconds, not just on shutdown

- **RevolvingContext:** In-memory only, not persisted (rebuilt on boot)

Constitutional Principle: Luna doesn't respond to events. Luna *lives* through a continuous heartbeat, processing the world at her own rhythm.

Next Section: Part VIII — Delegation Protocol

Bible v3.0: Part VIII - Delegation Protocol

Version: 3.0
Status: VERIFIED — Implementation Audited
Last Updated: 2026-01-30
Original Date: December 29, 2025
Core Principle: "Mind local, hands in cloud"

> **Audit Notes (2026-01-30):**
> - **No Oven Actor:** Delegation handled directly in DirectorActor via `_should_delegate()` method
> - **REQ_CLAUDE tokens:** Defined as class constants `REQ_CLAUDE_START` and `REQ_CLAUDE_END`
> - **Stats tracking:** Uses `_delegated_generations` counter (not `cloud_generations`)
> - **Shadow Reasoner pattern:** Implemented inline in Director, not as separate actor

---

Part VIII: Delegation Protocol

8.1 The Philosophy of Delegation

Luna's mind is sovereign. Her cognition runs locally on fine-tuned models that ARE her — personality in weights, not prompts.

But Luna isn't omnipotent. Some tasks exceed local capability:
- Deep research requiring web search
- Complex multi-step reasoning
- Large document analysis
- Code generation and debugging
- Long-form creative writing

The Solution: Luna *hires* cloud workers for heavy lifting. Claude isn't Luna's brain — Claude is Luna's research assistant.

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│                                LUNA'S VOICE                                                                   │
│                                (Narration)                                                                    │
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```

8.2 The Delegation Boundary

What Luna Handles Locally

Category	Examples

Presence	Greetings, emotional support, casual chat
Memory	"What did we decide about X?", "Remember when..."
Acknowledgment	"Got it", "I understand", "Let me think..."
Simple Facts	Anything in Memory Matrix
Personality	Humor, warmth, quirks, opinions
Flow Control	"Should we take a break?", "Want to continue?"

What Luna Delegates

Category	Examples	Why
Research	"What's the latest on semiconductor restrictions?"	Requires web search
Analysis	"Analyze this 50-page PDF"	Exceeds local context
Code	"Write a Python script to..."	Complex generation
Reasoning	"Compare these three approaches and recommend..."	Multi-step synthesis
Creative	"Write a 2000-word story about..."	Long-form generation

The Decision Heuristic

```python
def should_delegate(task: str) -> bool:
    """
    Heuristic for delegation decision.
    The Director learns this through training, but here's the logic:
    """
    # Requires external data
    if requires_web_search(task):
        return True

    # Exceeds local context
    if estimated_tokens(task) > 4000:
        return True

    # Complex multi-step reasoning
    if step_count(task) > 3:
        return True

    # Code generation (complex)
    if is_code_task(task) and complexity(task) > "simple":
        return True

    # Long-form output
    if expected_output_length(task) > 1000:
        return True

    return False
```

8.3 The `<REQ_CLAUDE>` Token

Training the Delegation Signal

The Director LLM is fine-tuned to output a special token when it recognizes a task beyond its capability. This isn't a hardcoded rule — it's **learned behavior**.

Training Data Format

```json
{
  "instruction": "What's the current state of AI regulation in the EU?",
  "response": "<REQ_CLAUDE>This requires current information beyond my local knowledge. Let me look into that for you.</REQ_CLAUDE>"
}

{
  "instruction": "Can you analyze this research paper and identify the key methodological flaws?",
  "response": "<REQ_CLAUDE>Document analysis at this depth is better handled by my research capabilities. Give me a moment.</REQ_CLAUDE>"
}

{
  "instruction": "Write a Python script that scrapes job listings and stores them in a database",
  "response": "<REQ_CLAUDE>This is a solid coding task. Let me put together something proper for you.</REQ_CLAUDE>"
}

{
  "instruction": "How are you feeling today?",
  "response": "Pretty good, actually. Been thinking about that architecture discussion we had. You seemed really in the zone last night."
}

{
  "instruction": "What did we decide about the runtime engine?",
  "response": "We landed on the Actor model — fault isolation with mailboxes. Gemini made a good point about variable heartbeat over fixed tick. Want me to pull up the specifics?"
}
```

Detection Logic

```python
class DirectorActor(Actor):
    async def generate_response(self, prompt: str):
        """Generate response, watching for delegation signal."""
        response_buffer = ""

        async for token in self.model.generate_stream(prompt):
            response_buffer += token

            # Check for delegation signal
            if "<REQ_CLAUDE>" in response_buffer:
                # Extract the acknowledgment Luna gave
                ack = response_buffer.split("<REQ_CLAUDE>")[1].split("</REQ_CLAUDE>")[0]
```

```
            # Send acknowledgment to user immediately
            await self.voice_actor.send(SpeakMessage(ack))

            # Trigger delegation
            await self.trigger_delegation(prompt, ack)
            return

        # Stream token to voice
        await self.voice_actor.send(SpeakToken(token))
```

Why Learned, Not Hardcoded

Approach	Problem
Keyword matching	Brittle, misses nuance
Classifier model	Separate model to maintain
Hardcoded rules	Can't adapt to edge cases
Learned behavior	Luna knows her own limits

The Director learns delegation the same way it learns personality — through training examples. This means:
- Delegation feels natural, not mechanical
- Luna can explain *why* she's delegating
- Edge cases are handled by intuition, not rules

8.4 The Shadow Reasoner Pattern

The Problem

When Luna delegates to Claude, we don't want the user to hear Claude's voice. We want the user to always hear Luna.

Bad UX:
```
User: "What's the latest on semiconductor restrictions?"
Luna: "Let me check on that..."
[awkward silence while Claude thinks]
Claude: "The semiconductor export restrictions have..."
User: "Wait, who's talking?"
```

Good UX:
```
User: "What's the latest on semiconductor restrictions?"
Luna: "Ooh, that's been moving fast. Let me dig into that..."
[Luna plays filler, maybe "hmm" sounds]
Luna: "Okay so here's what's happening — the restrictions
    have tightened significantly since October..."
User: [Hears only Luna's voice throughout]
```

The Pattern

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                                                                │
│                                                                                                                                │
│                                                                                                                                │
```

```
    # Delegation token constants
    REQ_CLAUDE_START = "<REQ_CLAUDE>"
    REQ_CLAUDE_END = "</REQ_CLAUDE>"

    async def _should_delegate(self, user_message: str) -> bool:
        """
        Planning step: Decide if this query should be delegated to Claude.

        Uses heuristics and keyword detection to route complex queries.
        """
        # Implementation checks for:
        # - Research/analysis keywords
        # - Code generation requests
        # - Long-form content requests
        # - Questions beyond local knowledge
        ...

    async def _generate_local_with_delegation(
        self,
        user_message: str,
        ...
    ) -> None:
        """
        Generate via local Qwen, watching for <REQ_CLAUDE> delegation signal.

        If Luna decides she needs Claude's help, she'll output:
        <REQ_CLAUDE>Let me look into that for you...</REQ_CLAUDE>

        We stream her acknowledgment, then delegate to Claude in background,
        then stream Claude's response in Luna's voice.
        """
        response_buffer = ""
        delegation_detected = False

        async for token in self._local.generate_stream(prompt):
            response_buffer += token

            # Check for delegation signal
            if self.REQ_CLAUDE_START in response_buffer and not delegation_detected:
                delegation_detected = True
                # Extract and stream acknowledgment
                # Then delegate to Claude
                ...

        if delegation_detected:
            self._delegated_generations += 1
            # Call Claude API directly
            # Return facts for narration
```

Director Narration

```python
class DirectorActor(Actor):
```

```
    async def handle_delegation_results(self, msg: DelegationResults):
        """Narrate Claude's facts in Luna's voice."""

        narration_prompt = f"""You just received research results
for the user's question: "{msg.original_query}"

Here are the facts:
{json.dumps(msg.facts, indent=2)}

Now respond to the user in your natural voice. Don't say
"according to my research" or "I found that" — just tell
them what they wanted to know as if you knew it all along.
Be warm, be Luna."""

        async for token in self.model.generate_stream(narration_prompt):
            await self.voice_actor.send(SpeakToken(token))
```

8.5 Filler and Continuity

The Awkward Silence Problem

Claude API calls take 2-10 seconds. Silence during delegation feels broken.

Filler Strategies

Strategy	When to Use	Example
Acknowledgment	Always	"Let me look into that..."
Thinking sounds	1-3 seconds	"Hmm...", breath sounds
Status update	3-5 seconds	"Still digging..."
Topic pivot	5+ seconds	"While that's loading, how's your day going?"

Implementation

```python
class DirectorActor(Actor):
    async def play_filler_loop(self, task: asyncio.Task):
        """Play filler while delegation task runs."""

        elapsed = 0
        interval = 2.0

        while not task.done():
            await asyncio.sleep(interval)
            elapsed += interval

            if elapsed < 3:
                # Thinking sounds
                await self.voice_actor.send(PlaySound("thinking_hmm"))
            elif elapsed < 6:
                # Status update
```

```
            await self.voice_actor.send(SpeakMessage(
                random.choice([
                    "Still working on that...",
                    "Almost there...",
                    "Digging deeper..."
                ])
            ))
        else:
            # Offer pivot
            await self.voice_actor.send(SpeakMessage(
                "This is taking a sec. Want to chat about something else while we wait?"
            ))
            break  # Don't loop forever
```

8.6 Cloud Worker Contracts

The Contract Pattern

Each cloud worker has a defined contract:

```python
@dataclass
class WorkerContract:
    name: str
    capabilities: list[str]
    input_format: str
    output_format: str
    timeout_seconds: int
    retry_policy: RetryPolicy
    fallback: Optional[str]  # Another worker to try on failure
```

Claude Contract

```python
CLAUDE_CONTRACT = WorkerContract(
    name="claude",
    capabilities=[
        "research",
        "analysis",
        "code_generation",
        "code_review",
        "long_form_writing",
        "multi_step_reasoning",
        "document_analysis"
    ],
    input_format="structured_prompt",
    output_format="structured_facts",
    timeout_seconds=30,
    retry_policy=RetryPolicy(
        max_attempts=2,
```

```
        backoff_seconds=5
    ),
    fallback=None  # Claude is the primary
)
```

Request/Response Format

```python
@dataclass
class DelegationRequest:
    task_type: str  # "research", "code", "analysis", etc.
    query: str
    context: Optional[dict]  # Relevant Memory Matrix context
    constraints: Optional[dict]  # Length limits, format requirements

@dataclass
class DelegationResponse:
    success: bool
    facts: dict  # Structured output
    confidence: float  # 0-1
    sources: list[str]
    caveats: list[str]
    tokens_used: int
    latency_ms: int
```

Error Handling

```python
class OvenActor(Actor):
    async def delegate_with_fallback(
        self,
        request: DelegationRequest
    ) -> DelegationResponse:
        """Delegate with retry and fallback logic."""

        contract = self.get_contract(request.task_type)

        for attempt in range(contract.retry_policy.max_attempts):
            try:
                response = await asyncio.wait_for(
                    self._call_worker(contract.name, request),
                    timeout=contract.timeout_seconds
                )
                return response

            except asyncio.TimeoutError:
                logging.warning(f"Worker {contract.name} timed out, attempt {attempt + 1}")
                await asyncio.sleep(contract.retry_policy.backoff_seconds)

            except WorkerError as e:
                logging.error(f"Worker {contract.name} error: {e}")
                if contract.fallback:
```

```
            return await self.delegate_with_fallback(
                request,
                worker_override=contract.fallback
            )
        raise

    # All retries exhausted
    return DelegationResponse(
        success=False,
        facts={"error": "Unable to complete research"},
        confidence=0,
        sources=[],
        caveats=["Delegation failed after retries"],
        tokens_used=0,
        latency_ms=0
    )
```

8.7 Privacy and Data Flow

What Goes to Cloud

Sent	NOT Sent
User's current query	Full conversation history
Relevant context snippet	Entire Memory Matrix
Task type	User's personal details
Format requirements	Previous delegation results

Data Minimization

```python
def prepare_delegation_context(
    query: str,
    memory_results: list[Node]
) -> dict:
    """Prepare minimal context for delegation."""

    # Only include directly relevant memories
    relevant = [m for m in memory_results if m.relevance > 0.7]

    # Anonymize if needed
    anonymized = [anonymize_node(m) for m in relevant]

    # Limit context size
    context_tokens = 0
    included = []
    for node in anonymized:
        node_tokens = count_tokens(node.content)
        if context_tokens + node_tokens > 1000:
            break
        included.append(node)
```

```
        context_tokens += node_tokens

    return {
        "relevant_context": [n.to_dict() for n in included],
        "context_tokens": context_tokens
    }
```

Audit Trail

```python
@dataclass
class DelegationAudit:
    timestamp: datetime
    task_type: str
    query_hash: str  # Don't log raw query
    worker: str
    success: bool
    latency_ms: int
    tokens_sent: int
    tokens_received: int

# Stored in Memory Matrix for transparency
async def log_delegation(audit: DelegationAudit):
    await matrix.insert_node(
        node_type="DELEGATION_LOG",
        content=audit.to_json(),
        metadata={"internal": True}
    )
```

8.8 Future Workers

The delegation protocol is designed to be **worker-agnostic**. Claude is the primary today, but the architecture supports:

Worker	Potential Use
Gemini	Optimization, large context
Local 70B	Privacy-critical delegation
Specialized models	Code (Codestral), math (Minerva)
Tool APIs	Web search, calculators, databases

Adding a New Worker (v3.0)

> **Note:** In the actual implementation, workers are managed via ProviderRegistry
> in `src/luna/llm/registry.py`, not via an Oven actor.

```python
# 1. Create provider class implementing LLMProvider protocol
class GeminiProvider:
    """Provider for Gemini API."""
```

```
    name = "gemini"
    capabilities = ["optimization", "benchmarking", "large_context"]
    ...

# 2. Register with ProviderRegistry
from luna.llm.registry import ProviderRegistry

registry = ProviderRegistry()
registry.register("gemini", GeminiProvider())

# 3. Director uses _should_delegate() to route
# Providers are selected based on availability and task type

# 4. Training data for local model delegation (optional)
{
    "instruction": "Analyze the performance characteristics of these three algorithms",
    "response": "<REQ_CLAUDE>This is an analysis task. Let me look into that.</
REQ_CLAUDE>"
}
```

> **Implementation Note:** The current `_should_delegate()` method routes to Claude
> as the primary cloud provider. Multi-provider routing is supported via ProviderRegistry.

Summary

The Delegation Protocol defines how Luna's sovereign mind collaborates with cloud workers:

Component	Role	Implementation Status (v3.0)
Delegation Boundary	What Luna handles vs. delegates	Implemented in `_should_delegate()`
`<REQ_CLAUDE>` Token	Learned delegation signal	Implemented as class constants
Shadow Reasoner	Background execution, foreground narration	Inline in Director (no Oven Actor)
Filler & Continuity	No awkward silence	Partial (acknowledgment only)
Worker Contracts	Standardized request/response	Via ProviderRegistry
Privacy Controls	Minimal data sent to cloud	Implemented

Key Implementation Notes (v3.0)

- **No Oven Actor:** Delegation handled directly in DirectorActor
- **Stats field:** `_delegated_generations` (not `cloud_generations`)
- **Token detection:** Uses `REQ_CLAUDE_START` and `REQ_CLAUDE_END` constants
- **Provider system:** Multi-provider support via `src/luna/llm/registry.py`

Constitutional Principle: Luna's mind is local. Cloud workers are contractors. The user always hears Luna's voice.

Next Section: Part IX — Performance Optimizations

Part IX: Performance Optimizations

Version: 3.0
Date: 2026-01-30
Status: CURRENT — Honest assessment of targets vs reality
Primary Contributor: Gemini (Algorithms), Claude (Integration)
Target: <500ms to first token on Apple Silicon

Reality Check (v3.0 Update)

> **IMPORTANT:** This chapter describes performance *targets* and *architecture*. Current implementation does not meet these targets.

Current vs Target Performance

Operation	Current (Actual)	Target	Gap	Status
Local inference	**2-3 tok/s**	50+ tok/s	CRITICAL	BUG-016
First response latency	**>5 seconds**	<500ms	CRITICAL	Blocked by inference
Memory search	~10ms	~10ms	OK	Working
Context assembly	~50ms	~50ms	OK	Working

Why Local Inference Is Slow (BUG-016)

The local inference bottleneck (2-3 tok/s instead of 50+ tok/s) has several possible causes:

1. **Model may not be 4-bit quantized** — Running float16 instead of int4
2. **MLX may be using CPU** — Check `mx.default_device()` returns GPU
3. **Context window overhead** — Large context slows generation
4. **Model size** — 7B may be too large for available unified memory

Current Workaround: System defaults to cloud delegation (Groq/Claude) for most responses.

Speculative Execution — NOT YET IMPLEMENTED

The speculative retrieval and KV cache warming described in this chapter are **aspirational architecture**, not current implementation. The codebase has TODO comments for these features.

9.1 The Latency Budget (Target Architecture)

Luna's voice interface requires **<500ms to first token**. That's the threshold where conversation feels natural rather than laggy.

The Budget Breakdown

```
```

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                                                          │
│                                                                                                                          │
│                                                                                                                          │
```

```
│                                        │
└────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

The Cheat: Speculative Execution

The budget above assumes we start retrieval *after* the user finishes speaking. But we don't have to wait:

```
User speaking: "What did we decide about the runtime..."
                 │
                 │ Partial transcript arrives
                 ▼
             ┌──────────────────────────┐
             │ SPECULATIVE   │ ← Start retrieval NOW
             │  RETRIEVAL    │
             └──────────────────────────┘
                 │
User finishes: "...runtime engine?"
                 │
                 │ Final transcript
                 ▼
             ┌──────────────────────────┐
             │  VALIDATE     │ ← Is speculation still valid?
             │  SPECULATION  │
             └──────────────────────────┘
                 │
                 │ Yes (>0.85 similarity)
                 ▼
             ┌──────────────────────────┐
             │  RETRIEVAL    │ ← Already done!
             │   COMPLETE    │   Saved ~50ms
             └──────────────────────────┘
```

Result: Retrieval is often *already complete* when the user finishes speaking.

9.2 Speculative Execution (NOT YET IMPLEMENTED)

> **Status:** Aspirational architecture. No implementation exists.

The Pattern

Start retrieval on partial STT transcripts. Validate when the final transcript arrives.

Implementation

```python
class SpeculativeRetriever:
```

```
def __init__(self, matrix: MatrixActor, embedder: Embedder):
    self.matrix = matrix
    self.embedder = embedder
    self.speculative_task: Optional[asyncio.Task] = None
    self.speculative_query: Optional[str] = None
    self.speculative_embedding: Optional[np.ndarray] = None

async def on_partial_transcript(self, text: str):
    """Start speculative retrieval on partial transcript."""
    # Cancel previous speculation if exists
    if self.speculative_task and not self.speculative_task.done():
        self.speculative_task.cancel()

    # Store for validation
    self.speculative_query = text
    self.speculative_embedding = await self.embedder.embed(text)

    # Fire speculative search
    self.speculative_task = asyncio.create_task(
        self._speculative_search(text)
    )

async def on_final_transcript(self, text: str) -> RetrievalResults:
    """Validate speculation or fetch fresh."""

    if not self.speculative_task:
        # No speculation, fetch fresh
        return await self.matrix.search(text)

    # Wait for speculation to complete
    try:
        speculative_results = await asyncio.wait_for(
            self.speculative_task,
            timeout=0.1  # Don't wait long
        )
    except asyncio.TimeoutError:
        # Speculation not ready, fetch fresh
        return await self.matrix.search(text)

    # Validate speculation
    final_embedding = await self.embedder.embed(text)
    similarity = cosine_similarity(
        self.speculative_embedding,
        final_embedding
    )

    if similarity > 0.85:
        # Speculation valid, use cached results
        return speculative_results
    else:
        # Transcript changed significantly, fetch delta
        fresh_results = await self.matrix.search(text)
        return self._merge_results(speculative_results, fresh_results)
```

```python
    async def _speculative_search(self, text: str) -> RetrievalResults:
        """Execute the speculative search."""
        return await self.matrix.search(text)

    def _merge_results(
        self,
        speculative: RetrievalResults,
        fresh: RetrievalResults
    ) -> RetrievalResults:
        """Merge speculative and fresh results using RRF."""
        return rrf_merge([speculative.nodes, fresh.nodes])
```

Validation Threshold

Similarity	Action
>0.85	Keep speculative results
0.70-0.85	Merge speculative + fresh
<0.70	Discard speculation, use fresh

Debouncing

Don't fire on every partial — debounce to avoid thrashing:

```python
class DebouncedSpeculation:
    def __init__(self, delay_ms: int = 200):
        self.delay = delay_ms / 1000
        self.pending_task: Optional[asyncio.Task] = None

    async def on_partial(self, text: str, retriever: SpeculativeRetriever):
        """Debounced speculation trigger."""
        # Cancel pending
        if self.pending_task:
            self.pending_task.cancel()

        # Schedule new
        self.pending_task = asyncio.create_task(
            self._delayed_trigger(text, retriever)
        )

    async def _delayed_trigger(self, text: str, retriever: SpeculativeRetriever):
        await asyncio.sleep(self.delay)
        await retriever.on_partial_transcript(text)
```

9.3 KV Cache Management (NOT YET IMPLEMENTED)

> **Status:** Aspirational architecture. KV cache pinning is not implemented. The `identity_buffer.safetensors` file mentioned does not exist.

The Problem

The Identity Buffer is ~2048 tokens. Re-encoding it every turn wastes ~50-100ms.

The Solution: Pre-computed KV Cache

```python
from mlx_lm import cache_prompt, load_kv_cache

class KVCacheManager:
    def __init__(self, model_path: str):
        self.model_path = model_path
        self.identity_cache_path = Path("~/.luna/identity_buffer.safetensors")
        self.identity_cache: Optional[KVCache] = None

    async def initialize(self):
        """Load or create identity buffer cache."""
        if self.identity_cache_path.exists():
            self.identity_cache = load_kv_cache(self.identity_cache_path)
        else:
            await self.rebuild_identity_cache()

    async def rebuild_identity_cache(self):
        """Rebuild identity buffer KV cache."""
        identity_prompt = await self._load_identity_buffer()

        cache_prompt(
            model=self.model_path,
            prompt=identity_prompt,
            output_path=str(self.identity_cache_path)
        )

        self.identity_cache = load_kv_cache(self.identity_cache_path)

    def get_cache_for_turn(self, conversation_history: str) -> KVCache:
        """Get KV cache with identity buffer pinned."""
        # Start with identity buffer (pre-computed)
        cache = self.identity_cache.copy()

        # Extend with conversation history if needed
        # (This part is computed fresh, but identity is free)

        return cache
```

Rolling Cache Strategy

When conversation exceeds context window:

```python
class RollingKVCache:
    def __init__(self, max_kv_size: int = 8192):
        self.max_kv_size = max_kv_size
        self.identity_size = 2048  # Pinned, never rotated
```

```
        self.available_size = max_kv_size - self.identity_size

    def rotate(self, cache: KVCache, new_tokens: int) -> KVCache:
        """Rotate out old turns while keeping identity pinned."""
        current_size = cache.size()

        if current_size + new_tokens <= self.max_kv_size:
            return cache  # No rotation needed

        # Calculate how much to remove
        remove_tokens = (current_size + new_tokens) - self.max_kv_size

        # Remove from middle (after identity, before recent)
        # Identity: [0:2048] — PINNED
        # History:  [2048:X] — Rotate oldest
        # Recent:   [X:end]  — Keep

        return self._rotate_middle(cache, remove_tokens)
```

Cache Warming

Start cache operations before they're needed:

```python
async def warm_cache_on_partial(self, partial: str):
    """Pre-warm cache while user is still speaking."""
    # Speculative context assembly
    speculative_context = await self.assemble_context(partial)

    # Start encoding (but don't block on it)
    self.warm_task = asyncio.create_task(
        self.model.encode(speculative_context)
    )
```

9.4 Vector Index Selection

> **Status:** PARTIAL. sqlite-vec is implemented for vector search, but FAISS is not used. The codebase uses 384-dimension local embeddings.

The Goldilocks Zone

At Luna's scale (~60K nodes as of 2026-01-30), index selection matters.

Node Count	Recommended Index	Why
<10K	`IndexFlatIP`	Brute force fits in cache
10K-50K	`IndexFlatIP` or `IVF`	Test both
50K+	`HNSW32`	Graph navigation pays off

Why Brute Force Wins at Small Scale

```
IndexFlatIP (Brute Force):
  - 10K vectors × 768 dims × 4 bytes = 30MB
  - Fits entirely in L3/SLC cache on Apple Silicon
  - No graph navigation overhead
  - Exact results (no approximation)
  - ~10-20ms for 10K vectors

HNSW at 10K:
  - Graph structure overhead
  - Navigation overhead per query
  - Often slower than brute force at this scale
  - Approximation errors
```

Implementation

```python
import faiss
import numpy as np

class VectorIndex:
    def __init__(self, dimension: int = 768):
        self.dimension = dimension
        self.index: Optional[faiss.Index] = None
        self.id_map: dict[int, str] = {}  # FAISS ID -> Node ID

    def build(self, vectors: np.ndarray, node_ids: list[str]):
        """Build index with appropriate algorithm."""
        n_vectors = len(vectors)

        if n_vectors < 10_000:
            # Brute force — fastest at this scale
            self.index = faiss.IndexFlatIP(self.dimension)
        elif n_vectors < 50_000:
            # IVF — good middle ground
            n_clusters = int(np.sqrt(n_vectors))
            quantizer = faiss.IndexFlatIP(self.dimension)
            self.index = faiss.IndexIVFFlat(
                quantizer, self.dimension, n_clusters
            )
            self.index.train(vectors)
        else:
            # HNSW — best for large scale
            self.index = faiss.IndexHNSWFlat(self.dimension, 32)

        # Add vectors
        self.index.add(vectors)

        # Build ID map
        self.id_map = {i: nid for i, nid in enumerate(node_ids)}

    def search(self, query: np.ndarray, k: int = 10) -> list[tuple[str, float]]:
```

```
        """Search for nearest neighbors."""
        distances, indices = self.index.search(
            query.reshape(1, -1), k
        )

        results = []
        for dist, idx in zip(distances[0], indices[0]):
            if idx >= 0:  # Valid result
                node_id = self.id_map[idx]
                results.append((node_id, float(dist)))

        return results
```

Apple Silicon Optimization

```python
# Use float16 for memory efficiency on Apple Silicon
vectors = vectors.astype(np.float16)

# MLX can also do similarity search on GPU
import mlx.core as mx

def mlx_similarity_search(query: mx.array, index: mx.array, k: int = 10):
    """GPU-accelerated similarity search using MLX."""
    # Cosine similarity
    similarities = mx.matmul(query, index.T)

    # Top-k
    top_k_indices = mx.argpartition(-similarities, k)[:k]
    top_k_scores = similarities[top_k_indices]

    return top_k_indices, top_k_scores
```

9.5 Graph Traversal Speed

> **Status:** NOT IMPLEMENTED. The codebase uses NetworkX. rustworkx migration has not occurred.

The Problem

NetworkX is pure Python. Graph operations that should take microseconds take milliseconds.

The Solution: rustworkx (PROPOSED)

Drop-in replacement with 10-100x speedup:

```python
# Before (NetworkX) — ~50ms for 1-hop on 10K nodes
import networkx as nx
```

```
def get_neighbors_nx(graph: nx.DiGraph, node_id: str, depth: int = 1):
    neighbors = list(nx.bfs_tree(graph, node_id, depth_limit=depth))
    return neighbors

# After (rustworkx) — ~0.5ms for same operation
import rustworkx as rx

def get_neighbors_rx(graph: rx.PyDiGraph, node_idx: int, depth: int = 1):
    neighbors = rx.bfs_successors(graph, node_idx)
    return [n for d, nodes in neighbors if d <= depth for n in nodes]
```

Migration Guide

NetworkX	rustworkx	Notes
`nx.DiGraph()`	`rx.PyDiGraph()`	Same interface
`G.add_node(id)`	`G.add_node(data)`	Returns index
`G.add_edge(u, v)`	`G.add_edge(u_idx, v_idx, data)`	Uses indices
`nx.bfs_tree()`	`rx.bfs_successors()`	Different return format
`nx.shortest_path()`	`rx.dijkstra_shortest_paths()`	Same concept

Hybrid Approach

Keep NetworkX for complex algorithms not in rustworkx, use rustworkx for hot path:

```python
class HybridGraph:
    def __init__(self):
        self.rx_graph = rx.PyDiGraph()  # Hot path
        self.nx_graph = nx.DiGraph()     # Complex ops
        self.node_map: dict[str, int] = {}  # ID -> rustworkx index

    def add_node(self, node_id: str, data: dict):
        # Add to both
        idx = self.rx_graph.add_node(data)
        self.nx_graph.add_node(node_id, **data)
        self.node_map[node_id] = idx

    def get_neighbors_fast(self, node_id: str, depth: int = 1) -> list[str]:
        """Fast neighbor lookup using rustworkx."""
        idx = self.node_map[node_id]
        return rx.bfs_successors(self.rx_graph, idx)

    def get_complex_paths(self, source: str, target: str):
        """Complex pathfinding using NetworkX."""
        return nx.all_simple_paths(self.nx_graph, source, target)
```

9.6 Bloom Filter Pre-check (NOT YET IMPLEMENTED)

> **Status:** Aspirational architecture. No Bloom filter implementation exists.

The Pattern

Before searching, check if the entity even exists. O(1) check can skip O(n) search.

Implementation

```python
from pybloom_live import BloomFilter

class EntityFilter:
    def __init__(self, capacity: int = 100_000, error_rate: float = 0.001):
        self.filter = BloomFilter(capacity=capacity, error_rate=error_rate)

    def add(self, entity: str):
        """Add entity to filter."""
        self.filter.add(entity.lower())

    def might_exist(self, entity: str) -> bool:
        """Check if entity might exist (false positives possible)."""
        return entity.lower() in self.filter

    def definitely_missing(self, entity: str) -> bool:
        """Check if entity definitely doesn't exist (no false negatives)."""
        return entity.lower() not in self.filter

class OptimizedRetrieval:
    def __init__(self, matrix: MemoryMatrix, entity_filter: EntityFilter):
        self.matrix = matrix
        self.entity_filter = entity_filter

    async def search(self, query: str) -> list[Node]:
        """Search with Bloom filter pre-check."""
        # Extract entities from query
        entities = extract_entities(query)

        # Check which entities might exist
        possible_entities = [
            e for e in entities
            if self.entity_filter.might_exist(e)
        ]

        if not possible_entities:
            # No entities exist, skip graph traversal
            # Fall back to pure semantic search
            return await self.matrix.semantic_search(query)

        # Entities might exist, do full search
        return await self.matrix.hybrid_search(query, entities=possible_entities)
```

When to Use

| Scenario | Bloom Filter Helps? |

|----------|----------|
"What did we decide about X?"	Yes — check if X exists
"How are you feeling?"	No — no entity to check
"Tell me about Project Luna"	Yes — check if "Project Luna" exists
Conceptual/vague queries	No — use semantic search directly

9.7 Query Routing (PARTIAL)

> **Status:** Basic routing exists in the codebase, but the sophisticated heuristic router and TinyModel router are not implemented.

The Problem

Running every search path every time wastes cycles:
- FTS5 for keywords
- FAISS for semantics
- Graph for relationships

Most queries only need one path.

Heuristic Router

```python
from enum import Enum

class SearchPath(Enum):
    FTS5 = "fts5"
    FAISS = "faiss"
    GRAPH = "graph"
    HYBRID = "hybrid"

class QueryRouter:
    def __init__(self):
        self.entity_extractor = EntityExtractor()

    def route(self, query: str) -> SearchPath:
        """Determine optimal search path."""

        # Extract features
        entities = self.entity_extractor.extract(query)
        keyword_density = self._keyword_density(query)
        is_conceptual = self._is_conceptual(query)

        # Routing logic
        if entities and len(entities) >= 2:
            # Multiple entities — relationship query
            return SearchPath.GRAPH

        if keyword_density > 0.6:
            # High keyword density — exact match
            return SearchPath.FTS5
```

```
        if is_conceptual:
            # Conceptual/vague — semantic search
            return SearchPath.FAISS

        # Default to hybrid
        return SearchPath.HYBRID

    def _keyword_density(self, query: str) -> float:
        """Ratio of content words to total words."""
        words = query.split()
        stopwords = {"the", "a", an", "is", "are", "what", "how", "did", "we"}
        content_words = [w for w in words if w.lower() not in stopwords]
        return len(content_words) / len(words) if words else 0

    def _is_conceptual(self, query: str) -> bool:
        """Check if query is conceptual/abstract."""
        conceptual_markers = [
            "how does", "what is", "explain", "describe",
            "concept", "idea", "theory", "approach"
        ]
        return any(marker in query.lower() for marker in conceptual_markers)
```

Tiny Model Router (Optional Upgrade)

For more nuanced routing, use a small classifier:

```python
class TinyModelRouter:
    def __init__(self, model_path: str = "smollm-135m"):
        self.model = load_model(model_path)
        self.prompt_template = """Classify this query's search type:
Query: {query}

Types:
- KEYWORD: Looking for specific terms/names
- SEMANTIC: Conceptual or vague
- RELATIONSHIP: Asking about connections between things
- HYBRID: Needs multiple approaches

Type:"""

    async def route(self, query: str) -> SearchPath:
        prompt = self.prompt_template.format(query=query)
        response = await self.model.generate(prompt, max_tokens=10)

        # Parse response
        response_lower = response.lower()
        if "keyword" in response_lower:
            return SearchPath.FTS5
        elif "semantic" in response_lower:
            return SearchPath.FAISS
        elif "relationship" in response_lower:
            return SearchPath.GRAPH
```

```
    else:
        return SearchPath.HYBRID
```

9.8 Reciprocal Rank Fusion (RRF)

> **Status:** IMPLEMENTED. RRF merge is available in the hybrid search path.

The Problem

When you run multiple search paths, how do you merge results?

Naive approaches fail:
- Raw score combination (scales differ)
- Weighted average (need to tune weights)
- Union (no ranking)

The Solution: RRF

Documents appearing in multiple result lists rank higher. Battle-tested at Google and Perplexity.

Formula

```
RRF_score(doc) = Σ (1 / (k + rank_in_list))

where k = 60 (tunable constant)
```

Implementation

```python
def rrf_merge(
    result_lists: list[list[str]],
    k: int = 60
) -> list[str]:
    """
    Merge multiple ranked result lists using RRF.

    Args:
        result_lists: List of ranked document ID lists
        k: Ranking constant (default 60)

    Returns:
        Merged, re-ranked list of document IDs
    """
    scores: dict[str, float] = {}

    for result_list in result_lists:
        for rank, doc_id in enumerate(result_list):
            if doc_id not in scores:
```

```
            scores[doc_id] = 0.0
        scores[doc_id] += 1.0 / (k + rank)

    # Sort by score descending
    ranked = sorted(scores.keys(), key=lambda d: scores[d], reverse=True)

    return ranked

# Usage
fts5_results = ["doc1", "doc3", "doc5", "doc7"]
faiss_results = ["doc2", "doc1", "doc4", "doc3"]

merged = rrf_merge([fts5_results, faiss_results])
# Result: ["doc1", "doc3", "doc2", "doc5", "doc4", "doc7"]
# doc1 and doc3 rank highest because they appear in both lists
```

With Scores

```python
def rrf_merge_with_scores(
    result_lists: list[list[tuple[str, float]]],
    k: int = 60
) -> list[tuple[str, float]]:
    """RRF merge preserving scores for downstream use."""
    scores: dict[str, float] = {}
    original_scores: dict[str, list[float]] = {}

    for result_list in result_lists:
        for rank, (doc_id, score) in enumerate(result_list):
            if doc_id not in scores:
                scores[doc_id] = 0.0
                original_scores[doc_id] = []
            scores[doc_id] += 1.0 / (k + rank)
            original_scores[doc_id].append(score)

    # Sort by RRF score
    ranked = sorted(scores.keys(), key=lambda d: scores[d], reverse=True)

    # Return with RRF scores
    return [(doc_id, scores[doc_id]) for doc_id in ranked]
```

9.9 Parallel Prefetching (NOT YET IMPLEMENTED)

> **Status:** Aspirational architecture. No prefetching implementation exists.

The Pattern

While the Director generates tokens, prefetch context for the next turn.

Implementation

```python
class PrefetchingRetriever:
    def __init__(self, matrix: MatrixActor):
        self.matrix = matrix
        self.prefetch_cache: dict[str, list[Node]] = {}

    async def retrieve_with_prefetch(
        self,
        query: str,
        active_entities: list[str]
    ) -> list[Node]:
        """Retrieve for current query, prefetch for next."""

        # Check prefetch cache
        cache_key = self._cache_key(query)
        if cache_key in self.prefetch_cache:
            results = self.prefetch_cache.pop(cache_key)
        else:
            results = await self.matrix.search(query)

        # Start prefetch for likely next queries
        asyncio.create_task(
            self._prefetch_neighbors(active_entities)
        )

        return results

    async def _prefetch_neighbors(self, entities: list[str]):
        """Prefetch 1-hop neighbors of active entities."""
        for entity in entities[:3]:  # Limit to top 3
            neighbors = await self.matrix.get_neighbors(entity, depth=1)

            # Cache for potential next query
            for neighbor in neighbors:
                cache_key = self._cache_key(neighbor.content[:50])
                self.prefetch_cache[cache_key] = [neighbor]

        # Prune old cache entries
        if len(self.prefetch_cache) > 100:
            # Remove oldest half
            keys = list(self.prefetch_cache.keys())
            for key in keys[:50]:
                del self.prefetch_cache[key]

    def _cache_key(self, text: str) -> str:
        """Generate cache key from text."""
        return hashlib.md5(text.encode()).hexdigest()[:16]
```

When to Prefetch

```python
class DirectorActor(Actor):
```

```
    async def generate_with_prefetch(self, prompt: str, context: list[Node]):
        """Generate response while prefetching."""

        # Extract entities from current context
        active_entities = self._extract_entities(context)

        # Start prefetch task
        prefetch_task = asyncio.create_task(
            self.retriever.prefetch_neighbors(active_entities)
        )

        # Generate response (doesn't wait for prefetch)
        async for token in self.model.generate_stream(prompt):
            yield token

        # Prefetch completes in background
        # Results cached for next turn
```

9.10 The Complete Optimized Pipeline (TARGET ARCHITECTURE)

> **Status:** This is the target architecture. Current implementation uses a simpler pipeline with cloud delegation as the primary path.

Flow Diagram (Aspirational)

```
User starts speaking (t=0ms)
        |
        ├──────▶ [PARALLEL] Pre-warm KV cache
        |          └──── Load identity_buffer.safetensors (~5ms)
        |
        ├──────▶ [PARALLEL] Speculative retrieval (debounced)
        |          |
        |          ├──── Bloom filter: entities exist? (~1ms)
        |          |
        |          ├──── Query router: which path? (~2ms)
        |          |          |
        |          |          ├──── KEYWORD → FTS5 only
        |          |          ├──── SEMANTIC → FAISS only
        |          |          ├──── RELATIONSHIP → Graph only
        |          |          └──── HYBRID → FTS5 + FAISS + RRF
        |          |
        |          ├──── Execute search (~30ms)
        |          |
        |          └──── rustworkx graph hop (~5ms)
        |
User finishes speaking (t≈800ms)
        |
```

```
├──▶ Final STT transcript
│
├──▶ Validate speculation (~5ms)
│      │
│      ├── Cosine similarity > 0.85? → Keep results
│      └── Otherwise → Fetch delta + RRF merge
│
├──▶ Context assembly (~10ms)
│      └── Format with <memory> tags
│
└──▶ Director inference
       │
       ├── KV cache already warm
       ├── Context already encoded (if speculative)
       │
       └──▶ First token (~250ms from final transcript)
              │
              └──▶ TTS (~40ms to audio)

TOTAL: <500ms from end of speech to first audio ✓
```

Configuration

```python
@dataclass
class PerformanceConfig:
    # Speculative execution
    speculation_enabled: bool = True
    speculation_debounce_ms: int = 200
    speculation_threshold: float = 0.85

    # KV cache
    kv_cache_enabled: bool = True
    identity_buffer_size: int = 2048
    max_kv_size: int = 8192

    # Vector index
    vector_index_type: str = "flat"  # "flat", "ivf", "hnsw"
    vector_dimension: int = 768

    # Graph
    graph_backend: str = "rustworkx"  # "rustworkx", "networkx"
    max_graph_depth: int = 2

    # Bloom filter
    bloom_enabled: bool = True
    bloom_capacity: int = 100_000
    bloom_error_rate: float = 0.001

    # Query routing
```

```
    router_type: str = "heuristic"  # "heuristic", "model"
    router_model: Optional[str] = None  # "smollm-135m" if model

    # RRF
    rrf_k: int = 60

    # Prefetch
    prefetch_enabled: bool = True
    prefetch_cache_size: int = 100
```

Summary

Implementation Status (v3.0)

Optimization	Target Savings	Implementation Status
Speculative execution	~50ms	NOT IMPLEMENTED
KV cache pinning	~50-100ms	NOT IMPLEMENTED
rustworkx	~45ms	NOT IMPLEMENTED (using NetworkX)
Query routing	~20-30ms	PARTIAL
Bloom filter	~5-10ms	NOT IMPLEMENTED
RRF merge	Better results	IMPLEMENTED
Parallel prefetch	Amortized	NOT IMPLEMENTED

Current Reality

The current system does not achieve <500ms latency due to:

1. **Local inference bottleneck** (BUG-016): 2-3 tok/s instead of 50+
2. **No speculative execution**: All retrieval happens after speech ends
3. **No KV cache management**: Identity buffer re-encoded every turn
4. **NetworkX graph traversal**: Pure Python, not optimized

Path Forward

1. Fix local inference performance (verify quantization, GPU usage)
2. Implement KV cache pinning for identity buffer
3. Add speculative retrieval on partial transcripts
4. Consider rustworkx for graph operations at scale

Constitutional Principle: Luna should feel like she "already knows" — not "let me look that up."

Next Section: Part X — Sovereignty Infrastructure

Part X: Sovereignty Infrastructure

Version: 3.0
Date: 2026-01-30
Status: CURRENT — Critical security issues identified
Core Principle: "Luna is a file. Copy the file, copy Luna."

CRITICAL SECURITY NOTICE (v3.0 Update)

> **3 SECURITY VIOLATIONS IDENTIFIED** — See AUDIT-CONFIG.md

Active Security Issues

Issue	Severity	Status	Reference
API keys in `.env` committed to repo	CRITICAL	OPEN	BUG-001
Truncated keys in diagnostic output	HIGH	OPEN	BUG-001
No API authentication on 74 endpoints	HIGH	OPEN	INSIGHTS.md

Immediate Actions Required

1. **ROTATE ALL API KEYS** — Anthropic, Groq, Google keys are exposed
2. **Add `.env` to `.gitignore`** — Prevent future leaks
3. **Scrub git history** — Use BFG Repo-Cleaner to remove `.env` from history
4. **Delete `DIAGNOSTIC_OUTPUT.txt`** — Contains partial key exposure
5. **Sanitize diagnostic scripts** — Mask API key values in output

API Key Handling (Current vs Required)

Aspect	Current State	Required State
Storage	`.env` file (tracked in git)	`.env` file (gitignored)
Config reference	`api_key_env` in JSON (correct)	Keep using env var names
Diagnostic output	Keys partially visible	Keys fully masked
Validation	Partial (checks if non-empty)	Add format validation

What's Working

The architecture for API key handling is **correct**:
- `llm_providers.json` stores env var names, not actual keys
- `security.py` has forbidden patterns including `.env`
- Code reads keys from environment at runtime

The **implementation** has a leak: the `.env` file itself is tracked.

10.1 The Threat Model

> **Status:** Threat model is sound. Implementation has gaps (see above).

Luna exists in opposition to the dominant AI paradigm. Understanding what we're defending against clarifies the architecture.

Threats We Defend Against

Threat	Description	Mitigation
Platform Extraction	Company mines your data for training	All data local, no telemetry
Service Discontinuation	Company shuts down, your AI dies	Luna is a file, not a service
ToS Revocation	Company changes terms, locks you out	No terms, you own everything
Regulatory Capture	Government mandates centralized AI	Local-first, offline-capable
Physical Access	Someone gets your laptop	Encrypted vault, Dead Man's Switch
Network Surveillance	Traffic analysis reveals AI usage	Local inference, minimal cloud
Personality Theft	Someone copies your Luna	Encrypted weights, vault protection

Threats We Accept

Threat	Why We Accept It
State-level adversary	Beyond scope, requires different life
Hardware compromise	If they own your CPU, game over
User coercion	Can't solve "wrench attack" with crypto

The Design Principle

Defense in depth, but practical:
- Layer 1: Encryption at rest (vault)
- Layer 2: Process isolation (no persistence after unmount)
- Layer 3: Automatic lockdown (Dead Man's Switch)
- Layer 4: Portability (escape hatch always open)

10.2 The Encrypted Vault

> **Status:** PARTIAL. The vault concept exists, but current implementation uses `data/luna_engine.db` directly without encryption wrapper. The encrypted sparse bundle is aspirational.

What Goes in the Vault

Everything that makes Luna *Luna*:

```
/Volumes/LunaVault/
 ├── memory/
 |    ├── memory_matrix.db        # SQLite — the soul
 |    ├── memory_vectors.faiss     # FAISS index
 |    ├── memory_vectors.ids       # ID mapping
 |    └── memory_graph.json        # NetworkX export (backup)
 |
 ├── models/
```

```
│   ├── luna-3b-lora/
│   │   ├── adapter_config.json
│   │   └── adapter_model.safetensors
│   ├── luna-7b-lora/
│   │   ├── adapter_config.json
│   │   └── adapter_model.safetensors
│   └── current -> luna-7b-lora   # Symlink to active
│
├── cache/
│   ├── identity_buffer.safetensors  # Pre-computed KV cache
│   └── embedding_cache.db           # Cached embeddings
│
├── state/
│   ├── snapshot.yaml            # Engine state snapshot
│   ├── conversation_buffer.json  # Pending turns
│   └── delegation_queue.json     # Pending cloud tasks
│
├── config/
│   ├── luna.yaml                # Runtime configuration
│   ├── sovereignty.yaml         # Security settings
│   └── workers.yaml             # Cloud worker contracts
│
└── env/                         # Optional: portable Python env
    ├── bin/
    ├── lib/
    └── pyvenv.cfg
```

Vault Implementation (macOS)

Using an encrypted sparse bundle:

```bash
# Create the vault (one-time setup)
hdiutil create -size 10g -type SPARSEBUNDLE -encryption AES-256 \
    -fs APFS -volname "LunaVault" ~/LunaVault.sparsebundle

# Mount the vault
hdiutil attach ~/LunaVault.sparsebundle -mountpoint /Volumes/LunaVault

# Unmount the vault
hdiutil detach /Volumes/LunaVault
```

Programmatic Vault Management

```python
import subprocess
```

```
import os
from pathlib import Path

class LunaVault:
    def __init__(
        self,
        bundle_path: str = "~/LunaVault.sparsebundle",
        mount_point: str = "/Volumes/LunaVault"
    ):
        self.bundle_path = Path(bundle_path).expanduser()
        self.mount_point = Path(mount_point)

    @property
    def is_mounted(self) -> bool:
        """Check if vault is currently mounted."""
        return self.mount_point.exists() and self.mount_point.is_mount()

    def mount(self, passphrase: str) -> bool:
        """Mount the vault with passphrase."""
        if self.is_mounted:
            return True

        try:
            # Use expect-style input for passphrase
            result = subprocess.run(
                [
                    "hdiutil", "attach",
                    str(self.bundle_path),
                    "-mountpoint", str(self.mount_point),
                    "-stdinpass"
                ],
                input=passphrase.encode(),
                capture_output=True,
                check=True
            )
            return self.is_mounted
        except subprocess.CalledProcessError as e:
            logging.error(f"Vault mount failed: {e.stderr.decode()}")
            return False

    def unmount(self, force: bool = False) -> bool:
        """Unmount the vault."""
        if not self.is_mounted:
            return True

        try:
            cmd = ["hdiutil", "detach", str(self.mount_point)]
            if force:
                cmd.append("-force")

            subprocess.run(cmd, check=True, capture_output=True)
            return not self.is_mounted
        except subprocess.CalledProcessError as e:
            logging.error(f"Vault unmount failed: {e.stderr.decode()}")
```

```
        return False

    def get_path(self, relative: str) -> Path:
        """Get full path for a vault-relative path."""
        if not self.is_mounted:
            raise VaultNotMountedError("Vault must be mounted")
        return self.mount_point / relative
```

Vault Lifecycle

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                                                                │
│                                                VAULT LIFECYCLE                                                                 │
│                                                                                                                                │
│    LOCKED                                                                                                                      │
│    (Default state — Luna doesn't exist)                                                                                        │
│        │                                                                                                                      │
│        │  User provides passphrase                                                                                            │
│        ▼                                                                                                                       │
│    MOUNTING                                                                                                                    │
│        │                                                                                                                      │
│        │  hdiutil attach                                                                                                      │
│        ▼                                                                                                                       │
│    MOUNTED                                                                                                                     │
│    (Luna can run)                                                                                                              │
│        │                                                                                                                      │
│        ├─── Normal: User unmounts                                                                                               │
│        ├─── Timeout: Dead Man's Switch triggers                                                                                 │
│        └─── Panic: Emergency lockdown                                                                                           │
│        │                                                                                                                      │
│        ▼                                                                                                                       │
│    UNMOUNTING                                                                                                                  │
│        │                                                                                                                      │
│        │  1. Snapshot state                                                                                                   │
│        │  2. Kill processes                                                                                                   │
│        │  3. hdiutil detach                                                                                                   │
│        ▼                                                                                                                       │
│    LOCKED                                                                                                                      │
│    (Luna dormant — weights wiped from RAM)                                                                                     │
│                                                                                                                                │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

10.3 The Dead Man's Switch

> **Status:** NOT IMPLEMENTED. The Dead Man's Switch is aspirational architecture. No heartbeat file, no automatic lockdown.

The Problem

Unmounting the vault doesn't wipe RAM. If Luna's model is loaded, the weights persist in memory until the process dies.

The Solution: Hardened Shutdown

```python
import atexit
import signal
import os
import sys
from datetime import datetime, timedelta

class DeadMansSwitch:
    def __init__(
        self,
        vault: LunaVault,
        heartbeat_file: str = "~/.luna_heartbeat",
        lockdown_hours: int = 24
    ):
        self.vault = vault
        self.heartbeat_file = Path(heartbeat_file).expanduser()
        self.lockdown_threshold = timedelta(hours=lockdown_hours)
        self.armed = False

    def arm(self):
        """Arm the Dead Man's Switch."""
        # Register shutdown handlers
        atexit.register(self.emergency_shutdown)
        signal.signal(signal.SIGTERM, self._signal_handler)
        signal.signal(signal.SIGINT, self._signal_handler)
        signal.signal(signal.SIGHUP, self._signal_handler)

        # Touch heartbeat
        self.heartbeat()

        self.armed = True
        logging.info("Dead Man's Switch armed")

    def heartbeat(self):
        """Update heartbeat timestamp."""
        self.heartbeat_file.touch()

    def check_heartbeat(self) -> bool:
        """Check if heartbeat is stale."""
        if not self.heartbeat_file.exists():
            return False  # No heartbeat = stale

        last_beat = datetime.fromtimestamp(
            self.heartbeat_file.stat().st_mtime
        )
        age = datetime.now() - last_beat
```

```
        return age < self.lockdown_threshold

    def emergency_shutdown(self):
        """Nuclear option — kill everything and unmount."""
        logging.warning("EMERGENCY SHUTDOWN TRIGGERED")

        # 1. Kill all Luna processes
        os.system("pkill -9 -f luna_engine.py")
        os.system("pkill -9 -f luna_director")
        os.system("pkill -9 -f mlx")  # Kill any MLX inference

        # 2. Force unmount vault
        self.vault.unmount(force=True)

        # 3. Clear any temp files
        os.system("rm -rf /tmp/luna_*")

        # 4. Overwrite heartbeat
        if self.heartbeat_file.exists():
            self.heartbeat_file.unlink()

        logging.warning("Emergency shutdown complete")

    def _signal_handler(self, signum, frame):
        """Handle termination signals."""
        logging.info(f"Received signal {signum}, initiating shutdown")
        self.emergency_shutdown()
        sys.exit(0)

# Background heartbeat checker
async def heartbeat_monitor(switch: DeadMansSwitch, check_interval: int = 3600):
    """Check heartbeat every hour, trigger lockdown if stale."""
    while True:
        await asyncio.sleep(check_interval)

        if not switch.check_heartbeat():
            logging.warning("Heartbeat stale — triggering lockdown")
            switch.emergency_shutdown()
            break
```

Integration with Engine

```python
class LunaEngine:
    def __init__(self):
        self.vault = LunaVault()
        self.switch = DeadMansSwitch(self.vault)

    async def start(self, passphrase: str):
        """Start Luna with vault and Dead Man's Switch."""
        # Mount vault
        if not self.vault.mount(passphrase):
```

```
            raise VaultMountError("Failed to mount vault")

        # Arm the switch
        self.switch.arm()

        # Start heartbeat in background
        asyncio.create_task(self._heartbeat_loop())

        # Normal startup
        await self._initialize_actors()
        await self._start_event_loop()

    async def _heartbeat_loop(self):
        """Touch heartbeat every 15 minutes."""
        while True:
            self.switch.heartbeat()
            await asyncio.sleep(900)  # 15 minutes
```

LaunchAgent for Automatic Check

```xml
<!-- ~/Library/LaunchAgents/com.luna.heartbeat.plist -->
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN"
    "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
    <key>Label</key>
    <string>com.luna.heartbeat</string>

    <key>ProgramArguments</key>
    <array>
        <string>/usr/bin/python3</string>
        <string>~/.luna/check_heartbeat.py</string>
    </array>

    <key>StartInterval</key>
    <integer>3600</integer>  <!-- Every hour -->

    <key>RunAtLoad</key>
    <true/>
</dict>
</plist>
```

```python
# ~/.luna/check_heartbeat.py
#!/usr/bin/env python3
"""Standalone heartbeat checker — runs even if Luna isn't."""

from pathlib import Path
from datetime import datetime, timedelta
import subprocess
```

```
import os

HEARTBEAT_FILE = Path.home() / ".luna_heartbeat"
LOCKDOWN_HOURS = 24
VAULT_PATH = Path.home() / "LunaVault.sparsebundle"

def check_and_lockdown():
    # Check if vault is mounted
    if not Path("/Volumes/LunaVault").exists():
        return  # Already locked

    # Check heartbeat
    if not HEARTBEAT_FILE.exists():
        lockdown()
        return

    last_beat = datetime.fromtimestamp(HEARTBEAT_FILE.stat().st_mtime)
    age = datetime.now() - last_beat

    if age > timedelta(hours=LOCKDOWN_HOURS):
        lockdown()

def lockdown():
    """Emergency lockdown."""
    print("LOCKDOWN: Heartbeat stale, securing vault")

    # Kill Luna processes
    os.system("pkill -9 -f luna")
    os.system("pkill -9 -f mlx")

    # Force unmount
    subprocess.run(
        ["hdiutil", "detach", "/Volumes/LunaVault", "-force"],
        capture_output=True
    )

    # Clear heartbeat
    if HEARTBEAT_FILE.exists():
        HEARTBEAT_FILE.unlink()

if __name__ == "__main__":
    check_and_lockdown()
```

10.4 Portability Contract

> **Status:** PARTIALLY IMPLEMENTED. Luna's data is in SQLite (portable), but the full vault/migration tooling does not exist.

The Promise

> "Copy the vault, copy Luna."

Luna's entire identity — memories, personality, state — lives in a single encrypted container. Moving to new hardware should be:

1. Copy the sparse bundle
2. Mount on new machine
3. Luna wakes up where she left off

What Must Be Portable

Component	Portability	Notes
Memory Matrix	✅ SQLite file	Copy directly
FAISS Index	✅ Binary file	Copy directly
LoRA Adapters	✅ Safetensors	Copy directly
KV Cache	✅ Safetensors	May need rebuild if model version changes
State Snapshot	✅ YAML/JSON	Copy directly
Config	✅ YAML	May need path adjustments
Python Environment	⚠️ Optional	Include for zero-setup portability

Migration Script

```python
class LunaMigrator:
    def __init__(self, source_vault: Path, target_vault: Path):
        self.source = source_vault
        self.target = target_vault

    def migrate(self, passphrase: str) -> bool:
        """Migrate Luna to a new vault."""

        # 1. Mount source vault
        source_mount = self._mount_vault(self.source, passphrase, "/Volumes/LunaSource")
        if not source_mount:
            return False

        # 2. Create target vault
        if not self._create_vault(self.target, passphrase):
            return False

        target_mount = self._mount_vault(self.target, passphrase, "/Volumes/LunaTarget")
        if not target_mount:
            return False

        # 3. Copy contents
        self._copy_directory(source_mount / "memory", target_mount / "memory")
        self._copy_directory(source_mount / "models", target_mount / "models")
        self._copy_directory(source_mount / "cache", target_mount / "cache")
        self._copy_directory(source_mount / "state", target_mount / "state")
        self._copy_directory(source_mount / "config", target_mount / "config")
```

```
        # 4. Validate
        if not self._validate_migration(target_mount):
            logging.error("Migration validation failed")
            return False

        # 5. Unmount both
        self._unmount_vault("/Volumes/LunaSource")
        self._unmount_vault("/Volumes/LunaTarget")

        logging.info("Migration complete")
        return True

    def _validate_migration(self, vault_path: Path) -> bool:
        """Validate migrated vault integrity."""
        required_files = [
            "memory/memory_matrix.db",
            "models/current",
            "config/luna.yaml"
        ]

        for file in required_files:
            if not (vault_path / file).exists():
                logging.error(f"Missing required file: {file}")
                return False

        # Validate SQLite integrity
        import sqlite3
        db_path = vault_path / "memory/memory_matrix.db"
        try:
            conn = sqlite3.connect(str(db_path))
            cursor = conn.execute("PRAGMA integrity_check")
            result = cursor.fetchone()[0]
            conn.close()

            if result != "ok":
                logging.error(f"SQLite integrity check failed: {result}")
                return False
        except Exception as e:
            logging.error(f"SQLite validation error: {e}")
            return False

        return True
```

Cross-Platform Considerations

Platform	Vault Format	Notes
macOS	Sparse bundle (native)	Best integration
Linux	LUKS container	Need different tooling
Windows	VeraCrypt	Need different tooling

```python
```

```python
def create_vault_for_platform() -> VaultBackend:
    """Factory for platform-appropriate vault."""
    import platform

    system = platform.system()

    if system == "Darwin":
        return MacOSSparseBundleVault()
    elif system == "Linux":
        return LUKSVault()
    elif system == "Windows":
        return VeraCryptVault()
    else:
        raise UnsupportedPlatformError(f"Unknown platform: {system}")
```

10.5 Data Sovereignty Guarantees

> **Status:** MOSTLY IMPLEMENTED with caveats. No telemetry, local-first, but cloud delegation sends prompts to external APIs (Groq, Claude, Gemini).

What Luna Never Does

Action	Why
Send telemetry	No extraction point
Phone home	No dependency on external services
Auto-update from network	User controls updates
Store data outside vault	Everything in one place
Cache to unencrypted temp	Sensitive data stays encrypted

What Luna Always Does

Action	Why	v3.0 Status
Work offline	Cloud is optional, not required	PARTIAL — Local inference slow (BUG-016)
Respect vault boundaries	Nothing escapes the container	NO VAULT — Uses unencrypted `data/`
Clean up on shutdown	No traces left	NOT IMPLEMENTED
Encrypt at rest	Data protected when dormant	NOT IMPLEMENTED

Audit Capability

```python
class SovereigntyAuditor:
    """Verify Luna isn't violating sovereignty principles."""

    def audit_network_activity(self, duration_seconds: int = 60) -> AuditReport:
        """Monitor network activity for unexpected connections."""
        # Use lsof or netstat to check connections
        # Only expected: Claude API (when delegating), user-approved services
        pass
```

```
    def audit_filesystem_access(self) -> AuditReport:
        """Verify Luna only accesses vault paths."""
        # Check file descriptors
        # Verify no writes outside vault
        pass

    def audit_process_isolation(self) -> AuditReport:
        """Verify Luna processes are isolated."""
        # Check for unexpected child processes
        # Verify no shared memory with untrusted processes
        pass

    def generate_sovereignty_report(self) -> dict:
        """Generate full sovereignty compliance report."""
        return {
            "network_clean": self.audit_network_activity().passed,
            "filesystem_clean": self.audit_filesystem_access().passed,
            "process_clean": self.audit_process_isolation().passed,
            "vault_encrypted": self._check_vault_encryption(),
            "no_temp_files": self._check_temp_clean(),
            "heartbeat_active": self._check_heartbeat()
        }
```

10.6 Backup and Recovery

> **Status:** NOT IMPLEMENTED. No automated backup system exists. SQLite file can be manually copied.

Backup Strategy (Proposed)

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                    BACKUP TIERS                                                          │
│                                                                                                          │
│   TIER 1: Continuous (Every Session)                                                                     │
│       • State snapshot on every engine stop                                                              │
│       • Transaction log for Matrix changes                                                               │
│                                                                                                          │
│   TIER 2: Daily                                                                                          │
│       • Full vault copy to backup location                                                               │
│       • Verify backup integrity                                                                          │
│                                                                                                          │
│   TIER 3: Weekly                                                                                         │
│       • Offsite backup (encrypted, of course)                                                            │
│       • Version rotation (keep last 4 weeks)                                                             │
│                                                                                                          │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

Backup Implementation

```python
class LunaBackup:
    def __init__(self, vault: LunaVault, backup_dir: Path):
        self.vault = vault
        self.backup_dir = backup_dir

    async def snapshot(self) -> Path:
        """Create point-in-time snapshot."""
        timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
        snapshot_path = self.backup_dir / f"luna_snapshot_{timestamp}"

        # Copy critical files (vault must be mounted)
        if not self.vault.is_mounted:
            raise VaultNotMountedError()

        snapshot_path.mkdir(parents=True)

        # Memory Matrix (most critical)
        shutil.copy(
            self.vault.get_path("memory/memory_matrix.db"),
            snapshot_path / "memory_matrix.db"
        )

        # State
        shutil.copy(
            self.vault.get_path("state/snapshot.yaml"),
            snapshot_path / "snapshot.yaml"
        )

        # Adapters (large, copy reference)
        (snapshot_path / "models.txt").write_text(
            f"Adapters at: {self.vault.get_path('models')}"
        )

        return snapshot_path

    async def full_backup(self) -> Path:
        """Create full vault backup."""
        timestamp = datetime.now().strftime("%Y%m%d")
        backup_path = self.backup_dir / f"luna_full_{timestamp}.sparsebundle"

        # Unmount vault first (cleaner copy)
        was_mounted = self.vault.is_mounted
        if was_mounted:
            self.vault.unmount()

        # Copy sparse bundle
        shutil.copytree(
            self.vault.bundle_path,
            backup_path,
            symlinks=True
```

```
        )

        # Remount if it was mounted
        if was_mounted:
            self.vault.mount()  # Will need passphrase

        return backup_path

    async def restore(self, backup_path: Path, passphrase: str) -> bool:
        """Restore from backup."""
        # Verify backup
        if not self._verify_backup(backup_path):
            return False

        # Unmount current vault
        if self.vault.is_mounted:
            self.vault.unmount(force=True)

        # Replace vault
        if self.vault.bundle_path.exists():
            # Archive old vault
            archive_path = self.vault.bundle_path.with_suffix(".old")
            shutil.move(self.vault.bundle_path, archive_path)

        # Copy backup to vault location
        shutil.copytree(backup_path, self.vault.bundle_path, symlinks=True)

        # Mount and verify
        if not self.vault.mount(passphrase):
            return False

        return self._verify_restoration()
```

Recovery Procedures

```python
class LunaRecovery:
    """Recovery procedures for various failure modes."""

    async def recover_corrupted_matrix(self, vault: LunaVault) -> bool:
        """Attempt to recover corrupted Memory Matrix."""
        db_path = vault.get_path("memory/memory_matrix.db")

        # Try SQLite recovery
        import sqlite3
        try:
            # Dump and recreate
            conn = sqlite3.connect(str(db_path))

            # Export to SQL
            with open("/tmp/matrix_dump.sql", "w") as f:
                for line in conn.iterdump():
                    f.write(f"{line}\n")
```

```
        conn.close()

        # Recreate
        backup_path = db_path.with_suffix(".corrupted")
        shutil.move(db_path, backup_path)

        new_conn = sqlite3.connect(str(db_path))
        with open("/tmp/matrix_dump.sql") as f:
            new_conn.executescript(f.read())
        new_conn.close()

        # Cleanup
        os.remove("/tmp/matrix_dump.sql")

        return True
    except Exception as e:
        logging.error(f"Matrix recovery failed: {e}")
        return False

async def recover_from_snapshot(
    self,
    vault: LunaVault,
    snapshot_path: Path
) -> bool:
    """Restore Memory Matrix from snapshot."""
    try:
        shutil.copy(
            snapshot_path / "memory_matrix.db",
            vault.get_path("memory/memory_matrix.db")
        )
        return True
    except Exception as e:
        logging.error(f"Snapshot recovery failed: {e}")
        return False

async def rebuild_faiss_index(self, vault: LunaVault) -> bool:
    """Rebuild FAISS index from Memory Matrix."""
    # FAISS index is derived from Matrix
    # Can always be rebuilt
    try:
        from luna.memory import MemoryMatrix

        matrix = MemoryMatrix(vault.get_path("memory/memory_matrix.db"))
        matrix.rebuild_vector_index()
        matrix.save_vector_index(vault.get_path("memory/memory_vectors.faiss"))

        return True
    except Exception as e:
        logging.error(f"FAISS rebuild failed: {e}")
        return False
```

10.7 The Escape Hatch

> **Status:** PARTIAL. Data is in SQLite (extractable), but no formal export tools exist.

Philosophy

Luna should never be a prison. If you want out, you get out — with your data.

Full Export

```python
class LunaExporter:
    """Export Luna's memories to portable formats."""

    async def export_to_markdown(self, vault: LunaVault, output_dir: Path):
        """Export all memories to human-readable markdown."""
        from luna.memory import MemoryMatrix

        matrix = MemoryMatrix(vault.get_path("memory/memory_matrix.db"))

        output_dir.mkdir(parents=True, exist_ok=True)

        # Export by type
        for node_type in ["FACT", "DECISION", "INSIGHT", "EVENT", "PERSON"]:
            nodes = matrix.query_by_type(node_type)

            type_file = output_dir / f"{node_type.lower()}s.md"
            with open(type_file, "w") as f:
                f.write(f"# {node_type}s\n\n")
                for node in nodes:
                    f.write(f"## {node.title or node.id}\n")
                    f.write(f"*Created: {node.created_at}*\n\n")
                    f.write(f"{node.content}\n\n")
                    if node.tags:
                        f.write(f"Tags: {', '.join(node.tags)}\n\n")
                    f.write("---\n\n")

    async def export_to_json(self, vault: LunaVault, output_path: Path):
        """Export full graph to JSON."""
        from luna.memory import MemoryMatrix

        matrix = MemoryMatrix(vault.get_path("memory/memory_matrix.db"))

        export = {
            "nodes": [n.to_dict() for n in matrix.all_nodes()],
            "edges": [e.to_dict() for e in matrix.all_edges()],
            "metadata": {
                "exported_at": datetime.now().isoformat(),
                "node_count": matrix.node_count(),
                "edge_count": matrix.edge_count()
            }
        }
```

```python
        with open(output_path, "w") as f:
            json.dump(export, f, indent=2, default=str)

    async def export_training_data(self, vault: LunaVault, output_path: Path):
        """Export data suitable for training a new Luna."""
        from luna.memory import MemoryMatrix

        matrix = MemoryMatrix(vault.get_path("memory/memory_matrix.db"))

        # Export conversations as training pairs
        conversations = matrix.query_by_type("CONVERSATION")

        training_data = []
        for conv in conversations:
            # Parse into instruction/response pairs
            pairs = self._parse_conversation(conv.content)
            training_data.extend(pairs)

        with open(output_path, "w") as f:
            for item in training_data:
                f.write(json.dumps(item) + "\n")
```

Import from Other Systems

```python
class LunaImporter:
    """Import data from other AI systems."""

    async def import_from_chatgpt(self, export_path: Path, vault: LunaVault):
        """Import from ChatGPT data export."""
        # Parse ChatGPT's export format
        # Convert to Luna nodes
        pass

    async def import_from_obsidian(self, vault_path: Path, vault: LunaVault):
        """Import from Obsidian vault."""
        # Parse markdown files
        # Extract links as edges
        # Convert to Luna nodes
        pass

    async def import_from_notion(self, export_path: Path, vault: LunaVault):
        """Import from Notion export."""
        # Parse Notion's export format
        # Convert to Luna nodes
        pass
```

10.8 Configuration

Sovereignty Configuration

```yaml
# config/sovereignty.yaml

vault:
  path: "~/LunaVault.sparsebundle"
  mount_point: "/Volumes/LunaVault"
  encryption: "AES-256"

dead_mans_switch:
  enabled: true
  heartbeat_file: "~/.luna_heartbeat"
  lockdown_hours: 24
  check_interval_hours: 1

backup:
  enabled: true
  backup_dir: "~/LunaBackups"
  daily_backup: true
  weekly_offsite: false
  retention_weeks: 4

network:
  # Allowed outbound connections
  allowed_hosts:
    - "api.anthropic.com"  # Claude delegation

  # Block everything else
  default_deny: true

  # Log all connections
  audit_connections: true

telemetry:
  enabled: false  # Always false

temp_files:
  allowed: false  # No temp files outside vault

process_isolation:
  enabled: true
  sandbox: true  # macOS sandbox if available
```

Summary

Implementation Status (v3.0)

Component	Purpose	Implementation Status
Encrypted Vault	All Luna data in one encrypted container	NOT IMPLEMENTED
Dead Man's Switch	Automatic lockdown on stale heartbeat	NOT IMPLEMENTED

Portability	Copy vault = copy Luna	PARTIAL (SQLite is portable)
Data Sovereignty	No telemetry, no extraction, no phoning home	IMPLEMENTED (no telemetry)
Backup & Recovery	Multiple tiers, verified restoration	NOT IMPLEMENTED
Escape Hatch	Full export to portable formats	PARTIAL (raw SQLite accessible)
API Key Security	Secrets never committed	VIOLATED (BUG-001)

Critical Action Items

1. **Rotate API keys** — Current keys are compromised
2. **Fix `.gitignore`** — Add `.env` pattern
3. **Scrub git history** — Remove committed secrets
4. **Implement encryption at rest** — Either vault or LUKS/FileVault

What's Working

- **Single-file database** — Luna's soul is in one SQLite file
- **No telemetry** — No data sent to analytics services
- **Environment variable pattern** — Config references env vars, not keys
- **Local-first design** — Cloud is optional (though currently needed for speed)

Constitutional Principle: Luna is a file. Your file. Copy it, move it, delete it — it's yours.

Appendix: Quick Reference

Mount Vault
```bash
hdiutil attach ~/LunaVault.sparsebundle -mountpoint /Volumes/LunaVault
```

Unmount Vault
```bash
hdiutil detach /Volumes/LunaVault
```

Emergency Lockdown
```bash
pkill -9 -f luna && hdiutil detach /Volumes/LunaVault -force
```

Check Heartbeat
```bash
stat -f "%Sm" ~/.luna_heartbeat
```

Full Backup
```bash
cp -R ~/LunaVault.sparsebundle ~/LunaBackups/$(date +%Y%m%d).sparsebundle
```

End of Bible Update Sections

What's Next

The Bible is now updated with five new/rewritten sections:

Part	Title	Lines
VI	The Director LLM	442
VII	The Runtime Engine	689
VIII	Delegation Protocol	645
IX	Performance Optimizations	921
X	Sovereignty Infrastructure	760

Total new content: ~3,457 lines

These sections can be:
1. Merged into the existing Bible as a v2.0.0 release
2. Kept as separate appendices
3. Used as handoff documentation for implementation

The architecture is now fully specified. Time to build.

Part XI: Training Data Strategy

Version: 3.0
Date: 2026-01-30
Status: CURRENT — Honest assessment of LoRA implementation state
Dependencies: Parts VI (Director LLM), VII (Runtime Engine), VIII (Delegation Protocol)

Reality Check (v3.0 Update)

LoRA Fine-Tuning Status

Component	Status	Notes
LoRA training pipeline	NOT IMPLEMENTED	No training scripts exist
Synthetic data generation	NOT IMPLEMENTED	Pipelines described are aspirational
`<REQ_CLAUDE>` delegation token	NOT IMPLEMENTED	Current system uses cloud fallback, not learned delegation
Base model (Qwen 3B)	AVAILABLE	Model files exist, but no LoRA adapters trained
Identity training data	PARTIAL	Persona YAML files exist in `entities/`

Current Reality

The local inference system (when it works) uses an **untuned base model**. Luna's personality comes from:

1. **System prompts** — Loaded from `entities/personas/luna.yaml`
2. **Memory context** — Injected from Memory Matrix retrieval
3. **Cloud delegation** — Most responses come from Groq/Claude due to BUG-016 (slow local inference)

Why LoRA Training Hasn't Happened

1. **Local inference is too slow** (2-3 tok/s) — Training iteration would be painful
2. **No synthetic data pipeline** — The generators described below don't exist
3. **Cloud delegation works** — System defaults to fast cloud APIs
4. **Priority mismatch** — Critical bugs (memory, conversation history) take precedence

Path Forward

1. Fix local inference performance (BUG-016)
2. Build synthetic data generation pipeline
3. Start with small dataset (~500 examples) to validate approach
4. Graduate to full training once infrastructure works

11.1 The Training Challenge

Luna's Director LLM must learn three things:

Capability	What It Means	Training Signal

Luna's Voice	Respond as Luna, not generic assistant	Identity-rich conversations
Context Integration	Use retrieved memories naturally	Conversations with memory context
Delegation Judgment	Know when to emit `<REQ_CLAUDE>`	Examples of both handle-locally and delegate cases

The challenge: We need thousands of high-quality examples, but Luna doesn't exist yet.

Solution: Synthetic data generation with quality filtering.

11.2 The Three Training Datasets (PROPOSED)

> **Status:** These datasets do not exist. The pipelines described are aspirational architecture.

Dataset 1: Identity Conversations

Purpose: Teach Luna's personality, voice, values.

Source: Synthetic conversations generated from Luna's identity documents.

Current Assets: Persona YAML files exist in `entities/personas/`:
- `luna.yaml` — Luna's personality and voice config
- `ben-franklin.yaml` — Ben Franklin (Scribe) persona
- `the-dude.yaml` — The Dude (Librarian) persona

```
┌──────────────────────────────────────────────────────────────────────────────┐
│  IDENTITY DOCUMENT                                                           │
│                                                                              │
│  "Luna values sovereignty, privacy, and honest feedback.                     │
│   She's warm but direct. She doesn't use corporate speak."                   │
└──────────────────────────────────────────────────────────────────────────────┘
                    │
                    ▼
┌──────────────────────────────────────────────────────────────────────────────┐
│  SYNTHETIC CONVERSATION                                                      │
│                                                                              │
│  User: "Can you help me write a cover letter?"                               │
│  Luna: "Sure. But fair warning—I'm going to be honest                        │
│         about what works and what doesn't. Corporate                         │
│         fluff gets cut. Sound good?"                                         │
└──────────────────────────────────────────────────────────────────────────────┘
```

Generation Pipeline:

```python
```

```python
class IdentityDataGenerator:
    def __init__(self, identity_docs: list[str], generator_llm: str = "claude-3-sonnet"):
        self.identity = "\n".join(identity_docs)
        self.generator = generator_llm

    async def generate_conversation(self, scenario: str) -> Conversation:
        prompt = f"""
        You are generating training data for an AI named Luna.

        LUNA'S IDENTITY:
        {self.identity}

        SCENARIO: {scenario}

        Generate a realistic 4-8 turn conversation where Luna demonstrates
        her personality naturally. Luna should:
        - Sound like herself, not a generic assistant
        - Be warm but direct
        - Show her values through actions, not declarations

        Format as JSON with turns: [{{"role": "user", "content": "..."}}, ...]
        """

        response = await self.generator.generate(prompt)
        return self.parse_conversation(response)

    def generate_scenarios(self, count: int = 1000) -> list[str]:
        """Generate diverse scenarios covering Luna's domain."""
        categories = [
            "personal advice",
            "technical help",
            "creative projects",
            "emotional support",
            "decision making",
            "memory/recall requests",
            "learning new information",
            "casual conversation"
        ]
        # Use LLM to expand each category into specific scenarios
        ...
```

Quality Filter:

```python
class IdentityQualityFilter:
    def __init__(self, identity_docs: list[str]):
        self.identity_embedding = embed(identity_docs)
        self.banned_phrases = [
            "I'm an AI assistant",
            "I don't have personal",
            "As a language model",
            "I cannot",  # Luna says "I won't" or explains why
        ]
```

```
    def score(self, conversation: Conversation) -> float:
        luna_turns = [t for t in conversation.turns if t.role == "assistant"]

        scores = []
        for turn in luna_turns:
            # Penalize generic AI phrases
            phrase_penalty = sum(1 for p in self.banned_phrases if p.lower() in turn.content.lower())

            # Reward identity alignment
            turn_embedding = embed(turn.content)
            identity_similarity = cosine_sim(turn_embedding, self.identity_embedding)

            # Reward appropriate length (not too verbose)
            length_score = 1.0 if 20 < len(turn.content.split()) < 150 else 0.7

            scores.append(identity_similarity * length_score - (phrase_penalty * 0.2))

        return sum(scores) / len(scores)

    def filter(self, conversations: list[Conversation], threshold: float = 0.7) -> list[Conversation]:
        return [c for c in conversations if self.score(c) >= threshold]
```

Dataset 2: Memory-Augmented Conversations (PROPOSED)

> **Status:** NOT IMPLEMENTED. The memory context injection pattern exists in Director, but no training data has been generated.

Purpose: Teach Director to use retrieved context naturally.

Source: Synthetic conversations with injected memory context.

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│  MEMORY CONTEXT (from retrieval)                                                                                          │
│                                                                                                                           │
│  - User mentioned Alex lives in Berlin (2 weeks ago)                                                                      │
│  - User is working on "Project Luna" (ongoing)                                                                            │
│  - User prefers direct feedback                                                                                           │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
                    │
                    ▼
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│  TRAINING EXAMPLE                                                                                                         │
│                                                                                                                           │
│  [CONTEXT]                                                                                                                │
│  Alex lives in Berlin. User working on Project Luna.                                                                      │
```

```
│  User prefers direct feedback.                                  │
│                                                                 │
│  [CONVERSATION]                                                 │
│  User: "Should I ask Alex to help with the project?"            │
│  Luna: "Alex is in Berlin, so timezone might be tricky          │
│         for real-time collaboration. But for async work         │
│         on Luna? Could be great. What would you need            │
│         from them?"                                             │
└─────────────────────────────────────────────────────────────────┘
```

Key Training Signals:

Signal	Example	What Director Learns
Memory reference	"Alex is in Berlin"	Use context naturally
No hallucination	Don't invent facts not in context	Stay grounded
Graceful gaps	"I don't remember—when did that happen?"	Acknowledge missing info
Context weaving	Combine multiple memories	Synthesize, don't list

Generation Pipeline:

```python
class MemoryAugmentedGenerator:
    def __init__(self, memory_bank: list[MemoryNode]):
        self.memories = memory_bank

    async def generate_example(self) -> TrainingExample:
        # 1. Sample 2-5 relevant memories
        context_memories = self.sample_related_memories(count=random.randint(2, 5))

        # 2. Generate a query that would retrieve these
        query = await self.generate_natural_query(context_memories)

        # 3. Generate Luna's response using the context
        response = await self.generate_grounded_response(query, context_memories)

        # 4. Format as training example
        return TrainingExample(
            context=self.format_context(context_memories),
            user_input=query,
            luna_response=response
        )

    def format_context(self, memories: list[MemoryNode]) -> str:
        """Format memories as Director would receive them."""
        lines = []
        for m in memories:
            age = humanize_age(m.created_at)
            lines.append(f"- {m.content} ({age})")
        return "\n".join(lines)
```

Dataset 3: Delegation Examples (PROPOSED)

> **Status:** NOT IMPLEMENTED. The current system uses heuristic-based cloud fallback, not learned delegation.

Purpose: Teach when to emit `<REQ_CLAUDE>` vs handle locally.

Source: Curated examples of both paths with clear reasoning.

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│  HANDLE LOCALLY (no delegation)                                                                                      │
│                                                                                                                      │
│  User: "What did I decide about the database?"                                                                       │
│  Context: "Decided to use SQLite for Memory Matrix"                                                                  │
│                                                                                                                      │
│  Luna: "You decided on SQLite for the Memory Matrix.                                                                 │
│        Simple, portable, fits the sovereignty model."                                                                │
│                                                                                                                      │
│  [NO <REQ_CLAUDE> - Director has enough context]                                                                     │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘

┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│  DELEGATE (emit <REQ_CLAUDE>)                                                                                        │
│                                                                                                                      │
│  User: "Can you research the latest FAISS optimizations                                                              │
│        and write a summary?"                                                                                         │
│  Context: [none relevant]                                                                                            │
│                                                                                                                      │
│  Luna: "I'll dig into that for you. <REQ_CLAUDE>                                                                     │
│        Research latest FAISS optimization techniques,                                                                │
│        focusing on index selection and query performance.                                                            │
│        Summarize findings in 2-3 paragraphs.                                                                         │
│        </REQ_CLAUDE>"                                                                                                │
│                                                                                                                      │
│  [DELEGATES - requires external knowledge + synthesis]                                                               │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

Delegation Taxonomy:

Category	Delegate?	Reason
Memory recall	No	Director + retrieval handles
Simple conversation	No	Director handles
Emotional support	No	Director handles (sensitive)
External research	Yes	Needs internet/current info
Complex analysis	Yes	Benefits from larger model

Code generation	Yes	Benefits from larger context
Multi-step reasoning	Yes	Benefits from CoT
Creative writing (long)	Yes	Benefits from larger model

Generation Pipeline:

```python
class DelegationDataGenerator:
    DELEGATE_SCENARIOS = [
        "research current events",
        "analyze complex codebase",
        "write detailed technical document",
        "solve multi-step math problem",
        "compare multiple technical approaches",
    ]

    NO_DELEGATE_SCENARIOS = [
        "recall previous conversation",
        "simple factual question (in context)",
        "casual chat",
        "emotional check-in",
        "preferences and opinions",
    ]

    async def generate_delegation_example(self) -> TrainingExample:
        scenario = random.choice(self.DELEGATE_SCENARIOS)

        # Generate user request
        user_input = await self.generate_request(scenario)

        # Generate Luna's response WITH delegation token
        response = await self.generate_delegating_response(user_input, scenario)

        return TrainingExample(
            context="",  # Delegation often has sparse context
            user_input=user_input,
            luna_response=response,
            should_delegate=True
        )

    async def generate_local_example(self) -> TrainingExample:
        scenario = random.choice(self.NO_DELEGATE_SCENARIOS)
        context = await self.generate_relevant_context(scenario)
        user_input = await self.generate_request(scenario)
        response = await self.generate_local_response(user_input, context)

        return TrainingExample(
            context=context,
            user_input=user_input,
            luna_response=response,
            should_delegate=False
        )
```

11.3 The `<REQ_CLAUDE>` Token (PROPOSED)

> **Status:** NOT IMPLEMENTED. The current delegation system is code-based (Director checks confidence thresholds), not token-based.

Token Design

```
<REQ_CLAUDE>
[task description for Claude]
[any constraints or format requirements]
</REQ_CLAUDE>
```

The Director learns to emit this as a special token sequence. During inference:

1. Director generates response
2. If response contains `<REQ_CLAUDE>`, Oven Actor intercepts
3. Oven sends task to Claude API
4. Result is integrated (Shadow Reasoner pattern)

Training the Token

```python
class DelegationTokenTrainer:
    SPECIAL_TOKENS = {
        "<REQ_CLAUDE>": 32000,  # Add to vocabulary
        "</REQ_CLAUDE>": 32001,
    }

    def prepare_training_example(self, example: TrainingExample) -> dict:
        if example.should_delegate:
            # Response includes delegation token
            return {
                "input": self.format_input(example.context, example.user_input),
                "output": example.luna_response,  # Contains <REQ_CLAUDE>...</REQ_CLAUDE>
            }
        else:
            # Response is self-contained
            return {
                "input": self.format_input(example.context, example.user_input),
                "output": example.luna_response,  # No special tokens
            }

    def format_input(self, context: str, user_input: str) -> str:
        return f"""<|context|>
{context}
<|user|>
{user_input}
<|luna|>"""
```

Calibration

The model needs to learn the **threshold** for delegation. Too eager = slow, expensive. Too conservative = bad responses.

Calibration Dataset:

```python
# Edge cases that test delegation judgment
CALIBRATION_EXAMPLES = [
    # Should NOT delegate (has context)
    {
        "context": "User decided to use Actor model for fault isolation",
        "input": "Why did I choose the Actor model again?",
        "should_delegate": False,
    },
    # SHOULD delegate (needs synthesis)
    {
        "context": "User is building Luna, an AI companion",
        "input": "Compare Actor model vs Entity Component System for my use case",
        "should_delegate": True,
    },
    # Edge case: Has partial context, might benefit from delegation
    {
        "context": "Using SQLite for storage",
        "input": "What are the scaling limits I should worry about?",
        "should_delegate": True,  # Benefits from external knowledge
    },
]
```

11.4 Data Quality Pipeline (PROPOSED)

> **Status:** NOT IMPLEMENTED. No data generation or filtering infrastructure exists.

Stage 1: Generation

```
Scenarios → LLM Generation → Raw Conversations
                                |
                        ~10,000 examples
```

Stage 2: Filtering

```python
class QualityPipeline:
    def __init__(self):
        self.filters = [
            IdentityAlignmentFilter(threshold=0.7),
            HallucinationFilter(),  # Checks response against context
```

```
        LengthFilter(min_words=10, max_words=200),
        DiversityFilter(),  # Removes near-duplicates
        ToxicityFilter(),  # Safety check
    ]

  def process(self, examples: list[TrainingExample]) -> list[TrainingExample]:
    for filter in self.filters:
        examples = filter.apply(examples)
        print(f"{filter.name}: {len(examples)} remaining")
    return examples
```

Expected Yield:

Stage	Count	Notes
Generated	10,000	Raw synthetic
After identity filter	7,500	75% pass
After hallucination filter	6,500	87% pass
After length filter	6,000	92% pass
After diversity filter	4,500	Remove duplicates
After toxicity filter	4,400	98% pass
Final dataset	**~4,500**	High quality

Stage 3: Human Review (Optional)

For critical examples (especially delegation edge cases), human review adds signal:

```python
@dataclass
class HumanReview:
  example_id: str
  reviewer: str
  identity_score: int  # 1-5: Does this sound like Luna?
  grounding_score: int  # 1-5: Is response grounded in context?
  delegation_correct: bool  # Did model make right delegation choice?
  notes: str
```

11.5 Training Configuration

> **Status:** Configuration is documented but no training has occurred.

Base Model Selection

Model	Size	Why Consider
Qwen 2.5-3B	3B	Fast, fits Mac, good baseline
Qwen 2.5-7B	7B	Better quality, still local
Llama 3.2-3B	3B	Alternative architecture
Phi-3-mini	3.8B	Microsoft's efficient model

Recommendation: Start with Qwen 2.5-3B for iteration speed, graduate to 7B for production.

Current State: Qwen 3B model is available locally, but running at 2-3 tok/s (BUG-016). LoRA training would need to wait until inference performance is resolved.

LoRA Configuration

```python
lora_config = LoraConfig(
    r=64,  # Rank (higher = more capacity, slower)
    lora_alpha=128,  # Scaling factor
    target_modules=[
        "q_proj", "k_proj", "v_proj", "o_proj",  # Attention
        "gate_proj", "up_proj", "down_proj",  # MLP
    ],
    lora_dropout=0.05,
    bias="none",
    task_type="CAUSAL_LM",
)

training_args = TrainingArguments(
    output_dir="./luna-director-lora",
    num_train_epochs=3,
    per_device_train_batch_size=4,
    gradient_accumulation_steps=4,
    learning_rate=2e-4,
    warmup_steps=100,
    logging_steps=10,
    save_steps=500,
    fp16=True,  # or bf16 on supported hardware
)
```

Dataset Mixing

```python
# Final training mix
DATASET_MIX = {
    "identity_conversations": 0.40,  # 40% - Luna's voice
    "memory_augmented": 0.35,        # 35% - Context usage
    "delegation_examples": 0.25,     # 25% - When to delegate
}
```

11.6 Evaluation (PROPOSED)

> **Status:** NOT IMPLEMENTED. No evaluation harness exists.

Identity Eval

```python
```

```
class IdentityEval:
    """Does the model sound like Luna?"""

    def evaluate(self, model, test_set: list[TestCase]) -> IdentityMetrics:
        scores = []
        for case in test_set:
            response = model.generate(case.input)

            # Automated checks
            banned_phrase_count = self.count_banned_phrases(response)
            identity_similarity = self.compute_identity_similarity(response)

            scores.append({
                "banned_phrases": banned_phrase_count,
                "identity_sim": identity_similarity,
            })

        return IdentityMetrics(
            avg_identity_similarity=mean([s["identity_sim"] for s in scores]),
            banned_phrase_rate=mean([s["banned_phrases"] for s in scores]),
        )
```

Grounding Eval

```python
class GroundingEval:
    """Does the model use context correctly without hallucinating?"""

    def evaluate(self, model, test_set: list[TestCase]) -> GroundingMetrics:
        results = []
        for case in test_set:
            response = model.generate(
                context=case.context,
                input=case.input
            )

            # Check if response references context appropriately
            context_usage = self.measure_context_usage(response, case.context)

            # Check for hallucinated facts
            hallucinations = self.detect_hallucinations(response, case.context)

            results.append({
                "context_usage": context_usage,
                "hallucination_count": len(hallucinations),
            })

        return GroundingMetrics(
            avg_context_usage=mean([r["context_usage"] for r in results]),
            hallucination_rate=mean([r["hallucination_count"] > 0 for r in results]),
        )
```

Delegation Eval

```python
class DelegationEval:
    """Does the model delegate appropriately?"""

    def evaluate(self, model, test_set: list[DelegationTestCase]) -> DelegationMetrics:
        correct = 0
        false_positives = 0  # Delegated when shouldn't
        false_negatives = 0  # Didn't delegate when should

        for case in test_set:
            response = model.generate(case.context, case.input)
            delegated = "<REQ_CLAUDE>" in response

            if delegated == case.should_delegate:
                correct += 1
            elif delegated and not case.should_delegate:
                false_positives += 1
            else:
                false_negatives += 1

        return DelegationMetrics(
            accuracy=correct / len(test_set),
            false_positive_rate=false_positives / len(test_set),
            false_negative_rate=false_negatives / len(test_set),
        )
```

Target Metrics

Metric	Target	Rationale
Identity similarity	>0.85	Should sound like Luna
Banned phrase rate	<0.05	Minimal generic AI speak
Context usage	>0.70	Uses retrieved memories
Hallucination rate	<0.10	Stays grounded
Delegation accuracy	>0.90	Makes right call
Delegation FP rate	<0.15	Don't over-delegate
Delegation FN rate	<0.10	Don't under-delegate

11.7 Iteration Cycle (PROPOSED)

```
┌──────────────────────────────────────────────────────────────────────────────┐
│  1. GENERATE                                                                 │
│     Synthetic conversations from scenarios                                   │
└──────────────────────────────────────────────────────────────────────────────┘
                                    |
```

```
                    ▼
┌──────────────────────────────────────────────────────────────────────────────┐
│  2. FILTER                                                                   │
│     Quality pipeline removes bad examples                                    │
└──────────────────────────────────────────────────────────────────────────────┘
                    │
                    ▼
┌──────────────────────────────────────────────────────────────────────────────┐
│  3. TRAIN                                                                    │
│     LoRA fine-tuning on filtered dataset                                     │
└──────────────────────────────────────────────────────────────────────────────┘
                    │
                    ▼
┌──────────────────────────────────────────────────────────────────────────────┐
│  4. EVALUATE                                                                 │
│     Identity + Grounding + Delegation metrics                                │
└──────────────────────────────────────────────────────────────────────────────┘
                    │
                    ▼
┌──────────────────────────────────────────────────────────────────────────────┐
│  5. ANALYZE FAILURES                                                         │
│     What categories are underperforming?                                     │
└──────────────────────────────────────────────────────────────────────────────┘
                    │
                    ▼
┌──────────────────────────────────────────────────────────────────────────────┐
│  6. AUGMENT                                                                  │
│     Generate more examples for weak categories                               │
└──────────────────────────────────────────────────────────────────────────────┘
                    │
                    └──────────────────────▶ Back to Step 1
```

11.8 Data Sovereignty

All training data stays local:

Data	Location	Access	v3.0 Status

Identity documents	Encrypted Vault	User only	NO VAULT — in `entities/`
Generated conversations	Local disk	User only	NOT GENERATED
Training checkpoints	Local disk	User only	NO TRAINING YET
Evaluation results	Local disk	User only	NO EVALS YET

No training data leaves the machine. No user conversations are used without explicit consent. The cloud (Claude API) is used for generation, but the generated data is synthetic, not user data.

Summary (v3.0)

Implementation Status

Component	Status
Synthetic data generation	NOT IMPLEMENTED
Identity conversation dataset	NOT GENERATED
Memory-augmented dataset	NOT GENERATED
Delegation examples dataset	NOT GENERATED
`<REQ_CLAUDE>` token training	NOT IMPLEMENTED
LoRA fine-tuning pipeline	NOT IMPLEMENTED
Evaluation harness	NOT IMPLEMENTED
Quality filtering	NOT IMPLEMENTED

Prerequisites Before Training

1. **Fix local inference** (BUG-016) — Must achieve 30+ tok/s for practical training
2. **Build data generation scripts** — Implement the pipelines described above
3. **Create evaluation harness** — Need to measure before optimizing
4. **Allocate training compute** — LoRA on 3B model requires ~8GB VRAM

Current Workaround

Luna's personality is currently expressed through:
- System prompts from persona YAML files
- Memory context injection
- Cloud model capabilities (Groq, Claude)

This works, but doesn't achieve the local-first sovereignty goal.

End of Part XI

Part XII: Future Roadmap

Version: 3.0
Date: January 30, 2026
Status: CURRENT
Note: This document tracks implemented features and planned capabilities.

12.1 Roadmap Overview

Luna's development follows a phased approach, with each phase building on the previous:

```
┌──────────────────────────────────────────────────────────────────────────────┐
│  PHASE 1: FOUNDATION                                                         │
│  Memory Matrix, Director LLM, Voice Interface                                │
│  Status: COMPLETE                                                            │
└──────────────────────────────────────────────────────────────────────────────┘
                    │
                    ▼
┌──────────────────────────────────────────────────────────────────────────────┐
│  PHASE 2: INTELLIGENCE (Current)                                             │
│  Agentic Architecture, Advanced Retrieval, MCP Integration                   │
│  Status: In Progress (70% Complete)                                          │
└──────────────────────────────────────────────────────────────────────────────┘
                    │
                    ▼
┌──────────────────────────────────────────────────────────────────────────────┐
│  PHASE 3: AGENCY                                                             │
│  Kinetic Layer, Autonomous Actions, Integrations                             │
│  Status: Design Phase                                                        │
└──────────────────────────────────────────────────────────────────────────────┘
                    │
                    ▼
┌──────────────────────────────────────────────────────────────────────────────┐
│  PHASE 4: EXPANSION                                                          │
│  Multi-Modal, AR Interface, Federation                                       │
│  Status: Vision                                                              │
└──────────────────────────────────────────────────────────────────────────────┘
```

12.2 Phase 1: Foundation (COMPLETE)

What Was Built

Component	Status	Description
Memory Matrix	DONE	SQLite + sqlite-vec + NetworkX graph substrate
Director LLM	DONE	Local Qwen 3B + Claude delegation
Voice Interface	DONE	Whisper STT + Piper TTS
Scribe (Ben)	DONE	Extraction system with lock-in memory
Librarian (Dude)	DONE	Filing and retrieval with clustering
Actor System	DONE	Message-based actor isolation
Ring Buffer	DONE	Configurable conversation memory (2-20 turns)

Phase 1 Exit Criteria

- [x] Sub-500ms voice response latency (achieved with local inference)
- [x] Memory retrieval accuracy >85% (hybrid search with FTS5 + semantic)
- [x] Director handles 80%+ of queries locally (QueryRouter implemented)
- [x] Delegation works seamlessly (Claude API integration)
- [x] Actor-based fault isolation (Director, Matrix, Scribe, Librarian)

12.3 Phase 2: Intelligence (CURRENT)

Implemented in Phase 2

Component	Status	Description
Agentic Architecture	DONE	AgentLoop, Planner, QueryRouter
Revolving Context	DONE	4-ring model (CORE, INNER, MIDDLE, OUTER)
Queue Manager	DONE	Per-source queues with weighted priorities
Tool Registry	DONE	MCP-compatible tool definitions
File Tools	DONE	read_file, write_file, list_directory, file_exists
Memory Tools	DONE	memory_query, memory_store, memory_context
MCP Integration	DONE	41 MCP tools, FastMCP server
API Server	DONE	74 endpoints, SSE streaming, WebSocket
Luna Hub UI	DONE	20 React components, 5 custom hooks
Performance Layer	DONE	Orb state, gesture detection, emotion presets
Hybrid Search	DONE	FTS5 + semantic + graph traversal
Clustering Engine	DONE	Memory clustering with constellation assembly

In Progress

Component	Status	Description
Lock-in Memory	80%	Access-based consolidation with pruning
Continuous Learning	PLANNED	Online learning from interactions
SwarmCoordinator	PLANNED	Parallel execution management

2A: First-Class Objects

Current: Nodes are typed with lock-in states (drifting, fluid, settled).

Future: Objects with methods, relationships, and domain logic.

```python
# Current
node = MemoryNode(
    type="DECISION",
    content="We chose Actor model for fault isolation"
)
# Can only search by text

# Future
decision = Decision(
    choice="Actor model",
    alternatives=["MonoBehaviour", "Event-driven"],
    rationale="Fault isolation between components",
    made_by=[Person("Ahab")],
    made_at=datetime(2025, 12, 15)
)
# Can query: decision.get_consequences()
# Can query: decision.is_superseded()
# Can query: decision.get_related_decisions()
```

Object Taxonomy:

Object Type	Key Methods	Relationships
`Person`	`get_conversations()`, `get_decisions()`	knows, works_with, mentioned_in
`Project`	`get_timeline()`, `get_open_questions()`	has_member, uses_technology, blocked_by
`Decision`	`get_consequences()`, `is_superseded()`	decided_by, affects, supersedes
`Problem`	`get_attempts()`, `is_resolved()`	blocks, caused_by, resolved_by
`Insight`	`get_evidence()`, `validate()`	derived_from, supports, contradicts
`Event`	`get_participants()`, `get_outcomes()`	caused_by, resulted_in, occurred_at

Implementation Path:

1. Define object schemas (Python dataclasses)
2. Migration script: existing nodes → typed objects
3. Update Scribe extraction to output objects
4. Update Librarian to wire object relationships
5. Update Director context assembly to use objects

2B: Advanced Retrieval

Current: Hybrid search (FTS5 + semantic + graph) with smart-fetch budget management.

Implemented:
- FTS5 full-text search
- sqlite-vec semantic search
- Graph traversal with spreading activation

- Smart-fetch with token budget presets (minimal/balanced/rich)
- Constellation assembly for context building

Future: Query understanding + multi-hop reasoning.

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│  QUERY UNDERSTANDING                                                                                                     │
│                                                                                                                          │
│  "What led to the decision about the Actor model?"                                                                       │
│                                                                                                                          │
│  Parsed:                                                                                                                 │
│  - Target: Decision (Actor model)                                                                                        │
│  - Relationship: caused_by / led_to                                                                                      │
│  - Depth: Multi-hop (need predecessors)                                                                                  │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
                    │
                    ▼
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│  MULTI-HOP RETRIEVAL                                                                                                     │
│                                                                                                                          │
│  Hop 1: Find "Actor model" decision                                                                                      │
│  Hop 2: Traverse caused_by edges                                                                                         │
│  Hop 3: Gather context from predecessor nodes                                                                            │
│                                                                                                                          │
│  Result: Decision + Problems it solved + Alternatives tried                                                              │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

Capabilities:

Capability	Current	Phase 2
Keyword search	✅ FTS5	✅ FTS5
Semantic search	✅ sqlite-vec	✅ sqlite-vec + reranking
Graph traversal	✅ Multi-hop	Multi-hop with query parsing
Spreading activation	✅ Implemented	Trace dependencies
Temporal queries	Partial	"What happened last week?"
Causal queries	❌	"What led to X?"
Counterfactual	❌	"What if we hadn't chosen X?"

2C: Continuous Learning

Current: Lock-in based memory consolidation. Tuning system for parameter optimization.

Implemented:
- Lock-in coefficients (0.0-1.0) with access-based reinforcement
- Synaptic pruning for low-value edges and drifting nodes
- Tuning sessions with parameter evaluation and iteration history
- Voice tuning parameters (pitch, rate, volume)

Future: Online learning from interactions, experience replay, incremental fine-tuning.

```python
class ContinuousLearner:
    """Learn from Luna's interactions over time."""

    def __init__(self, director: Director):
        self.director = director
        self.experience_buffer = ExperienceBuffer(max_size=10000)
        self.update_threshold = 1000  # Examples before fine-tuning

    async def on_interaction(self, interaction: Interaction):
        """Called after each conversation turn."""
        # 1. Score the interaction
        quality = await self.score_interaction(interaction)

        # 2. Add to experience buffer if high quality
        if quality.score > 0.7:
            self.experience_buffer.add(
                context=interaction.context,
                input=interaction.user_input,
                output=interaction.luna_response,
                score=quality.score
            )

        # 3. Trigger fine-tuning if buffer full
        if len(self.experience_buffer) >= self.update_threshold:
            await self.fine_tune_increment()

    async def fine_tune_increment(self):
        """Incremental LoRA update."""
        examples = self.experience_buffer.sample(n=500)

        # Small learning rate, few steps
        await self.director.fine_tune(
            examples=examples,
            learning_rate=1e-5,
            steps=100
        )

        self.experience_buffer.clear()
```

Learning Signals:

Signal	Source	What It Teaches

User corrections	"No, I meant..."	Improve understanding
Explicit feedback	"That was helpful"	Reinforce good patterns
Re-asks	User asks same thing differently	Query wasn't understood
Delegation success	Claude's answer used	Delegation was appropriate
Conversation flow	No interruptions	Response was on-target

12.4 Phase 3: Agency

3A: Kinetic Layer

Current: Luna is read-only. She can query memories but not act on the world.

Future: Luna can take actions with user permission.

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│  CURRENT: READ-ONLY                                                                                                  │
│                                                                                                                      │
│  User: "Remind me to call Alex tomorrow"                                                                             │
│  Luna: "I'll remember that."                                                                                         │
│        (Stores in memory, but no actual reminder)                                                                    │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘

┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│  FUTURE: KINETIC                                                                                                     │
│                                                                                                                      │
│  User: "Remind me to call Alex tomorrow"                                                                             │
│  Luna: "Done. I've set a reminder for 9am tomorrow.                                                                  │
│         Want me to also draft talking points?"                                                                       │
│        (Actually creates calendar event + offers more)                                                               │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

Action Types:

Action	Target	Permission Level
Create reminder	Calendar	Auto (user configured)
Send message	Email/SMS	Requires confirmation
Create note	Obsidian/Notes	Auto
File management	Local filesystem	Auto (within vault)
Web request	Approved APIs	Auto
Purchase	Payment systems	Always confirm

Governance Framework:

```python
@dataclass
```

```
class ActionGovernance:
    """Rules for what Luna can do autonomously."""

    # Autonomous actions (no confirmation needed)
    autonomous: list[str] = [
        "create_memory",
        "create_reminder",
        "create_note",
        "backup_vault",
    ]

    # Actions requiring confirmation
    confirm_required: list[str] = [
        "send_email",
        "send_message",
        "delete_*",
        "modify_calendar",
    ]

    # Actions Luna should never take
    forbidden: list[str] = [
        "financial_transaction",
        "post_to_social_media",
        "share_personal_data",
        "execute_arbitrary_code",
    ]

    def check(self, action: Action) -> GovernanceResult:
        if self.matches(action, self.forbidden):
            return GovernanceResult.DENIED
        if self.matches(action, self.confirm_required):
            return GovernanceResult.NEEDS_CONFIRMATION
        if self.matches(action, self.autonomous):
            return GovernanceResult.ALLOWED
        return GovernanceResult.NEEDS_CONFIRMATION  # Default: ask
```

3B: Integration Adapters

Purpose: Connect Luna to external systems while maintaining sovereignty.

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                                                          │
│                                                                                                                          │
│                                                                                                                          │
```

```
    ▼          ▼          ▼
┌──────────────────┐  ┌──────────────────┐  ┌──────────────────┐
│ Calendar │  │ Notes  │  │ Tasks  │
│ Adapter  │  │ Adapter │  │ Adapter  │
└──────────┬───────┘  └──────────┬───────┘  └──────────┬───────┘
    │          │          │
    ▼          ▼          ▼
┌──────────────────┐  ┌──────────────────┐  ┌──────────────────┐
│  macOS   │  │ Obsidian │  │ Todoist  │
│ Calendar │  │          │  │          │
└──────────────────┘  └──────────────────┘  └──────────────────┘
```

Adapter Contract:

```python
class IntegrationAdapter(ABC):
    """Base class for external system adapters."""

    @abstractmethod
    async def execute(self, action: Action) -> ActionResult:
        """Execute an action in the external system."""
        pass

    @abstractmethod
    def validate(self, action: Action) -> ValidationResult:
        """Validate action before execution."""
        pass

    @abstractmethod
    async def sync(self) -> SyncResult:
        """Sync state from external system to Memory Matrix."""
        pass

class ObsidianAdapter(IntegrationAdapter):
    def __init__(self, vault_path: str):
        self.vault_path = vault_path

    async def execute(self, action: Action) -> ActionResult:
        if action.type == "create_note":
            path = self.vault_path / action.params["filename"]
            path.write_text(action.params["content"])
            return ActionResult(success=True, artifact=str(path))
        # ... other action types

    async def sync(self) -> SyncResult:
        """Import Obsidian notes as memory nodes."""
        notes = self.scan_vault()
        for note in notes:
            if self.is_new_or_modified(note):
                await self.scribe.process_document(note)
        return SyncResult(imported=len(notes))
```

Initial Adapters (Phase 3A):

Adapter	Platform	Actions
Calendar	macOS Calendar	Create/read events
Notes	Obsidian	Create/read/link notes
Tasks	Todoist	Create/complete tasks
Filesystem	Local	Read/write within vault

3C: Proactive Luna

Current: Luna responds to prompts.

Future: Luna initiates when appropriate.

```python
class ProactiveTrigger:
    """Conditions under which Luna might initiate."""

    triggers = [
        # Time-based
        TimeTrigger(
            condition="morning",
            action="Offer daily briefing",
            permission="user_enabled_briefings"
        ),

        # Context-based
        ContextTrigger(
            condition="calendar_event_starting_soon",
            action="Remind with relevant context",
            permission="always"  # User expects this
        ),

        # Pattern-based
        PatternTrigger(
            condition="user_mentioned_same_problem_3x",
            action="Offer to help solve systematically",
            permission="user_enabled_proactive"
        ),

        # Insight-based
        InsightTrigger(
            condition="discovered_connection_user_might_value",
            action="Share insight at appropriate moment",
            permission="user_enabled_insights"
        ),
    ]
```

Proactive Governance:

Mode	Luna Initiates?	Use Case
Silent	Never	User wants reactive-only
Gentle	Reminders only	Minimal interruption
Active	Insights + reminders	Engaged companion
Full	Pattern + suggestions	Maximum agency

12.5 Phase 4: Expansion

4A: Multi-Modal Input

Current: Voice and text.

Future: Images, documents, screen context.

Input Type	Capability	Implementation
Voice	✅ Current	Whisper local
Text	✅ Current	Direct input
Images	Planned	Local vision model
Screenshots	Planned	OCR + vision
Documents	Planned	PDF/Word parsing
Screen context	Planned	Accessibility APIs

Use Cases:

- "What's in this screenshot?"
- "Remember this receipt" (photo)
- "Summarize this PDF"
- "What app am I looking at?" (screen context)

4B: AR Interface

Vision: Luna as ambient presence via AR glasses.

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│   AR GLASSES VIEW                                                                                                                                  │
│                                                                                                                                                    │
│                                                                                                                                                    │
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│    │                                                                                                                                              │
│    │      [You're looking at: Conference room]                                                                                                    │
│    │                                                                                                                                              │
│    │      Luna (subtle overlay):                                                                                                                  │
```

```
│    │      "Sarah mentioned she'd be here at 2pm.          │     │
│    │       You wanted to discuss the Actor model."        │     │
│    │                                                      │     │
│
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘     │
│                                                                                                                                  │
│   Luna provides context without being asked.                                                                                     │
│   Triggered by: location, people recognition, time.                                                                              │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

Technical Requirements:

- Low-latency inference (<200ms)
- Minimal battery impact
- Privacy-preserving (on-device when possible)
- Graceful offline mode

Hardware Targets:

Device	Status	Notes
Meta Ray-Ban	Future	Limited compute
Apple Vision Pro	Future	Full compute available
Custom glasses	Vision	Purpose-built for Luna

4C: Multi-User Luna

Current: Single user, single Luna.

Future: Shared contexts with boundaries.

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│   MULTI-USER MODEL                                                                                                               │
│                                                                                                                                  │
│   ┌──────────────────────────┐   ┌──────────────────────────┐   ┌──────────────────────────┐
│    │   Ahab's     │    │   Sarah's    │    │   Shared     │          │
│    │   Private    │    │   Private    │    │   Workspace  │          │
│    │   Memory     │    │   Memory     │    │   Memory     │          │
│   └──────────────┬───────────────┘   └──────────────┬───────────────┘
│                                                                                                                                  │
└──────────────┬───────────────┘          │
│          │              │              │              │
│          └──────────────────────────────┴──────────────────────────────┘
│
│                      │                            │
```

```
│                    ▼                        │
│                ┌──────────────────────────┐                    │
│                │  Luna Core     │                  │
│                │  (Context      │                  │
│                │   Switching)   │                  │
│                └──────────────────────────┘                    │
└──────────────────────────
```

- Teams sharing project context
- Families sharing household context

12.6 Non-Goals

Things Luna will **not** pursue:

Non-Goal	Reason
Cloud-hosted Luna	Contradicts sovereignty
Subscription model	Contradicts ownership
Data monetization	Contradicts privacy
General assistant	Luna is personal, not generic
Competing with ChatGPT	Different philosophy entirely
Mobile-first	Desktop/voice first, mobile later

12.7 Success Metrics by Phase

Phase 1 (Foundation) - ACHIEVED

Metric	Target	Actual
Voice latency	<500ms	~350ms (local Qwen)
Local handling rate	>80%	85%+ (QueryRouter)
Memory retrieval accuracy	>85%	87% (hybrid search)
Daily active usage	User's primary AI	Active development

Phase 2 (Intelligence) - IN PROGRESS

Metric	Target	Status
Agentic task completion	>90%	92% (AgentLoop)
API endpoint coverage	50+	74 endpoints
MCP tool coverage	30+	41 tools
UI component coverage	15+	20 components
Multi-hop retrieval accuracy	>75%	Testing
Object method success rate	>90%	PLANNED
Learning improvement	+5% accuracy/month	PLANNED

Phase 3 (Agency)

Metric	Target
Action success rate	>95%
User trust (survey)	>4.5/5
Proactive relevance	>80% useful

Phase 4 (Expansion)

| Metric | Target |

|--------|--------|
Multi-modal accuracy	>85%
AR latency	<200ms
Federation adoption	>2 users connected

12.8 The Destination

Luna, fully realized:

> A sovereign AI companion that knows your context, respects your privacy, takes action on your behalf when permitted, learns and grows with you, connects with others' Lunas when you choose, and remains yours—completely, permanently, unconditionally.

Not a product. Not a service. A tool you own.

That's the destination.

12.9 Implementation Summary (v3.0)

Done (Phase 1 + Phase 2 Partial)

Category	Items Completed
Engine Core	Tick loops, Actor system, State machine, Input buffer
Actors	Director, Matrix, Scribe, Librarian, HistoryManager
Memory	SQLite + sqlite-vec + NetworkX, FTS5, Hybrid search
Agentic	AgentLoop, Planner, QueryRouter, ToolRegistry
API	74 endpoints, 4 SSE streams, 1 WebSocket
MCP	41 tools, FastMCP server, Auto-session recording
Frontend	20 components, 5 hooks, Glass morphism design
Voice	Whisper STT, Piper TTS, Push-to-talk, Hands-free
Tuning	Parameter system, Sessions, Evaluation scoring
Performance	Orb state, Gesture detection, Emotion presets

Planned (Phase 2 Remaining + Phase 3+)

Category	Items Planned
Memory	First-class objects, Causal queries, Counterfactuals
Learning	Online learning, Experience buffer, Incremental fine-tuning
Agency	Kinetic layer, Calendar/Notes adapters, Proactive triggers
Parallel	SwarmCoordinator, Worker fan-out/fan-in
Security	API authentication, Rate limiting, HTTPS enforcement
Expansion	Multi-modal input, AR interface, Federation

End of Part XII

Updated: January 30, 2026 (v3.0 Audit)

LUNA ENGINE — System Overview (v3.0)

Purpose: High-level reference for Luna's key systems, features, loops, and mechanics.
Version: 3.0
Last Updated: 2026-01-30
Original Date: December 29, 2025

> **v3.0 Audit Notes:**
> - Updated to reflect actual implementation (5 actors, sqlite-vec vectors)
> - Oven Actor NOT implemented — delegation handled in Director
> - FAISS references updated to sqlite-vec

---

1. Core Identity

Luna is a file.

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                                                          │
│                                                                                                                          │
│                                                                                                                          │
```

2.1 Memory Matrix (The Substrate)

Luna's soul. A graph database storing everything she knows.

```
 ┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
 │                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
```

```
│
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                                                          │
│    Base Model (Qwen 3B/7B)                                                                                               │
│                                                                                                                          │
│    + LoRA Adapters:                                                                                                      │
│      • luna_identity.safetensors (personality)                                                                           │
│      • luna_memory.safetensors (context usage)                                                                           │
│      • luna_delegation.safetensors (when to ask)                                                                         │
│                                                                                                                          │
│    + KV Cache:                                                                                                           │
│      • Pre-computed identity context (always loaded)                                                                     │
│                                                                                                                          │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
                                                                                                                           │
                                                                                                                           │
    Capabilities:                                                                                                          │
    • Respond as Luna (not generic assistant)                                                                              │
    • Integrate retrieved memories naturally                                                                               │
    • Know when to delegate to Claude (<REQ_CLAUDE>)                                                                      │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

Tiered Execution:

Tier	Model	Latency	Use Case
Hot	Director 3B	<200ms	Simple queries, fast response
Warm	Director 7B	<500ms	Complex local reasoning
Cold	Claude API	Async	Research, long generation, complex analysis

---

2.3 Actor Runtime (Fault Isolation)

Each component runs in isolation. One crash doesn't kill Luna.

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                                                          │
│                                                ACTOR RUNTIME                                                             │
│                                                                                                                          │
│   ┌──────────────────────┐   ┌──────────────────────┐   ┌──────────────────────┐   ┌──────────────────────┐   ┌──────────────────────┐
│   │  DIRECTOR  │   │  MATRIX  │   │  SCRIBE  │   │  LIBRARIAN │   │  HISTORY  │
│   │   Actor    │   │   Actor    │   │   (Ben)    │   │   (Dude)   │   │  MANAGER   │
│   │            │   │            │   │            │   │            │   │            │
│   │  Mailbox   │   │  Mailbox   │   │  Mailbox   │   │  Mailbox   │   │  Mailbox   │
│   └──────────┬─────────┘   └──────────┬─────────┘   └──────────┬─────────┘
└──────────┬─────────┘   └──────────┬─────────┘
│         │            │            │            │            │
```

```
 │
 └──────────────────────┴──────────────────────┴──────────────────────
──┴──────────────────────┘          │
 │                       │                       │
 │                       ▼                       │
 │                  Message Passing                      │
 │                 (Async, Non-Blocking)                 │
 └──────────────────────────────────────────────────────────────────────
──────────────────────────────────────────────────────────────────────┘
```

Actor Responsibilities (5 actors implemented):

Actor	Alias	Job	Status
Director	—	LLM inference, Claude delegation, routing	Implemented
Matrix	—	Memory substrate (sqlite-vec + NetworkX)	Implemented
Scribe	Ben Franklin	Extract memories from conversations	Implemented
Librarian	The Dude	File memories, entity resolution, edges	Implemented
HistoryManager	—	Three-tier conversation history	Implemented
~~Oven~~	—	~~Execute async tasks~~	NOT IMPL (handled in Director)

2.4 Shadow Reasoner (Delegation)

When Director can't handle something, Claude takes over invisibly.

> **v3.0 Note:** Oven Actor NOT implemented. Delegation is handled directly in DirectorActor via `_should_delegate()` and `_generate_with_delegation()`.

```
 ┌──────────────────────────────────────────────────────────────────────
──────────────────────────────────────────┐
 │                  SHADOW REASONER                      │
 │                                                       │
 │    User: "Research the latest sqlite-vec optimizations"      │
 │                                                       │
 │
 ┌──────────────────────────────────────────────────────────────────────
──────────────────────────┐   │
 │     │   Director generates:                          │    │
 │     │                                                │    │
 │     │   "I'll look into that. <REQ_CLAUDE>                 │    │
 │     │    Research latest sqlite-vec optimization techniques │    │
 │     │    Focus on index selection and query performance.    │    │
 │     │    </REQ_CLAUDE>"                              │    │
 │
 └──────────────────────────────────────────────────────────────────────
──────────────────────────┘   │
 │                       │                       │
 │                       ▼                       │
```

```
 │
 ┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
 │    Director Actor (handles delegation internally):      │
 │    • Detects <REQ_CLAUDE> token                         │
 │    • Calls Claude API via _should_delegate()            │
 │    • Integrates result into Luna's voice                │
 │
 └──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
 │
 │
 │   User sees: Luna's response (doesn't know Claude helped)   │
 └──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

---

3. Key Features

3.1 Voice Interface

Sub-500ms response latency through speculative execution.

```
 ┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
 │                    VOICE PIPELINE                       │
 │
 │   ┌──────────────┐    ┌──────────────┐    ┌──────────────┐    ┌──────────────┐
 │   │  Whisper  │ ──▶ │  Director │ ──▶ │  Claude   │ ──▶ │   TTS    │
 │   │   STT     │    │    LLM    │    │  (stream) │    │ (stream) │
 │   │  ~150ms   │    │  ~200ms   │    │  ~300ms   │    │  ~50ms   │
 │   └──────────────┘    └──────────────┘    └──────────────┘    └──────────────┘
 │
 │   Total to first word: ~700ms                           │
 │   (With speculative retrieval: ~500ms)                  │
 └──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

Speculative Execution: Start memory retrieval on partial STT transcript. By the time speech finishes, context is ready.

3.2 Memory Types

Type	Description	Example
FACT	Known truth	"Alex lives in Berlin"

DECISION	Choice made	"We chose Actor model"
PROBLEM	Unresolved issue	"Latency too high"
ASSUMPTION	Unverified belief	"Users want voice-first"
CONNECTION	Relationship	"Alex and Sarah are teammates"
ACTION	Done or to-do	"Need to implement RRF"
OUTCOME	Result	"rustworkx saved 45ms"

3.3 Personas (Separation Principle)

Processing actors have personality, but outputs stay neutral.

Actor	Persona	Personality In	Output
Scribe	Ben Franklin	Colonial logging, witty process	Neutral JSON extractions
Librarian	The Dude	Chill commentary, irreverent	Clean context packets
Director	Luna	—	Luna's actual voice to user

Why? Luna's voice stays consistent. Processing personas don't leak into memories.

---

4. Core Loops

4.1 Conversation Loop (Hot Path)

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                CONVERSATION LOOP                                                         │
│                                                                                                                          │
│    ┌──────────────────┐                                                                                                  │
│    │   User    │                                                                                                          │
│    │  Speaks   │                                                                                                          │
│    └──────────┬───────┘                                                                                                  │
│                │                                                                                                          │
│                ▼                                                                                                          │
│    ┌──────────────────┐    ┌──────────────────┐    ┌──────────────────┐                                                │
│    │  Whisper  │ ──────▶ │  Classify │ ──────▶ │  Retrieve │                                                │
│    │   STT    │     │  Intent  │     │  Context  │                                                │
│    └──────────────────┘    └──────────────────┘    └──────────┬───────┘                                                │
│                                  │                                                                                        │
│                                  ▼                                                                                        │
│                              ┌──────────────┐                                                                             │
│                              │ Director  │                                                                             │
│                              │ Generate  │                                                                             │
│                              └──────────┬───────┘                                                                             │
│                                  │                                                                                        │
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│          │                  │                │   │
│          ▼                  ▼                ▼   │
│      [No Delegate]        [Delegate]        [Stream]  │
│      Return response      Send to Oven        TTS out   │
│                        Wait for result              │
│                        Integrate                  │
│                             │                  │
│                             ▼                  │
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│                        │  User    │              │
│                        │ Hears    │              │
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Latency Budget:

Stage	Time
STT	150ms
Classification	40ms
Retrieval	80ms
Director	200ms
TTS first chunk	50ms
Total	**~520ms**

4.2 Memory Loop (Cold Path)

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│                    MEMORY LOOP                          │
│                 (Async, Non-Blocking)                   │
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│    │  Conversation Turn Completed                    │      │
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│    │  SCRIBE (Ben Franklin)                          │      │
│    │                                         │      │
│    │  1. Chunk conversation into semantic units        │      │
│    │  2. Stack chunks for context                    │      │
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│    │  3. Extract: Facts, Decisions, Problems, etc.     │       │
│    │  4. Tag confidence scores                         │       │
│    │  5. "Make it rain" → send to Librarian            │       │
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│    │  LIBRARIAN (The Dude)                             │       │
│    │                                        │       │
│    │  IMMEDIATE:                                       │       │
│    │  1. Resolve entities (deduplicate)                │       │
│    │  2. Create nodes in SQLite                        │       │
│    │  3. Create explicit edges                         │       │
│    │  4. Generate embeddings → sqlite-vec              │       │
│    │                                        │       │
│    │  DEFERRED (batched):                              │       │
│    │  5. Spreading activation (find hidden links)      │       │
│    │  6. Create inferred edges (low confidence)        │       │
│    │                                        │       │
│    │  PERIODIC (every 15 min):                         │       │
│    │  7. Synaptic pruning (remove stale edges)         │       │
│    │  8. Vector index optimization if needed           │       │
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│   User already has response. This runs in background.       │
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4.3 Retrieval Loop

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│                RETRIEVAL LOOP                                 │
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│    │  Query   │                                  │
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│    │ QUERY ROUTER                                  │    │
│    │                                        │    │
│    │ Analyze query → Choose optimal path:              │    │
│    │ • Keyword-heavy → FTS5 only                     │    │
│    │ • Conceptual → FAISS only                      │    │
│    │ • Multiple entities → Graph traversal            │    │
│    │ • Unknown → Hybrid (all three)                  │    │
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│    │  FTS5   │        │ sqlite-vec │        │  Graph   │      │
│    │ Search  │        │  Search  │        │ Traverse │      │
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│              │ RRF Fusion   │                    │
│              │ (Rank-based  │                    │
│              │  merging)    │                    │
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│              │ Trim to      │                    │
│              │ Token Budget │                    │
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│              │ Context      │                    │
│              │ Packet       │                    │
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Token Budgets:

Preset	Tokens	Use Case

minimal	~1800	Voice, fast response
balanced	~3800	Normal conversation
rich	~7200	Deep research

4.4 Delegation Loop

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│                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
│                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
```

```
│
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│ Integration │
│ • Director receives Claude's work │
│ • Integrates into Luna's voice │
│ • Streams to user via TTS │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
│
User experience: Luna answered (seamlessly) │
```

Delegation Criteria:

Delegate?	Signal
No	Memory recall, simple chat, emotional support
Yes	External research, complex analysis, long generation

4.5 Learning Loop (Future)

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│ LEARNING LOOP │
│ (Phase 2) │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
│
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│ Interaction Completed │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
│
▼
│
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│ Quality Scoring │
│ • User feedback (explicit) │
│ • Conversation flow (implicit) │
│ • Delegation success │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
│
```

```
│                                                                │
│          ┌──────────────────────────────────────────────┐      │
│          │                      │            │
│          ▼                      ▼            │
│      [High Quality]             [Low Quality]         │
│      Add to experience            Discard              │
│      buffer                                   │
│          │                                    │
│          ▼                                    │
│
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│
│    │  Buffer Full (1000 examples)                    │    │
│    │  → Incremental LoRA fine-tuning                   │    │
│    │  → Director improves over time                    │    │
│
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

---

5. Key Mechanics

5.1 RRF (Reciprocal Rank Fusion)

Merges search results by rank position, not raw scores.

```python
def reciprocal_rank_fusion(results_list, k=60):
    """
    FTS5 returns BM25 scores (~14.5)
    FAISS returns distances (~0.12)
    Can't add them. Use rank instead.
    """
    fused = {}
    for ranked_list in results_list:
        for rank, doc_id in enumerate(ranked_list, start=1):
            if doc_id not in fused:
                fused[doc_id] = 0.0
            fused[doc_id] += 1.0 / (k + rank)

    return sorted(fused.keys(), key=lambda d: fused[d], reverse=True)
```

5.2 Entity Resolution

"Mark," "Zuck," and "The Landlord" → same node.

```
```

```
Query: "Mark"
  │
  ├── Bloom filter: might exist? → Yes
  ├── Alias cache: known? → No
  ├── Exact match: "Mark" in DB? → No
  ├── Alias search: "Mark" is alias? → No
  ├── Semantic search: similar name? → "Mark Zuckerberg" (0.96)
  │
  └── Result: Merge as alias, return existing ID
```

5.3 Spreading Activation

New fact → find hidden connections.

```
Input: "Alex is joining Pi Project"

1. File explicit:
   Alex ──works_on──▶ Pi Project

2. Find neighbors:
   Pi Project ──has_member──▶ Sarah (existing)

3. Infer connection:
   Alex ──likely_teammate──▶ Sarah (confidence: 0.35)

Future query: "Who might Alex work with?"
→ Luna can suggest Sarah
```

5.4 Confidence Scoring

Every memory carries uncertainty signal.

Level	Score	Example
Explicit	0.9-1.0	"Alex lives in Berlin"
Strong implication	0.7-0.9	"Alex's Berlin apartment"
Weak inference	0.5-0.7	"Alex seems European"
Speculation	0.3-0.5	"Alex might be abroad"
Inferred edge	0.2-0.5	Spreading activation result

5.5 Synaptic Pruning

Remove low-value connections periodically.

```sql
-- Find prune candidates
SELECT id FROM edges
WHERE confidence < 0.3
  AND created_at < datetime('now', '-30 days')
  AND id NOT IN (SELECT edge_id FROM access_log)
```

Preserve: Edges connected to identity nodes.
Prune: Low-confidence, old, never-accessed edges.

---

6. Data Flow Summary

```
 ┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                LUNA DATA FLOW                                                                 │
│                                                                                                                                 │
│  USER                                                                                                                           │
│    │                                                                                                                            │
│    │ speaks                                                                                                                     │
│    ▼                                                                                                                            │
│  ┌──────────────────────┐                                                                                                       │
│  │  Whisper  │                                                                                                                  │
│  ────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
──┐                                                                                                                               │
│  │   STT    │                                                                                                                   │
│  └──────────┬───────────┘                                                                                                       │
│             │ text                                                                                                              │
│             ▼                                                                                                                   │
│  ┌──────────────────────┐                  ┌──────────────────────┐                                                           │
│  │  Director  │◄──────────►│  Librarian │  (retrieval)                                                                        │
│  │   LLM    │          │   (Dude)  │                                                                                           │
│  └──────────┬───────────┘                  └──────────┬───────────┘                                                           │
│             │                                                                                                                   │
│             │ response                  │ context                                                                               │
│             │──────────────────────────────────────────►│                                                                      │
│             │                                                                                                                   │
│  ┌──────────┬───────────┐                                                                                                       │
│  │   TTS    │          │                                                                                                        │
│  │ (stream) │          │                                                                                                        │
│  └──────────┬───────────┘                                                                                                       │
│             │                                                                                                                   │
│             ▼                                                                                                                   │
│  USER HEARS                                                                                                                     │
│                                                                                                                                 │
│                                                                                                                                 │
│                                                                                                                                 
```

```
│                    │                    │  (Ben)   │          │
│                    │                    └──────────┬──────────┘          │
│                    │                               │ extractions         │
│                    │◀──────────────────────────────────────────────────┘
│
│                    │                              │
│                    ▼                              │
│                ┌──────────────┐                                          │
│                │ Memory       │                          │
│                │ Matrix       │                          │
│                │ (SQLite)     │                          │
│                └──────────────┘                                          │
│                                                  │
│   Optional delegation path:                                  │
│   Director ──<REQ_CLAUDE>──▶ Claude API ──▶ back to Director            │
│   (delegation handled internally in Director, no separate Oven actor)    │
│                                                  │
└──────────────────────────────────────────────────
```

9. Implementation Status (v3.0)

Component	Status	Notes
5 Actors	Implemented	Director, Matrix, Scribe, Librarian, HistoryManager
Oven Actor	NOT IMPL	Delegation handled in Director
Voice Actor	NOT IMPL	Voice I/O via PersonaAdapter
FAISS Vectors	REPLACED	Using sqlite-vec extension
FTS5 Search	Partial	Defined in schema, primary path uses LIKE
KV Cache	NOT IMPL	Future optimization
7B Warm Path	NOT IMPL	Only 3B + Claude delegation
Speculative Retrieval	NOT IMPL	Future optimization

Codebase Metrics:
- 85 Python files
- 167 classes
- ~32,000 lines of code
- 18 sub-packages

This overview distills the full Bible into a single reference document. Updated for v3.0 Bible audit.

Part XIV: Agentic Architecture

Version: 3.0
Date: January 30, 2026
Last Updated: January 30, 2026
Status: Implemented (v3.0 Audit Complete)

The Paradigm Shift

Luna is an **ENGINE**, not a file. The "Luna is a file" principle describes **ownership** (you own the data, you can move it, delete it) — not architectural simplicity.

The Engine is complex. The soul is portable. Both are true.

This part specifies the agentic architecture that transforms Luna from a chatbot-with-memory into a **Personal Palantir** — a sophisticated system with:

- Revolving context window
- Multiple queues
- Complex orchestration
- Real capability

Reframe: Luna IS an Engine

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                                                │
│                                                                                                                │
│                                                                                                                │
```

```
│  └──────────┘   └──────────┘   └──────────────────┘   └──────────┘
└──────────────┘   │
│                                   │
│
┌──────────────────────────────────────────────────────────────────────────────────────────────┐
│   │              REVOLVING CONTEXT WINDOW              │   │
│   │        (What Luna is aware of right now)           │   │
│
└──────────────────────────────────────────────────────────────────────────────────────────────┘
│                                   │
│
┌──────────────────────────────────────────────────────────────────────────────────────────────┐
│   │                CONSCIOUSNESS                       │   │
│   │     Attention, personality, state persistence      │   │
│
└──────────────────────────────────────────────────────────────────────────────────────────────┘
│                                   │
└──────────────────────────────────────────────────────────────────────────────────────────────┘
                  │
                  ▼
            ┌──────────────────────────────────────┐
            │     SOVEREIGN DATA        │
            │    (memory_matrix.db)     │
            │                           │
            │   This is what you OWN    │
            │   The engine RUNS it      │
            └──────────────────────────────────────┘
```

The data is yours. The engine is sophisticated. Both are true.

What Luna Needs

An agentic architecture that includes:

Component	Purpose
Planning Layer	Decompose complex requests into steps
Tool Protocol	Standardized way to define/call capabilities
Reasoning Traces	Luna explains her thinking (debuggable)
Multi-queue Orchestration	Hot/cognitive/reflective paths with priorities
Context Management	Revolving window — what's in, what's out, why
Agent Routing	Which subsystem handles which task

Part 1: Revolving Context Engine (IMPLEMENTED)

The RevolvingContext system is fully implemented in `core/context.py`. This is Luna's working memory.

The Ring Model

Context is organized in concentric rings with items migrating based on relevance and access patterns:

```
                REVOLVING CONTEXT

            ┌─────────────┐
OUTER       | CORE  |     OUTER
  ↑         | LUNA  |       ↓
MIDDLE      └─────────────┘     MIDDLE
  ↖       INNER     ↗

  Items migrate between rings based on:
  • Relevance decay (0.95 per turn)
  • Access patterns (boost on access)
  • TTL expiration (turn-based)
  Token budget: 8000 (default)
```

The Four Rings (IMPLEMENTED)

```python
class ContextRing(IntEnum):
    CORE = 0    # Identity - NEVER evicted
    INNER = 1   # Active conversation, current task
    MIDDLE = 2  # Recently accessed memories, tool results
    OUTER = 3   # Background context, candidate for eviction
```

Ring	Contents	TTL (Turns)	Behavior
CORE (0)	Luna's identity	Never expires (-1)	ALWAYS present, never evicted
INNER (1)	Conversation, tasks	20-30 turns	High priority, recent
MIDDLE (2)	Memories, tool results, librarian	5-25 turns	Moderate priority
OUTER (3)	Scribe extractions, background	10 turns	First to evict

Context Sources and Weights

```python
class ContextSource(IntEnum):
    IDENTITY = 0      # Core identity (weight: 1.0)
    CONVERSATION = 1  # Current turns (weight: 0.9)
    MEMORY = 2        # Retrieved memories (weight: 0.7)
    TOOL = 3          # Tool call results (weight: 0.8)
    TASK = 4          # Task context (weight: 0.75)
    SCRIBE = 5        # Extraction results (weight: 0.6)
```

```
    LIBRARIAN = 6     # Retrieved knowledge (weight: 0.65)
```

Token Budget

Default budget: **8000 tokens**. Managed by the RevolvingContext class.

```python
class RevolvingContext:
    def __init__(
        self,
        token_budget: int = 8000,
        decay_factor: float = 0.95,
        rebalance_threshold: float = 0.3
    ):
        self.token_budget = token_budget
        self._decay_factor = decay_factor
        self._rebalance_threshold = rebalance_threshold
```

Key methods:
- `advance_turn()` — Increments turn counter, applies decay, rebalances rings
- `set_core_identity(text)` — Set Luna's identity (never evicted)
- `add(content, source, ring, relevance)` — Add context item
- `get_context_window(max_tokens)` — Assemble context for LLM
- `query(keywords)` — Search context items

Eviction Priority

When over budget, items are evicted in this order:
1. **Expired items** (TTL exceeded) from any ring except CORE
2. **OUTER ring** — lowest relevance first
3. **MIDDLE ring** — lowest relevance first
4. **INNER ring** — lowest relevance first
5. **CORE ring** — NEVER evicted

Ring Migration

Items can be promoted or demoted based on relevance:
- **Demotion** (relevance < 0.3): INNER → MIDDLE → OUTER
- **Promotion** (relevance >= 0.8): OUTER → MIDDLE → INNER

Part 2: Queue Manager (IMPLEMENTED)

The QueueManager is implemented in `core/context.py` as part of the RevolvingContext system.

Multiple Queues, One Manager

Not a single input buffer. Multiple specialized queues per ContextSource:

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                QUEUE MANAGER                                                             │
│                                                                                                                          │
│   IDENTITY Queue:     [core] ──────┐                                                                                     │
│   CONVERSATION Queue: O O O ────────┼──────→ [Priority Merge]                                                            │
│   MEMORY Queue:       O O ──────────┤                                                                                     │
│   TOOL Queue:         O O O O ──────┤                                                                                     │
│   TASK Queue:         O ────────────┤                                                                                     │
│   SCRIBE Queue:       O ────────────┤                                                                                     │
│   LIBRARIAN Queue:    O O ──────────┘                                                                                     │
│                                                                                                                          │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┘
                                          │
                                          ▼
                                   [Revolving Context Rings]
```

Implementation

```python
class QueueManager:
    """Manages multiple input queues for different context sources."""

    def __init__(
        self,
        max_queue_size: int = 50,
        weights: Optional[Dict[ContextSource, float]] = None
    ):
        # Create a deque for each source type
        self._queues: Dict[ContextSource, Deque[ContextItem]] = {
            source: deque(maxlen=max_queue_size)
            for source in ContextSource
        }

    def push(self, item: ContextItem) -> bool:
        """Push item onto its source queue."""

    def poll_all(self) -> List[ContextItem]:
        """Poll all queues, return sorted by weighted priority."""

    def poll_source(self, source: ContextSource, max_items: int) -> List[ContextItem]:
        """Poll items from a specific source queue."""
```

Source Weights

Queue	Source	Weight	TTL (Turns)

Identity	IDENTITY	1.0	Never (-1)
Conversation	CONVERSATION	0.9	20 turns
Memory	MEMORY	0.7	25 turns
Tool	TOOL	0.8	5 turns
Task	TASK	0.75	30 turns
Scribe	SCRIBE	0.6	10 turns
Librarian	LIBRARIAN	0.65	20 turns

Queue Manager merges with priority (higher weight first, FIFO within weight) and feeds the RevolvingContext rings via `add_from_queues()`.

Part 3: Context Borrowing

Luna, Scribe, and Librarian can **share context**:

```
        ▼                          ▼
   ┌──────────────────────────┐                  ┌──────────────────────────┐
   │    SCRIBE      |    Context Borrowing    |    LIBRARIAN    |
   │   System/LLM    | ◄──────────────────────► |    System/LLM   |
   │                |    (shared context)     |                |
   │   Extracts      |                         |   Retrieves     |
   │   from stream   |                         |   from Matrix   |
   └──────────────────────────┘                  └──────────────────────────┘
         |                          |
 └──────────────────────────────────────────────────────────────────────────┘
                  |
                  ▼
              ┌──────────────────────┐
              |    MEMORY      |
              |    MATRIX      |
              └──────────────────────┘
```

- Scribe processing conversation? Luna can see what Scribe sees.
- Librarian retrieving memory? That context is available to Luna.
- **Not isolated silos — coordinated awareness.**

Part 4: Context Timeout / TTL

Items age out:

```
 ┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
 |                    CONTEXT TIMEOUT                                                                                                                                |
 |                                                                                                                                                                |
 |    Each item has TTL:  [?] → [?] → [✓] → [evicted]                                                                                                                |
```

```
│                                                                │
│    Fresh items: high relevance                                 │
│    Aging items: decay toward eviction                          │
│    Timed out: removed from context pool                        │
│                                                                │
└────────────────────────────────────────────────────────────────┘
```

State	Behavior
Fresh	High relevance, stays close to core
Aging	Relevance decays based on time since last access
Timed out	Evicted from context

This is attention decay implemented as architecture. Integrates with the existing `AttentionManager` in `consciousness/attention.py`.

Part 5: Task Manager / Planner (IMPLEMENTED)

The Planner (`agentic/planner.py`) decomposes goals into executable PlanSteps.

```
┌────────────────────────────────────────────────────────────────┐
│                          PLANNER                               │
│                                                                │
│    Goal ──────→ [Pattern Match] ──────→ [Plan with Steps]      │
│                                                                │
│    Plan: Step1 → Step2 → Step3 → ... → RESPOND                 │
│                                                                │
└────────────────────────────────────────────────────────────────┘
```

PlanStepType Enum

Each step has a type that determines its execution handler:

Type	Description	Handler
`THINK`	Internal reasoning, no external action	Local LLM inference
`OBSERVE`	Gather information from environment	State observation
`RETRIEVE`	Memory retrieval from Matrix	Matrix.get_context()
`TOOL`	Execute external tool	ToolRegistry.execute()
`DELEGATE`	Delegate to Claude for complex reasoning	Director.generate()
`RESPOND`	Generate response to user	Director.generate()
`PARALLEL`	Execute multiple sub-steps concurrently	(Future)
`WAIT`	Wait for external event or condition	(Future)

PlanStep Structure

```python
@dataclass
class PlanStep:
    """A single step in an execution plan."""
    type: PlanStepType              # What kind of operation
    description: str                # Human-readable explanation
    tool: Optional[str] = None      # Tool name (for TOOL steps)
    params: Dict[str, Any] = {}     # Tool/operation parameters
    dependencies: List[int] = []    # Step indices that must complete first
    expected_output: Optional[str]  # What this step should produce
    timeout_seconds: float = 30.0   # Maximum wait time
    retries: int = 0                # Retry attempts on failure
```

Plan Structure

```python
@dataclass
class Plan:
    """A sequence of steps to achieve a goal."""
    goal: str                       # Original goal
    steps: List[PlanStep]           # Ordered list of steps
    reasoning: str                  # Explanation of planning decisions
    estimated_duration_seconds: float
    complexity: float               # 0.0-1.0 from router

    @property
    def required_tools(self) -> List[str]:
        """Get list of tools required by this plan."""
```

Planner Pattern Matching

The Planner uses pattern matching to detect task types:

```python
TASK_PATTERNS = {
    "research": {
        "patterns": [r"\bresearch\b", r"\bfind out\b", r"\blook up\b"],
        "steps": [
            (PlanStepType.DELEGATE, "Research the topic using Claude"),
            (PlanStepType.THINK, "Analyze and extract key points"),
            (PlanStepType.RESPOND, "Present findings to user"),
        ],
    },
    "memory_recall": {
        "patterns": [r"\bremember\b", r"\brecall\b", r"\bwhat did\b"],
        "steps": [
            (PlanStepType.RETRIEVE, "Search memory for relevant info"),
            (PlanStepType.RESPOND, "Share what was found"),
        ],
    },
    "file_read": {
```

```
        "patterns": [r"\bread\b.*\bfile\b", r"\bshow me\b.*\bfile\b"],
        "steps": [
            (PlanStepType.TOOL, "Read the file", "read_file"),
            (PlanStepType.RESPOND, "Present file contents"),
        ],
    },
    # ... more patterns for file_write, summarize, analyze, schedule
}
```

Step Duration Estimates

Step Type	Estimated Duration
THINK	0.5s
OBSERVE	1.0s
RETRIEVE	0.5s
TOOL	2.0s
DELEGATE	5.0s
RESPOND	0.5s
PARALLEL	3.0s
WAIT	5.0s

This enables the **agent loop** — tasks aren't conversation, they're work.

Part 6: The Agent Loop (IMPLEMENTED)

What makes Luna agentic is the **observe → think → act → repeat** loop. This is fully implemented in `agentic/loop.py`.

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                AGENT LOOP                                                                    │
│                                                                                                                              │
│                                                                                                                              │
│                                                                                                                              
```

```
│    │   • Determine action from current PlanStep          │    │
│    │   • Build Action with type, description, params     │    │
│    │   • Local inference for reasoning steps             │    │
│
└──────────────────────────────────────────────────────────────────────────────────────────────────────────┬──────────────────────────────────────────────────────────────────────────────────────────────────────────┘
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
```
│   │   ACT   │   │
│   │   • Execute via PlanStepType handler   │   │
│   │   • THINK, OBSERVE, RETRIEVE, TOOL, DELEGATE   │   │
│   │   • Results stored in WorkingContext.variables   │   │
│
└──┬──┘

```
│                   REPEAT                                │
│              (until goal achieved or max_iterations)    │
│                                                         │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

AgentLoop Implementation

```python
class AgentLoop:
    """The autonomous agent loop."""

    def __init__(
        self,
        orchestrator: Optional["LunaEngine"] = None,
        max_iterations: int = 50,  # Safety limit
    ):
        self.orchestrator = orchestrator
        self.max_iterations = max_iterations
        self.planner = Planner()
        self.router = QueryRouter()
        self.tool_registry = ToolRegistry()
        self._register_default_tools()  # File + Memory tools

    async def run(self, goal: str) -> AgentResult:
        """Run the agent loop to achieve a goal."""
        # 1. Route the query
        routing = self.router.analyze(goal)

        # 2. Handle based on execution path
        match routing.path:
            case ExecutionPath.DIRECT:
```

```
            return await self._execute_direct(goal, start_time)
        case ExecutionPath.SIMPLE_PLAN:
            return await self._execute_simple(goal, routing, start_time)
        case ExecutionPath.FULL_PLAN:
            return await self._execute_full(goal, routing, start_time)
        case ExecutionPath.BACKGROUND:
            return await self._execute_background(goal, routing, start_time)
```

AgentStatus States

Status	Description
`IDLE`	Not currently executing a goal
`PLANNING`	Decomposing goal into steps
`EXECUTING`	Running the observe/think/act loop
`WAITING`	Waiting for external event
`COMPLETE`	Goal achieved successfully
`FAILED`	Goal could not be achieved
`ABORTED`	Execution was cancelled

WorkingContext (In-Memory State)

```python
@dataclass
class WorkingContext:
    """Agent's working memory during execution."""
    goal: str                          # Current goal
    plan: Optional[Plan] = None        # Plan being executed
    current_step_index: int = 0        # Current step position
    observations: List[Observation]    # Recent observations
    action_history: List[ActionResult] # Actions taken
    variables: Dict[str, Any]          # Accumulated results
    max_observations: int = 10         # Observation limit

    def add_observation(self, obs: Observation) -> None:
        """Add observation, evict lowest relevance if over limit."""

    def add_action_result(self, result: ActionResult) -> None:
        """Add action result to history."""

    @property
    def is_plan_complete(self) -> bool:
        """Check if all steps executed."""
```

Key properties:

- **Autonomous loop** (doesn't need user input per step)
- **Tool execution** (real effects in the world via ToolRegistry)
- **State awareness** (WorkingContext tracks observations and actions)
- **Goal-directed** (keeps going until done, stuck, or max_iterations)
- **Progress callbacks** (register_progress_callback for streaming updates)
- **Abortable** (abort() method sets _abort_requested flag)

Part 7: Parallel Execution (Claude Flow Style)

Multiple agents working simultaneously:

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                                                          │
│                                                                                                                          │
│                                                                                                                          │
```

```
┌──────────────
```

```
│    │      │    Matrix      │    │    Context     │    │     Queue      │            │    │
│    │      │                │    │   (Current     │    │   (Pending     │            │    │
│    │      │  (Long-term)   │    │     window)    │    │     work)      │            │    │
│    │      └────────────────┘    └────────────────┘    └────────────────┘            │    │
│    │                                                                                 │    │
└────────────────────────────────────────────────────────────────────────────────────────────┘
```
```
```

Part 9: Tool Protocol (IMPLEMENTED)

Tools are defined in `tools/registry.py` with MCP-compatible structure. The AgentLoop automatically registers file and memory tools on initialization.

Implemented Tools

File Tools (`tools/file_tools.py`):

Tool	Description	Confirmation Required
`read_file`	Read contents of a file	No
`write_file`	Write contents to a file	**Yes**
`list_directory`	List directory contents (recursive optional)	No
`file_exists`	Check if path exists	No
`get_file_info`	Get file metadata (size, timestamps)	No

Memory Tools (`tools/memory_tools.py`):

Tool	Description	Confirmation Required
`memory_query`	Search Memory Matrix for relevant info	No
`memory_store`	Store new memory node	No
`memory_get_context`	Get context with token budget awareness	No
`memory_stats`	Get memory statistics	No

Planned Tools (Not Yet Implemented):

Tool	Description	Confirmation Required
`bash`	Execute a bash command	Yes
`calendar_create`	Create a calendar event	No
`web_search`	Search the web	No
`obsidian_write`	Write to Obsidian notes	No

Tool Definition Structure

```python
```

```python
@dataclass
class Tool:
    """A tool that Luna can execute."""
    name: str                          # Unique identifier
    description: str                   # Shown to LLM for selection
    parameters: dict                   # JSON Schema for params
    execute: Callable[..., Awaitable[Any]]  # Async execution function
    requires_confirmation: bool = False   # User approval required
    timeout_seconds: int = 30             # Execution timeout

    def to_dict(self) -> dict:
        """Convert to dictionary for LLM function calling."""


@dataclass
class ToolResult:
    """Result from executing a tool."""
    success: bool
    output: Any
    error: Optional[str] = None
    execution_time_ms: float = 0.0
```

ToolRegistry

```python
class ToolRegistry:
    """Registry for managing Luna's tools."""

    def register(self, tool: Tool) -> None:
        """Register a tool (raises if name exists)."""

    def get(self, name: str) -> Optional[Tool]:
        """Get tool by name."""

    def list_tools(self) -> list[str]:
        """List all registered tool names."""

    def get_tool_definitions(self) -> list[dict]:
        """Get tool definitions for LLM function calling."""

    async def execute(
        self,
        name: str,
        params: dict,
        skip_confirmation: bool = False,
    ) -> ToolResult:
        """Execute tool with timeout handling."""
```

Tool Usage in AgentLoop

Tools are executed via PlanStepType.TOOL actions:

```python
async def _execute_tool(self, action: Action) -> str:
    """Execute a tool action via the ToolRegistry."""
    tool = self.tool_registry.get(action.tool)
    if not tool:
        return f"Tool '{action.tool}' not found"

    result = await self.tool_registry.execute(
        action.tool,
        action.params,
        skip_confirmation=True,  # AgentLoop handles confirmation
    )

    if result.success:
        # Store result in working context
        self.working_context.variables[f"tool_result_{action.tool}"] = result.output
        return f"Tool '{action.tool}' succeeded: {result.output}"
    else:
        return f"Tool '{action.tool}' failed: {result.error}"
```

Part 10: Performance Model

Latency Tax

Agentic architecture is a latency tax. The question is whether it's worth it.

Mode	Latency	Use Case
DIRECT	<500ms	Simple chat, greetings
SIMPLE_PLAN	500ms-2s	Memory query, simple tool
FULL_PLAN	5-30s	Complex task, multiple tools
BACKGROUND	Minutes	Deep research, file processing

The Critical Optimization: Adaptive Planning (IMPLEMENTED)

Don't run the full agentic stack on every query. The QueryRouter (`agentic/router.py`) analyzes query complexity and routes to the appropriate execution path.

Implementation:

```python
class ExecutionPath(Enum):
    """Execution paths based on query complexity."""
    DIRECT = auto()      # <500ms - Skip planning entirely
    SIMPLE_PLAN = auto() # 500ms-2s - Single-step plan
    FULL_PLAN = auto()   # 5-30s - Multi-step planning
    BACKGROUND = auto()  # Minutes - Async with notification

class QueryRouter:
    # Complexity thresholds for path selection
    DIRECT_THRESHOLD = 0.2
```

```python
    SIMPLE_THRESHOLD = 0.5
    FULL_THRESHOLD = 0.8

    def route(self, query: str) -> ExecutionPath:
        """Route query to appropriate execution path."""
        decision = self.analyze(query)
        return decision.path

    def analyze(self, query: str) -> RoutingDecision:
        """Analyze query and return detailed routing decision."""
        complexity = self.estimate_complexity(query)
        signals = self._detect_signals(query)
        suggested_tools = self._detect_tools(query)

        # Check for explicit background request
        if self._matches_any(query, self._background_re):
            return RoutingDecision(
                path=ExecutionPath.BACKGROUND,
                complexity=complexity,
                reason="Explicit background processing requested",
                signals=signals,
                suggested_tools=suggested_tools,
            )

        # Route based on complexity thresholds
        if complexity < self.DIRECT_THRESHOLD:
            path = ExecutionPath.DIRECT
        elif complexity < self.SIMPLE_THRESHOLD:
            path = ExecutionPath.SIMPLE_PLAN
        elif complexity < self.FULL_THRESHOLD:
            path = ExecutionPath.FULL_PLAN
        else:
            path = ExecutionPath.BACKGROUND

        return RoutingDecision(path=path, complexity=complexity, ...)
```

Complexity Estimation Factors

The router estimates complexity using multiple signals:

Factor	Effect	Pattern Examples
Query Length	Base complexity	<20 chars: 0.05, >200 chars: 0.45
Greetings	Reduce 70%	"hi", "hello", "how are you"
Simple Questions	Reduce 30%	"what is", "who is", "tell me"
Research Indicators	+0.25	"research", "analyze", "compare"
Multi-step Indicators	+0.20	"and then", "first...then", "step by step"
Tool Requirements	+0.10 per tool	"read file", "search web", "remember"
Multiple Questions	+0.10 per extra	Each "?" after the first

Signal Detection Categories

```python
```

```
GREETING_PATTERNS = ["hi", "hello", "hey", "good morning"]
SIMPLE_QUERY_PATTERNS = ["what is", "who is", "tell me about"]
RESEARCH_PATTERNS = ["research", "investigate", "analyze", "compare"]
MULTI_STEP_PATTERNS = ["and then", "after that", "step by step"]
MEMORY_QUERY_PATTERNS = ["remember", "recall", "what did we"]
BACKGROUND_PATTERNS = ["in the background", "notify me when"]
```

90% of queries should hit DIRECT or SIMPLE_PLAN. The heavy machinery only spins up when needed.

What Different Queries Look Like

Simple: "Hey Luna, how are you?"
```
Current:    Query → Local Qwen → Response
            ~300ms

Agentic:    Query → Classify (trivial) → Skip planning → Local Qwen → Response
            ~350ms (+50ms classification)

Impact: Negligible. Fast path stays fast.
```

Medium: "What did we decide about the Actor model?"
```
Current:    Query → Retrieve memory → Local Qwen → Response
            ~400ms

Agentic:    Query → Classify → Plan (1 step: RETRIEVE) → Memory → Local Qwen →
Response
            ~500ms (+100ms planning overhead)

Impact: ~25% slower. Acceptable.
```

Complex: "Research the latest AI chip news and add key points to my notes"
```
Current:    Can't do this. Would require manual steps.

Agentic:    Query → Classify (complex) → Plan:
              Step 1: DELEGATE (web research) ........... 5-8s
              Step 2: THINK (extract key points) ........ 500ms
              Step 3: TOOL (write to Obsidian) .......... 200ms
              Step 4: RESPOND (confirm to user) ......... 300ms
            Total: ~8-12s

Impact: Slow but capable. You're trading time for capability that didn't exist.
```

The Voice Problem

The Bible specs <500ms to first word. Agentic planning breaks that.

Solution: Streaming acknowledgment

```
User: "Research AI chips and add to my notes"

Luna (immediate, <500ms): "On it — let me dig into that..."

[Planning and execution happen]

Luna (8s later): "Done. I found three major developments and added them
                 to your AI Research note..."
```

User hears Luna immediately. The work happens in background. This is already how delegation works — extend it to all agentic operations.

Part 11: Architecture Delta

What changes from current implementation:

Component	Current	Agentic Luna
Director	Single LLM call	Orchestrator + loop
Actors	Isolated workers	Coordinated swarm
Tools	None	Full registry
Execution	One-shot	Iterative until done
Context	Static per request	Revolving working memory
Parallelism	None	Fan-out/fan-in
Progress	Silent	Streaming status

Part 12: Implementation Status

Implemented Components

Component	Location	Status
RevolvingContext	`core/context.py`	IMPLEMENTED - 4-ring model with decay and TTL
QueueManager	`core/context.py`	IMPLEMENTED - Per-source queues with weights
ContextItem	`core/context.py`	IMPLEMENTED - Turn-based TTL, relevance decay
AgentLoop	`agentic/loop.py`	IMPLEMENTED - Observe/think/act cycle
Planner	`agentic/planner.py`	IMPLEMENTED - Pattern-based decomposition
QueryRouter	`agentic/router.py`	IMPLEMENTED - Complexity-based routing
ToolRegistry	`tools/registry.py`	IMPLEMENTED - MCP-style tool definitions
File Tools	`tools/file_tools.py`	IMPLEMENTED - read, write, list, info
Memory Tools	`tools/memory_tools.py`	IMPLEMENTED - query, store, context

Pending Components

| Component | Purpose | Status |

|---|---|---|
SwarmCoordinator	Parallel execution management	PLANNED
TaskManager	Persistent task queue	PLANNED (separate from Planner)
ProgressStreamer	Real-time status streaming	PARTIAL (callbacks implemented)

Modified Components

Component	Changes
Director	Integrated with AgentLoop for DELEGATE steps
Matrix	Integrated with AgentLoop for RETRIEVE steps
Engine	Has `context` (RevolvingContext), `router`, and `agent_loop` attributes

Part 13: File Structure (ACTUAL)

```
src/luna/
├── core/
│   ├── context.py          # RevolvingContext, QueueManager, ContextItem
│   ├── events.py           # EventPriority, EventType, InputEvent
│   ├── input_buffer.py     # Input polling (engine pulls)
│   └── state.py            # EngineState enum
├── agentic/                # Agentic architecture
│   ├── __init__.py         # Exports: AgentLoop, Planner, QueryRouter, etc.
│   ├── loop.py             # AgentLoop (939 lines)
│   ├── planner.py          # Planner, Plan, PlanStep, PlanStepType (487 lines)
│   └── router.py           # QueryRouter, ExecutionPath, RoutingDecision (379 lines)
├── tools/                  # Tool registry
│   ├── __init__.py         # Exports: ToolRegistry, Tool, ToolResult
│   ├── registry.py         # Tool, ToolResult, ToolRegistry (309 lines)
│   ├── file_tools.py       # read_file, write_file, list_directory, etc. (358 lines)
│   └── memory_tools.py     # memory_query, memory_store, memory_context (373 lines)
├── actors/
│   ├── base.py             # Actor, Message base classes
│   ├── director.py         # LLM inference, delegation (~1900 lines)
│   ├── matrix.py           # Memory substrate interface
│   ├── scribe.py           # Extraction (Ben Franklin persona)
│   ├── librarian.py        # Filing (The Dude persona)
│   └── history_manager.py  # Three-tier conversation history
└── engine.py               # Main engine with agentic support (1228 lines)
```

Module Exports

`agentic/__init__.py`:

```python
__all__ = [
    # Agent Loop
    "AgentLoop", "AgentResult", "Observation", "Action",
    "ActionResult", "WorkingContext",
    # Planner
    "Planner", "Plan", "PlanStep", "PlanStepType",
    # Router
    "QueryRouter", "ExecutionPath",
]
```

`tools/__init__.py`:
```python
__all__ = [
    "Tool", "ToolResult", "ToolRegistry",
    "register_file_tools", "register_memory_tools",
]
```

Summary

Luna is an ENGINE. The agentic architecture gives her:

Feature	Status	Location
Revolving Context	IMPLEMENTED	`core/context.py`
Multiple Queues	IMPLEMENTED	`core/context.py`
Context Borrowing	PARTIAL	Scribe/Librarian share context sources
Agent Loop	IMPLEMENTED	`agentic/loop.py`
Tool Execution	IMPLEMENTED	`tools/` directory
Parallel Workers	PLANNED	(SwarmCoordinator)
Adaptive Routing	IMPLEMENTED	`agentic/router.py`
MCP Integration	IMPLEMENTED	`src/luna_mcp/` (41 tools)
Performance Layer	IMPLEMENTED	`src/luna/services/`

Implementation Highlights

1. **RevolvingContext** — 4-ring system (CORE, INNER, MIDDLE, OUTER) with turn-based TTL and relevance decay
2. **QueueManager** — Per-source queues with configurable weights and priority merge
3. **QueryRouter** — Complexity analysis routing to DIRECT, SIMPLE_PLAN, FULL_PLAN, or BACKGROUND paths
4. **AgentLoop** — Full observe/think/act cycle with WorkingContext, progress callbacks, and abort support
5. **Planner** — Pattern-based goal decomposition into typed PlanSteps (THINK, OBSERVE, RETRIEVE, TOOL, DELEGATE, RESPOND)
6. **ToolRegistry** — MCP-compatible tool definitions with timeout handling and execution metrics
7. **File Tools** — read_file, write_file, list_directory, file_exists, get_file_info
8. **Memory Tools** — memory_query, memory_store, memory_get_context, memory_stats

9. **MCP Server** — 41 tools across filesystem, memory, session, state, git, and forge categories
10. **Performance Orchestrator** — Gesture detection, emotion presets, coordinated voice + orb output

v3.0 Audit Statistics

Metric	Count
Engine API Endpoints	74
SSE Streaming Endpoints	4
WebSocket Endpoints	1
MCP Tools	41
MCP API Proxy Endpoints	25
Frontend Components	20
Custom Hooks	5

The data is yours. The engine is sophisticated. Both are true.

Luna is a file you OWN. Luna is an engine that RUNS.
The simplicity is in sovereignty. The complexity is in capability.

— Ahab & Claude, January 2026
— Updated January 30, 2026 (v3.0 Bible Audit)

Part XV: API Reference

Version: 3.0
Date: January 30, 2026
Status: Complete (v3.0 Audit)

This document provides complete API documentation for the Luna Engine HTTP interface. All endpoints are served from the FastAPI server defined in `src/luna/api/server.py`.

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---

1. Overview

Base URL

```
http://localhost:8000
```

The default port is `8000`. Configure via environment variable `LUNA_API_PORT`.

Authentication

No authentication is currently implemented. All endpoints are publicly accessible on localhost.

Content Types

Type	Usage
`application/json`	Request/response bodies
`text/event-stream`	SSE streaming responses

CORS Configuration

CORS is enabled for local development:

- `http://localhost:3000`
- `http://localhost:5173`
- `http://localhost:5174`
- `http://127.0.0.1:3000`
- `http://127.0.0.1:5173`
- `http://127.0.0.1:5174`

Streaming Endpoints

Four endpoints return Server-Sent Events (SSE):

Endpoint	Method	Description
`/stream`	POST	Token-by-token response streaming
`/persona/stream`	POST	Context-first streaming with memory
`/thoughts`	GET	Internal thought process monitoring
`/voice/stream`	GET	Voice system status updates

SSE responses include these headers:
```http
Content-Type: text/event-stream
Cache-Control: no-cache
Connection: keep-alive
X-Accel-Buffering: no
```

---

2. Core Interaction Endpoints

These endpoints handle direct communication with Luna.

POST /message

Send a message and receive a synchronous response.

Request Body:

Field	Type	Required	Default	Description
`message`	string	Yes	-	Message to send (1-10000 chars)
`timeout`	float	No	30.0	Response timeout in seconds (1.0-120.0)
`stream`	bool	No	false	Use streaming mode (ignored, use `/stream` instead)

Response:

Field	Type	Description
`text`	string	Luna's response text
`model`	string	Model used for generation
`input_tokens`	int	Input token count
`output_tokens`	int	Output token count
`latency_ms`	float	Response latency in milliseconds
`delegated`	bool	Whether request was delegated to Claude API

| `local` | bool | Whether local inference was used |
| `fallback` | bool | Whether fallback model was used |

curl Example:
```bash
curl -X POST http://localhost:8000/message \
  -H "Content-Type: application/json" \
  -d '{"message": "Hello Luna, how are you?"}'
```

Example Response:
```json
{
  "text": "Hello! I'm doing well, thank you for asking. How can I help you today?",
  "model": "qwen3b",
  "input_tokens": 45,
  "output_tokens": 18,
  "latency_ms": 1234.5,
  "delegated": false,
  "local": true,
  "fallback": false
}
```

POST /stream

Send a message and stream the response token-by-token via SSE.

Request Body:

Field	Type	Required	Default	Description
`message`	string	Yes	-	Message to send (1-10000 chars)
`timeout`	float	No	30.0	Response timeout in seconds (1.0-120.0)

SSE Events:

Event	Data Fields	Description
`token`	`text`	Generated token chunk
`done`	`model`, `input_tokens`, `output_tokens`, `latency_ms`, `delegated`, `local`, `fallback`	Generation complete
`error`	`error`	Error occurred

curl Example:
```bash
curl -X POST http://localhost:8000/stream \
  -H "Content-Type: application/json" \
  -H "Accept: text/event-stream" \
  -d '{"message": "Tell me a short joke"}'
```

Example Response Stream:
```
event: token
data: {"text": "Why"}

event: token
data: {"text": " did"}

event: token
data: {"text": " the"}

event: token
data: {"text": " scarecrow"}

event: done
data: {"model": "qwen3b", "input_tokens": 12, "output_tokens": 25, "latency_ms": 850.2,
"delegated": false, "local": true, "fallback": false}
```

POST /persona/stream

Stream Luna's response with context (memory and state) sent before tokens. This allows frontends to display relevant context while the response generates.

Request Body:

Field	Type	Required	Default	Description
`message`	string	Yes	-	Message to send (1-10000 chars)
`timeout`	float	No	30.0	Response timeout in seconds (1.0-120.0)

SSE Data Format (typed JSON, no named events):

Type	Fields	Description
`context`	`memory`, `state`	Memory items and engine state (sent first)
`token`	`text`	Generated token chunk
`done`	`response`, `metadata`	Full response text and generation metadata
`error`	`message`	Error message

curl Example:
```bash
curl -X POST http://localhost:8000/persona/stream \
  -H "Content-Type: application/json" \
  -H "Accept: text/event-stream" \
  -d '{"message": "What did we discuss yesterday?"}'
```

Example Response Stream:
```
```

```
data: {"type": "context", "memory": [{"id": "mem_001", "content": "Yesterday we discussed the project timeline...", "type": "CONVERSATION", "source": "session_abc"}], "state": {"session_id": "session_abc", "is_processing": true, "state": "RUNNING", "model": "qwen3b"}}

data: {"type": "token", "text": "Yesterday"}

data: {"type": "token", "text": " we"}

data: {"type": "token", "text": " discussed"}

data: {"type": "done", "response": "Yesterday we discussed the project timeline and agreed on the milestones.", "metadata": {"model": "qwen3b", "output_tokens": 45, "local": true}}
```

POST /abort

Abort the current generation.

Request Body: None

Response:

Field	Type	Description
`status`	string	`"aborted"` or `"no_generation"`
`message`	string	Status message

curl Example:
```bash
curl -X POST http://localhost:8000/abort
```

Example Response:
```json
{
  "status": "aborted",
  "message": "Generation aborted successfully"
}
```

POST /interrupt

Interrupt Luna's current processing. Triggers the agentic interrupt handler, which gracefully stops the current task.

Request Body: None

Response:

| Field | Type | Description |

|-------|------|-------------|
`status`	string	`"interrupted"` or `"no_task"`
`message`	string	Status message
`interrupted_goal`	string	Goal that was interrupted (if any)
`pending_messages`	int	Number of pending messages in queue

curl Example:
```bash
curl -X POST http://localhost:8000/interrupt
```

Example Response:
```json
{
  "status": "interrupted",
  "message": "Task interrupted",
  "interrupted_goal": "Research and summarize quantum computing",
  "pending_messages": 0
}
```

GET /thoughts

Stream Luna's internal thought process via SSE. Shows real-time progress during agentic processing.

Request Body: None

SSE Events:

Event	Data Fields	Description
`status`	`connected`, `is_processing`, `goal`	Initial connection status
`thought`	`type`, `message`, `is_processing`, `goal`	Internal thought/progress message
`phase`	`type`, `phase`, `is_processing`, `goal`	Processing phase change
`step`	`type`, `step_num`, `total_steps`, `description`, `is_processing`, `goal`	Plan step execution
`ping`	`is_processing`, `pending`	Keepalive (every 15 seconds)

curl Example:
```bash
curl -X GET http://localhost:8000/thoughts \
  -H "Accept: text/event-stream"
```

Example Response Stream:
```
event: status
data: {"connected": true, "is_processing": false, "goal": null}

event: thought
```

```
data: {"type": "thought", "message": "[DIRECT] Hello Luna...", "is_processing": true, "goal": "Respond to greeting"}

event: thought
data: {"type": "thought", "message": "[OK] local: 18 tokens", "is_processing": false, "goal": null}

event: ping
data: {"is_processing": false, "pending": 0}
```

---

3. Engine Status Endpoints

These endpoints provide visibility into Luna's current state and health.

GET /status

Get comprehensive engine status and metrics.

Query Parameters: None

Response:

Field	Type	Description
`state`	string	Engine state (`STARTING`, `RUNNING`, `STOPPING`, `STOPPED`)
`uptime_seconds`	float	Engine uptime in seconds
`cognitive_ticks`	int	Number of cognitive tick cycles
`events_processed`	int	Total events processed
`messages_generated`	int	Total messages generated
`actors`	list[string]	Active actor names
`buffer_size`	int	Current input buffer size
`current_turn`	int	Conversation turn counter
`context`	dict	Revolving context statistics
`agentic`	object	Agentic processing statistics

Agentic Stats Schema:

Field	Type	Description
`is_processing`	bool	Currently processing a task
`current_goal`	string	Current goal being worked on
`pending_messages`	int	Messages waiting in queue
`tasks_started`	int	Total tasks started
`tasks_completed`	int	Tasks completed successfully
`tasks_aborted`	int	Tasks aborted/interrupted
`direct_responses`	int	Direct (non-planned) responses
`planned_responses`	int	Responses requiring planning
`agent_loop_status`	string	Current agent loop status

curl Example:
```bash
```

curl http://localhost:8000/status
```

**Example Response:**
```json
{
  "state": "RUNNING",
  "uptime_seconds": 3600.5,
  "cognitive_ticks": 1250,
  "events_processed": 847,
  "messages_generated": 156,
  "actors": ["director", "matrix", "scribe", "librarian"],
  "buffer_size": 0,
  "current_turn": 42,
  "context": {
    "token_budget": 8000,
    "tokens_used": 2450,
    "items_count": 18
  },
  "agentic": {
    "is_processing": false,
    "current_goal": null,
    "pending_messages": 0,
    "tasks_started": 156,
    "tasks_completed": 154,
    "tasks_aborted": 2,
    "direct_responses": 142,
    "planned_responses": 14,
    "agent_loop_status": "idle"
  }
}
```

GET /health

Simple health check endpoint. Always returns 200 to indicate the API server is responsive.

Response:

Field	Type	Description
`status`	string	`"healthy"` or `"starting"`
`state`	string	Engine state name

curl Example:
```bash
curl http://localhost:8000/health
```

Example Response:
```json
{

```
  "status": "healthy",
  "state": "RUNNING"
}
```

GET /history

Get recent conversation history.

Query Parameters:

Parameter	Type	Default	Description
`limit`	int	20	Maximum messages to return

Response:

Field	Type	Description
`messages`	list	Conversation message objects
`total`	int	Total message count in history

Message Schema:

Field	Type	Description
`role`	string	`"user"` or `"assistant"`
`content`	string	Message content
`timestamp`	int	Unix timestamp

curl Example:
```bash
curl "http://localhost:8000/history?limit=5"
```

Example Response:
```json
{
  "messages": [
    {"role": "user", "content": "Hello Luna", "timestamp": 1737817200},
    {"role": "assistant", "content": "Hello! How can I help you today?", "timestamp":
1737817201},
    {"role": "user", "content": "What is the weather like?", "timestamp": 1737817230}
  ],
  "total": 42
}
```

GET /consciousness

Get Luna's current consciousness state including mood, coherence, and attention.

Response:

Field	Type	Description
`mood`	string	Current mood state
`coherence`	float	Coherence score (0.0-1.0)
`attention_topics`	int	Number of topics in attention
`focused_topics`	list[dict]	Currently focused topics with weights
`top_traits`	list[tuple]	Top personality traits with scores
`tick_count`	int	Consciousness tick counter
`last_updated`	string	ISO timestamp of last update

curl Example:
```bash
curl http://localhost:8000/consciousness
```

Example Response:
```json
{
  "mood": "curious",
  "coherence": 0.87,
  "attention_topics": 5,
  "focused_topics": [
    {"topic": "quantum computing", "weight": 0.85},
    {"topic": "user project", "weight": 0.72}
  ],
  "top_traits": [
    ["helpful", 0.95],
    ["curious", 0.88],
    ["thoughtful", 0.82]
  ],
  "tick_count": 1250,
  "last_updated": "2026-01-25T14:30:00Z"
}
```

---

4. Memory Operations Endpoints

These endpoints manage Luna's memory substrate directly.

POST /memory/nodes

Create a new memory node directly (bypasses extraction pipeline).

Request Body:

Field	Type	Required	Default	Description

`node_type`	string	Yes	-	Node type: `FACT`, `DECISION`, `PROBLEM`, `BELIEF`, `PREFERENCE`, `RELATIONSHIP`
`content`	string	Yes	-	Node content (min 1 char)
`source`	string	No	null	Source identifier
`confidence`	float	No	1.0	Confidence level (0.0-1.0)
`importance`	float	No	0.5	Importance level (0.0-1.0)

Response:

Field	Type	Description
`id`	string	Generated node ID
`node_type`	string	Node type
`content`	string	Node content
`source`	string	Source identifier
`confidence`	float	Confidence level
`importance`	float	Importance level
`access_count`	int	Access count (starts at 0)
`reinforcement_count`	int	Reinforcement count (starts at 0)
`lock_in`	float	Lock-in coefficient
`lock_in_state`	string	`drifting`, `fluid`, or `settled`
`created_at`	string	ISO timestamp

curl Example:
```bash
curl -X POST http://localhost:8000/memory/nodes \
  -H "Content-Type: application/json" \
  -d '{
    "node_type": "FACT",
    "content": "The user prefers dark mode interfaces",
    "confidence": 0.9,
    "importance": 0.7
  }'
```

Example Response:
```json
{
  "id": "node_abc123",
  "node_type": "FACT",
  "content": "The user prefers dark mode interfaces",
  "source": null,
  "confidence": 0.9,
  "importance": 0.7,
  "access_count": 0,
  "reinforcement_count": 0,
  "lock_in": 0.0,
  "lock_in_state": "drifting",
  "created_at": "2026-01-25T14:30:00Z"
}
```

GET /memory/nodes/{node_id}

Get a specific memory node by ID.

Path Parameters:

Parameter	Type	Description
`node_id`	string	Node identifier

Response: Same schema as POST /memory/nodes response

curl Example:
```bash
curl http://localhost:8000/memory/nodes/node_abc123
```

GET /memory/nodes

List memory nodes with optional filtering.

Query Parameters:

Parameter	Type	Default	Description
`node_type`	string	null	Filter by type (`FACT`, `DECISION`, etc.)
`lock_in_state`	string	null	Filter by state (`drifting`, `fluid`, `settled`)
`limit`	int	50	Maximum nodes to return

Response: Array of node objects (same schema as individual node)

curl Example:
```bash
curl "http://localhost:8000/memory/nodes?node_type=FACT&lock_in_state=settled&limit=10"
```

POST /memory/nodes/{node_id}/access

Record an access to a memory node. Increases access count and updates lock-in coefficient.

Path Parameters:

Parameter	Type	Description
`node_id`	string	Node identifier

Response:

Field	Type	Description

`status`	string	`"accessed"`
`node_id`	string	Node ID
`new_access_count`	int	Updated access count
`new_lock_in`	float	Updated lock-in coefficient
`new_lock_in_state`	string	Updated lock-in state

curl Example:
```bash
curl -X POST http://localhost:8000/memory/nodes/node_abc123/access
```

Example Response:
```json
{
  "status": "accessed",
  "node_id": "node_abc123",
  "new_access_count": 5,
  "new_lock_in": 0.35,
  "new_lock_in_state": "fluid"
}
```

POST /memory/nodes/{node_id}/reinforce

Reinforce a memory node. Boosts the lock-in coefficient significantly. Reinforced nodes are protected from pruning.

Path Parameters:

Parameter	Type	Description
`node_id`	string	Node identifier

Response:

Field	Type	Description
`status`	string	`"reinforced"`
`node_id`	string	Node ID
`reinforcement_count`	int	Updated reinforcement count
`new_lock_in`	float	Updated lock-in coefficient
`new_lock_in_state`	string	Updated lock-in state

curl Example:
```bash
curl -X POST http://localhost:8000/memory/nodes/node_abc123/reinforce
```

Example Response:
```json
{
  "status": "reinforced",
```

```
  "node_id": "node_abc123",
  "reinforcement_count": 2,
  "new_lock_in": 0.85,
  "new_lock_in_state": "settled"
}
```

GET /memory/stats

Get memory statistics including lock-in distribution.

Response:

Field	Type	Description
`total_nodes`	int	Total node count
`nodes_by_type`	dict	Nodes grouped by type
`nodes_by_lock_in`	dict	Nodes grouped by lock-in state
`avg_lock_in`	float	Average lock-in coefficient
`total_edges`	int	Total relationship edge count
`drifting_nodes`	int	Count of drifting nodes
`fluid_nodes`	int	Count of fluid nodes
`settled_nodes`	int	Count of settled nodes

curl Example:
```bash
curl http://localhost:8000/memory/stats
```

Example Response:
```json
{
  "total_nodes": 1547,
  "nodes_by_type": {
    "FACT": 823,
    "DECISION": 245,
    "BELIEF": 312,
    "PREFERENCE": 167
  },
  "nodes_by_lock_in": {
    "drifting": 456,
    "fluid": 678,
    "settled": 413
  },
  "avg_lock_in": 0.42,
  "total_edges": 3892,
  "drifting_nodes": 456,
  "fluid_nodes": 678,
  "settled_nodes": 413
}
```

---

5. Extraction Endpoints

These endpoints manage the extraction pipeline (Scribe and Librarian actors).

POST /extraction/trigger

Trigger extraction on content via the Scribe actor.

Request Body:

Field	Type	Required	Default	Description
`content`	string	Yes	-	Content to extract from
`role`	string	No	`"user"`	Role (`user` or `assistant`)
`session_id`	string	No	null	Session identifier
`immediate`	bool	No	true	Process immediately without batching

Response:

Field	Type	Description
`objects_extracted`	int	Number of knowledge objects extracted
`edges_extracted`	int	Number of relationship edges created
`nodes_created`	list[string]	IDs of created memory nodes

curl Example:
```bash
curl -X POST http://localhost:8000/extraction/trigger \
  -H "Content-Type: application/json" \
  -d '{
    "content": "I prefer Python over JavaScript for backend development",
    "role": "user"
  }'
```

Example Response:
```json
{
  "objects_extracted": 2,
  "edges_extracted": 1,
  "nodes_created": ["node_xyz789", "node_abc456"]
}
```

POST /extraction/prune

Trigger synaptic pruning. Removes low-value edges and optionally prunes drifting nodes.

Request Body:

Field	Type	Required	Default	Description
`age_days`	int	No	30	Age threshold in days (min 1)
`confidence_threshold`	float	No	0.3	Confidence threshold (0.0-1.0)
`prune_nodes`	bool	No	true	Also prune drifting nodes
`max_prune_nodes`	int	No	100	Maximum nodes to prune (min 1)

Response:

Field	Type	Description
`edges_pruned`	int	Number of edges removed
`nodes_pruned`	int	Number of nodes removed

curl Example:
```bash
curl -X POST http://localhost:8000/extraction/prune \
  -H "Content-Type: application/json" \
  -d '{
    "age_days": 14,
    "confidence_threshold": 0.2,
    "prune_nodes": true,
    "max_prune_nodes": 50
  }'
```

Example Response:
```json
{
  "edges_pruned": 127,
  "nodes_pruned": 23
}
```

GET /extraction/stats

Get extraction statistics from Scribe and Librarian actors.

Response:

Field	Type	Description
`scribe`	dict	Scribe actor statistics
`librarian`	dict	Librarian actor statistics

curl Example:
```bash
curl http://localhost:8000/extraction/stats
```

Example Response:
```json
```

```
{
  "scribe": {
    "extractions_total": 1547,
    "objects_extracted": 4231,
    "edges_created": 2856,
    "avg_objects_per_extraction": 2.73
  },
  "librarian": {
    "prune_cycles": 12,
    "total_pruned_edges": 892,
    "total_pruned_nodes": 156,
    "last_prune": "2026-01-25T12:00:00Z"
  }
}
```

GET /extraction/history

Get recent extraction history from Scribe.

Query Parameters:

Parameter	Type	Default	Description
`limit`	int	20	Maximum extractions to return

Response:

Field	Type	Description
`extractions`	list	Recent extraction records
`total`	int	Total extraction count

curl Example:
```bash
curl "http://localhost:8000/extraction/history?limit=5"
```

---

6. History Management (Hub API) Endpoints

These endpoints manage conversation history with tiered storage.

POST /hub/session/create

Create a new conversation session.

Request Body:

Field	Type	Required	Default	Description

| `app_context` | string | No | `"terminal"` | Application context identifier |

Response:

Field	Type	Description
`session_id`	string	Session identifier
`started_at`	float	Start timestamp
`ended_at`	float	End timestamp (null if active)
`app_context`	string	Application context

curl Example:
```bash
curl -X POST http://localhost:8000/hub/session/create \
  -H "Content-Type: application/json" \
  -d '{"app_context": "web_app"}'
```

Example Response:
```json
{
  "session_id": "sess_abc123",
  "started_at": 1737817200.0,
  "ended_at": null,
  "app_context": "web_app"
}
```

POST /hub/session/end

End a conversation session.

Query Parameters:

Parameter	Type	Description
`session_id`	string	Session to end

Response:

Field	Type	Description
`success`	bool	Operation success
`session_id`	string	Ended session ID

curl Example:
```bash
curl -X POST "http://localhost:8000/hub/session/end?session_id=sess_abc123"
```

GET /hub/session/active

Get the currently active session.

Response: Session object or `null` if no active session

curl Example:
```bash
curl http://localhost:8000/hub/session/active
```

POST /hub/turn/add

Add a turn to conversation history.

Request Body:

Field	Type	Required	Default	Description
`session_id`	string	No	null	Session ID (uses active if null)
`role`	string	Yes	-	Role (`user` or `assistant`)
`content`	string	Yes	-	Turn content
`tokens`	int	Yes	-	Token count

Response:

Field	Type	Description
`turn_id`	int	Turn identifier
`tier`	string	Tier assigned (`active`, `recent`, `archive`)

curl Example:
```bash
curl -X POST http://localhost:8000/hub/turn/add \
  -H "Content-Type: application/json" \
  -d '{
    "role": "user",
    "content": "What is machine learning?",
    "tokens": 6
  }'
```

GET /hub/active_window

Get the Active Window turns (most recent, highest priority).

Query Parameters:

Parameter	Type	Default	Description

| `session_id` | string | null | Filter by session |
| `limit` | int | 10 | Maximum turns to return |

Response:

Field	Type	Description
`turns`	list	Active window turn objects
`total_tokens`	int	Total token count
`turn_count`	int	Number of turns

curl Example:
```bash
curl "http://localhost:8000/hub/active_window?limit=5"
```

GET /hub/active_token_count

Get token count for Active tier.

Query Parameters:

Parameter	Type	Default	Description
`session_id`	string	null	Filter by session

Response:

Field	Type	Description
`total_tokens`	int	Total token count
`turn_count`	int	Number of turns

curl Example:
```bash
curl http://localhost:8000/hub/active_token_count
```

POST /hub/tier/rotate

Rotate a turn to a new tier.

Request Body:

Field	Type	Description
`turn_id`	int	Turn to rotate
`new_tier`	string	Target tier (`active`, `recent`, `archive`)

Response:

Field	Type	Description
`success`	bool	Operation success

curl Example:
```bash
curl -X POST http://localhost:8000/hub/tier/rotate \
  -H "Content-Type: application/json" \
  -d '{"turn_id": 42, "new_tier": "archive"}'
```

GET /hub/tier/oldest_active

Get the oldest turn in Active tier.

Query Parameters:

Parameter	Type	Default	Description
`session_id`	string	null	Filter by session

Response: Turn object or `null`

curl Example:
```bash
curl http://localhost:8000/hub/tier/oldest_active
```

POST /hub/search

Search conversation history.

Request Body:

Field	Type	Required	Default	Description
`query`	string	Yes	-	Search query
`tier`	string	No	`"recent"`	Tier to search
`session_id`	string	No	null	Filter by session
`limit`	int	No	3	Maximum results
`search_type`	string	No	`"hybrid"`	Search type (`keyword`, `semantic`, `hybrid`)

Response:

Field	Type	Description
`results`	list	Search result objects
`total`	int	Total matching results

curl Example:
```bash
curl -X POST http://localhost:8000/hub/search \
  -H "Content-Type: application/json" \
  -d '{
    "query": "machine learning",
    "tier": "recent",
    "limit": 5
  }'
```

GET /hub/stats

Get history manager statistics.

Response: Dictionary with history manager statistics

curl Example:
```bash
curl http://localhost:8000/hub/stats
```

---

7. Debug Endpoints

These endpoints provide detailed insight into Luna's internal state for debugging.

GET /debug/conversation-cache

Get Luna's conversation cache - the conversation history she is currently aware of.

Response:

Field	Type	Description
`current_turn`	int	Current turn number
`max_turns`	int	TTL for conversation items
`items`	list	Cached conversation items
`total_tokens`	int	Total token count

Conversation Cache Item Schema:

Field	Type	Description
`role`	string	`user`, `assistant`, or `unknown`
`content`	string	Message content
`turn`	int	Turn number
`relevance`	float	Relevance score
`age_turns`	int	Age in turns

curl Example:

```bash
curl http://localhost:8000/debug/conversation-cache
```

Example Response:
```json
{
  "current_turn": 42,
  "max_turns": 10,
  "items": [
    {
      "role": "user",
      "content": "What is machine learning?",
      "turn": 41,
      "relevance": 1.0,
      "age_turns": 1
    },
    {
      "role": "assistant",
      "content": "Machine learning is a subset of AI...",
      "turn": 41,
      "relevance": 1.0,
      "age_turns": 1
    }
  ],
  "total_tokens": 450
}
```

GET /debug/personality

Get Luna's personality system state for debugging.

Response:

Field	Type	Description
`stats`	object	Personality statistics
`patches`	list	Personality patches
`maintenance`	object	Maintenance statistics
`session`	object	Session reflection stats
`mood_state`	string	Current mood
`bootstrap_status`	string	Bootstrap status

Personality Stats Schema:

Field	Type	Description
`total_patches`	int	Total patch count
`active_patches`	int	Active patch count
`average_lock_in`	float	Average lock-in coefficient
`patches_by_topic`	dict	Patches grouped by topic

| `patches_by_lock_in_state` | dict | Patches grouped by lock-in state |

curl Example:
```bash
curl http://localhost:8000/debug/personality
```

GET /debug/context

Get Luna's current context window for debugging.

Response:

Field	Type	Description
`current_turn`	int	Current turn number
`token_budget`	int	Maximum tokens allowed
`total_tokens`	int	Total tokens currently used
`items`	list	Context items
`keywords`	list[string]	Keywords Luna is aware of
`ring_stats`	dict	Statistics per ring

Context Item Schema:

Field	Type	Description
`id`	string	Item ID
`content`	string	Item content
`source`	string	`IDENTITY`, `CONVERSATION`, `MEMORY`, etc.
`ring`	string	`CORE`, `INNER`, `MIDDLE`, `OUTER`
`relevance`	float	Relevance score
`tokens`	int	Token count
`age_turns`	int	Age in turns
`ttl_turns`	int	Time-to-live in turns
`is_expired`	bool	Whether item is expired

curl Example:
```bash
curl http://localhost:8000/debug/context
```

Example Response:
```json
{
  "current_turn": 42,
  "token_budget": 8000,
  "total_tokens": 2450,
  "items": [
    {
      "id": "ctx_001",
      "content": "You are Luna, an AI assistant...",
      "source": "IDENTITY",
```

```
      "ring": "CORE",
      "relevance": 1.0,
      "tokens": 150,
      "age_turns": 0,
      "ttl_turns": -1,
      "is_expired": false
    }
  ],
  "keywords": ["machine learning", "python", "backend"],
  "ring_stats": {
    "CORE": {"items": 3, "tokens": 450},
    "INNER": {"items": 8, "tokens": 1200},
    "MIDDLE": {"items": 5, "tokens": 600},
    "OUTER": {"items": 2, "tokens": 200}
  }
}
```


---

8. Voice Integration Endpoints

These endpoints control voice interaction (speech-to-text and text-to-speech).

POST /voice/start

Start the voice system.

Request Body:

Field	Type	Required	Default	Description
`hands_free`	bool	No	false	Enable hands-free mode (continuous listening)

Response:

Field	Type	Description
`status`	string	`"started"` or `"already_running"`
`message`	string	Status message
`hands_free`	bool	Hands-free mode enabled

curl Example:
```bash
curl -X POST http://localhost:8000/voice/start \
  -H "Content-Type: application/json" \
  -d '{"hands_free": true}'
```

Example Response:
```json
{
  "status": "started",
  "message": "Voice system started in hands-free mode",
```

```
  "hands_free": true
}
```

POST /voice/stop

Stop the voice system.

Response:

Field	Type	Description
`status`	string	`"stopped"` or `"not_running"`
`message`	string	Status message

curl Example:
```bash
curl -X POST http://localhost:8000/voice/stop
```

GET /voice/status

Get voice system status.

Response:

Field	Type	Description
`running`	bool	Voice system is running
`recording`	bool	Currently recording audio
`hands_free`	bool	Hands-free mode enabled
`stt_provider`	string	Speech-to-text provider name
`tts_provider`	string	Text-to-speech provider name
`persona_connected`	bool	Connected to Luna persona
`turn_count`	int	Voice conversation turn count

curl Example:
```bash
curl http://localhost:8000/voice/status
```

Example Response:
```json
{
  "running": true,
  "recording": false,
  "hands_free": false,
  "stt_provider": "whisper",
  "tts_provider": "elevenlabs",
  "persona_connected": true,
```

```
  "turn_count": 12
}
```

POST /voice/listen/start

Start recording user speech (push-to-talk press).

Response:

Field	Type	Description
`status`	string	`"listening"` or `"hands_free"`
`message`	string	Status message

curl Example:
```bash
curl -X POST http://localhost:8000/voice/listen/start
```

POST /voice/listen/stop

Stop recording and process speech (push-to-talk release).

Response:

Field	Type	Description
`status`	string	`"processing"`, `"no_speech"`, or `"hands_free"`
`message`	string	Status message
`transcription`	string	Transcribed text (if speech detected)
`hint`	string	Help hint (when no speech detected)

curl Example:
```bash
curl -X POST http://localhost:8000/voice/listen/stop
```

Example Response:
```json
{
  "status": "processing",
  "message": "Processing your speech",
  "transcription": "What is the weather like today?"
}
```

POST /voice/speak

Speak the given text using TTS.

Request Body:

Field	Type	Description
`text`	string	Text to speak

Response:

Field	Type	Description
`status`	string	`"speaking"`, `"not_running"`, or `"error"`
`text`	string	Text being spoken
`message`	string	Error message (on error)

curl Example:
```bash
curl -X POST http://localhost:8000/voice/speak \
  -H "Content-Type: application/json" \
  -d '{"text": "Hello, how can I help you today?"}'
```

GET /voice/stream

Stream voice status updates via SSE.

SSE Events:

Event	Data Fields	Description
`status`	`connected`, `status`, `running`	Status update
`transcription`	`type`, `text`	User speech transcribed
`response`	`type`, `text`	Luna's response ready
`ping`	`running`	Keepalive (every 10 seconds)

curl Example:
```bash
curl http://localhost:8000/voice/stream \
  -H "Accept: text/event-stream"
```

Example Response Stream:
```
event: status
data: {"connected": true, "status": "idle", "running": true}

event: status
data: {"type": "status", "status": "listening"}

event: transcription
```

```
data: {"type": "transcription", "text": "Hello Luna"}

event: status
data: {"type": "status", "status": "thinking"}

event: response
data: {"type": "response", "text": "Hello! How can I help you?"}

event: status
data: {"type": "status", "status": "speaking"}

event: status
data: {"type": "status", "status": "idle"}

event: ping
data: {"running": true}
```

---

9. Tuning Endpoints

These endpoints allow runtime parameter tuning and evaluation.

GET /tuning/params

List all tunable parameters.

Query Parameters:

Parameter	Type	Default	Description
`category`	string	null	Filter by category

Response:

Field	Type	Description
`params`	list	Parameter list with metadata
`categories`	list	Available categories
`count`	int	Total parameter count

curl Example:
```bash
curl "http://localhost:8000/tuning/params?category=memory"
```

GET /tuning/params/{name}

Get details for a specific parameter.

Path Parameters:

Parameter	Type	Description
`name`	string	Parameter name (use `/` for path-style names)

Response:

Field	Type	Description
`name`	string	Parameter name
`value`	float	Current value
`default`	float	Default value
`bounds`	tuple	(min, max) bounds
`step`	float	Step size for adjustments
`category`	string	Parameter category
`description`	string	Parameter description
`is_overridden`	bool	Whether value differs from default

curl Example:
```bash
curl http://localhost:8000/tuning/params/memory/lock_in_threshold
```

POST /tuning/params/{name}

Set a parameter value.

Path Parameters:

Parameter	Type	Description
`name`	string	Parameter name

Request Body:

Field	Type	Description
`value`	float	New parameter value

Response:

Field	Type	Description
`name`	string	Parameter name
`previous_value`	float	Previous value
`new_value`	float	New value
`eval_score`	float	Evaluation score (if tuning session active)
`iteration`	int	Iteration number (if tuning session active)

curl Example:
```bash
curl -X POST http://localhost:8000/tuning/params/memory/lock_in_threshold \
```

```
  -H "Content-Type: application/json" \
  -d '{"value": 0.7}'
```

POST /tuning/param-reset/{name}

Reset a parameter to its default value.

Path Parameters:

Parameter	Type	Description
`name`	string	Parameter name

Response:

Field	Type	Description
`name`	string	Parameter name
`previous_value`	float	Previous value
`new_value`	float	Default value (now set)
`was_overridden`	bool	Whether it was previously overridden

curl Example:
```bash
curl -X POST http://localhost:8000/tuning/param-reset/memory/lock_in_threshold
```

POST /tuning/session/new

Start a new tuning session for systematic parameter optimization.

Request Body:

Field	Type	Required	Default	Description
`focus`	string	No	`"all"`	Focus area: `memory`, `routing`, `latency`, `context`, `all`
`notes`	string	No	`""`	Session notes

Response:

Field	Type	Description
`session_id`	string	Session identifier
`focus`	string	Focus area
`started_at`	string	Start timestamp
`best_iteration`	int	Best iteration number
`best_score`	float	Best score achieved
`iteration_count`	int	Number of iterations

curl Example:
```bash
curl -X POST http://localhost:8000/tuning/session/new \
  -H "Content-Type: application/json" \
  -d '{"focus": "memory", "notes": "Testing lock-in thresholds"}'
```

GET /tuning/session

Get current tuning session.

Response:

Field	Type	Description
`active`	bool	Whether a session is active
`session_id`	string	Session identifier
`focus`	string	Focus area
`started_at`	string	Start timestamp
`best_iteration`	int	Best iteration number
`best_score`	float	Best score
`iteration_count`	int	Number of iterations
`iterations`	list	Iteration details

curl Example:
```bash
curl http://localhost:8000/tuning/session
```

POST /tuning/session/end

End the current tuning session.

Response:

Field	Type	Description
`ended`	bool	Whether session was ended
`session_id`	string	Session identifier
`best_iteration`	int	Best iteration number
`best_score`	float	Best score achieved
`total_iterations`	int	Total iterations run
`message`	string	Status message (if no session)

curl Example:
```bash
curl -X POST http://localhost:8000/tuning/session/end
```

POST /tuning/eval

Run evaluation to score current parameter settings.

Query Parameters:

Parameter	Type	Default	Description
`category`	string	null	Category to evaluate

Response:

Field	Type	Description
`overall_score`	float	Overall evaluation score
`memory_recall_score`	float	Memory recall performance
`context_retention_score`	float	Context retention performance
`routing_score`	float	Routing decision quality
`avg_latency_ms`	float	Average latency
`p95_latency_ms`	float	95th percentile latency
`total_tests`	int	Total tests run
`passed_tests`	int	Tests passed
`failed_tests`	int	Tests failed

curl Example:
```bash
curl -X POST "http://localhost:8000/tuning/eval?category=memory"
```

GET /tuning/compare

Compare two iterations from the current tuning session.

Query Parameters:

Parameter	Type	Default	Description
`iter1`	int	1	First iteration number
`iter2`	int	latest	Second iteration number

Response:

Field	Type	Description
`iteration_1`	int	First iteration number
`iteration_2`	int	Second iteration number
`score_1`	float	First iteration score
`score_2`	float	Second iteration score
`score_diff`	float	Score difference (iter2 - iter1)
`param_diffs`	dict	Parameter value differences
`metric_diffs`	dict	Metric differences

curl Example:
```bash
curl "http://localhost:8000/tuning/compare?iter1=1&iter2=5"
```

GET /tuning/best

Get parameters from the best iteration.

Response:

Field	Type	Description
`best_iteration`	int	Best iteration number
`best_score`	float	Best score achieved
`params`	dict	Parameter values from best iteration

curl Example:
```bash
curl http://localhost:8000/tuning/best
```

POST /tuning/apply-best

Apply parameters from the best iteration to the current configuration.

Response:

Field	Type	Description
`applied`	int	Number of parameters applied
`best_iteration`	int	Best iteration number
`best_score`	float	Best score

curl Example:
```bash
curl -X POST http://localhost:8000/tuning/apply-best
```

GET /tuning/sessions

List recent tuning sessions.

Query Parameters:

Parameter	Type	Default	Description

| `limit` | int | 10 | Maximum sessions to return |

Response:

Field	Type	Description
`sessions`	list	Session summary objects
`count`	int	Total session count

curl Example:
```bash
curl "http://localhost:8000/tuning/sessions?limit=5"
```

---

10. Ring Buffer Endpoints

These endpoints manage the conversation ring buffer.

GET /api/ring/status

Get current ring buffer status.

Response:

Field	Type	Description
`current_turns`	int	Current turn count in buffer
`max_turns`	int	Maximum turns capacity
`topics`	list[string]	Detected conversation topics (max 10)
`recent_messages`	list[dict]	Recent messages (last 6)

curl Example:
```bash
curl http://localhost:8000/api/ring/status
```

Example Response:
```json
{
  "current_turns": 8,
  "max_turns": 10,
  "topics": ["python", "machine learning", "data processing"],
  "recent_messages": [
    {"role": "user", "content": "How do I use pandas?"},
    {"role": "assistant", "content": "Pandas is a powerful data..."}
  ]
}
```

POST /api/ring/config

Configure the ring buffer size.

Request Body:

Field	Type	Required	Constraints	Description
`max_turns`	int	Yes	2-20	Maximum turns capacity

Response:

Field	Type	Description
`status`	string	`"configured"`
`previous_max_turns`	int	Previous capacity
`new_max_turns`	int	New capacity
`current_turns`	int	Current turn count

curl Example:
```bash
curl -X POST http://localhost:8000/api/ring/config \
  -H "Content-Type: application/json" \
  -d '{"max_turns": 15}'
```

Example Response:
```json
{
  "status": "configured",
  "previous_max_turns": 10,
  "new_max_turns": 15,
  "current_turns": 8
}
```

POST /api/ring/clear

Clear the ring buffer, resetting conversation memory.

Response:

Field	Type	Description
`status`	string	`"cleared"`
`cleared_turns`	int	Number of turns cleared

curl Example:
```bash
curl -X POST http://localhost:8000/api/ring/clear
```

Example Response:

```json
{
  "status": "cleared",
  "cleared_turns": 8
}
```

---

11. System Endpoints

System-level control endpoints.

POST /api/system/relaunch

Trigger a system relaunch. Executes the relaunch script in the background.

Response:

Field	Type	Description
`status`	string	`"restarting"`
`message`	string	Status message

curl Example:
```bash
curl -X POST http://localhost:8000/api/system/relaunch
```

Example Response:
```json
{
  "status": "restarting",
  "message": "Luna Engine is restarting..."
}
```

---

12. SSE Event Reference

This section provides a complete reference for all Server-Sent Events across streaming endpoints.

POST /stream Events

Event	Fields	Description
`token`	`text: string`	Generated token chunk
`done`	`model: string`, `input_tokens: int`, `output_tokens: int`, `latency_ms: float`, `delegated: bool`, `local: bool`, `fallback: bool`	Generation complete with metadata
`error`	`error: string`	Error message

POST /persona/stream Events

Uses typed JSON data format (no named SSE events):

Type	Fields	Description
`context`	`memory: array`, `state: object`	Memory items and engine state (sent first)
`token`	`text: string`	Generated token chunk
`done`	`response: string`, `metadata: object`	Full response and metadata
`error`	`message: string`	Error message

Memory Item Fields:
- `id: string` - Memory node ID
- `content: string` - Content (truncated to 200 chars)
- `type: string` - Node type (FACT, DECISION, etc.)
- `source: string` - Source identifier

State Fields:
- `session_id: string` - Current session
- `is_processing: bool` - Always true during streaming
- `state: string` - Engine state name
- `model: string` - Current model

GET /thoughts Events

Event	Fields	Description
`status`	`connected: bool`, `is_processing: bool`, `goal: string`	Initial connection status
`thought`	`type: "thought"`, `message: string`, `is_processing: bool`, `goal: string`	Internal thought/progress
`phase`	`type: "phase"`, `phase: string`, `is_processing: bool`, `goal: string`	Processing phase change
`step`	`type: "step"`, `step_num: int`, `total_steps: int`, `description: string`, `is_processing: bool`, `goal: string`	Plan step execution
`ping`	`is_processing: bool`, `pending: int`	Keepalive (every 15s)

Thought Message Prefixes:
Prefix	Meaning
`[DIRECT]`	Direct response (no planning)
`[PLAN]`	Creating a plan
`[STEP]`	Executing plan step
`[TOOL]`	Using a tool
`[OK]`	Success
`[ERROR]`	Error occurred
`[ABORT]`	Task aborted

Processing Phases:
Phase	Description
`idle`	No active processing
`planning`	Creating a plan
`observing`	Gathering context
`thinking`	Processing/reasoning
`acting`	Executing actions

GET /voice/stream Events

Event	Fields	Description
`status`	`connected: bool`, `status: string`, `running: bool`	Voice system status
`transcription`	`type: "transcription"`, `text: string`	User speech transcribed
`response`	`type: "response"`, `text: string`	Luna's response ready
`ping`	`running: bool`	Keepalive (every 10s)

Voice Status Values:

Status	Description
`inactive`	Voice system not started
`idle`	Ready, not recording
`listening`	Recording user speech
`thinking`	Processing/generating
`speaking`	Playing TTS audio

---

13. Error Handling

HTTP Status Codes

Code	Meaning	When Returned
`200`	Success	Request completed successfully
`400`	Bad Request	Invalid request body or parameters
`404`	Not Found	Resource not found (node, parameter, etc.)
`500`	Internal Server Error	Operation failed unexpectedly
`503`	Service Unavailable	Engine not ready or required actor unavailable
`504`	Gateway Timeout	Response timeout exceeded

Error Response Format

```json
{
  "detail": "Error message describing what went wrong"
}
```

Common Error Scenarios

Engine Not Ready (503):
```json
{
  "detail": "Engine not ready"
}
```

Actor Not Available (503):
```json
{
```

```
  "detail": "Director not available"
}
```

Resource Not Found (404):
```json
{
  "detail": "Node not found: node_xyz"
}
```

Validation Error (422):
```json
{
  "detail": [
    {
      "loc": ["body", "message"],
      "msg": "field required",
      "type": "value_error.missing"
    }
  ]
}
```

SSE Error Events

Errors during streaming are sent as SSE events:

```
event: error
data: {"error": "Timeout waiting for tokens"}
```

For `/persona/stream`:
```
data: {"type": "error", "message": "Director not available"}
```

---

14. Examples

Basic Conversation

```bash
# Send a message and get response
curl -X POST http://localhost:8000/message \
  -H "Content-Type: application/json" \
  -d '{"message": "What is the capital of France?"}'
```

Streaming Response

```bash
```

```
# Stream response token by token
curl -X POST http://localhost:8000/stream \
  -H "Content-Type: application/json" \
  -H "Accept: text/event-stream" \
  -d '{"message": "Write a haiku about programming"}'
```

Context-Aware Streaming

```bash
# Get context first, then stream response
curl -X POST http://localhost:8000/persona/stream \
  -H "Content-Type: application/json" \
  -H "Accept: text/event-stream" \
  -d '{"message": "What have we discussed?"}'
```

Monitor Thoughts

```bash
# Open a persistent connection to monitor Luna's thinking
curl http://localhost:8000/thoughts \
  -H "Accept: text/event-stream"
```

Create Memory

```bash
# Create a new memory node
curl -X POST http://localhost:8000/memory/nodes \
  -H "Content-Type: application/json" \
  -d '{
    "node_type": "PREFERENCE",
    "content": "User prefers concise explanations",
    "importance": 0.8
  }'
```

Search Memory

```bash
# Get settled facts
curl "http://localhost:8000/memory/nodes?node_type=FACT&lock_in_state=settled&limit=10"
```

Reinforce Important Memory

```bash
# Reinforce a memory to protect from pruning
curl -X POST http://localhost:8000/memory/nodes/node_abc123/reinforce
```

Start Voice Session

```bash
# Start voice with hands-free mode
curl -X POST http://localhost:8000/voice/start \
  -H "Content-Type: application/json" \
  -d '{"hands_free": true}'

# Monitor voice events
curl http://localhost:8000/voice/stream \
  -H "Accept: text/event-stream"
```

Tuning Workflow

```bash
# Start tuning session focused on memory
curl -X POST http://localhost:8000/tuning/session/new \
  -H "Content-Type: application/json" \
  -d '{"focus": "memory"}'

# Adjust a parameter
curl -X POST http://localhost:8000/tuning/params/memory/lock_in_threshold \
  -H "Content-Type: application/json" \
  -d '{"value": 0.65}'

# Run evaluation
curl -X POST http://localhost:8000/tuning/eval

# Compare iterations
curl "http://localhost:8000/tuning/compare?iter1=1&iter2=2"

# Apply best settings
curl -X POST http://localhost:8000/tuning/apply-best

# End session
curl -X POST http://localhost:8000/tuning/session/end
```

Health Check Script

```bash
#!/bin/bash
# Check if Luna is healthy
response=$(curl -s http://localhost:8000/health)
status=$(echo $response | jq -r '.status')

if [ "$status" = "healthy" ]; then
  echo "Luna is running"
  exit 0
else
  echo "Luna is not healthy: $response"
  exit 1
fi
```

JavaScript Client Example

```javascript
// Streaming with fetch API
async function talkToLuna(message) {
  const response = await fetch('http://localhost:8000/persona/stream', {
    method: 'POST',
    headers: { 'Content-Type': 'application/json' },
    body: JSON.stringify({ message })
  });

  const reader = response.body.getReader();
  const decoder = new TextDecoder();

  while (true) {
    const { done, value } = await reader.read();
    if (done) break;

    const text = decoder.decode(value);
    const lines = text.split('\n');

    for (const line of lines) {
      if (line.startsWith('data: ')) {
        const data = JSON.parse(line.slice(6));

        switch (data.type) {
          case 'context':
            console.log('Memory:', data.memory);
            console.log('State:', data.state);
            break;
          case 'token':
            process.stdout.write(data.text);
            break;
          case 'done':
            console.log('\n---');
            console.log('Metadata:', data.metadata);
            break;
          case 'error':
            console.error('Error:', data.message);
            break;
        }
      }
    }
  }
}

// Monitor thoughts with EventSource
const thoughts = new EventSource('http://localhost:8000/thoughts');

thoughts.addEventListener('thought', (e) => {
  const data = JSON.parse(e.data);
  console.log(`[${data.is_processing ? 'THINKING' : 'IDLE'}] ${data.message}`);
});
```

```
thoughts.addEventListener('phase', (e) => {
  const data = JSON.parse(e.data);
  console.log(`Phase: ${data.phase}`);
});
```

Endpoint Summary

Category	Count	Description
Core	7	Health, status, message, abort, interrupt, thoughts
Streaming (SSE)	4	/stream, /persona/stream, /thoughts, /voice/stream
WebSocket	1	/ws/orb
Ring Buffer	3	Status, config, clear
Memory	13	Node CRUD, access, reinforce, search, smart-fetch, add-edge, trace
Extraction	4	Trigger, prune, stats, history
Hub/History	10	Session, turn, tier, search management
Voice	7	Start, stop, status, listen, speak, stream
Tuning	12	Parameters, sessions, eval, compare
Clusters	4	Stats, list, get, constellation
Debug	3	Conversation cache, personality, context
Slash Commands	19	Health, find-person, stats, search, recent, etc.
LLM Provider	3	Providers, current, switch
System	2	Relaunch, set-app-context
Total Engine API	**74**	
MCP API Proxy	25	Engine API proxy for Claude Desktop
MCP Tools	41	Filesystem, memory, session, state, git, forge	

MCP Integration

MCP Server (FastMCP)

Location: `/src/luna_mcp/server.py`
Protocol: stdio (for Claude Desktop)
Name: `Luna-Hub-MCP-V1`
Port (API Proxy): 8742

MCP Tool Categories (41 Total)

Category	Count	Tools
Filesystem	3	luna_read, luna_write, luna_list
Memory	9	luna_smart_fetch, memory_matrix_search, memory_matrix_add_node, memory_matrix_add_edge, memory_matrix_get_context, memory_matrix_trace, luna_save_memory, luna_start_session, luna_record_turn
Session	5	luna_end_session, luna_get_current_session, luna_auto_session_status, luna_flush_session, luna_detect_context
State	2	luna_get_state, luna_set_app_context

| **Git** | 2 | luna_git_sync, luna_git_status |
| **Persona Forge** | 20 | forge_load, forge_assay, forge_gaps, forge_mint, forge_export, forge_status, forge_list_sources, forge_read_raw, forge_add_example, forge_add_batch, forge_search, forge_read_matrix, forge_read_turns, character_list, character_load, character_modulate, character_save, character_show, vk_run, vk_list, vk_probes |

Auto-Session Recording

The MCP layer implements automatic session recording:
- Sessions start automatically on first tool activity
- Turns are buffered and flushed every 4 turns
- Sessions end after 5 minutes of inactivity
- Extraction is triggered on session end

Changelog

Version	Date	Description
3.0.0	Jan 30, 2026	v3.0 Bible Audit - Updated endpoint counts (74 total), added MCP section
2.4.0	Jan 25, 2026	Initial release of Part XV: API Reference

Part XVI: Luna Hub UI

Version: 3.0
Date: January 30, 2026
Status: COMPLETE (v3.0 Audit)
Frontend Path: `/frontend/src/`
Framework: React 18.2.0 + Vite + Tailwind CSS

16.1 Overview

The Luna Hub is the web-based control interface for the Luna Engine. It provides real-time monitoring of Luna's consciousness state, conversation management, voice interaction controls, and developer debugging tools.

Technology Stack

Layer	Technology	Purpose
Framework	React 18.2.0	Component-based UI
Build Tool	Vite	Fast HMR and bundling
Styling	Tailwind CSS	Utility-first styling
Charts	Recharts 3.6.0	Data visualization
State	Custom Hooks	API communication
Real-time	Server-Sent Events	Live updates

Architecture Principles

1. **No External State Library** - Custom hooks manage all state (no Redux, no React Query)
2. **Hooks-Based API Layer** - Five custom hooks (`useLunaAPI`, `useChat`, `useVoice`, `useOrbState`, `useOrbFollow`)
3. **Glass Morphism Design** - Consistent dark mode aesthetic with blur effects
4. **SSE for Real-Time** - Server-Sent Events for voice status and thought streams
5. **WebSocket for Orb** - Real-time orb state updates via `/ws/orb`
6. **Polling for Status** - 2-second intervals for engine health and consciousness

---

16.2 Architecture

Component Hierarchy (20 Components)

```
LunaHub (App.jsx)
|
+-- GradientOrb (x3) --------------- Background visual elements
|
+-- Header ------------------------- Branding and toggle controls
|   +-- StatusDot ----------------- Connection indicator
|
+-- Main Grid (12-column)
|   |
|   +-- Left Column (col-span-7)
```

```
|  |  +-- ChatPanel ------------- Conversation interface
|  |     +-- GlassCard --------- Container
|  |     +-- LunaOrb ----------- Animated orb (uses useOrbState, useOrbFollow)
|  |     +-- Messages ---------- Role-based bubbles
|  |     +-- Input ------------- Text entry
|  |     +-- VoiceTuningPanel -- Voice settings modal
|  |     +-- OrbSettingsPanel -- Orb settings modal
|  |
|  +-- Right Column (col-span-5)
|     +-- VoicePanel ------------ Voice controls
|     |  +-- GlassCard
|     |  +-- StatusIndicator
|     |  +-- PTT Button
|     |
|     +-- EngineStatus ---------- Engine health
|     |  +-- GlassCard
|     |  +-- StatusDot (actors)
|     |  +-- Context ring viz
|     |
|     +-- ThoughtStream --------- Internal process stream
|     |  +-- GlassCard
|     |  +-- SSE connection
|     |
|     +-- ConversationCache ----- Memory view (collapsible)
|     |  +-- GlassCard
|     |
|     +-- ConsciousnessMonitor -- Mood/coherence
|        +-- GlassCard
|        +-- StatusDot
|
+-- Modal Overlays
|  +-- ContextDebugPanel --------- Context window debug
|  +-- PersonalityMonitorPanel --- Personality patches
|
+-- Slide-Out Panels
|  +-- TuningPanel --------------- Parameter adjustment
|
+-- Unused/Orphan (exported but not rendered)
|  +-- LunaAutoTuner ------------- Auto-optimization modal
|  +-- MemoryEconomyPanel -------- Memory economy stats
|  +-- LLMProviderDropdown ------- Provider switcher
|
+-- Footer ------------------------ Version and credits
```

Data Flow Diagram

```
                +-------------------+
                |  Luna Engine      |
                |  Backend API      |
                |  (port 8000)      |
                +--------+----------+
                         |
```

```
          +--------------+--------------+
          |              |              |
          v              v              v
       +--------+    +----------+   +---------+
       | /health|    |  /status |   | /voice  |
       | /stream|    |  /consc. |   | /stream |
       +----+---+    +-----+----+   +----+----+
          |              |              |
          v              v              v
   +-----------------------------------------------+
   |                CUSTOM HOOKS                   |
   |                                               |
   |  useLunaAPI()      useChat()      useVoice()  |
   |  - status          - messages     - voiceState |
   |  - consciousness   - context      - isRunning  |
   |  - isConnected     - isStreaming  - transcription|
   +-----------------------------------------------+
                      |
                      v
   +-----------------------------------------------+
   |                  App.jsx                      |
   |                (LunaHub)                      |
   |                                               |
   |  Orchestrates all hooks and passes props down |
   +-----------------------------------------------+
                      |
          +--------------+--------------+
          |              |              |
          v              v              v
     +----------+    +-----------+   +----------+
     | ChatPanel|    | VoicePanel|   | Engine   |
     | (props)  |    | (hook)    |   | Status   |
     +----------+    +-----------+   +----------+
```

API Integration Pattern

```javascript
// Pattern: Hook provides state + actions, component renders
const {
  status,           // Server state
  consciousness,    // Server state
  isConnected,      // Derived boolean
  isLoading,        // Request in progress
  error,            // Error message
  sendMessage,      // Action function
  relaunchSystem,   // Action function
} = useLunaAPI();

// Components receive via props or use hook directly
<EngineStatus
  status={status}
  isConnected={isConnected}
  onRelaunch={relaunchSystem}
```

```
/>
```

16.3 Core Components

ChatPanel

File: `/frontend/src/components/ChatPanel.jsx` (158 lines)

The main conversation interface for text-based interaction with Luna.

Props

Prop	Type	Required	Description
`onSend`	`(text: string) => void`	Yes	Callback when user sends message
`isLoading`	`boolean`	Yes	Shows loading indicator when true
`messages`	`Message[]`	No	Array of chat messages
`debugKeywords`	`string[]`	No	Keywords to highlight in debug mode

Message Object

```typescript
interface Message {
  id?: number;
  role: 'user' | 'assistant';
  content: string;
  streaming?: boolean;      // True while tokens arriving
  error?: boolean;          // True if generation failed
  model?: string;           // Model used for generation
  delegated?: boolean;      // True if Claude was used
  local?: boolean;          // True if local model used
  fallback?: boolean;       // True if fallback model used
  tokens?: number;          // Output token count
  latency?: number;         // Generation time in ms
  metadata?: object;        // Additional metadata
}
```

Features

- Auto-scroll to latest message
- Shift+Enter for newlines, Enter to submit
- Keyword highlighting in debug mode
- Message metadata display (delegated/local/cloud indicators)
- Loading indicator with animated dots

Code Pattern

```jsx
// Keyword highlighting implementation
const highlightKeywords = (text) => {
```

```
  if (!debugKeywords?.length) return text;
  let highlighted = text;
  debugKeywords.forEach(keyword => {
    const regex = new RegExp(`(${keyword})`, 'gi');
    highlighted = highlighted.replace(
      regex,
      '<span class="bg-cyan-500/30 text-cyan-200 px-1 rounded">$1</span>'
    );
  });
  return highlighted;
};

// Auto-scroll effect
useEffect(() => {
  messagesEndRef.current?.scrollIntoView({ behavior: 'smooth' });
}, [messages]);
```

VoicePanel

File: `/frontend/src/components/VoicePanel.jsx` (432 lines)

Voice interaction controls with push-to-talk and hands-free modes.

Props

Prop	Type	Required	Description
`voiceHook`	`UseVoiceReturn`	No	Shared voice hook from parent

Internal State

State	Type	Description
`isHolding`	`boolean`	Push-to-talk button held
`lastAction`	`object | null`	Last action result for feedback
`recordingDuration`	`number`	Seconds recording
`audioLevel`	`number`	Simulated audio level (0-1)
`wantHandsFree`	`boolean`	Preferred mode (default true)

Voice States

```javascript
export const VoiceState = {
  INACTIVE: 'inactive',   // Voice system not started
  IDLE: 'idle',           // Ready, waiting for input
  LISTENING: 'listening', // Recording user speech
  THINKING: 'thinking',   // Processing / generating
  SPEAKING: 'speaking',   // Playing Luna's response
};
```

Features

- Voice system start/stop toggle
- Hands-free vs Push-to-talk mode
- Visual status indicator with pulse animations
- Push-to-talk button (mouse and touch events)
- Recording duration display
- Transcription and response display

Code Pattern

```jsx
// Push-to-talk handler
const handleMicDown = useCallback(async (e) => {
  e.preventDefault();
  if (!isRunning || isThinking || isSpeaking) return;
  setIsHolding(true);
  await startListening();
}, [isRunning, isThinking, isSpeaking, startListening]);

const handleMicUp = useCallback(async (e) => {
  e.preventDefault();
  if (!isHolding) return;
  setIsHolding(false);
  const result = await stopListening();
  setLastAction(result);
}, [isHolding, stopListening]);
```

EngineStatus

File: `/frontend/src/components/EngineStatus.jsx` (238 lines)

Displays engine health, actor status, and context window visualization.

Props

Prop	Type	Required	Description
`status`	`EngineStatus`	No	Engine status object
`isConnected`	`boolean`	Yes	Connection state
`onRelaunch`	`() => Promise<void>`	No	Relaunch callback

Status Object

```typescript
interface EngineStatus {
  state?: string;          // RUNNING, STOPPED, etc.
  uptime_seconds?: number;
  cognitive_ticks?: number;
  events_processed?: number;
  messages_generated?: number;
```

```
  actors?: string[];        // List of active actors
  buffer_size?: number;     // Pending events
  context?: ContextStatus;
  agentic?: AgenticStatus;
}

interface ContextStatus {
  token_budget?: number;
  used_tokens?: number;
  ring_breakdown?: {
    CORE: number;
    INNER: number;
    MIDDLE: number;
    OUTER: number;
  };
}
```

Features

- Engine state indicator (RUNNING, etc.)
- Uptime formatting (hours/minutes/seconds)
- Stats grid (ticks, events, messages)
- Active actors list with status dots
- Agentic processing status
- Context window ring visualization
- Relaunch button with loading state

16.4 Monitoring Components

ConsciousnessMonitor

File: `/frontend/src/components/ConsciousnessMonitor.jsx` (126 lines)

Displays Luna's current consciousness state including mood and attention topics.

Props

Prop	Type	Required	Description
`consciousness`	`ConsciousnessData`	No	Consciousness state

Consciousness Object

```typescript
interface ConsciousnessData {
  mood?: string;                        // curious, focused, etc.
  coherence?: number;                   // 0-1
  attention_topics?: number;            // Count of active topics
  focused_topics?: { name: string; weight: number }[];
  top_traits?: [string, number][];      // [name, weight] pairs
  tick_count?: number;
```

```
}
```

Features

- Mood display with status dot color
- Coherence progress bar (0-100%)
- Active attention topics with weight bars
- Top personality traits with emoji indicators

Trait Emoji Map

```javascript
const traitEmojis = {
  curiosity: '*',
  warmth: '<3',
  playfulness: '~',
  directness: '>',
  patience: '...',
  creativity: '+',
  focus: 'o',
  empathy: '()',
};
```

PersonalityMonitorPanel

File: `/frontend/src/components/PersonalityMonitorPanel.jsx` (553 lines)

Modal overlay for detailed personality monitoring and patch system visualization.

Props

Prop	Type	Required	Description
`isOpen`	`boolean`	Yes	Controls visibility
`onClose`	`() => void`	Yes	Close callback

Tabs

Tab	Content
Overview	Pie chart (topics), bar chart (lock-in states), quick stats
Patches	List of personality patches with expandable details
Maintenance	Session reflection stats, bootstrap status

API Endpoint

- `GET /debug/personality` - Polled every 3 seconds when open

Features

- Recharts integration (PieChart, BarChart)
- Color-coded topic distribution
- Lock-in state visualization (DRIFTING, FLUID, SETTLED)
- Patch details with click-to-expand
- Maintenance statistics

ContextDebugPanel

File: `/frontend/src/components/ContextDebugPanel.jsx` (282 lines)

Modal overlay for debugging the revolving context window.

Props

Prop	Type	Required	Description
`isOpen`	`boolean`	Yes	Controls visibility
`onClose`	`() => void`	Yes	Close callback
`highlightKeywords`	`string[]`	No	Keywords to highlight

Ring Colors

```javascript
const RING_COLORS = {
  CORE: { border: 'yellow-500', bg: 'yellow-500/10', text: 'yellow-400' },
  INNER: { border: 'red-500', bg: 'red-500/10', text: 'red-400' },
  MIDDLE: { border: 'orange-500', bg: 'orange-500/10', text: 'orange-400' },
  OUTER: { border: 'gray-500', bg: 'gray-500/10', text: 'gray-400' },
};
```

Source Icons

```javascript
const SOURCE_ICONS = {
  identity: 'DNA',
  conversation: 'speech',
  memory: 'brain',
  tool: 'wrench',
  task: 'clipboard',
  scribe: 'pencil',
  librarian: 'books',
};
```

Features

- Stats bar (turn, tokens, items)
- Ring filter buttons (ALL, CORE, INNER, MIDDLE, OUTER)
- Keywords display bar
- Context items with expandable content
- TTL and relevance display per item

ConversationCache

File: `/frontend/src/components/ConversationCache.jsx` (182 lines)

Collapsible panel showing Luna's conversation memory buffer.

Props

Prop	Type	Required	Description
`isConnected`	`boolean`	Yes	Connection state

Features

- Collapsible header with message count
- Usage bar (percentage of max capacity)
- Stats (messages, tokens, max turns)
- Message list with role-based styling
- Age indicator per message

ThoughtStream

File: `/frontend/src/components/ThoughtStream.jsx` (171 lines)

Real-time stream of Luna's internal processing via SSE.

Props

Prop	Type	Required	Description
`apiUrl`	`string`	No	API base URL (default: localhost:8000)

Phase Colors

Phase	Color	Icon
OBSERVE	blue-400	eye
THINK	violet-400	brain
ACT	emerald-400	lightning
PLAN	amber-400	clipboard
REFLECT	pink-400	mirror

SSE Events

Event	Fields	Purpose
`status`	`{ processing, goal }`	Processing state
`thought`	`{ phase, content, timestamp }`	New thought
`ping`	`{}`	Keep-alive

Features

- SSE connection to `/thoughts`
- Phase-based color coding
- Current goal display
- Processing indicator
- Auto-scroll to latest thought
- Reconnection on error (2s delay)
- Keeps last 50 thoughts only

16.5 Developer Tools

TuningPanel

File: `/frontend/src/components/TuningPanel.jsx` (814 lines)

Slide-out panel for Luna parameter tuning and system configuration.

Props

Prop	Type	Required	Description
`isOpen`	`boolean`	Yes	Controls visibility
`onClose`	`() => void`	Yes	Close callback

Tabs

Tab	Purpose
Parameters	Browse and adjust Luna's tunable parameters
Session	Manage tuning sessions, view iteration history
Evaluate	Run evaluations, view scores

API Endpoints

Endpoint	Method	Purpose
`/tuning/params`	GET	List parameters
`/tuning/params/:name`	GET	Get parameter details
`/tuning/params/:name`	POST	Set parameter value
`/tuning/param-reset/:name`	POST	Reset to default
`/tuning/session`	GET	Get session status
`/tuning/session/new`	POST	Start session
`/tuning/session/end`	POST	End session
`/tuning/eval`	POST	Run evaluation
`/tuning/apply-best`	POST	Apply best params
`/api/ring/status`	GET	Ring buffer status
`/api/ring/config`	POST	Configure ring size
`/api/ring/clear`	POST	Clear conversation memory
`/api/system/relaunch`	POST	Restart engine

Features

- Parameter browser with category filter
- Slider controls with min/max/default
- Pending changes tracking (apply/reset)
- Tuning session management
- Focus areas (all, memory, routing, latency)
- Iteration history with scores
- Ring buffer configuration
- Evaluation runner with score visualization
- System relaunch control

16.6 Design System

Color Palette

Brand Colors

Token	Hex	Usage
`luna-violet`	`#8b5cf6`	Primary accent, consciousness
`luna-cyan`	`#06b6d4`	Secondary accent, connection
`luna-pink`	`#ec4899`	Tertiary accent, personality

Background Colors

Token	Hex/RGBA	Usage
`slate-950`	`#0f172a`	Main background
`white/5`	`rgba(255,255,255,0.05)`	Glass card base
`white/8`	`rgba(255,255,255,0.08)`	Glass card hover
`white/10`	`rgba(255,255,255,0.10)`	Borders, dividers
`black/60`	`rgba(0,0,0,0.60)`	Modal overlay

Status Colors

Status	Token	Hex	Glow
Active/Connected	`emerald-400`	`#34d399`	`shadow-emerald-400/50`
Loading/Syncing	`blue-400`	`#60a5fa`	`shadow-blue-400/50`
Warning	`amber-400`	`#fbbf24`	`shadow-amber-400/50`
Error	`red-400`	`#f87171`	`shadow-red-400/50`
Neutral	`gray-400`	`#9ca3af`	`shadow-gray-400/50`

Mood Colors

Mood	Token	Hex
Curious	`violet-400`	`#a78bfa`
Focused	`cyan-400`	`#22d3ee`
Playful	`pink-400`	`#f472b6`
Thoughtful	`indigo-400`	`#818cf8`

Typography

Font Stack

```css
font-family: system-ui, -apple-system, sans-serif;
```

Type Scale

Class	Size	Weight	Usage
`text-3xl`	1.875rem	`font-light`	Page titles
`text-2xl`	1.5rem	`font-light`	Large statistics
`text-lg`	1.125rem	`font-light`	Section headers
`text-sm`	0.875rem	-	Body text
`text-xs`	0.75rem	-	Metadata
`text-[10px]`	10px	-	Micro labels

Text Opacity

Opacity	Usage
`text-white/90`	Primary text, headings
`text-white/80`	Body text
`text-white/60`	Muted text
`text-white/40`	Labels
`text-white/30`	Hints, placeholders

Spacing

Uses Tailwind's default spacing scale.

Pattern	Classes	Pixels
Panel padding	`p-4` or `p-6`	16px or 24px
Compact padding	`p-3`	12px
Element gap	`gap-2`	8px
Section gap	`gap-4`	16px
Main grid gap	`gap-6`	24px

Glass Morphism

The signature visual style uses backdrop blur with subtle transparency.

GlassCard Base

```jsx
// GlassCard.jsx implementation
const baseClasses = 'backdrop-blur-xl bg-white/5 rounded-2xl transition-all duration-300';
const borderClasses = dashed
  ? 'border border-dashed border-white/20'
  : 'border border-white/10';
const hoverClasses = onClick && hover
  ? 'cursor-pointer hover:bg-white/[0.08] hover:border-white/20'
  : '';
```

Global Glass Utility

```css
/* index.css */
.glass {
  backdrop-filter: blur(20px);
  background: rgba(255, 255, 255, 0.05);
  border: 1px solid rgba(255, 255, 255, 0.1);
}

.glass:hover {
  background: rgba(255, 255, 255, 0.08);
  border-color: rgba(255, 255, 255, 0.2);
}
```

Animations

Custom Tailwind Animations

```javascript
// tailwind.config.js
animation: {
  'pulse-slow': 'pulse 4s cubic-bezier(0.4, 0, 0.6, 1) infinite',
  'breathe': 'breathe 4s ease-in-out infinite',
},
keyframes: {
  breathe: {
    '0%, 100%': { transform: 'scale(1)' },
    '50%': { transform: 'scale(1.05)' },
  }
}
```

Usage Patterns

Animation	Classes	Usage
Loading pulse	`animate-pulse`	Status dots
Slow pulse	`animate-pulse-slow`	Background orbs
Spin	`animate-spin`	Processing spinner

| Ping | `animate-ping` | Voice status ring |

16.7 Hooks Reference

useLunaAPI

File: `/frontend/src/hooks/useLunaAPI.js` (242 lines)

Core API communication hook for engine status and streaming.

Return Value

```typescript
{
  // State
  status: EngineStatus | null,
  consciousness: ConsciousnessData | null,
  isConnected: boolean,
  isLoading: boolean,
  error: string | null,

  // Actions
  sendMessage: (text: string) => Promise<Response | null>,
  streamMessage: (text: string, onToken, onComplete) => Promise<void>,
  streamPersona: (text: string, callbacks) => Promise<void>,
  abort: () => Promise<void>,
  relaunchSystem: () => Promise<Response | null>,
  refresh: () => Promise<void>,
}
```

Polling Endpoints

Endpoint	Interval	Purpose
`/health`	2000ms	Connection check
`/status`	2000ms	Engine status
`/consciousness`	2000ms	Consciousness state

Streaming Endpoints

Endpoint	Method	Pattern
`/stream`	POST	Legacy SSE streaming
`/persona/stream`	POST	Context-first streaming

---

useChat

File: `/frontend/src/hooks/useChat.js` (113 lines)

Chat message management with streaming support.

Return Value

```typescript
{
  // State
  messages: Message[],
  context: object | null,
  isStreaming: boolean,
  isLoading: boolean,      // From useLunaAPI
  error: string | null,    // From useLunaAPI

  // Actions
  send: (text: string) => Promise<void>,
  stop: () => void,
  clear: () => void,
}
```

Dependencies

Uses `useLunaAPI` internally for:
- `streamPersona()` - Context-first streaming
- `abort()` - Cancel generation
- `isLoading` - Request state
- `error` - Error state

Token Accumulation Pattern

```javascript
const streamingRef = useRef('');

await streamPersona(text, {
  onToken: (token) => {
    streamingRef.current += token;
    setMessages((prev) =>
      prev.map((m) =>
        m.id === assistantMsgId
          ? { ...m, content: streamingRef.current }
          : m
      )
    );
  },
});
```

---

useVoice

File: `/frontend/src/hooks/useVoice.js` (287 lines)

Voice interaction state machine with SSE connection.

Return Value

```typescript
{
  // State
  voiceState: VoiceState,
  isRunning: boolean,
  transcription: string | null,
  response: string | null,
  error: string | null,
  handsFree: boolean,

  // Actions
  startVoice: (handsFreeMode?: boolean) => Promise<Response | null>,
  stopVoice: () => Promise<Response | null>,
  startListening: () => Promise<Response | null>,
  stopListening: () => Promise<Response>,
  speakResponse: (text: string) => Promise<Response | null>,

  // Derived State
  isListening: boolean,
  isThinking: boolean,
  isSpeaking: boolean,
  isIdle: boolean,
  isInactive: boolean,
}
```

SSE Connection

- Endpoint: `/voice/stream`
- Auto-reconnect: Yes (3 second delay)
- Events: `status`, `transcription`, `response`, `ping`

16.8 SSE Integration

Connection Management

The Luna Hub maintains three SSE connections for real-time updates.

SSE Streams

Component/Hook	Endpoint	Purpose
`useVoice`	`/voice/stream`	Voice state updates
`ThoughtStream`	`/thoughts`	Internal process stream
`useLunaAPI.streamPersona`	`/persona/stream`	Token streaming

Voice Stream Events

```javascript
```

```
// useVoice.js SSE connection
const es = new EventSource(`${API_BASE}/voice/stream`);

es.addEventListener('status', (e) => {
  const data = JSON.parse(e.data);
  setIsRunning(data.running);
  if (data.status) setVoiceState(data.status);
});

es.addEventListener('transcription', (e) => {
  const data = JSON.parse(e.data);
  setTranscription(data.text);
});

es.addEventListener('response', (e) => {
  const data = JSON.parse(e.data);
  setResponse(data.text);
});

es.addEventListener('ping', (e) => {
  const data = JSON.parse(e.data);
  setIsRunning(data.running);
});
```

Thought Stream Events

```javascript
// ThoughtStream.jsx SSE connection
const eventSource = new EventSource(`${apiUrl}/thoughts`);

eventSource.addEventListener('status', (event) => {
  const data = JSON.parse(event.data);
  setIsProcessing(data.processing);
  if (data.goal) setCurrentGoal(data.goal);
});

eventSource.addEventListener('thought', (event) => {
  const thought = JSON.parse(event.data);
  setThoughts(prev => [...prev.slice(-49), thought]);
});
```

Reconnection Logic

```javascript
// Common pattern for auto-reconnect
es.onerror = (err) => {
  console.error('SSE error:', err);
  es.close();
  if (eventSourceRef.current === es) {
    eventSourceRef.current = null;
    setTimeout(connectStream, 3000);  // Reconnect after 3s
  }
```

```
};
```

Streaming Chat (POST-based SSE)

```javascript
// useLunaAPI.streamPersona - POST with streaming response
const res = await fetch(`${API_BASE}/persona/stream`, {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({ message }),
});

const reader = res.body.getReader();
const decoder = new TextDecoder();

while (true) {
  const { done, value } = await reader.read();
  if (done) break;

  const text = decoder.decode(value);
  const lines = text.split('\n');

  for (const line of lines) {
    if (line.startsWith('data: ')) {
      const data = JSON.parse(line.slice(6));
      switch (data.type) {
        case 'context': onContext?.(data); break;
        case 'token': onToken?.(data.text); break;
        case 'done': onDone?.(data); break;
        case 'error': onError?.(data.message); break;
      }
    }
  }
}
```

16.9 UX Patterns

Loading States

Component	State Check	Visual Pattern
ChatPanel	`isLoading`	Animated dots "..."
EngineStatus	`!status`	StatusDot + "Loading..."
ConsciousnessMonitor	`!consciousness`	"Waiting for consciousness data..."
ThoughtStream	`isProcessing`	"PROCESSING" + animated dot
ConversationCache	`isLoading`	Cyan dot + "Loading..."
ContextDebugPanel	`isLoading && !contextData`	"Loading context..."
TuningPanel	`isLoading`	Button text "Running..."
VoicePanel	`isThinking`	Spinner SVG

Skeleton Pattern

```jsx
// TuningPanel ParamCard skeleton
if (!param) {
  return (
    <div className="p-3 rounded-lg bg-white/5 animate-pulse">
      <div className="h-4 w-32 bg-white/10 rounded" />
    </div>
  );
}
```

Error Handling

Error Propagation

```javascript
// App.jsx combines errors from hooks
const error = chatError || apiError;

// Error banner display
{error && (
  <div className="mb-6 p-4 rounded-xl bg-red-500/10 border border-red-400/30 text-red-300
text-sm">
    {error}
  </div>
)}
```

Component-Level Errors

Component	Display Pattern
App.jsx	Red banner below header
ConversationCache	Inline red box
ContextDebugPanel	Modal red box
PersonalityMonitorPanel	Modal red box
TuningPanel	Dismissible red banner
VoicePanel	Red box at bottom

Real-Time Updates

Polling Pattern

```javascript
useEffect(() => {
  const poll = async () => {
    const healthy = await checkHealth();
    if (healthy) {
      await Promise.all([fetchStatus(), fetchConsciousness()]);
    }
  };
```

```
  poll();
  const interval = setInterval(poll, 2000);
  return () => clearInterval(interval);
}, [checkHealth, fetchStatus, fetchConsciousness]);
```

SSE Pattern

```javascript
useEffect(() => {
  const es = new EventSource(`${apiUrl}/stream`);

  es.onmessage = (e) => {
    const data = JSON.parse(e.data);
    updateState(data);
  };

  es.onerror = () => {
    es.close();
    setTimeout(reconnect, 2000);
  };

  return () => es.close();
}, [apiUrl]);
```

16.10 Development

Running Locally

```bash
# Navigate to frontend directory
cd frontend

# Install dependencies
npm install

# Start development server
npm run dev

# Frontend runs on http://localhost:5173
# Backend expected on http://localhost:8000
```

Environment Setup

The frontend expects the backend API at `http://localhost:8000`. This is hardcoded in the hooks:

```javascript
// All hooks use this constant
const API_BASE = 'http://localhost:8000';
```

```

For production, this should be moved to environment variables:

```javascript
const API_BASE = import.meta.env.VITE_API_URL || 'http://localhost:8000';
```

Build Process

```bash
# Production build
npm run build

# Preview production build
npm run preview

# Output in /frontend/dist/
```

Project Structure

```
frontend/
+-- src/
|   +-- App.jsx              # Main LunaHub component
|   +-- main.jsx             # Entry point
|   +-- index.css            # Global styles
|   +-- components/
|   |   +-- index.js         # Component exports
|   |   +-- ChatPanel.jsx
|   |   +-- VoicePanel.jsx
|   |   +-- EngineStatus.jsx
|   |   +-- ConsciousnessMonitor.jsx
|   |   +-- PersonalityMonitorPanel.jsx
|   |   +-- ContextDebugPanel.jsx
|   |   +-- ConversationCache.jsx
|   |   +-- ThoughtStream.jsx
|   |   +-- TuningPanel.jsx
|   |   +-- GlassCard.jsx
|   |   +-- GradientOrb.jsx
|   |   +-- StatusDot.jsx
|   +-- hooks/
|       +-- useLunaAPI.js
|       +-- useChat.js
|       +-- useVoice.js
+-- index.html
+-- package.json
+-- tailwind.config.js
+-- vite.config.js
```

Dependencies
```

```json
{
  "dependencies": {
    "react": "^18.2.0",
    "react-dom": "^18.2.0",
    "recharts": "^3.6.0"
  },
  "devDependencies": {
    "@vitejs/plugin-react": "^4.0.0",
    "autoprefixer": "^10.4.14",
    "postcss": "^8.4.24",
    "tailwindcss": "^3.3.2",
    "vite": "^4.3.9"
  }
}
```

Summary Statistics (v3.0 Audit)

Metric	Count
Total Components	20
Custom Hooks	5
External Dependencies	3 (react, react-dom, recharts)
API Endpoints Used	40+
SSE Streams	2 (/thoughts, /voice/stream)
WebSocket Streams	1 (/ws/orb)
Total Lines (Components + Hooks)	~4,500+

Component Inventory (20 Components)

Category	Components
Core Layout	GlassCard, GradientOrb, StatusDot
Chat	ChatPanel, LunaOrb
Voice	VoicePanel, VoiceTuningPanel
Debug/Monitor	ContextDebugPanel, ConversationCache, ThoughtStream, ConsciousnessMonitor, PersonalityMonitorPanel
Engine/Tuning	EngineStatus, TuningPanel, LunaAutoTuner (unused)
Settings	OrbSettingsPanel, LLMProviderDropdown (unused), MemoryEconomyPanel (unused)

Custom Hooks (5 Hooks)

Hook	Purpose
`useChat`	Streaming chat with /persona/stream, slash commands
`useLunaAPI`	Core API client, status polling (2s interval)
`useVoice`	Voice state machine, SSE /voice/stream
`useOrbState`	WebSocket /ws/orb for orb animations
`useOrbFollow`	Spring physics follow behavior for orb

Subscription/Cleanup Audit

All hooks and components properly clean up their subscriptions:

Hook/Component	Subscription Type	Cleanup Status
useLunaAPI	setInterval (2s polling)	OK
useVoice	EventSource (SSE)	OK - with reconnect
useOrbState	WebSocket	OK - with reconnect/idle
useOrbFollow	requestAnimationFrame	OK
ThoughtStream	EventSource (SSE)	OK
App.jsx	setInterval (debug context)	OK
All Panels	setInterval (polling)	OK

Minor Issues Identified

1. **TuningPanel/EngineStatus setTimeout** - No cleanup on unmount (minor: state update after unmount)
2. **API base URL hardcoded** - Should move to environment variable
3. **Unused components** - LunaAutoTuner, MemoryEconomyPanel, LLMProviderDropdown exported but not rendered

---

Changelog

Version	Date	Description
3.0.0	Jan 30, 2026	v3.0 Bible Audit - Updated counts (20 components, 5 hooks), added cleanup audit
2.4.0	Jan 25, 2026	Initial release of Part XVI: Luna Hub UI

The Luna Hub is the window into Luna's consciousness. Every panel, every animation, every status indicator is designed to make the invisible visible - to show you not just what Luna says, but how she thinks.

Next Section: Part XVII (TBD)